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# SHEAF HOMOLOGY

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**Bachelor thesis**

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## Abstract

We give an alternate construction of the abelian pro-group valued cosheaf homology defined by Prasalov in [25] as the natural homology theory corresponding to sheaf cohomology, suggesting it should rather be called *sheaf* homology. Using this perspective we apply known results about sheaf cohomology to show that shape, Čech and (co)sheaf pro-homology agree on paracompact Hausdorff spaces and that singular and (co)sheaf homology agree on spaces which are in addition locally contractible. In particular this proves a conjecture of Prasalov. Basic properties of sheaf cohomology (functoriality in the space, open and closed Mayer–Vietoris, homotopy invariance, etc.) all imply immediate analogues for (co)sheaf homology. Using these results we calculate examples which illustrate that, even for locally anomalous topological spaces, (co)sheaf homology can often be (fully faithfully) represented by topological groups which accurately reflect the geometry of the space.

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# 1 Introduction

The singular homology and cohomology groups are foundational concepts in algebraic topology. Singular homology gives particularly intuitive descriptions of the ‘holes’ in a space due to its covariance in the space, but the less intuitive singular cohomology has been found to generalise much better. One such generalisation is *sheaf cohomology*.

Sheaf cohomology associates to a topological space  $X$ , a sheaf of abelian groups  $\mathcal{F}$  on  $X$  and a  $p \in \mathbb{Z}$  an abelian group  $H^p(X, \mathcal{F})$ : the  $p$ -th *sheaf cohomology* of  $X$  with coefficients in  $\mathcal{F}$ . Sheaf cohomology is incredibly important in modern algebraic geometry. To apply this to algebraic topology we consider for an abelian group  $T$  the *constant sheaf*  $T_X$  whose sections on each open  $U \subseteq X$  are the locally constant functions  $U \rightarrow T$ . If  $X$  is a ‘nice space’ (e.g. a CW-complex) the sheaf cohomology of  $T_X$  gives exactly the singular cohomology of  $X$  with coefficients in  $T$ . However as soon as a space has complicated local structure singular and sheaf cohomology with constant coefficients may differ. For example the subspace  $X = \{1/n : n \in \mathbb{Z}_{>0}\} \cup \{0\}$  of  $\mathbb{R}$  has 0th singular cohomology  $\prod_{i=0}^{\infty} T$  and sheaf cohomology  $\bigoplus_{i=0}^{\infty} T$ .

However this raises the question: is there a corresponding sheaf *homology* theory? To answer this question recall that the construction of the singular homology and cohomology of a topological space starts by constructing a chain complex  $C_{\bullet}$  of abelian groups. The  $p$ -th singular homology group of  $X$  is then the  $p$ -th homology group of  $C_{\bullet}$ , denoted  $h_p(C_{\bullet})$ . The  $p$ -th singular cohomology group with coefficients in an abelian group  $T$  is then  $h^p(C_{\bullet}, T)$ ,  $p$ -th cohomology of  $C_{\bullet}$  in  $T$ , which is the  $p$ -th cohomology of the cochain complex  $\text{Hom}(C_{\bullet}, T)$ . The key to relating the homology and cohomology of  $C_{\bullet}$  is to think of the cohomology groups  $h^p(C_{\bullet}, T)$  as a functor in  $T$  from the category of abelian groups  $\mathbf{Ab}$  to itself. If  $I \in \mathbf{Ab}$  is *injective* then (by definition) the functor  $\text{Hom}(-, I) : \mathbf{Ab}^{\text{op}} \rightarrow \mathbf{Ab}$  is exact, so there is an isomorphism

$$h^p(C_{\bullet}, I) = h^p(\text{Hom}(C_{\bullet}, I)) \simeq \text{Hom}(h_p(C_{\bullet}), I).$$

Hence the functors  $h^p(C_{\bullet}, -)$  and  $\text{Hom}(h_p(C_{\bullet}), -)$  are isomorphic on injective objects. The key observation is that this completely determines  $\text{Hom}(h_p(C_{\bullet}), -)$  given  $h^p(C_{\bullet}, -)$ . This is because in general given an additive functor  $F : \mathbf{Ab} \rightarrow \mathbf{Ab}$  there exists a unique left exact functor  $\tilde{F} : \mathbf{Ab} \rightarrow \mathbf{Ab}$  isomorphic to  $F$  on injective objects, called its *left-exactification*. Since  $\text{Hom}(h_p(C_{\bullet}), -) : \mathbf{Ab} \rightarrow \mathbf{Ab}$  is left exact, it must be the left-exactification of  $h^p(C_{\bullet}, -)$ . Combined with the Yoneda Lemma, which implies that  $h_p(C_{\bullet})$  is completely determined by the functor  $\text{Hom}(h_p(C_{\bullet}), -)$ , we conclude that  $h_p(C_{\bullet})$  is completely determined by  $h^p(C_{\bullet}, -)$ . In Proposition 2.9 we will extend this argument to show that for any coefficient group  $G \in \mathbf{Ab}$  the hom-functor of  $h_p(C_{\bullet} \otimes G)$  is the left-exactification of  $T \mapsto h^p(C_{\bullet}, \text{Hom}(G, T))$ . Hence the homology of  $C_{\bullet}$  with coefficients in  $G$  can also be recovered from  $h^p(C_{\bullet}, -)$ .

The idea of sheaf homology is that we know from comparison theorems and common properties (Mayer–Vietoris, etc.) that  $H^p(X, -_X)$  behaves a lot like the singular cohomology functors of  $X$ , and hence like the  $p$ -th cohomology functor of a chain complex in  $\mathbf{Ab}$ . If we pretend that  $H^p(X, -_X)$  is indeed the  $p$ -th cohomology functor of a complex, then applying the process above should give the hom-functor of the corresponding homology groups. This motivates the following definition.

**Definition 1.1.** Let  $X$  be a topological space. Then for  $p \in \mathbb{Z}$  we define the  $p$ -th *sheaf homology* of  $X$ , denoted  $\tilde{H}_p(X)$ , to be the left-exactification of  $H^p(X, -_X) : \mathbf{Ab} \rightarrow \mathbf{Ab}$ . More generally if  $G \in \mathbf{Ab}$ , then we define the  $p$ -th *sheaf homology with coefficients in  $G$*  of  $X$ , denoted  $\tilde{H}_p(X; G)$ , as the left-exactification of  $H^p(X, \text{Hom}(G, -)_X)$ .

It follows immediately from the motivation of this construction that if  $X$  is nice enough for the functor  $H^p(X, -_X)$  to be isomorphic to the  $p$ -th singular cohomology functor, then  $\tilde{H}_p(X; G)$  is represented by the  $p$ -th singular homology group with coefficients in  $G$ . In section 5.4 we will take a closer look

under what conditions on  $X$  this holds. However even if  $\vec{H}_p(X; G)$  is not representable, it is always a left exact functor  $\mathbf{Ab} \rightarrow \mathbf{Ab}$ . The opposite category  $\vec{\mathbf{Ab}}$  of left exact functors  $\mathbf{Ab} \rightarrow \mathbf{Ab}$  turns out to be quite well behaved: it is abelian and through the abelian Yoneda embedding it contains  $\mathbf{Ab}$  as an exact full subcategory. In chapters 3 and 4 we develop some theory on the following full subcategories

$$\mathbf{Ab} \hookrightarrow \{\text{topologically representable functors}\} \subseteq \{\text{strict abelian pro-groups}\} \subseteq \mathbf{pro}\text{-}\mathbf{Ab} \subseteq \vec{\mathbf{Ab}}.$$

Here  $\mathbf{pro}\text{-}\mathbf{Ab}$  is the category of abelian pro-groups: functors which can be expressed as a (small) cofiltered limit of representable functors. Abelian pro-groups can be defined differently, but viewing them as functors will be most convenient for our purposes. It turns out that to show that a functor is a pro-group it suffices to check a simpler condition called *boundedness*. In section 5.2 we use this to prove that  $\vec{H}_p(X; G)$  is bounded (and hence a pro-group) for all  $X$ ,  $p$  and  $G$ .

The *strict* abelian pro-groups are those which can be expressed as the limit of a cofiltered diagram consisting of abelian groups and surjective maps. As shown in [22], these are exactly the functors which can be represented by localic groups (a generalisation of topological groups) and hence strictness is an important necessary condition for being representable by a topological group. In Theorem 5.29 we will see that  $\vec{H}_1(X)$  is strict for all locally connected spaces  $X$ .

The topologically representable functors are those which can be expressed as the hom-functor of a topological abelian group to discrete groups. In chapter 4 we will see that the so called *pro-discrete* topological abelian groups embed fully faithfully into  $\vec{\mathbf{Ab}}$  as their hom-functor to discrete groups. In section 6.2 we show that  $\vec{H}_0(X)$  is topologically representable for *all* spaces  $X$ . In Theorem 5.40 we will show that for a compact metric space  $\vec{H}_p(X; \mathbb{Q})$  is always topologically representable. If  $X$  is in addition locally connected then  $\vec{H}_1(X)$  is topologically representable.

A key observation is that the way we derived sheaf homology from sheaf cohomology (precomposing with an additive functor  $\mathbf{Ab} \rightarrow \mathbf{Ab}$  and taking left-exactification) gives an *exact* (contravariant) functor from the category  $[\mathbf{Ab}, \mathbf{Ab}]_{\text{add}}$  of additive functors  $\mathbf{Ab} \rightarrow \mathbf{Ab}$  to  $\vec{\mathbf{Ab}}$ . In chapter 6 we use this to obtain functoriality in the space, the Vietoris-Begle mapping theorem, open and closed Mayer-Vietoris, the continuity theorem and relative variants for sheaf homology from known properties of sheaf cohomology. Throughout this chapter we will use these tools to calculate the sheaf homology of various spaces that give examples and non-examples of the above results.

This approach is useful in singular algebraic topology as well: Take your favourite long exact sequence (or even spectral sequence) using singular cohomology, interpret it as a long exact sequence in  $[\mathbf{Ab}, \mathbf{Ab}]_{\text{add}}$ , precompose with  $\text{Hom}(G, -)$  and take left-exactification to obtain an exact sequence in  $\vec{\mathbf{Ab}}$ . Since the embedding of  $\mathbf{Ab}$  in  $\vec{\mathbf{Ab}}$  is exact, this gives a long exact sequence for singular homology with coefficients in  $G$ .

In chapter 5 we compare sheaf homology with existing abelian pro-group valued homology theories. We begin chapter 5 by showing that sheaf homology is the same as the cosheaf homology defined by A. V. Prasolov in [25]. More precisely we show that the category of cosheaves on a topological space  $X$  of pro-objects in  $\mathbf{Ab}$  can be viewed as a full subcategory of the opposite category of left exact functors from  $\mathbf{Ab}$  to the category of sheaves of abelian groups on  $X$ . This allows us to derive many of the properties of the category of cosheaves from basic properties of left exact functor categories and of sheaves of abelian groups on  $X$ . The homology of a cosheaf is then the same as the sheaf homology with coefficients in the corresponding functor (see Definition 5.1).

In section 5.3 we give comparison morphisms between sheaf homology, cosheaf Čech pro-homology, nerve based Čech pro-homology and shape pro-homology. Most importantly all of these morphisms are isomorphisms if  $X$  is Hausdorff paracompact. In particular the equivalence of (co)sheaf homology and shape homology on paracompact Hausdorff spaces proves Conjecture 1.0.3.1 of [25]. We do this by showing that all of these pro-homology theories are obtained in the same way from their corresponding cohomology theories. We then apply known results to compare these cohomology theories. In 5.4 we

give examples of non-Hausdorff finite spaces where the comparison morphisms are *not* isomorphisms.

The results above suggest that, at least on a theoretical level, all the above mentioned (pro-)homology theories are simply derivatives of their corresponding cohomology theories and therefore contain no extra information. In fact in most of the calculations in chapter 6 we work with sheaf cohomology most of the time and only move to homology at the end. At the same time a universal coefficient theorem (Theorem 2.17) allows us to (to a large extent) recover sheaf cohomology from sheaf homology with integer coefficients. In Example 6.59 we give an example of a compact Hausdorff space with sheaf homology represented by  $\mathbb{Z}, \mathbb{Z}_2, 0, \dots$  where  $\mathbb{Z}_2$  is the pro-finite group of 2-adic integers. Using the universal coefficient theorem one can recover the sheaf cohomology of this space which is considerably more complicated. In Example 6.37 we see that the generators of  $\prod_{i \geq 0}^{\text{Top}} \mathbb{Z}$ , the topological group representing the first sheaf homology of the Hawaiian earring, correspond to each loop in the earring and that their convergence behaviour in this topological group reflects the fact that the loops in the Hawaiian earring shrink to a point. We hope that these examples illustrate to the reader that, while sheaf homology contains no more information than sheaf cohomology, it can be used to summarise the sheaf cohomology functor of a space with concrete mathematical objects (topological groups and inverse systems) that accurately reflect the geometry of the space.

**Reading guide.** We will assume throughout that the reader is familiar with the basic language of category theory (categories, functors, natural transformations, limits and colimits), see for example [18]. A review of the necessary homological algebra is given in section 2.1. Even though the word sheaf is in the title of this thesis, pretty much no sheaf theory is necessary for chapters 2-4. In chapter 6 some experience with sheaves is necessary to understand the proofs, but the examples can be understood without this. Finally chapter 4 requires only basic category theory and topology.

Chapters 2-4 serve mainly to develop the theory necessary in chapters 5 and 6 with the only notable results being the two universal coefficient theorems (Theorem 2.17 and 3.48). From chapter 2 we recommend all readers to read 2.2 and 2.4 since these tools are used throughout the thesis. From chapter 3 most essential are 3.1 and 3.2, although the  $*$ -notation introduced in 3.1 is only used in chapters 3 and 4. If one wishes to jump straight to chapter 6 for the concrete examples and sheaf theory, then the rest of chapter 3 and the details of chapter 4 can be skipped (although it may be a good idea to take a look at some of the examples in these chapters). If the reader wishes to dig into the technical details of sections 5.1 and 5.2 a good understanding of section 3.3 is necessary. This is not necessary for the results in section 5.3, but to understand the corollaries that give sufficient conditions for the topological representability of sheaf homology one does need to read 3.4, 3.5 and 4.3 carefully and understand Proposition 3.23 in the context of pro-finitely generated functors.

**Category theoretic conventions.** In this thesis we will often have to work with ‘large’ categories (e.g. the category of *all* sets) and categories of functors between them. Unfortunately such objects cannot be described in Zermelo–Fraenkel set theory. One way of solving this is to assume the existence of a *Grothendieck universe*: roughly speaking a set whose elements form a model of ZFC. Then one can restrict to considering only the category of groups, rings, etc. whose underlying set lies in this universe. We will adopt the intentionally ambiguous terminology of working with ‘collections’ of mathematical objects and calling collections which fit in set theory *sets* or *small*. We leave it to the reader to make these notions precise in their preferred foundations. Using Grothendieck universes a ‘set’ is a set in the universe and a ‘collection’ is a set.

We allow categories to have large collections of objects and of morphisms between two objects. A category  $\mathcal{C}$  is *locally small* if the collection of morphisms between two objects is small. A category or collection of objects in a category  $\mathcal{A}$  is *essentially small* if there is some small subcollection  $\mathcal{A}'$  so that each object of  $\mathcal{A}$  is isomorphic to one in  $\mathcal{A}'$ . A *diagram* is a functor with an essentially small category as domain.

## 2 Left-exactification

In this chapter we define the most important concept of this thesis: *left-exactification*. This is a natural way of turning an additive functor  $F : \mathcal{C} \rightarrow \mathcal{D}$  between abelian categories into a left-exact functor  $\tilde{F} : \mathcal{C} \rightarrow \mathcal{D}$ . This construction exists in large generality (see [10, II.2]), but in the case that  $\mathcal{C}$  has enough injectives it can be characterised as the unique left-exact functor isomorphic to  $F$  on injective objects.

In 2.1 we briefly review the basic definitions surrounding abelian categories. In section 2.2 we define left-exactification and elaborate on the motivation of sheaf homology in the introduction. In 2.3 we show how in some cases half-exact functors can be recovered from their left-exactification and the left-exactification of half-exact functors connected to them. This allows us to prove a universal coefficient theorem relating sheaf cohomology and sheaf homology. In section 2.4 we show that left-exactification is the reflector of the category of additive functors  $\mathcal{C} \rightarrow \mathcal{D}$  into the subcategory of left exact ones. We then use this to conclude that the category of left exact functors is an abelian category with a very concrete description of exact sequences.

### 2.1 Abelian categories

We begin by reviewing the basic definitions surrounding abelian categories and additive functors following [18, VIII] and [13, IV.1]. This is by no means a thorough introduction and mainly serves to fix our terminology. For a more complete introduction to the subject see for example [30].

An *additive* category is a category  $\mathcal{C}$  in which each  $\text{Hom}_{\mathcal{C}}(A, B)$  is given the structure of a (not necessarily small) abelian group so that (1) composition is bilinear and (2) all *finite* products (or equivalently all finite coproducts) exist. It follows that  $\mathcal{C}$  has a *zero object* (an object which is both final and initial) and that for  $A, B \in \mathcal{C}$  there exists an object  $A \oplus B$  together with morphisms

$$A \xrightleftharpoons[\pi_1]{\iota_1} A \oplus B \xleftarrow[\pi_2]{\iota_2} B$$

so that  $\iota_1 \circ \pi_1 + \iota_2 \circ \pi_2 = \text{id}_{A \oplus B}$ . Here  $A \oplus B$  is called the *biproduct* of  $A$  and  $B$  and  $(A \oplus B, \pi_1, \pi_2)$  and  $(A \oplus B, \iota_1, \iota_2)$  form the product and coproduct respectively of  $A$  and  $B$ . Between any two objects  $A, B \in \mathcal{C}$  there is a distinguished morphism  $0 : A \rightarrow B$  which is the identity for addition. Using this we define for  $f : A \rightarrow B$  the kernel (resp. cokernel) of  $f$  as the equaliser (resp. coequaliser) of  $0$  and  $f$  (if it exists). A functor  $F : \mathcal{C} \rightarrow \mathcal{D}$  between additive categories is called *additive* if it induces a group homomorphism between each hom-set.

An additive category  $\mathcal{C}$  is *abelian* if (3) it has all finite limits and colimits and (4) for any morphism  $f : A \rightarrow B$  the natural map

$$\text{coker}(\ker(f) \rightarrow A) \rightarrow \ker(B \rightarrow \text{coker } f)$$

(which can be constructed using the universal properties) is an isomorphism. (3) is equivalent to requiring that all morphisms of  $\mathcal{C}$  have a kernel and a cokernel. Roughly speaking an abelian category is a category which behaves exactly like the category **Ab** of abelian groups for finitely many (arbitrary) objects and morphisms. This is exemplified by the fact that ‘diagram chasing’-style proofs apply in all abelian categories (see [18, VIII.4]). In particular the kernel, cokernel and image can all be defined in abelian categories using finite limits and colimits. Examples include **Ab**, for a ring  $R$  the category  $R\text{-Mod}$  of  $R$ -modules and for a topological space  $X$  the category  $\text{Sh}(X)$  of sheaves of abelian groups on  $X$  (see Example 2.21).

If  $F : \mathcal{C} \rightarrow \mathcal{D}$  is a functor between categories, then  $F$  is *left exact* (resp. *right exact*) if  $F$  preserves all finite limits (resp. colimits) and *exact* if it is both left and right exact. If  $\mathcal{C}$  and  $\mathcal{D}$  are additive then if

$F$  is left exact or right exact it is necessarily additive. If  $\mathcal{C}$  and  $\mathcal{D}$  are abelian and  $F$  is additive, then  $F$  is left exact, right-exact and exact if for any short exact sequence  $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$  the induced sequence

$$0 \rightarrow F(A) \rightarrow F(B) \rightarrow F(C) \rightarrow 0$$

is exact in the first four, last four and all five objects respectively. Note however that any additive functor (between abelian categories) preserves *split* exact sequences.

A *cochain complex* in an abelian category  $\mathcal{C}$ , is a sequence of objects  $C^p$  for  $p \in \mathbb{Z}$  together with for each  $p \in \mathbb{Z}$  a morphism  $d^p : C^p \rightarrow C^{p+1}$  so that  $d^{p+1} \circ d^p = 0$  for all  $p$ . We often name all of these morphisms  $d$  and write simply  $C^\bullet$  for the entire complex. A *chain map*  $f^\bullet : C^\bullet \rightarrow D^\bullet$  between cochain complexes is a sequence of morphisms  $f^p : C^p \rightarrow D^p$  for  $p \in \mathbb{Z}$  compatible with the coboundary morphisms of  $C^\bullet$  and  $D^\bullet$ . A *homotopy* between two chain maps  $f^\bullet, g^\bullet : C^\bullet \rightarrow D^\bullet$  is a sequence of morphisms  $h^p : C^p \rightarrow D^{p-1}$  for  $p \in \mathbb{Z}$  so that  $f - g = dh + hd$  (here we suppress the indices). The  $p$ -th *cohomology object* of  $C^\bullet$ , denoted  $h^p(C^\bullet)$ , is the object  $\ker(d : C^p \rightarrow C^{p+1})/\text{im}(d : C^{p-1} \rightarrow C^p)$ . This construction is functorial in the complex and homotopic maps induce the same map on cohomology. In particular homotopy equivalences induce isomorphisms on cohomology.

**Lemma 2.1.** *If  $F : \mathcal{C} \rightarrow \mathcal{D}$  is an exact functor between abelian categories, then for all chain complexes  $C^\bullet$  in  $\mathcal{C}$  there is an isomorphism  $F(h^p(C^\bullet)) \simeq h^p(F(C^\bullet))$  natural in  $F$ .*

**Proof.** Since  $h^p(C^\bullet)$  is constructed entirely using finite limits and colimits, it suffices to argue that for a finite diagram  $D : \mathcal{I} \rightarrow \mathcal{C}$  there is an isomorphism  $F(\lim_i D(i)) \simeq \lim_i F(D(i))$  natural in  $F$  (the colimit case is dual). Here we view  $F \mapsto F(\lim_i D(i))$  as a functor by it mapping a natural transformation  $\eta : F \Rightarrow G$  to the map  $\eta_{\lim_i D(i)} : F(\lim_i D(i)) \rightarrow G(\lim_i D(i))$ . We view  $F \mapsto \lim_i F(D(i))$  as a functor in  $F$  by mapping a natural transformation  $\eta : F \Rightarrow G$  to the induced map on limits from the morphism of diagrams  $F \circ D \Rightarrow G \circ D$  given on each  $i \in \mathcal{I}$  by  $\eta_{D(i)}$ . Let  $L \in \mathcal{C}$  be the limit of  $D$  with cone morphisms  $f_i : L \rightarrow D(i)$ . By the exactness of  $F$  the object  $F(L)$  together with the cone morphisms  $F(f_i)$  form the limit of  $F \circ D$ . Hence we may choose  $F(L)$  together with the  $f_i$ 's as a representative for  $\lim_i D(i)$  and choose  $\text{id}_L$  as the canonical isomorphism  $F(\lim_i D(i)) \simeq \lim_i F(D(i))$ . Now if  $\eta : F \Rightarrow G$  is a natural transformation between exact functors  $\mathcal{C} \rightarrow \mathcal{D}$ , then by the naturality of  $\eta$  we have for each  $i \in \mathcal{I}$  a commutative diagram

$$\begin{array}{ccc} F(L) & \xrightarrow{\eta_L} & G(L) \\ \downarrow F(f_i) & & \downarrow G(f_i) \\ F(D(i)) & \xrightarrow{\eta_{D(i)}} & G(D(i)) \end{array}$$

Now, under the identifications of  $F(L) = \lim_i F(D(i))$  and  $G(L) = \lim_i G(D(i))$  as above, the map  $F(L) \rightarrow G(L)$  induced by the morphism of diagrams  $i \mapsto \eta_{D(i)}$  is defined as the unique map  $F(L) \rightarrow G(L)$  so that the above diagram commutes for all  $i \in \mathcal{I}$ . Hence  $\eta_L$  is this map.  $\square$

**Remark 2.2.** As the above proof shows, checking that certain constructed morphisms are natural requires more unfolding of definitions than actual non-trivial proofs. Hence we will in the future often omit the proof that a certain morphism is natural and note that it remains to be checked. This unfolding of definitions and keeping track of how the functoriality of each operation is defined is something that the author hopes future computer proof assistants can help with.

An object  $I$  of an abelian category  $\mathcal{C}$  is called *injective* if the functor  $\text{Hom}(-, I) : \mathcal{C}^{\text{op}} \rightarrow \mathbf{Ab}$  is exact. We write  $\text{inj}(\mathcal{C})$  for the full subcategory of injective objects. A *resolution* of an object  $A \in \mathcal{C}$  is a cochain complex  $C^\bullet$  in  $\mathcal{C}$  which is 0 in negative degrees and exact on positive degrees together with a morphism  $k : A \rightarrow C^0$  which is the kernel of the coboundary morphism  $C^0 \rightarrow C^1$  (or equivalently induces an isomorphism  $A \simeq h^0(C^\bullet)$ ). An injective resolution is a resolution consisting of injective objects. An



abelian category  $\mathcal{C}$  is said to have *enough injectives* if each object  $A \in \mathcal{C}$  admits a monomorphism into an injective object. In this case all objects of  $\mathcal{C}$  have an injective resolution. The *injective dimension* is the smallest integer  $n \in \mathbb{Z}_{\geq -1}$  (if it exists) so that all  $A \in \mathcal{C}$  have an injective resolution  $I^\bullet$  with  $I^p = 0$  for  $p > n$ . For example **Ab** has injective dimension 1.

Let  $F : \mathcal{C} \rightarrow \mathcal{D}$  be a left exact functor between abelian categories with  $\mathcal{C}$  having enough injectives. For  $p \in \mathbb{Z}$  the  $p$ -th *derived functor*  $R^p F : \mathcal{C} \rightarrow \mathcal{D}$  is defined by mapping an object  $A \in \mathcal{C}$  to the  $p$ -th cohomology object  $h^p(F(I^\bullet))$  of  $F(I^\bullet)$  where  $I^\bullet$  is an injective resolution of  $A$ . Note that  $R^0 F \simeq F$  and that if  $\mathcal{C}$  has injective dimension  $n \in \mathbb{Z}_{\geq 0}$  then  $R^p F = 0$  for  $p > n$ . If  $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$  is a short exact sequence in  $\mathcal{C}$ , then there is a long exact sequence

$$0 \rightarrow F(A) \rightarrow F(B) \rightarrow F(C) \rightarrow R^1 F(A) \rightarrow R^1 F(B) \rightarrow R^1 F(C) \rightarrow R^2 F(A) \rightarrow \dots$$

so  $R^p F$  ‘measures the failure’ of  $F$  to be exact. The proof of the long exact sequence and the well-definedness of  $R^p F$  depends on the following two lemma’s.

**Lemma 2.3.** *Let  $\mathcal{C}$  be an abelian category,  $f : A \rightarrow B$  a morphism,  $E^\bullet$  and  $I^\bullet$  resolutions of  $A$  and  $B$  respectively with  $I^\bullet$  injective. Then there exists a up to chain homotopy unique chain map  $\varphi : E^\bullet \rightarrow I^\bullet$  so that  $\varphi$  induces  $f$  on 0-th cohomology (under the isomorphisms  $A \simeq h^0(E^\bullet)$  and  $B \simeq h^0(I^\bullet)$ ).*

**Proof.** See [30, Theorem 2.3.7]. □

**Lemma 2.4.** (Horseshoe Lemma) *Let  $\mathcal{C}$  be an abelian category,  $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$  a short exact sequence and  $I^\bullet, J^\bullet$  injective resolutions of  $A$  and  $C$  respectively. Then there exist morphisms  $d^i : I^i \oplus J^i \rightarrow I^{i+1} \oplus J^{i+1}$  and a morphism  $k : B \rightarrow I^0 \oplus J^0$  making  $I^\bullet \oplus J^\bullet$  into a resolution of  $B$  so that the canonical morphisms  $I^\bullet \rightarrow I^\bullet \oplus J^\bullet \rightarrow J^\bullet$  are chain maps inducing the morphisms  $A \rightarrow B \rightarrow C$  on 0-th cohomology.*

**Proof.** This is dual to [30, Lemma 2.2.8]. □

## 2.2 Existence and uniqueness of left-exactification

**Lemma 2.5.** *Let  $F, G : \mathcal{C} \rightarrow \mathcal{D}$  be left exact functors between abelian categories with  $\mathcal{C}$  having enough injectives. For any natural transformation  $\eta : F|_{\text{inj}(\mathcal{C})} \Rightarrow G|_{\text{inj}(\mathcal{C})}$ , there exists a unique  $\varepsilon : F \Rightarrow G$  so that  $\varepsilon|_{\text{inj}(\mathcal{C})} = \eta$ . If  $\eta$  is an isomorphism, then so is  $\varepsilon$ .*

**Proof.** Let  $A \in \mathcal{C}$  and let  $0 \rightarrow A \xrightarrow{k} I^0 \xrightarrow{i} I^1$  the start of an injective resolution  $I^\bullet$  of  $A$ . Suppose  $\varepsilon : F \Rightarrow G$  is a natural transformation with  $\varepsilon|_{\text{inj}(\mathcal{C})} = \eta$ , then we have the following commutative diagram:

$$\begin{array}{ccccccc} 0 & \longrightarrow & F(A) & \xrightarrow{F(k)} & F(I^0) & \xrightarrow{F(i)} & F(I^1) \\ & & \downarrow \varepsilon_A & & \downarrow \eta_{I^0} & & \downarrow \eta_{I^1} \\ 0 & \longrightarrow & G(A) & \xrightarrow{G(k)} & G(I^0) & \xrightarrow{G(i)} & G(I^1) \end{array} \quad (2.1)$$

Now because  $F$  and  $G$  are left exact  $F(k)$  and  $G(k)$  are the kernels of  $F(i)$  and  $F(j)$  respectively. It follows that  $\varepsilon_A$  must be the morphism on kernels induced by  $\eta_{I^0}$  and  $\eta_{I^1}$ , so  $\varepsilon_A$  is uniquely determined. Since this holds for all  $A$  the entirety of  $\varepsilon$  is uniquely determined. For existence we define  $\varepsilon_A$  to be the morphism on kernels induced by  $\eta_{I^0}$  and  $\eta_{I^1}$ . Now of course we need to check that  $\varepsilon_A$  is independent of the choice of injective resolution. If  $0 \rightarrow A \xrightarrow{l} J^0 \xrightarrow{j} J^1$  is the beginning of another injective resolution

$J^\bullet$  of  $A$ , then by Lemma 2.3 there exists a chain map  $f^\bullet : I^\bullet \rightarrow J^\bullet$  so that the diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & A & \xrightarrow{k} & I^0 & \xrightarrow{i} & I^1 \\ & & \parallel & & \downarrow f^0 & & \downarrow f^1 \\ 0 & \longrightarrow & A & \xrightarrow{l} & J^0 & \xrightarrow{j} & J^1 \end{array}$$

commutes and this map is unique up to homotopy. Because  $\eta$  is a natural transformation we have the following commutative diagram:

$$\begin{array}{ccccccc} 0 & \longrightarrow & F(A) & \xrightarrow{F(k)} & F(I^0) & \xrightarrow{F(i)} & F(I^1) \\ & & \parallel & & \downarrow \eta_{I^0} & & \downarrow \eta_{I^1} \\ 0 & \longrightarrow & G(A) & \xrightarrow{G(k)} & G(I^0) & \xrightarrow{G(i)} & G(I^1) \\ & & \parallel & & \downarrow F(f^0) & & \downarrow F(f^1) \\ 0 & \longrightarrow & F(A) & \xrightarrow{F(l)} & F(J^0) & \xrightarrow{F(j)} & F(J^1) \\ & & \parallel & & \downarrow \eta_{J^0} & & \downarrow \eta_{J^1} \\ 0 & \longrightarrow & G(A) & \xrightarrow{G(l)} & G(J^0) & \xrightarrow{G(j)} & G(J^1) \end{array}$$

It follows using a little diagram chasing that the induced maps on kernels are the same, so  $\varepsilon_A$  is well defined. To see that  $\varepsilon|_{\text{inj}(\mathcal{C})} = \eta$  notice that an injective object  $I \in \mathcal{C}$  has injective resolution  $0 \rightarrow I \xrightarrow{k} I \xrightarrow{i} 0$  with  $k = \text{id}_I$ . Plugging this resolution into diagram (2.1) then shows that  $\varepsilon_I = \eta_I$ . Finally if  $\eta$  is an isomorphism, then from diagram (2.1) it follows immediately that  $\varepsilon_A$  is an isomorphism for all  $A$ .  $\square$

**Proposition 2.6.** *Let  $F : \mathcal{C} \rightarrow \mathcal{D}$  be an additive functor between abelian categories with  $\mathcal{C}$  having enough injectives. Then there exists a unique left exact functor  $\tilde{F} : \mathcal{C} \rightarrow \mathcal{D}$  together with a natural isomorphism  $\varepsilon : F|_{\text{inj}(\mathcal{C})} \simeq \tilde{F}|_{\text{inj}(\mathcal{C})}$ . This  $\varepsilon$  extends canonically to a natural transformation  $\eta : F \Rightarrow \tilde{F}$ .*

**Proof.** We first prove existence. Let  $A \in \mathcal{C}$  and let  $0 \rightarrow A \rightarrow I^0 \xrightarrow{i} I^1$  be the beginning of an injective resolution of  $A$ . We define  $\tilde{F}(A)$  as  $\ker(F(i))$ . We now prove that  $\tilde{F}$  is well-defined and functorial in one go. Let  $f : A \rightarrow B$  be a morphism in  $\mathcal{C}$  and let  $I^\bullet, J^\bullet$  be injective resolutions of  $A$  and  $B$  respectively. Then  $F(A)$  and  $F(B)$  are defined as  $h^0(F(I^\bullet))$  and  $h^0(F(J^\bullet))$  respectively. By Lemma 2.3 there exists a chain map  $\varphi : I^\bullet \rightarrow J^\bullet$  inducing  $f$  on 0-th cohomology and this  $f$  is unique up to homotopy. Applying  $F$  gives a canonical homotopy class of chain maps  $F(I^\bullet) \rightarrow F(J^\bullet)$ . Thus we have a canonical morphism  $h^0(F(\varphi)) : h^0(F(I^\bullet)) \rightarrow h^0(F(J^\bullet))$  which we define to be  $\tilde{F}(f)$ . In the case that  $A = B$  and  $f = \text{id}_A$ , then  $\varphi$  is a homotopy equivalence, so  $h^0(F(\varphi))$  is a canonical isomorphism  $h^0(F(I^\bullet)) \simeq h^0(F(J^\bullet))$ , so  $\tilde{F}$  is well-defined. We omit the verification that  $\tilde{F}$  satisfies the functor axioms and that it is additive.

Now notice that an injective object  $I \in \mathcal{C}$  has injective resolution  $0 \rightarrow I \xrightarrow{\text{id}_I} I \xrightarrow{i} 0$ , so  $\tilde{F}(I) = \ker F(i) = \ker(F(I) \rightarrow 0) \simeq F(I)$ . It is not hard to see that this isomorphism is indeed an isomorphism of functors  $\varepsilon : F|_{\text{inj}(\mathcal{C})} \simeq \tilde{F}|_{\text{inj}(\mathcal{C})}$ . Now let  $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$  be exact in  $\mathcal{C}$ . By Lemma 2.4 there exist simultaneous injective resolutions of  $A, B, C$  as in the diagram

$$\begin{array}{ccccccc} & & 0 & & 0 & & 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \rightarrow & A & \rightarrow & B & \rightarrow & C \rightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \rightarrow & I^0 & \rightarrow & I^0 \oplus J^0 & \rightarrow & J^0 \rightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \rightarrow & I^1 & \rightarrow & I^1 \oplus J^1 & \rightarrow & J^1 \rightarrow 0 \end{array}$$

where the lower two rows are split exact. Applying  $F$  to the lower two rows and taking kernels gives

$$\begin{array}{ccccccc}
& 0 & & 0 & & 0 & \\
& \downarrow & & \downarrow & & \downarrow & \\
0 & \rightarrow & \tilde{F}(A) & \longrightarrow & \tilde{F}(B) & \longrightarrow & \tilde{F}(C) \\
& \downarrow & & \downarrow & & \downarrow & \\
0 & \rightarrow & F(I^0) & \rightarrow & F(I^0 \oplus J^0) & \rightarrow & F(J^0) \rightarrow 0 \\
& \downarrow & & \downarrow & & \downarrow & \\
0 & \rightarrow & F(I^1) & \rightarrow & F(I^1 \oplus J^1) & \rightarrow & F(J^1) \rightarrow 0
\end{array}$$

Since the bottom two horizontal rows are split exact in the first diagram, they are still split exact in the second diagram. Since kernels are left exact the upper row is exact. We conclude that  $\tilde{F}$  is left exact. This proves the existence of  $\tilde{F}$  and  $\varepsilon$  as required. For uniqueness note that if  $\tilde{F}' : \mathcal{C} \rightarrow \mathcal{D}$  is another left exact functor with  $\varepsilon' : F|_{\text{inj}(\mathcal{C})} \simeq \tilde{F}'|_{\text{inj}(\mathcal{C})}$ , then by Lemma 2.5 the composition  $\varepsilon' \circ \varepsilon^{-1}$  extends uniquely to an isomorphism  $\tilde{F} \simeq \tilde{F}'$ . To construct  $\eta$ , let  $A \in \mathcal{C}$  with  $0 \rightarrow A \xrightarrow{k} I^0 \xrightarrow{i} I^1$  be the beginning of an injective resolution of  $A$ . Because  $F(i) \circ F(k) = F(i \circ k) = F(0) = 0$ , there is a unique morphism  $\eta_A : F(A) \rightarrow \ker F(i) = \tilde{F}(A)$  which composed with the canonical morphism  $\ker F(i) \rightarrow F(I^0)$  gives  $F(k)$ . We omit the verification that  $\eta$  is natural. Repeating this construction for injective  $I$  with the resolution  $0 \rightarrow I \xrightarrow{\text{id}_I} I \rightarrow 0$  shows that  $\eta$  agrees with  $\varepsilon$  on  $\text{inj}(\mathcal{C})$ .  $\square$

**Definition 2.7.** We will call the functor  $\tilde{F}$  from Proposition 2.6 the *left-exactification* of  $F$ .

**Remark 2.8.** Notice that the construction of  $\tilde{F}(A)$  is the exact same as the construction of  $R^0 F(A)$  if  $F$  were left exact: Take an injective resolution  $I^\bullet$  of  $A$  and take the 0-th cohomology of  $F(I^\bullet)$ . In particular the proof that  $\tilde{F}$  is well-defined is a carbon copy of the proof that  $R^p G$  is well-defined for left exact  $G$ . This motivates the notation  $R^0 F$  for  $\tilde{F}$  which is the notation used in [10]. We will not use the notation  $R^p F$  for  $h^p(F(I^\bullet))$  if  $F$  is not left exact, since  $F$  and  $\tilde{F}$  are isomorphic on injective objects and hence  $h^p(F(I^\bullet)) \simeq h^p(\tilde{F}(I^\bullet)) = R^p \tilde{F}(A)$ .

**Proposition 2.9.** Let  $C_\bullet$  be a chain complex in an abelian category  $\mathcal{C}$  having enough injectives. Then for any  $p \in \mathbb{Z}$ , the left-exactification of the  $p$ -th cohomology functor  $h^p(C_\bullet, -) = h^p(\text{Hom}(C_\bullet, -))$  of  $C_\bullet$  is represented by  $h_p(C_\bullet)$ . If  $\mathcal{C} = \mathbf{Ab}$  and  $G \in \mathbf{Ab}$  then the left-exactification of the functor  $\mathbf{Ab} \rightarrow \mathbf{Ab}$  given by  $T \mapsto h^p(C_\bullet, \text{Hom}(G, T))$  is represented by  $h_p(C_\bullet \otimes_{\mathbb{Z}} G)$ .

**Proof.** If  $T$  is injective then the functor  $\text{Hom}(-, T)$  is exact, so  $h^p(C_\bullet, T) = h^p(\text{Hom}(C_\bullet, T)) \simeq \text{Hom}(h_p(C_\bullet), T)$ . This isomorphism is natural in  $T$  by Lemma 2.1 (a morphism  $f : S \rightarrow T$  between injective objects induces a natural transformation  $\text{Hom}(-, S) \Rightarrow \text{Hom}(-, T)$  between exact functors). By Lemma 2.5 it follows that the left-exactification of  $h^p(C_\bullet, -)$  is isomorphic to  $\text{Hom}(h_p(C_\bullet), -)$ . In the case that  $\mathcal{C} = \mathbf{Ab}$  we can apply the above argument to  $C_\bullet \otimes G$  to show that  $h_p(C_\bullet \otimes G)$  represents the left-exactification of the functor  $h^p(C_\bullet \otimes G, -) : \mathbf{Ab} \rightarrow \mathbf{Ab}$ . However notice that by the tensor-hom adjunction we have

$$h^p(C_\bullet \otimes G, -) = h^p(\text{Hom}(C_\bullet \otimes G, -)) \simeq h^p(\text{Hom}(C_\bullet, \text{Hom}(G, -))) = h^p(C_\bullet, \text{Hom}(G, -))$$

(Note that we use the naturality of the tensor-hom adjunction to see that this is a natural isomorphism.) We conclude that  $h_p(C_\bullet \otimes G)$  represents the left-exactification of  $h^p(C_\bullet, -) \circ \text{Hom}(G, -)$ .  $\square$

**Remark 2.10.** As mentioned in the introduction this shows that the homology of  $C_\bullet$  and  $C_\bullet \otimes G$  are completely determined by the cohomology functor  $h^p(C_\bullet, -)$ .

### 2.3 Half exact functors

**Lemma 2.11.** *Let  $\mathcal{A} \xrightarrow{F} \mathcal{B} \xrightarrow{G} \mathcal{C}$  be additive functors between abelian categories with  $\mathcal{A}$  and  $\mathcal{B}$  having enough injectives, then  $\widetilde{G \circ F} \simeq \widetilde{G} \circ \widetilde{F}$ . In particular if  $G$  is left exact, then  $\widetilde{G \circ F} \simeq \widetilde{G} \circ \widetilde{F}$ .*

**Proof.** For injective  $I \in \mathcal{C}$  we have  $G(F(I)) \simeq G(\widetilde{F}(I))$ .  $\square$

**Remark 2.12.** Relating the composition  $\widetilde{G} \circ F$  to  $\widetilde{G \circ F}$  is much harder. This is partially done in the upcoming Lemma 2.33. This and the upcoming universal coefficient theorem (Theorem 2.17) for sheaf homology depends on the following proposition, which allows us to recover a half exact functor from its left-exactification and the left-exactification of another half exact functor connected to it (see Definition 2.13 below).

**Definition 2.13.** We say that a functor  $F : \mathcal{C} \rightarrow \mathcal{D}$  between abelian categories is *half exact* if it is additive and if for all short exact sequences  $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$  the induced sequence  $F(A) \rightarrow F(B) \rightarrow F(C)$  is exact. If  $F, G : \mathcal{C} \rightarrow \mathcal{D}$  are half exact then a *connection* from  $F$  to  $G$  is a collection of morphisms  $\delta : F(C) \rightarrow G(A)$  for all short exact sequences  $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$  in  $\mathcal{C}$  so that the sequence

$$F(A) \rightarrow F(B) \rightarrow F(C) \xrightarrow{\delta} G(A) \rightarrow G(B) \rightarrow G(C)$$

is exact and so that  $\delta$  is natural with respect to morphisms of short exact sequences. Note that if  $H : \mathcal{B} \rightarrow \mathcal{C}$  is an exact functor with  $\mathcal{B}$  abelian, then  $F \circ H$  is half exact if  $F$  is half exact and a connection between  $F$  and  $G$  yields a connection between  $F \circ H$  and  $G \circ H$ .

**Example 2.14.** If  $F : \mathcal{C} \rightarrow \mathcal{D}$  is left exact with  $\mathcal{C}, \mathcal{D}$  abelian and  $\mathcal{C}$  with enough injectives, then the derived functors  $R^p F$  are half exact. Furthermore the long exact sequence for derived functors gives connections between  $R^p F$  and  $R^{p+1} F$ .

**Proposition 2.15.** *Let  $F, G : \mathcal{C} \rightarrow \mathcal{D}$  be half exact functors between abelian categories with  $\mathcal{C}$  of injective dimension  $\leq 1$  and  $\delta$  a connection between  $F$  and  $G$ . Then there is a natural transformation  $\varepsilon : R^1 \widetilde{F} \Rightarrow G$  such that for all  $A \in \mathcal{C}$  the sequence*

$$0 \rightarrow R^1 \widetilde{F}(A) \xrightarrow{\varepsilon_A} G(A) \xrightarrow{\eta_A} \widetilde{G}(A) \rightarrow 0$$

is exact, where  $\eta$  is as in Proposition 2.6.

**Proof.** Let  $A \in \mathcal{C}$ . Since  $\mathcal{C}$  has injective dimension  $\leq 1$  there exists a short exact sequence  $0 \rightarrow A \rightarrow I^0 \xrightarrow{i} I^1 \rightarrow 0$  with  $I^0, I^1$  injective. Applying the connection  $\delta$  to this gives:

$$F(A) \rightarrow F(I^0) \xrightarrow{i} F(I^1) \xrightarrow{\delta} G(A) \rightarrow G(I^0) \rightarrow G(I^1)$$

Taking the short exact sequence of this centred at the object  $G(A)$  gives:

$$0 \rightarrow \text{coker } F(i) \rightarrow G(A) \rightarrow \ker G(i) \rightarrow 0$$

Because the morphism in the resolution of  $A$  between indices 1 and 2 is the zero map, the first term is actually  $R^1 \widetilde{F}(A)$  (see Remark 2.8). Secondly by definition the third term is  $\widetilde{G}(A)$ , so by setting  $\varepsilon_A$  to be the morphism from  $R^1 \widetilde{F}(A)$  to  $G(A)$  in this diagram we have the required exact sequence.

We will now show that  $\varepsilon$  is well-defined (i.e. independent of the choice of injective resolution) and a natural transformation in one go. Let  $f : A \rightarrow B$  be a morphism in  $\mathcal{C}$  and pick injective resolutions  $0 \rightarrow A \rightarrow I^0 \xrightarrow{i} I^1 \rightarrow 0$  and  $0 \rightarrow B \rightarrow J^0 \xrightarrow{j} J^1 \rightarrow 0$ . Then by Lemma 2.3 we can extend  $f$  to a morphism of exact sequences

$$\begin{array}{ccccccc} 0 & \longrightarrow & A & \longrightarrow & I^0 & \longrightarrow & I^1 \longrightarrow 0 \\ & & \downarrow f & & \downarrow g^0 & & \downarrow g^1 \\ 0 & \longrightarrow & B & \longrightarrow & J^0 & \longrightarrow & J^1 \longrightarrow 0 \end{array}$$

Applying the naturality of  $\delta$  and then immediately taking the short exact sequence centered at  $G(A)$  and  $G(B)$  gives:

$$\begin{array}{ccccccc} 0 & \longrightarrow & \operatorname{coker} F(i) & \longrightarrow & G(A) & \longrightarrow & \ker G(i) \longrightarrow 0 \\ & & \downarrow & & \downarrow G(f) & & \downarrow \\ 0 & \longrightarrow & \operatorname{coker} F(j) & \longrightarrow & G(B) & \longrightarrow & \ker G(j) \longrightarrow 0 \end{array}$$

Recalling how the functoriality of right derived functors was defined we see that the morphism  $\operatorname{coker} F(i) \rightarrow \operatorname{coker} F(j)$  is exactly  $R^1\tilde{F}(f)$ . It follows that  $\varepsilon$  (if it is well-defined) is a natural transformation. Finally in the case that  $A = B$  and  $f = \operatorname{id}$  the diagram becomes

$$\begin{array}{ccc} \operatorname{coker} F(i) & \longrightarrow & G(A) \\ R^1\tilde{F}(\operatorname{id}) \downarrow & & \downarrow G(\operatorname{id})=\operatorname{id} \\ \operatorname{coker} F(j) & \longrightarrow & G(B) \end{array}$$

where the top and bottom horizontal arrows are  $\varepsilon_A$  but computed with different injective resolutions. This diagram shows that they coincide, so  $\varepsilon$  is well-defined.  $\square$

**Corollary 2.16.** *Let  $\mathcal{A} \xrightarrow{F} \mathcal{B} \xrightarrow{G} \mathcal{C}$  be abelian categories and additive functors with  $\mathcal{A}$  and  $\mathcal{B}$  having enough injectives,  $\mathcal{A}$  of injective dimension  $\leq 1$ ,  $F$  exact and  $G$  left exact. Then for each  $p \in \mathbb{Z}$  there is a natural transformation  $\varepsilon : R^1((R^{p-1}G) \circ F) \Rightarrow (R^pG) \circ F$  such that for each  $T \in \mathcal{A}$  the sequence*

$$0 \rightarrow R^1(\widetilde{(R^{p-1}G) \circ F})(T) \xrightarrow{\varepsilon_T} (R^pG)(F(T)) \xrightarrow{\eta_T} (\widetilde{(R^pG) \circ F})(T) \rightarrow 0$$

is exact, where  $\eta$  is as in Proposition 2.6.

**Proof.** Apply Proposition 2.15 to the functors  $R^{p-1}G \circ F$  and  $R^pG \circ F$ . The required connecting homomorphism comes from applying the long exact sequence of  $G$  to the result of applying  $F$  to the original exact sequence.  $\square$

**Theorem 2.17.** (Universal Coefficient Theorem for Sheaf Cohomology) *Let  $X$  be a topological space and  $p \in \mathbb{Z}$ , then for all  $T \in \mathbf{Ab}$  we have the following exact sequence which is natural in  $T$ :*

$$0 \rightarrow R^1\vec{H}_{p-1}(X)(T) \rightarrow H^p(X, T_X) \rightarrow \vec{H}_p(X)(T) \rightarrow 0$$

**Proof.** Apply Corollary 2.16 to  $\mathcal{A} = \mathcal{C} = \mathbf{Ab}$ ,  $\mathcal{B} = \operatorname{Sh}(X)$  the category of sheaves of abelian groups on  $X$ ,  $F = (-)_X$  the constant sheaf functor and  $G = \Gamma(X, -)$  the global section functor.  $\square$

**Remark 2.18.** Notice that if in the above theorem  $\vec{H}_{p-1}(X)$  is represented by some group  $H_{p-1}(X) \in \mathbf{Ab}$  (i.e.  $\vec{H}_{p-1}(X) \simeq \operatorname{Hom}(H_{p-1}(X), -)$  as functors  $\mathbf{Ab} \rightarrow \mathbf{Ab}$ ), then  $R^1\vec{H}_{p-1}(X)(T) \simeq \operatorname{Ext}^1(H_{p-1}(X), T)$  by the definition of the Ext functors. Thus if  $\vec{H}_p(X)$  is similarly represented by a  $H_p(X) \in \mathbf{Ab}$ , then the universal coefficient theorem has the familiar form

$$0 \rightarrow \operatorname{Ext}^1(H_{p-1}(X), T) \rightarrow H^p(X, T_X) \rightarrow \operatorname{Hom}(H_p(X), T) \rightarrow 0.$$

Note though that it is not clear whether the sequence in this theorem is always split, so the isomorphism class of  $H^p(X, T_X)$  need not be able to be reconstructed from  $\vec{H}_p(X)$  and  $\vec{H}_{p-1}(X)$ .

## 2.4 Left exact functor categories

We begin this section by recalling some basic properties of reflective subcategories. These can be found in any text on category theory such as in [18, IV.3].

**Definition 2.19.** Let  $\mathcal{C}$  be a category and  $\mathcal{D}$  a full subcategory. If  $C \in \mathcal{C}$  then a *reflection* of  $C$  in  $\mathcal{D}$  is an object  $\bar{C} \in \mathcal{D}$  and a morphism  $r : C \rightarrow \bar{C}$  so that for any  $D \in \mathcal{D}$  and  $f : C \rightarrow D$  there exists a unique morphism  $\bar{f} : \bar{C} \rightarrow D$  so that the diagram

$$\begin{array}{ccc} C & & \\ r \downarrow & \searrow f & \\ \bar{C} & \xrightarrow{\bar{f}} & D \end{array}$$

commutes. If a reflection of  $C$  exists it is unique up to canonical isomorphism. We say that  $\mathcal{D}$  is a *reflective subcategory* of  $\mathcal{C}$  if every object of  $\mathcal{C}$  has a reflection in  $\mathcal{D}$ . In this case the map  $\mathcal{C} \rightarrow \mathcal{D}$  sending an object to its reflection is called the *reflector*. The dual notions are called *coreflections*, *coreflective subcategories* and *coreflector* respectively.

**Example 2.20.** The category **Grp** of groups has a reflective subcategory **Ab** of abelian groups with the reflector given by abelianisation.

**Example 2.21.** Let  $X$  be a topological space. Consider the category  $\text{PSh}(X)$  of *presheaves of abelian groups* on  $X$ , i.e. the category of contravariant functors from the category  $\text{Op}(X)$  of open sets of  $X$  (with there being a morphism  $V \rightarrow U$  for each inclusion  $V \subseteq U$ ) to **Ab**. If  $\mathcal{F} \in \text{PSh}(X)$ ,  $V \subseteq U$  opens in  $X$  and  $s \in \mathcal{F}(U)$  we write  $s|_V \in \mathcal{F}(V)$  for the image of  $s$  under the map induced by  $V \rightarrow U$  and  $\mathcal{F}$ . A presheaf  $\mathcal{F}$  is a *sheaf* if it satisfies the following gluing axiom: Let  $U \in \text{Op}(X)$  and  $(U_i)_{i \in I}$  an open cover of  $U$  together with for each  $i$  an element  $s_i \in \mathcal{F}(U_i)$  so that for all  $i, j \in I$  we have  $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$ . Then there exists a unique  $s \in \mathcal{F}(U)$  so that  $s|_{U_i} = s_i$  for all  $i \in I$ . We write  $\text{Sh}(X)$  for the full subcategory of  $\text{PSh}(X)$  consisting of sheaves. Given a presheaf  $\mathcal{F}$  and  $x \in X$  we define the *stalk*  $\mathcal{F}_x$  of  $\mathcal{F}$  at  $x$  as the colimit of  $\mathcal{F}(U)$  as  $U$  ranges over the open neighbourhoods of  $x$ . If  $\mathcal{F} \in \text{PSh}(X)$ , then the *sheafification* of  $\mathcal{F}$ , is the sheaf  $\mathcal{F}^+$  with for each  $U \subseteq X$  open  $\mathcal{F}^+(U)$  to be the subset of  $\prod_{x \in U} \mathcal{F}_x$  of families  $(s_x)_{x \in U}$  which are locally given by a section of  $\mathcal{F}$ . That is, for all  $x \in U$  there exists an open  $V \subseteq U$  and  $t \in \mathcal{F}(V)$  so that  $s_y = t_y$  for all  $y \in V$ . Then  $\mathcal{F}^+$ , together with the natural morphism  $\mathcal{F} \rightarrow \mathcal{F}^+$ , is the reflection of  $\mathcal{F}$  in  $\text{Sh}(X)$  (see [14, Prop. II.2.4]).

**Lemma 2.22.** Let  $\mathcal{C}$  be a category and  $\mathcal{D}$  a reflective subcategory. Then the reflector can be made into a functor  $\mathcal{C} \rightarrow \mathcal{D}$  by sending a morphism  $f : C \rightarrow C'$  in  $\mathcal{C}$  to the morphism  $\bar{C} \rightarrow \bar{C}'$  resulting from the reflection universal property applied to the composite  $C \rightarrow C' \rightarrow \bar{C}'$ . Furthermore the reflector is left adjoint to the inclusion functor  $\mathcal{D} \hookrightarrow \mathcal{C}$ .

**Lemma 2.23.** Let  $\mathcal{C}$  be a category and  $\mathcal{D}$  a reflective subcategory.

1. A diagram in  $\mathcal{D}$  has a limit in  $\mathcal{D}$  if and only if it has a limit in  $\mathcal{C}$  which lies in  $\mathcal{D}$ . In this case both limits agree.
2. If a diagram in  $\mathcal{D}$  has a colimit in  $\mathcal{C}$ , then the reflection of this colimit is the colimit in  $\mathcal{D}$ .

**Notation 2.24.** Let  $\mathcal{C}, \mathcal{D}$  be categories, then we write  $[\mathcal{C}, \mathcal{D}]$  for the category of functors  $\mathcal{C} \rightarrow \mathcal{D}$  with natural transformations as morphisms. If  $\mathcal{C}$  and  $\mathcal{D}$  are additive, then we write  $[\mathcal{C}, \mathcal{D}]_{\text{add}}$  for the full subcategory of additive functors. If  $\mathcal{C}$  and  $\mathcal{D}$  are abelian then we write  $[\mathcal{C}, \mathcal{D}]_{\text{LE}}$  for the full subcategory of left exact functors  $\mathcal{C} \rightarrow \mathcal{D}$ .

**Lemma 2.25.** Let  $\mathcal{C}$  and  $\mathcal{D}$  be abelian categories with  $\mathcal{C}$  having enough injectives. Then  $[\mathcal{C}, \mathcal{D}]_{\text{LE}}$  is a reflective subcategory of  $[\mathcal{C}, \mathcal{D}]_{\text{add}}$  with the reflector given by left-exactification. Hence left-exactification gives a functor  $[\mathcal{C}, \mathcal{D}]_{\text{add}} \rightarrow [\mathcal{C}, \mathcal{D}]_{\text{LE}}$ .

**Proof.** Fix an  $F \in [\mathcal{C}, \mathcal{D}]_{\text{add}}$ . We will show that  $\eta : F \Rightarrow \tilde{F}$  from Proposition 2.6 satisfies the reflection universal property. Fix  $G \in [\mathcal{C}, \mathcal{D}]_{\text{LE}}$  together with a morphism  $\varepsilon : F \Rightarrow G$ . If  $\nu : \tilde{F} \Rightarrow G$  is a morphism

of functors so that  $\varepsilon = \nu \circ \eta$ , then the fact that  $\varepsilon_I = \nu_I \circ \eta_I$  for injective  $I \in \mathcal{C}$  completely determines  $\nu_I$  because  $\eta_I$  is an isomorphism. By Lemma 2.5 this means that the entirety of  $\nu$  is determined. To show that such a  $\nu$  exists, define  $\nu$  to be the unique extension  $\tilde{F} \Rightarrow G$  of  $\varepsilon|_{\text{inj}(\mathcal{C})} \circ (\eta|_{\text{inj}(\mathcal{C})})^{-1}$  using Lemma 2.5. Let  $A \in \mathcal{C}$  and let  $0 \rightarrow A \xrightarrow{k} I$  be an embedding of  $A$  in an injective object. Note that by construction  $\varepsilon_I = \nu_I \circ \eta_I$ , so by the naturality of  $\varepsilon, \eta, \nu$  we have

$$G(k) \circ \varepsilon_A = \varepsilon_I \circ F(k) = \nu_I \circ \eta_I \circ F(k) = \nu_I \circ \tilde{F}(k) \circ \eta_A = G(k) \circ \nu_A \circ \eta_A.$$

Since  $G(k)$  is monic (because  $G$  is left exact) it follows that  $\varepsilon_A = \nu_A \circ \eta_A$ . We conclude that  $\nu \circ \eta = \varepsilon$ , so  $F$  satisfies the reflection universal property.  $\square$

**Lemma 2.26.** *Let  $\mathcal{B}$  be an abelian category and  $\mathcal{A}$  a reflective subcategory. Suppose that the reflector seen as a functor  $\mathcal{B} \rightarrow \mathcal{B}$  is left exact, then  $\mathcal{A}$  is an abelian category and the reflector  $\mathcal{B} \rightarrow \mathcal{A}$  is exact.*

**Proof.** By Lemma 2.23.2 the reflector  $R : \mathcal{B} \rightarrow \mathcal{A}$  preserves all colimits. Since we assumed that the functor  $\iota \circ R : \mathcal{B} \rightarrow \mathcal{B}$  is left exact (here  $\iota : \mathcal{A} \rightarrow \mathcal{B}$  is the inclusion functor), this means that for any finite diagram  $D : \mathcal{I} \rightarrow \mathcal{B}$  with limit  $L$  the composition  $\iota \circ R \circ D$  has limit  $\iota(R(L))$ . By Lemma 2.23.1 it follows that  $R(L)$  is the limit of  $R \circ D$ , so  $R$  also preserves the limit of  $D$ . We conclude that  $R$  is exact. First of all  $\mathcal{A}$  is additive since it induces the abelian group structure on hom-sets from  $\mathcal{B}$ . Secondly if  $D : \mathcal{I} \rightarrow \mathcal{A}$  is a finite diagram with limit (resp. colimit)  $L$  in  $\mathcal{B}$  then  $R(L)$  is the limit (resp. colimit) of  $D$  in  $\mathcal{A}$  by the exactness of  $R$ . Hence  $\mathcal{A}$  is additive and it has all finite limits. For axiom (4) note that by the exactness of  $R$  for any  $f : A \rightarrow B$  in  $\mathcal{A}$  the canonical morphism  $\text{coker}(\ker(f) \rightarrow A) \rightarrow \ker(B \rightarrow \text{coker } f)$  where the kernels and cokernels are taken in  $\mathcal{B}$  gets mapped by  $R$  to the canonical morphism  $\text{coker}(\ker(f) \rightarrow A) \rightarrow \ker(B \rightarrow \text{coker } f)$  where the kernels and cokernels are taken in  $\mathcal{A}$ . Since functors map isomorphisms to isomorphisms this completes the proof.  $\square$

**Remark 2.27.** If  $\mathcal{C}$  and  $\mathcal{D}$  are categories, then a cone in  $[\mathcal{C}, \mathcal{D}]$  satisfies the limit (resp. colimit) universal property if and only if it satisfies it on each object. It follows that if  $\mathcal{D}$  has all limits (resp. colimits) of a fixed shape, then so does  $[\mathcal{C}, \mathcal{D}]$ : The limit of a diagram  $D : \mathcal{I} \rightarrow [\mathcal{C}, \mathcal{D}]$  is the functor which maps each  $T \in \mathcal{C}$  to the limit (resp. colimit) of the diagram  $i \mapsto D(i)(T)$ . It also follows that if  $F : \mathcal{B} \rightarrow \mathcal{C}$  is a functor, then the precomposition functor  $(-) \circ F : [\mathcal{C}, \mathcal{D}] \rightarrow [\mathcal{B}, \mathcal{D}]$  preserves all limits and colimits (in particular it is exact). If  $\mathcal{D}$  is abelian, then  $[\mathcal{C}, \mathcal{D}]$  inherits the additive structure on hom-collections from  $\mathcal{D}$  and  $[\mathcal{C}, \mathcal{D}]$  has all finite (co)limits since  $\mathcal{D}$  does. Since axiom (4) of an abelian category is phrased entirely using limits and colimits, the fact that  $\mathcal{D}$  satisfies axiom (4) can be applied object-wise to show that  $[\mathcal{C}, \mathcal{D}]$  satisfies axiom (4). Hence  $[\mathcal{C}, \mathcal{D}]$  is abelian if  $\mathcal{D}$  is. Similarly by unfolding the definition of exact sequences in an abelian category one can show that a sequence  $F \rightarrow G \rightarrow H$  is exact in  $[\mathcal{C}, \mathcal{D}]$  if and only if  $F(T) \rightarrow G(T) \rightarrow H(T)$  is exact for all  $T \in \mathcal{C}$ . If  $\mathcal{C}$  and  $\mathcal{D}$  are additive then the above arguments apply just as well to  $[\mathcal{C}, \mathcal{D}]_{\text{add}}$  since limits and colimits in  $[\mathcal{C}, \mathcal{D}]$  of additive functors are additive.

**Example 2.28.** (Continuing Example 2.21) Recall that  $\text{PSh}(X)$  is the functor category  $[\text{Op}(X)^{\text{op}}, \mathbf{Ab}]$ , so by the above remark it is abelian. By Lemma 2.26 to show that  $\text{Sh}(X)$  is abelian it suffices to show that sheafification, considered as a functor  $\text{PSh}(X) \rightarrow \text{PSh}(X)$ , is left-exact, which follows relatively easily from the exactness of each stalk functor  $\text{PSh}(X) \rightarrow \mathbf{Ab}$ .

**Proposition 2.29.** *Let  $\mathcal{C}, \mathcal{D}$  be abelian categories with  $\mathcal{C}$  having enough injectives. Then the category  $[\mathcal{C}, \mathcal{D}]_{\text{LE}}$  is an abelian category in which limits can be computed object-wise and where a sequence  $F \rightarrow G \rightarrow H$  is exact if and only if  $F(I) \rightarrow G(I) \rightarrow H(I)$  is exact for all injective  $I \in \mathcal{C}$ . In particular left-exactification gives an exact functor  $[\mathcal{C}, \mathcal{D}]_{\text{add}} \rightarrow [\mathcal{C}, \mathcal{D}]_{\text{LE}}$ .*

**Proof.** Firstly  $[\mathcal{C}, \mathcal{D}]_{\text{add}}$  is abelian by Remark 2.27. Now we know from Lemma 2.25 that  $[\mathcal{C}, \mathcal{D}]_{\text{LE}}$  is a reflective subcategory of  $[\mathcal{C}, \mathcal{D}]_{\text{add}}$ , so to apply Lemma 2.26 we need to show that left-exactification as a functor  $[\mathcal{C}, \mathcal{D}]_{\text{add}} \rightarrow [\mathcal{C}, \mathcal{D}]_{\text{add}}$  is left exact. To this end let  $0 \Rightarrow F \Rightarrow G \Rightarrow H$  be exact in  $[\mathcal{C}, \mathcal{D}]_{\text{add}}$

and let  $T \in \mathcal{C}$  with  $0 \rightarrow T \rightarrow I^0 \rightarrow I^1$  the start of an injective resolution, then we have a commutative diagram

$$\begin{array}{ccccccc}
& 0 & & 0 & & 0 & \\
& \downarrow & & \downarrow & & \downarrow & \\
0 & \rightarrow & \tilde{F}(T) & \rightarrow & F(I^0) & \rightarrow & F(I^1) \\
& \downarrow & & \downarrow & & \downarrow & \\
0 & \rightarrow & \tilde{G}(T) & \rightarrow & G(I^0) & \rightarrow & G(I^1) \\
& \downarrow & & \downarrow & & \downarrow & \\
0 & \rightarrow & \tilde{H}(T) & \rightarrow & H(I^0) & \rightarrow & H(I^1)
\end{array} \tag{2.2}$$

with all rows and the two right most columns exact. A little diagram chasing or the general fact that limits commute with limits shows that the first column is also exact. Since this holds for all  $T \in \mathcal{C}$  the sequence  $0 \Rightarrow \tilde{F} \Rightarrow \tilde{G} \Rightarrow \tilde{H}$  is exact in  $[\mathcal{C}, \mathcal{D}]_{\text{add}}$ . We conclude that we can apply Lemma 2.26 to see that  $[\mathcal{C}, \mathcal{D}]_{\text{LE}}$  is abelian and that left-exactification is an exact functor  $[\mathcal{C}, \mathcal{D}]_{\text{add}} \rightarrow [\mathcal{C}, \mathcal{D}]_{\text{LE}}$ . Since limits in  $[\mathcal{C}, \mathcal{D}]_{\text{add}}$  are computed object-wise it follows from Lemma 2.23 that limits in  $[\mathcal{C}, \mathcal{D}]_{\text{LE}}$  are also computed object-wise. Let  $F_0 \Rightarrow F_1 \Rightarrow F_2$  be a (short) sequence in  $[\mathcal{C}, \mathcal{D}]_{\text{LE}}$ . First of all notice that because these functors are left exact, the composite  $F_0 \Rightarrow F_2$  is 0 if and only if the composite  $F_0(I) \Rightarrow F_2(I)$  is zero for all injective  $I \in \mathcal{C}$  (see Lemma 2.5). Therefore we can assume that  $F_0 \Rightarrow F_1 \Rightarrow F_2$  is a chain complex and prove that it has trivial cohomology if and only if  $F_0(I) \rightarrow F_1(I) \rightarrow F_2(I)$  has trivial cohomology for all injective  $I \in \mathcal{C}$ . Write  $\text{LE} : [\mathcal{C}, \mathcal{D}]_{\text{add}} \rightarrow [\mathcal{C}, \mathcal{D}]_{\text{LE}}$  for left-exactification and  $h_{\mathcal{A}}^p$  for the  $p$ -th cohomology of a chain complex in an abelian category  $\mathcal{A}$ . Then because  $\text{LE}$  is exact we have

$$h_{[\mathcal{C}, \mathcal{D}]_{\text{LE}}}^1(F_{\bullet}) = h_{[\mathcal{C}, \mathcal{D}]_{\text{LE}}}^1(\text{LE}(F_{\bullet})) \simeq \text{LE}(h_{[\mathcal{C}, \mathcal{D}]_{\text{add}}}^1(F_{\bullet}))$$

so  $F_{\bullet}$  is exact if and only if  $\text{LE}(h_{[\mathcal{C}, \mathcal{D}]_{\text{add}}}^1(F_{\bullet})) = 0$ , which holds if and only if  $h_{[\mathcal{C}, \mathcal{D}]_{\text{add}}}^1(F_{\bullet})(I) = 0$  for all injective  $I \in \mathcal{C}$ , which holds if and only if  $F_{\bullet}(I)$  has trivial cohomology for all injective  $I \in \mathcal{C}$ .  $\square$

**Remark 2.30.** The existence of a reflector  $[\mathcal{C}, \mathcal{D}]_{\text{add}} \rightarrow [\mathcal{C}, \mathcal{D}]_{\text{LE}}$  assuming  $\mathcal{C}$  has enough injectives has been proven before in [10, I.9]. Note that in [10, II.2] it is shown that this reflection also exists if  $\mathcal{C}$  is small and  $\mathcal{D}$  is AB5. Gabriel assumes that  $\mathcal{C}$  is small in both cases, but when assuming  $\mathcal{C}$  has enough injectives it is not necessary (except for avoiding size issues). A version of Proposition 2.29 in this second case can also be found in [10, Proposition II.2.5].

**Lemma 2.31.** *Let  $\mathcal{C}, \mathcal{D}$  be abelian categories with  $\mathcal{C}$  having enough injectives. Then for any short exact sequence  $0 \rightarrow F \rightarrow G \rightarrow H \rightarrow 0$  in  $[\mathcal{C}, \mathcal{D}]_{\text{LE}}$  there is in  $[\mathcal{C}, \mathcal{D}]_{\text{add}}$  a long exact sequence*

$$0 \rightarrow F \rightarrow G \rightarrow H \rightarrow R^1 F \rightarrow R^1 G \rightarrow R^1 H \rightarrow R^2 F \rightarrow \dots$$

**Proof.** Let  $A \in \mathcal{C}$  with  $I^{\bullet}$  an injective resolution. Since the given short exact sequence is exact on injectives, evaluating in  $I^{\bullet}$  gives a short exact sequence of complexes  $0 \rightarrow F(I^{\bullet}) \rightarrow G(I^{\bullet}) \rightarrow H(I^{\bullet}) \rightarrow 0$ . Taking the long exact sequence of this short exact sequence of complexes gives a long exact sequence

$$0 \rightarrow F(A) \rightarrow G(A) \rightarrow H(A) \rightarrow R^1 F(A) \rightarrow R^1 G(A) \rightarrow R^1 H(A) \rightarrow R^2 F(A) \rightarrow \dots$$

Independence in the choice of resolution and naturality in  $A$  both follow from the naturality of the long exact sequence of a short exact sequence of complexes.  $\square$

**Remark 2.32.** This lemma suggests (although doesn't fully prove) that 'the  $p$ -derived functor of forgetting a functor is left exact is the  $p$ -th derived functor functor', which is a somewhat amusing sentence.

**Lemma 2.33.** *Let  $\mathcal{C}, \mathcal{D}, F, G$  be as in Proposition 2.15. If  $\mathcal{B}$  is another abelian category with enough injectives and  $H : \mathcal{B} \rightarrow \mathcal{C}$  left exact, then there is in  $[\mathcal{B}, \mathcal{D}]_{\text{LE}}$  a short exact sequence*

$$0 \rightarrow (\widetilde{R^1 \tilde{F}}) \circ H \rightarrow \widetilde{G \circ H} \rightarrow \tilde{G} \circ H \rightarrow 0$$



**Proof.** By Proposition 2.15 there is in  $[\mathcal{C}, \mathcal{D}]_{\text{add}}$  a short exact sequence  $0 \rightarrow R^1\tilde{F} \rightarrow G \rightarrow \tilde{G} \rightarrow 0$ . Then applying the exact functors  $(-) \circ H : [\mathcal{C}, \mathcal{D}]_{\text{add}} \rightarrow [\mathcal{B}, \mathcal{D}]_{\text{add}}$  (see Remark 2.27) and  $\underline{(-)} : [\mathcal{B}, \mathcal{D}]_{\text{add}} \rightarrow [\mathcal{B}, \mathcal{D}]_{\text{LE}}$  (see Proposition 2.29) gives the desired short exact sequence (note that  $\tilde{G} \circ H \simeq \tilde{G} \circ H$  since  $H$  is left exact).  $\square$

### 3 Pro-objects as functors

In this chapter we develop the theory of the category  $\vec{\mathcal{C}}$  of left exact functors from a locally small abelian category  $\mathcal{C}$  to  $\mathbf{Ab}$  and in particular how the category of pro-objects  $\mathbf{pro}\text{-}\mathcal{C}$  may be viewed as a subcategory of this. In this thesis we choose to view  $\mathbf{pro}\text{-}\mathcal{C}$  as a subcategory of  $\vec{\mathcal{C}}$  for several reasons. Firstly sheaf homology, the central object of this thesis, gives objects in  $\vec{\mathcal{C}}$  by construction, but it is not immediately clear that it gives objects in  $\mathbf{pro}\text{-}\mathcal{C}$ . Secondly because we know from chapter 2 that  $\vec{\mathcal{C}}$  is abelian we can use this to prove that  $\mathbf{pro}\text{-}\mathcal{C}$  is abelian and give concrete descriptions of exact sequences in  $\mathbf{pro}\text{-}\mathcal{C}$ . Finally in section 5.1 we will see how the category of cosheaves with values in  $\vec{\mathcal{C}}$  is equivalent to a certain left exact functor category. This then explains many properties of the category of cosheaves in  $\mathbf{pro}\text{-}\mathcal{C}$ .

In section 3.1 we define the abelian Yoneda embedding  $\mathcal{C} \rightarrow \vec{\mathcal{C}}$  and introduce some notation for it. In section 3.2 we describe limits, colimits and in particular exact sequences in  $\vec{\mathcal{C}}$ . In section 3.3 we define the category  $\mathbf{pro}\text{-}\mathcal{C}$  of pro-objects in  $\mathcal{C}$  as a full subcategory of  $\vec{\mathcal{C}}$  and develop a necessary and sufficient condition for a functor in  $\vec{\mathcal{C}}$  to be a pro-object. We then use this to show that it is also an abelian category with the inclusion  $\mathbf{pro}\text{-}\mathcal{C} \rightarrow \vec{\mathcal{C}}$  exact. While of theoretical importance, the distinction between  $\mathbf{pro}\text{-}\mathcal{C}$  and  $\vec{\mathcal{C}}$  is of no importance when calculating examples since  $\vec{\mathcal{C}}$  is already an abelian category with explicit exact sequences. In section 3.4 we define *strict* objects of  $\vec{\mathcal{C}}$  which are roughly objects with well-behaved quotients in  $\mathcal{C}$ . The strict pro-objects are exactly those who are the limit of a cofiltered diagram in  $\mathcal{C}$  with epic transition morphisms. If  $\mathcal{C} = \mathbf{Ab}$  these are exactly the pro-objects representable by localic groups (generalisations of topological groups) and hence strictness is an important necessary condition for representing functors with topological groups. Finally in section 3.5 we extend the tensor product on  $\mathbf{Ab}$  to  $\vec{\mathbf{Ab}}$  and use it to generalise sheaf homology to coefficients in  $\vec{\mathbf{Ab}}$ . This then allows us to formulate a universal coefficient theorem for sheaf homology with constant coefficients.

For the rest of this section  $\mathcal{C}$  is an abelian category which is locally small, i.e. for which the collection of morphisms  $\mathrm{Hom}_{\mathcal{C}}(A, B)$  is a set for all  $A, B \in \mathcal{C}$ .

#### 3.1 The abelian Yoneda embedding

**Definition 3.1.** We write  $\vec{\mathcal{C}}$  for the opposite category of the category  $[\mathcal{C}, \mathbf{Ab}]_{\mathrm{LE}}$  of left exact functors from  $\mathcal{C}$  to  $\mathbf{Ab}$ . Thus a morphism  $\eta : F \rightarrow G$  in  $\vec{\mathcal{C}}$  is a natural transformation  $\eta : G \Rightarrow F$ . We define the *abelian Yoneda embedding*  $(\vec{-}) : \mathcal{C} \rightarrow \vec{\mathcal{C}}$  to be the covariant functor

$$\begin{aligned} \mathcal{C} &\longrightarrow \vec{\mathcal{C}} \\ A &\longmapsto \left( \begin{array}{ccc} \vec{A} : & \mathcal{C} & \rightarrow \mathbf{Ab} \\ & X & \mapsto \mathrm{Hom}_{\mathcal{C}}(A, X) \\ (f : X \rightarrow Y) & \mapsto & \left( \begin{array}{c} \vec{A}(X) \rightarrow \vec{A}(Y) \\ \psi \mapsto f \circ \psi \end{array} \right) \end{array} \right) \\ (f : A \rightarrow B) &\longmapsto \vec{f} := \left( \begin{array}{ccc} \vec{B} & \Rightarrow & \vec{A} \\ X & \mapsto & \left( \begin{array}{c} \vec{B}(X) \rightarrow \vec{A}(X) \\ \psi \mapsto \psi \circ f \end{array} \right) \end{array} \right) \end{aligned}$$

The notation  $(\vec{-})$  is chosen because then for  $A, B \in \mathcal{C}$  the abelian group  $\vec{A}(B)$  is the abelian group of morphisms  $A \rightarrow B$ . A functor isomorphic to  $\vec{A}$  for some  $A \in \mathcal{C}$  is called *representable*.

**Lemma 3.2.** (Abelian Yoneda Lemma) *Let  $A \in \mathcal{C}$  and  $F \in \vec{\mathcal{C}}$ . Then there exists an isomorphism*

$$F(A) \simeq \text{Hom}_{\vec{\mathcal{C}}}(F, \vec{A})$$

$$s \mapsto \vec{s}$$

*of abelian groups. In particular if  $F = \vec{B}$  for some  $B \in \mathcal{C}$  this bijection is equal to the map from  $\text{Hom}_{\mathcal{C}}(B, A)$  to  $\text{Hom}_{\vec{\mathcal{C}}}(\vec{B}, \vec{A})$  induced by the abelian Yoneda embedding. (i.e. the vec-notations coincide.)*

**Proof.** This proof is identical to the proof of the regular Yoneda Lemma (see for example [18, p. 61]). One only needs to check that the isomorphism from the regular Yoneda Lemma respects the group structures of  $F(A)$  and  $\text{Hom}_{\vec{\mathcal{C}}}(F, \vec{A})$ .  $\square$

The Yoneda Lemma tells us that we can think of an element  $s \in F(A)$  as a morphism from  $F$  to  $\vec{A}$ . It turns out that under this correspondence applying  $F(f)$  for a morphism  $f : A \rightarrow B$  corresponds to post-composition with  $\vec{f}$  in  $\vec{\mathcal{C}}$  and that applying  $\eta_A$  for a morphism  $\eta : G \rightarrow F$  corresponds to precomposing with  $\eta$  in  $\vec{\mathcal{C}}$ . We therefore give notation for these operations that reflects this correspondence better.

**Definition 3.3.** Let  $F \in \vec{\mathcal{C}}$ ,  $f : X \rightarrow Y$  in  $\mathcal{C}$  and  $s \in F(X)$ . If  $F$  is clear from the context we write  $f * s = f *_F s := F(f)(s) \in F(Y)$ .

Let  $F, G \in \vec{\mathcal{C}}$ ,  $\eta : F \rightarrow G$  a morphism in  $\vec{\mathcal{C}}$ ,  $X \in \mathcal{C}$  and  $s \in G(X)$ . If  $X$  and  $F$  are clear from the context, we write  $s * \eta := \eta_X(s) \in F(X)$ .

**Lemma 3.4.** *Let  $F, G \in \vec{\mathcal{C}}$ ,  $\eta : F \rightarrow G$ ,  $A, B \in \mathcal{C}$ ,  $f : A \rightarrow B$ , and  $s \in G(A)$ , then*

1.  $\overrightarrow{f *_G s} = \vec{f} \circ \vec{s}$ ,
2.  $\overrightarrow{s * \eta} = \vec{s} \circ \eta$ ,
3.  $f *_F (s * \eta) = (f *_G s) * \eta$ ,
4. *and if  $g : B \rightarrow C$ , then  $g * \vec{f} = g *_A f = g \circ f$ .*

**Proof.** Left to the reader.  $\square$

**Remark 3.5.** In what follows we will often be working with multiple objects and morphisms in  $\vec{\mathcal{C}}$ , objects and morphisms in  $\mathcal{C}$  and elements of  $F(A)$  for  $F \in \vec{\mathcal{C}}$  and  $A \in \mathcal{C}$ . One can keep track of all these by drawing a diagram with the nodes being both objects of  $\vec{\mathcal{C}}$  and  $\mathcal{C}$  and arrows being morphisms in  $\mathcal{C}$  and  $\vec{\mathcal{C}}$ . Additionally, for  $F \in \vec{\mathcal{C}}$  and  $A \in \mathcal{C}$  we draw arrows  $F \rightarrow A$  for elements of  $F(A)$ . We then say that such a diagram ‘commutes’ if placing an arrow above the necessary symbols yields a commutative diagram in  $\vec{\mathcal{C}}$ . For example if  $F, G \in \vec{\mathcal{C}}$  and  $A, B \in \mathcal{C}$  then a diagram

$$\begin{array}{ccc} F & \xrightarrow{\eta} & G \\ r \downarrow & \swarrow s & \downarrow t \\ A & \xrightarrow{f} & B \end{array}$$

means that  $r \in F(A)$ ,  $s \in G(A)$ ,  $t \in G(B)$  and  $f : A \rightarrow B$ . Furthermore this diagram ‘commutes’ when  $f *_G s = t$  and  $r = s * \eta$ . When reading chapters 3 and 4 we recommend the reader to frequently sketch such diagrams to keep track of all morphisms and functor-output elements involved.

## 3.2 Limits and colimits

In this section we give descriptions of limits and colimits in  $\vec{\mathcal{C}}$ . In particular cofiltered limits will be important as (as we will see in section 3) many objects of  $\vec{\mathcal{C}}$  can be expressed as cofiltered limits of representable objects. We begin by applying the theory from section 2.4 to  $\vec{\mathcal{C}}$ .

**Proposition 3.6.** *Assume  $\mathcal{C}$  has enough injectives, then  $\vec{\mathcal{C}}$  is an abelian category in which*

1. a sequence  $F \xrightarrow{\eta} G \xrightarrow{\epsilon} H$  is exact if and only if the sequence  $H(I) \xrightarrow{\epsilon_I} G(I) \xrightarrow{\eta_I} F(I)$  is exact for all injective  $I \in \mathcal{C}$ ,
2. if  $F \rightarrow G \rightarrow H \rightarrow 0$  is exact then  $0 \rightarrow H(T) \xrightarrow{\epsilon_T} G(T) \xrightarrow{\eta_T} F(T)$  is exact for all  $T \in \mathcal{C}$ ,
3. the cokernel of a morphism  $\eta : F \rightarrow G$  is given by the functor  $T \mapsto \ker \eta_T$ ,
4. the kernel of a morphism  $\eta : F \rightarrow G$  is given by the left-exactification of the functor  $T \mapsto \operatorname{coker} \eta_T$ ,
5. a morphism  $\eta : F \rightarrow G$  is epic if and only if  $\eta_I$  is injective for all injective  $I \in \mathcal{C}$ . In this case  $\eta_T$  is injective for all  $T \in \mathcal{C}$ .
6. a morphism  $\eta : F \rightarrow G$  is monic if and only if  $\eta_I$  is surjective for all injective  $I \in \mathcal{C}$ .

**Proof.** 1 and 2 follow from applying Proposition 2.29 with  $\mathcal{D} = \mathbf{Ab}$  and swapping the direction of morphisms accordingly. 3-6 follow immediately from applying 1 and 2 to specific exact sequences.  $\square$

**Theorem 3.7.** If  $\mathcal{C}$  has enough injectives, then the abelian Yoneda embedding  $(\vec{-}) : \mathcal{C} \rightarrow \vec{\mathcal{C}}$  is exact.

**Proof.** Since  $\vec{\mathcal{C}}$  is abelian and  $(\vec{-})$  is additive it suffices to show that  $(\vec{-})$  maps exact sequences to exact sequences. If  $A \rightarrow B \rightarrow C$  is exact in  $\mathcal{C}$ , then for any injective  $I \in \mathcal{C}$  the functor  $\operatorname{Hom}_{\mathcal{C}}(-, I) : \mathcal{C}^{\operatorname{op}} \rightarrow \mathbf{Ab}$  is exact, so the sequence  $\operatorname{Hom}_{\mathcal{C}}(C, I) \rightarrow \operatorname{Hom}_{\mathcal{C}}(B, I) \rightarrow \operatorname{Hom}_{\mathcal{C}}(A, I)$  is exact. Using Proposition 3.6.1 we conclude that the sequence  $\vec{A} \rightarrow \vec{B} \rightarrow \vec{C}$  is exact in  $\vec{\mathcal{C}}$ .  $\square$

**Remark 3.8.** It is actually not that hard to prove the above without the assumption that  $\mathcal{C}$  has enough injectives by proving that the abelian Yoneda embedding preserves all finite limits and all colimits directly using the universal properties and Lemma 3.2.

**Remark 3.9.** In Proposition 6 of [10] it is shown that  $\vec{\mathcal{C}}$  is an abelian category if  $\mathcal{C}$  is small but does not necessarily have enough injectives. This more general statement is essential to the proof of Mitchell's embedding theorem (see Theorem 1.12 of [29]).

**Example 3.10.** Take  $\mathcal{C} = \mathbf{Ab}$  and consider the exact sequence  $0 \rightarrow \mathbb{Z}/2\mathbb{Z} \rightarrow \mathbb{Z}/4\mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z} \rightarrow 0$ . Notice that for any  $n > 0$  the functor  $\operatorname{Hom}(\mathbb{Z}/n\mathbb{Z}, -) : \mathbf{Ab} \rightarrow \mathbf{Ab}$  is isomorphic to the functor which maps a group  $T$  to its subgroup  $T[n]$  of  $n$ -torsion elements. The morphism  $\mathbb{Z}/4\mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z}$  in the exact sequence then corresponds to a natural transformation which maps each  $T \in \mathbf{Ab}$  to the inclusion  $T[2] \hookrightarrow T[4]$ . Now we can define a functor  $F : \mathbf{Ab} \rightarrow \mathbf{Ab}$  which maps each  $T \in \mathbf{Ab}$  to the cokernel of this inclusion. By Theorem 3.7 and Proposition 3.6 the left-exactification of this functor should be represented by  $\mathbb{Z}/2\mathbb{Z}$ . Indeed if  $I \in \mathbf{Ab}$  is injective it is divisible, so the necessarily injective map

$$F(I) = I[4]/I[2] \longrightarrow I[2], \bar{x} \mapsto 2x$$

is surjective. Thus the functors  $\operatorname{Hom}(\mathbb{Z}/2\mathbb{Z}, -)$  and  $F$  are isomorphic on injectives, so  $\operatorname{Hom}(\mathbb{Z}/2\mathbb{Z}, -)$  is the left-exactification of  $F$ .

**Definition 3.11.** A category  $\mathcal{I}$  is *filtered* if (1) for any two objects  $i, j \in \mathcal{I}$  there exists a  $k \in \mathcal{I}$  and morphisms  $i \rightarrow k \leftarrow j$  and (2) for any two morphisms  $i \rightrightarrows j$  there exists a morphism  $j \rightarrow k$  so that the composites  $i \rightarrow k$  are equal. Similarly  $\mathcal{I}$  is *cofiltered* if  $\mathcal{I}^{\operatorname{op}}$  is filtered. If  $I$  is a partially ordered set, then view it as a category where between two elements  $i, j \in I$  there is exactly 1 morphism  $i \rightarrow j$  if  $i \leq j$  and no morphisms if otherwise. A partially ordered set is directed (resp. codirected) if and only if it is filtered (resp. cofiltered).

**Lemma 3.12.**  $\vec{\mathcal{C}}$  has all colimits and all cofiltered limits and these can be calculated by taking limits and filtered colimits objectwise respectively. The abelian Yoneda embedding  $\mathcal{C} \rightarrow \vec{\mathcal{C}}$  maps colimits to colimits. If  $\mathcal{C}$  has enough injectives, then  $\vec{\mathcal{C}}$  has all limits.

**Proof.** Recall from Remark 2.27 that the category  $[\mathcal{C}, \mathbf{Ab}]_{\operatorname{add}}$  has all limits and colimits and that they are given by taking the (co)limit objectwise. Now consider the full subcategory  $[\mathcal{C}, \mathbf{Ab}]_{\operatorname{LE}}$ . Notice that limits and filtered colimits of diagrams in  $[\mathcal{C}, \mathbf{Ab}]_{\operatorname{LE}}$  taken in  $[\mathcal{C}, \mathbf{Ab}]_{\operatorname{add}}$  are again objects of  $[\mathcal{C}, \mathbf{Ab}]_{\operatorname{LE}}$

because limits and filtered colimits in  $\mathbf{Ab}$  are left exact. It follows that  $[\mathcal{C}, \mathbf{Ab}]_{\text{LE}}$  has all limits and filtered colimits. Additionally if  $\mathcal{C}$  has enough injectives, then by Lemma 2.25  $[\mathcal{C}, \mathbf{Ab}]_{\text{LE}}$  is a reflective subcategory of  $[\mathcal{C}, \mathbf{Ab}]_{\text{add}}$ , so by Lemma 2.23  $[\mathcal{C}, \mathbf{Ab}]_{\text{LE}}$  has all colimits. Swapping the direction of arrows accordingly gives the required statements for  $\vec{\mathcal{C}}$ . The proof that the abelian Yoneda embedding preserves colimits is left to the reader.  $\square$

**Lemma 3.13.** *Let  $D : \mathcal{I} \rightarrow \mathcal{C}$  be a cofiltered diagram in  $\mathcal{C}$ . Let  $F \in \vec{\mathcal{C}}$  together with for each  $i \in \mathcal{I}$  an element  $s_i \in F(D(i))$  so that for each morphism  $p : i \rightarrow j$  in  $\mathcal{I}$  we have  $s_i *_F D(p) = s_j$ . (That is, the morphisms  $(\vec{s}_i)_{i \in \mathcal{I}}$  form a cone of  $F$  over  $(\vec{-}) \circ D$  in  $\vec{\mathbf{Ab}}$ .) Then  $F$  together with the morphisms  $\vec{s}_i$  for  $i \in \mathcal{I}$  are the limit of  $D$  in  $\vec{\mathcal{C}}$  (i.e. the limit of  $(\vec{-}) \circ D$ ) if and only if for each  $T \in \mathcal{C}$  the following holds:*

1. *For each  $t \in F(T)$  there exists an  $i \in \mathcal{I}$  and a  $f \in \text{Hom}(D(i), T)$  so that  $f *_F s_i = t$ .*
2. *For each  $i \in \mathcal{I}$  and  $f \in \text{Hom}(D(i), T)$  so that  $f *_F s_i = 0$  there exists  $p : j \rightarrow i$  in  $\mathcal{I}$  so that  $f \circ D(p) = 0$ .*

**Proof.** Let  $G$  be the limit of  $(\vec{-}) \circ D$  in  $\vec{\mathcal{C}}$ . The above described cone gives a morphism  $\varphi : F \rightarrow G$ . Now  $\varphi$  is an isomorphism if and only if  $\varphi_T : G(T) \rightarrow F(T)$  is an isomorphism for each  $T \in \mathcal{C}$ . Fix a  $T \in \mathbf{Ab}$ . We will show that condition 1 is equivalent to  $\varphi_T$  being surjective and condition 2 to  $\varphi_T$  being injective. By Lemma 3.12,  $G(T)$  can be defined as the colimit of the filtered diagram  $\mathcal{I}^{\text{op}} \rightarrow \mathbf{Ab}$  which maps each  $i \in \mathcal{I}$  to  $\vec{D}(i)(T)$  and each morphism  $p : i \rightarrow j$  to the morphism  $\vec{D}(p) : \vec{D}(j) \rightarrow \vec{D}(i)$ . By the definition of the Yoneda embedding this is equivalent to the diagram  $\mathcal{I}^{\text{op}} \rightarrow \mathbf{Ab}$  which maps each  $i \in \mathcal{I}$  to  $\text{Hom}(D(i), T)$  and each morphism  $p : i \rightarrow j$  to  $- \circ D(p) : \text{Hom}(D(j), T) \rightarrow \text{Hom}(D(i), T)$ . Thus  $G(T)$  consists of equivalence classes of pairs  $(i, f)$  where  $i \in \mathcal{I}$  and  $f \in \text{Hom}(D(i), T)$  where two pairs  $(i, f), (j, g)$  are considered the same if there exist  $i \xleftarrow{p} k \xrightarrow{q} j$  in  $\mathcal{I}$  so that  $f \circ D(p) = g \circ D(q)$ . Now the cone  $(\vec{s}_i)_{i \in \mathcal{I}}$  gives rise to a cocone over this diagram consisting of  $F(T)$  and for each  $i \in \mathcal{I}$  the map  $- *_F s_i : \text{Hom}(D(i), T) \rightarrow F(T)$ , i.e. the map  $- *_F s_i : \text{Hom}(D(i), T) \rightarrow F(T)$ . Then  $\varphi_T$  is given by the homomorphism induced by this cocone, so  $\varphi_T$  maps a class  $[(i, f)] \in G(T)$  to  $f *_F s_i \in F(T)$ . Now clearly condition 1 is equivalent to the surjectivity of this map and after taking another look at the equivalence relation defining  $G(T)$  it is also easy to see that condition 2 is equivalent to the injectivity of this homomorphism.  $\square$

**Lemma 3.14.** *Assume  $\mathcal{C}$  has enough injectives. Left-exactification, considered as a functor  $[\mathcal{C}, \mathbf{Ab}]_{\text{add}} \rightarrow [\mathcal{C}, \mathbf{Ab}]_{\text{add}}$ , maps filtered colimits to filtered colimits and considered as a (contravariant) functor  $[\mathcal{C}, \mathbf{Ab}]_{\text{add}} \rightarrow \vec{\mathcal{C}}$  maps colimits to limits.*

**Proof.** Since left-exactification is a reflection to the subcategory  $[\mathcal{C}, \mathbf{Ab}]_{\text{LE}}$  of  $[\mathcal{C}, \mathbf{Ab}]_{\text{add}}$  it maps colimits in  $[\mathcal{C}, \mathbf{Ab}]_{\text{add}}$  to colimits in  $[\mathcal{C}, \mathbf{Ab}]_{\text{LE}}$  (Lemma 2.23), i.e. limits in  $\vec{\mathcal{C}}$ . By Lemma 3.12 filtered colimits in  $[\mathcal{C}, \mathbf{Ab}]_{\text{LE}}$  are the same as in  $[\mathcal{C}, \mathbf{Ab}]_{\text{add}}$ , so left-exactification considered as a functor  $[\mathcal{C}, \mathbf{Ab}]_{\text{add}} \rightarrow [\mathcal{C}, \mathbf{Ab}]_{\text{add}}$  maps filtered colimits to filtered colimits.  $\square$

### 3.3 Pro-objects

**Definition 3.15.** We say that a functor  $F \in \vec{\mathcal{C}}$  is a *pro-object* of  $\mathcal{C}$  if  $F$  is a limit of a cofiltered diagram of representable functors in  $\vec{\mathcal{C}}$ . We write  $\mathbf{pro}\text{-}\mathcal{C}$  for the full subcategory of  $\vec{\mathcal{C}}$  of pro-objects. A pro-object of  $\mathbf{Ab}$  is called an *abelian pro-group*.

**Example 3.16.** Let  $p$  be a prime and consider the left exact functor  $F \in \vec{\mathbf{Ab}}$  which maps a  $T \in \mathbf{Ab}$  to its subgroup  $T[p^\infty]$ : the subgroup of elements  $t \in T$  so that  $p^n t = 0$  for some  $n \geq 0$ . For a fixed  $T \in \mathbf{Ab}$  the group  $T[p^\infty]$  can be expressed as the union of the  $p^n$  subgroups  $T[p^n]$  for  $n \geq 0$ , i.e. as the colimit

of the (filtered) diagram

$$T[p^0] \rightarrow T[p^1] \rightarrow T[p^2] \rightarrow T[p^3] \rightarrow \dots$$

where each map is the inclusion. By Lemma 3.12,  $F$  is the limit of the diagram  $(-)[p^0] \leftarrow (-)[p^1] \leftarrow (-)[p^2] \leftarrow \dots$  in  $\vec{\mathbf{Ab}}$  where each morphism is the natural transformation given by the inclusion on each object. For a fixed  $n \geq 0$  the functor  $T \mapsto T[p^n]$  is represented by  $\mathbb{Z}/p^n\mathbb{Z}$  since there is a natural isomorphism  $\text{Hom}(\mathbb{Z}/p^n\mathbb{Z}, T) \simeq T[p^n]$  given by  $f \mapsto f(\bar{1})$ . Under the abelian Yoneda embedding the natural transformations  $(-)[p^n] \Rightarrow (-)[p^{n+1}]$  are induced by the quotient maps  $\mathbb{Z}/p^{n+1}\mathbb{Z} \rightarrow \mathbb{Z}/p^n\mathbb{Z}$ . We conclude that  $F$  is the limit in  $\vec{\mathbf{Ab}}$  of the diagram

$$0 \leftarrow \mathbb{Z}/p\mathbb{Z} \leftarrow \mathbb{Z}/p^2\mathbb{Z} \leftarrow \mathbb{Z}/p^3\mathbb{Z} \leftarrow \dots \quad (3.1)$$

and hence  $F$  is a pro-object.

**Example 3.17.** Consider the left exact functor  $F \in \vec{\mathbf{Ab}}$  which maps a  $T \in \mathbf{Ab}$  to the abelian group of tails of sequences in  $T$ , i.e. the group of sequences in  $T$  modulo eventually zero sequences. For  $T \in \mathbf{Ab}$  we can express  $F(T)$  as the colimit of the diagram

$$\prod_{i \geq 0} T \rightarrow \prod_{i \geq 1} T \rightarrow \prod_{i \geq 2} T \rightarrow \prod_{i \geq 3} T \rightarrow \dots$$

where each morphism removes the first coordinate. For each  $n \geq 0$  we have a natural isomorphism  $\text{Hom}(\bigoplus_{i \geq n} \mathbb{Z}, T) \simeq \prod_{i \geq n} \text{Hom}(\mathbb{Z}, T) \simeq \prod_{i \geq n} T$  with each projection map  $\prod_{i \geq n} T \rightarrow \prod_{i \geq n+1} T$  being induced by the inclusion  $\bigoplus_{i \geq n+1} \mathbb{Z} \hookrightarrow \bigoplus_{i \geq n} \mathbb{Z}$ . By Lemma 3.12 it follows that  $F$  is the limit in  $\vec{\mathbf{Ab}}$  of the diagram

$$\bigoplus_{i \geq 0} \mathbb{Z} \leftarrow \bigoplus_{i \geq 1} \mathbb{Z} \leftarrow \bigoplus_{i \geq 2} \mathbb{Z} \leftarrow \bigoplus_{i \geq 3} \mathbb{Z} \leftarrow \dots \quad (3.2)$$

and hence  $F$  is a pro-object.

**Remark 3.18.** By Lemma 1.1 of [16] any pro-object can also be expressed as the limit of a diagram of representable objects indexed by a *codirected set*, i.e. an inverse system of objects in  $\mathcal{C}$ .

**Remark 3.19.** For an arbitrary category  $\mathcal{C}$  there are two main ways of defining the category  $\mathbf{pro}\text{-}\mathcal{C}$ . Firstly one can define  $\mathbf{pro}\text{-}\mathcal{C}$  to be the category with inverse systems  $(A_i)_{i \in I}$  (diagrams indexed by codirected sets) in  $\mathcal{C}$  as objects and the set of morphisms between inverse systems  $(A_i)_{i \in I}$  and  $(B_i)_{i \in I}$  defined as

$$\lim_{\substack{\longleftarrow \\ j \in J}} \text{colim}_{i \in I} \text{Hom}(A_i, B_j), \quad (3.3)$$

where these limits and colimits are taken in  $\mathbf{Set}$ . See for example [1, p. A.2.1]. This has the advantage of avoiding foundational issues, but it takes much more effort to prove that constructions using pro-objects are invariant under isomorphism.

Secondly if  $\mathcal{C}$  is locally small, then one can consider the standard covariant Yoneda embedding  $\mathcal{C} \rightarrow [\mathcal{C}, \mathbf{Set}]^{\text{op}}$ . The category  $[\mathcal{C}, \mathbf{Set}]^{\text{op}}$  is complete and cocomplete and  $\mathbf{pro}\text{-}\mathcal{C}$  is then defined as the full subcategory consisting of functors which are cofiltered limits of representable functors, see for example [15, Def. 6.1.1]. Using the universal property of limits and the construction of limits in  $[\mathcal{C}, \mathbf{Set}]^{\text{op}}$  one can see that the hom-sets between the limits in  $[\mathcal{C}, \mathbf{Set}]^{\text{op}}$  of inverse systems are the same as in equation (3.3). If  $\mathcal{C}$  is again abelian, then the forgetful functor  $[\mathcal{C}, \mathbf{Ab}]_{\text{add}}^{\text{op}} \rightarrow [\mathcal{C}, \mathbf{Set}]^{\text{op}}$  is fully faithful [15, Prop. 8.2.12]. This functor also preserves cofiltered limits since colimits in  $[\mathcal{C}, \mathbf{Ab}]_{\text{add}}$  and  $[\mathcal{C}, \mathbf{Set}]$  are both computed object-wise and the forgetful functor  $\mathbf{Ab} \rightarrow \mathbf{Set}$  preserves filtered colimits. Thus we can also define pro-objects as the cofiltered limits of representable functors in  $[\mathcal{C}, \mathbf{Ab}]_{\text{add}}^{\text{op}}$ . (This is dual to the discussion in [15, p. 193].) Finally since the inclusion  $\vec{\mathcal{C}} = [\mathcal{C}, \mathbf{Ab}]_{\text{LE}}^{\text{op}} \rightarrow [\mathcal{C}, \mathbf{Ab}]_{\text{add}}^{\text{op}}$  preserves cofiltered limits (since filtered colimits in  $[\mathcal{C}, \mathbf{Ab}]_{\text{add}}$  of left-exact functors are left-exact), we can also define pro-objects as the cofiltered limits of representable functors in  $\vec{\mathcal{C}}$ .

The main purpose of this section is to develop an equivalent condition for a functor  $F \in \vec{\mathcal{C}}$  to be a pro-object, namely *boundedness*. This will be used in section 5.2 to prove that sheaf homology is always a pro-object and it allows us to show that **pro- $\mathcal{C}$**  is closed under taking kernels and cokernels. A special case boundedness, *pro-finitely generatedness*, will be an important tool in showing the sheaf homology of a compact Hausdorff space with coefficients in  $\mathbb{Q}$  is topologically representable.

**Definition 3.20.** Let  $F : \mathcal{C} \rightarrow \mathbf{Ab}$  be a functor and  $\mathcal{A}$  an essentially small collection of objects in  $\mathcal{C}$  closed under finite products and taking subobjects. We say that  $F$  is  $\mathcal{A}$ -*bounded* if for all  $T \in \mathcal{C}$  and  $t \in F(T)$  there exists an  $A \in \mathcal{A}$ ,  $f : A \rightarrow T$  and  $s \in F(A)$  so that  $F(f)(s) = t$ . Finally we say that  $F$  is *bounded* if it is  $\mathcal{A}$ -bounded for such an  $\mathcal{A}$ . If  $\mathcal{C} = \mathbf{Ab}$  or  $R\text{-}\mathbf{Mod}$  for a ring  $R$  and  $\kappa$  is an infinite cardinal, then we say that  $F$  is  $\kappa$ -*bounded* if it is  $\mathcal{A}_\kappa$ -bounded where  $\mathcal{A}_\kappa$  is the class of objects of  $\mathcal{C}$  of cardinality  $\leq \kappa$ . This definition is based on [16, p. 186]. Since each essentially small collection  $\mathcal{A}$  of objects in  $R\text{-}\mathbf{Mod}$  has a cardinal bounding the cardinality of the objects of  $\mathcal{A}$ ,  $F$  is bounded if and only if it is  $\kappa$ -bounded for an infinite cardinal  $\kappa$ . Finally if  $\mathcal{C} = \mathbf{Ab}$  and  $F \in \vec{\mathbf{Ab}}$  we say that  $F$  is *pro-finitely generated* if it is  $\mathcal{A}$ -bounded where  $\mathcal{A}$  is the class of finitely generated abelian groups.

**Lemma 3.21.** Let  $F \in \vec{\mathcal{C}}$ , then the following are equivalent.

1.  $F$  is  $\mathcal{A}$ -bounded.
2. For any  $T \in \mathcal{C}$  and  $t \in F(T)$  there exists an  $A \in \mathcal{A}$ ,  $s \in F(A)$  and  $f : A \rightarrow T$  so that  $f *_F s = t$ .
3. Any morphism from  $F$  to a representable functor factors through a functor represented by an object in  $\mathcal{A}$ .

**Proof.** (2) is simply a rephrasing of (1) using the notation from Definition 3.3. (3) is equivalent by (2) using Lemma 3.4 and the fully faithfulness of the abelian Yoneda embedding.  $\square$

**Example 3.22.** (Continuing Example 3.17) Let  $T \in \mathbf{Ab}$  and  $(t_i)_{i=0}^\infty$  a sequence of elements in  $T$  which represents an element  $\alpha \in F(T)$ . Let  $A \subseteq T$  be the subgroup of  $T$  generated by the  $t_i$  with  $\iota : A \hookrightarrow T$  the inclusion. Now let  $\beta \in F(A)$  be the class of the sequence  $(t_i)_{i=0}^\infty$  considered as a sequence in  $A$ , then  $\iota *_F \beta = \alpha$  since  $\iota$  maps each  $t_i$  to  $t_i$ . Since  $A$  is countable we conclude using the above lemma that  $H$  is  $\aleph_0$ -bounded.

On the other hand  $F$  is not pro-finitely generated. Indeed, consider the group  $T := \bigoplus_{n=0}^\infty \mathbb{Z}$ . If we write  $e_i$  for the  $i$ 'th standard basis vector of  $T$ , then we can define an element  $\alpha \in F(T)$  as the class of the sequence  $(e_n)_{n \geq 0}$ . Assume that there exists a finitely generated abelian group  $A$  and a sequence  $(a_n)_{n \geq 0}$  with class  $\beta \in F(A)$  so that there exists a group homomorphism  $f : A \rightarrow T$  so that  $f *_H \beta = \alpha$ . Notice that  $f *_H \beta$  is represented by the sequence  $(f(a_n))_{n \geq 0}$ . Because  $f[A]$  is finitely generated there exists an  $N \in \mathbb{N}$  so that  $f(a_n)$  is generated by  $e_1, \dots, e_N$  for all  $n \in \mathbb{N}$ , so the tails of the sequences  $(f(a_n))_{n \geq 0}$  and  $(e_n)_{n \geq 0}$  can never be equal. We conclude that  $H$  is not pro-finitely generated.

**Proposition 3.23.** Let  $\mathcal{A}$  be an essentially small collection of objects of  $\mathcal{C}$  closed under finite products and taking subobjects. Then any  $F \in \vec{\mathcal{C}}$  is  $\mathcal{A}$ -bounded if and only if it is the limit in  $\vec{\mathcal{C}}$  of a cofiltered diagram of objects from  $\mathcal{A}$ .

**Proof.** First assume that  $F$  is the limit in  $\vec{\mathbf{Ab}}$  of a cofiltered diagram  $D : \mathcal{I} \rightarrow \mathcal{A}$ , Lemma 3.13.1 then immediately shows that  $F$  is  $\mathcal{A}$ -bounded. For the other direction assume  $F$  is  $\mathcal{A}$ -bounded. Write  $\mathcal{I}$  for the category of pairs  $(A, s)$  where  $A \in \mathcal{A}$  and  $s \in F(A)$  with morphisms between  $(A, s)$  and  $(B, t)$  being morphisms  $f : A \rightarrow B$  in  $\mathcal{C}$  so that  $f *_F s = t$ . Notice that because  $\mathcal{A}$  is essentially small  $\mathcal{I}$  is essentially small. We will show that  $\mathcal{I}$  is cofiltered. First of all let  $0$  be the terminal object in  $\mathcal{C}$ , then because  $F$  is left exact  $F(0)$  will be the terminal object in  $\mathbf{Ab}$ , namely  $\{0\}$ . Then  $(0, 0)$  will be an object in  $\mathcal{I}$  ( $0 \in \mathcal{A}$  as it is the empty product), so  $\mathcal{I}$  contains at least 1 object. Secondly of all let  $(A, s), (B, t) \in \mathcal{I}$ . Now let  $A \times B$  be the product of  $A$  and  $B$  in  $\mathcal{C}$  with projections  $\pi_1, \pi_2$ . Because  $F$  is left exact the set  $F(A \times B)$  will be the product of  $F(A)$  and  $F(B)$  with projections  $\pi_1 *_F -$  and  $\pi_2 *_F -$ . This means that there exists

an element  $r \in F(A \times B)$  so that  $\pi_1 *_F r = s$  and  $\pi_2 *_F r = t$ , so  $\pi_1$  is a morphism from  $(A \times B, r)$  to  $(A, s)$  and  $\pi_2$  is a morphism from  $(A \times B, r)$  to  $(B, t)$  in  $\mathcal{A}$  (here we use that  $\mathcal{A}$  is closed under finite products). Finally let  $f, g : (A, s) \rightarrow (B, t)$  be two parallel morphisms in  $\mathcal{I}$ . Let  $\iota : E \rightarrow A$  be the equaliser of  $f, g$  in  $\mathcal{C}$ . Since  $\iota$  is monic we have  $E \in \mathcal{A}$ . Because  $F$  is left exact the set  $F(E)$  together with the maps  $\iota *_F - : F(E) \rightarrow F(A)$  will be the equaliser of  $f *_F -$  and  $g *_F -$ . This means that the fact that  $f *_F s = t = g *_F s$  implies that there exists some  $r \in F(E)$  so that  $\iota *_F r = s$ . This means that  $\iota$  is a morphism  $(E, r) \rightarrow (A, s)$  in  $\mathcal{I}$  such that  $f \circ \iota = g \circ \iota$ . We conclude that  $\mathcal{I}$  is cofiltered.

We have a clear diagram  $D : \mathcal{I} \rightarrow \mathcal{C}$  given by  $(A, s) \mapsto A$  and for each  $(A, s) \in \mathcal{I}$  an element  $s \in F(A) = F(D((A, s)))$ . Furthermore for each morphism  $f : (A, s) \rightarrow (B, t)$  we have  $f *_F s = t$ . Thus we can use Lemma 3.13 to prove that  $F$  is the limit of  $D$  in  $\vec{\mathcal{C}}$ . Let  $T \in \mathcal{C}$ . Condition 1 is exactly the assumption that  $F$  is  $\mathcal{A}$ -bounded. For condition 2 let  $(A, s) \in \mathcal{I}$  and  $f : A \rightarrow T$  so that  $f *_F s = 0$ . Let  $k : K \rightarrow A$  be the kernel of  $f$  in  $\mathcal{C}$ , then  $K \in \mathcal{A}$ . Because  $F$  is left exact  $k *_F - : F(K) \rightarrow F(A)$  is the kernel of  $f *_F - : F(A) \rightarrow F(T)$ , so from  $f *_F s = 0$  it follows that there exists a  $t \in F(K)$  so that  $k *_F t = s$ . We conclude that we have a morphism  $k : (K, t) \rightarrow (A, s)$  so that  $k \circ f = 0$ , so condition 2 is satisfied. This concludes the proof.  $\square$

**Proposition 3.24.** *Assume that  $\mathcal{C}$  is well-powered, that is each object has an essentially small collection of subobjects. Then a functor  $F \in \vec{\mathcal{C}}$  is a pro-object if and only if it is bounded.*

**Proof.** First suppose  $F$  is the limit of a cofiltered diagram  $D : \mathcal{I} \rightarrow \mathcal{C}$ . Now let  $\mathcal{A}$  be the collection of subobjects of finite products of objects in the image of  $D$ , then by Lemma 3.13.1  $F$  is  $\mathcal{A}$ -bounded. Note that  $\mathcal{A}$  is essentially small because  $\mathcal{C}$  is well-powered. On the other hand if  $F$  is  $\mathcal{A}$ -bounded for some  $\mathcal{A}$ , then by Proposition 3.23 it is the limit of a cofiltered diagram of representable objects.  $\square$

**Lemma 3.25.** *Let  $\eta : F \Rightarrow G$  be a natural transformation between additive functors  $\mathcal{C} \rightarrow \mathbf{Ab}$ . If  $G$  is bounded then the functor  $H : T \mapsto \text{coker } \eta_T$  is bounded.*

**Proof.** Suppose  $G$  is  $\mathcal{A}$ -bounded and let  $T \in \mathbf{Ab}$  with  $\bar{t} \in H(T) = \text{coker } \eta_T$  with  $t \in G(T)$  a representative. Applying the  $\mathcal{A}$ -boundedness of  $G$  gives an  $A \in \mathcal{A}$ ,  $f : A \rightarrow T$  and  $s \in G(A)$  so that  $G(f)(s) = t$ . Then  $H(f)(\bar{s}) = \overline{G(f)(s)} = \bar{t}$ , so  $H$  is  $\mathcal{A}$ -bounded.  $\square$

**Lemma 3.26.** *Assume  $\mathcal{C}$  has enough injectives and let  $F : \mathcal{C} \rightarrow \mathbf{Ab}$  be additive. If  $F$  is  $\mathcal{A}$ -bounded then so is its left-exactification  $\tilde{F}$ .*

**Proof.** Let  $T \in \mathcal{C}$  and  $t \in \tilde{F}(T)$ . First pick an embedding  $i : T \hookrightarrow I$  of  $T$  into an injective object. If we write  $\eta : F \Rightarrow \tilde{F}$  for the reflection morphism, then  $\eta_I$  is an isomorphism. Applying the  $\mathcal{A}$ -boundedness of  $F$  to  $\eta_I^{-1}(\tilde{F}(i)(t)) \in F(I)$  gives an  $A \in \mathcal{A}$ ,  $s \in F(A)$  and  $f : A \rightarrow I$  so that  $F(f)(s) = \eta_I^{-1}(\tilde{F}(i)(t))$ . Using the naturality of  $\eta$  it follows that  $\tilde{F}(i)(t) = \eta_I(F(f)(s)) = \tilde{F}(f)(\eta_A(s))$ , so setting  $\tilde{s} = \eta_A(s)$  we have  $i *_F t = f *_F \tilde{s}$ . Let  $A \times_I T$  be the pullback of  $A$  and  $T$  over  $I$ , then because  $\tilde{F}$  is left exact it follows that there exists an element  $r \in F(A \times_I T)$  as in the following commutative diagram

$$\begin{array}{ccccc}
 \tilde{F} & & & & \\
 \searrow & & & & \\
 & A \times_I T & \xrightarrow{\pi_2} & T & \\
 \swarrow \tilde{s} & \downarrow \pi_1 & & \downarrow i & \\
 & A & \xrightarrow{f} & I & 
 \end{array}$$

(here we use the diagram style of Remark 3.5). Now because monics are stable under pullback  $\pi_1$  is monic, so  $A \times_I T \in \mathcal{A}$ . Thus the upper right triangle in this diagram shows that  $\tilde{F}$  is  $\mathcal{A}$ -bounded.  $\square$

**Theorem 3.27.** *Assume that  $\mathcal{C}$  is well-powered and has enough injectives. Then finite products, quotients and subobjects in  $\vec{\mathcal{C}}$  of pro-objects are pro-objects. It follows that  $\mathbf{pro}\text{-}\mathcal{C}$  is a locally small abelian*



category with the inclusion  $\mathbf{pro}\text{-}\mathcal{C} \rightarrow \vec{\mathcal{C}}$  exact.

**Proof.** Since  $\vec{\mathcal{C}}$  is abelian each monic is the kernel of some morphism. Thus by Proposition 3.24 it suffices to show that for  $\eta : F \rightarrow G$  a morphism with  $F$  bounded that  $\ker \eta$  is bounded. Recall from Proposition 3.6 that  $\ker \eta$  is the left-exactification of the functor  $T \mapsto \operatorname{coker} \eta_T$ , so combining Lemma 3.25 and Lemma 3.26 shows that  $\ker \eta$  is bounded. Now suppose  $\eta : F \twoheadrightarrow G$  is an epimorphism where  $F$  is  $\mathcal{A}$ -bounded. Let  $T \in \mathcal{C}$  and  $t \in G(T)$ . Applying the  $\mathcal{A}$ -boundedness of  $F$  to  $t * \eta$  gives an  $A \in \mathcal{A}$ ,  $s \in F(A)$  and  $f : A \rightarrow T$  so that  $f *_F s = t * \eta$ . Let  $\operatorname{im}(f) \in \mathcal{C}$  be the image of  $f$  with  $p : A \rightarrow \operatorname{im}(f)$  and  $i : \operatorname{im}(f) \rightarrow T$  the natural morphisms. We then have the following commutative diagram in  $\vec{\mathcal{C}}$ :

$$\begin{array}{ccc} F & \xrightarrow{\eta} & G \\ p *_F s \downarrow & \swarrow \text{dashed} & \downarrow t \\ \operatorname{im}(f) & \xrightarrow{i} & T \end{array}$$

Note that by the exactness of the Yoneda embedding  $\vec{\mathcal{C}}$  is also monic. Since in abelian categories all epimorphisms are strong epimorphisms there exists a morphism  $G \rightarrow \overrightarrow{\operatorname{im}(f)}$  so that the diagram commutes, i.e. an element  $r \in G(T)$  so that  $i *_G r = t$ . Notice that  $\operatorname{im}(f)$  is a quotient of  $A$ , so if we could increase  $\mathcal{A}$  to an essentially small class  $\mathcal{A}'$  containing all quotients of objects of  $\mathcal{A}$  then  $F$  would be  $\mathcal{A}'$ -bounded. Indeed because  $\mathcal{C}$  is abelian there is a bijection between the subobjects and quotients of objects in  $\mathcal{C}$ , so  $\mathcal{C}$  being well-powered also implies that each object in  $\mathcal{C}$  has a small class of quotient objects. Thus we can construct the essentially small class  $\mathcal{A}'$  as the class of subobjects of finite products of quotients of objects of  $\mathcal{A}$ . Showing finite products of bounded functors are bounded is left as an exercise to the reader. Finally to see that  $\mathbf{pro}\text{-}\mathcal{C}$  is locally small notice that for  $F, G \in \vec{\mathcal{C}}$  with  $G$  the limit of a cofiltered diagram  $D : \mathcal{I} \rightarrow \mathcal{C}$  in  $\vec{\mathcal{C}}$ , then by the definition of a limit each morphism  $G \rightarrow F$  is determined by the composite morphisms  $G \rightarrow \overrightarrow{D(i)}$  for  $i \in \mathcal{I}$ . Since  $\mathcal{I}$  is small and since  $\operatorname{Hom}_{\vec{\mathcal{C}}}(G, \overrightarrow{D(i)}) \simeq G(D(i))$  is small for each  $i \in \mathcal{I}$  it follows that  $\operatorname{Hom}_{\vec{\mathcal{C}}}(G, F)$  is small.  $\square$

**Remark 3.28.** Note that by defining pro-objects as inverse systems with morphisms as in (3.3) it can be shown that  $\mathbf{pro}\text{-}\mathcal{C}$  is abelian without the assumption that  $\mathcal{C}$  has enough injectives. See for example [1, p. A.]. The main point of this theorem is that exact sequences in  $\mathbf{pro}\text{-}\mathcal{C}$  are the same as in  $\vec{\mathcal{C}}$ .

**Proposition 3.29.** Let  $F \in \vec{\mathcal{C}}$  be a pro-object which is the limit in  $\vec{\mathcal{C}}$  of a cofiltered diagram  $D : \mathcal{I} \rightarrow \mathcal{C}$  which has limit  $A$  in  $\mathcal{C}$ . Then  $\vec{A}$  together with the canonical morphism  $\vec{A} \rightarrow F$  is the coreflection of  $F$  in the subcategory of representable functors. In particular  $A$  is independent of the choice of  $D$ .

**Proof.** For each  $i \in \mathcal{I}$  write  $p_i : A \rightarrow D(i)$  and  $\vec{s}_i : F \rightarrow \overrightarrow{D(i)}$  with  $s_i \in F(D(i))$  for the cone morphisms. Notice that then  $\vec{A}$  together with the morphisms  $\vec{p}_i : \vec{A} \rightarrow \overrightarrow{D(i)}$  form a cone over  $(-) \circ D$ , so by the universal property of the limit we obtain a morphism  $\eta : \vec{A} \rightarrow F$  so that  $\vec{s}_i \circ \eta = \overrightarrow{s_i * \eta} = \vec{p}_i$ , i.e.  $s_i * \eta = p_i$ . Now let  $B \in \mathcal{C}$  and  $\epsilon : \vec{B} \rightarrow F$  a morphism. Then for each  $i \in \mathcal{I}$  we have  $s_i * \epsilon \in \vec{B}(D(i)) = \operatorname{Hom}(B, D(i))$  and for a morphism  $p : i \rightarrow j$  we have  $D(p) *_F s_i = s_j$ , so  $D(p) \circ (s_i * \epsilon) = (D(p) *_F s_i) * \epsilon = s_j * \epsilon$ . Thus  $B$  together with the morphisms  $s_i * \epsilon$  for  $i \in \mathcal{I}$  form a cone over  $D$  in  $\mathcal{C}$ , so there is a morphism  $f : B \rightarrow A$  so that  $p_i \circ f = s_i * \epsilon$  for  $i \in \mathcal{I}$ . Now for all  $i \in \mathcal{I}$  we have  $\vec{s}_i \circ \eta \circ \vec{f} = \vec{p}_i \circ \vec{f} = \vec{s}_i \circ \epsilon$ , so  $\eta \circ \vec{f} = \epsilon$ . Now assume there exists some other morphism  $g : B \rightarrow A$  so that  $\eta \circ \vec{g} = \epsilon$ . Then for all  $i \in \mathcal{I}$  we have  $\vec{p}_i \circ \vec{g} = \vec{s}_i \circ \eta \circ \vec{g} = \vec{s}_i \circ \epsilon = \vec{p}_i \circ \vec{f}$ , so  $p_i \circ g = p_i \circ f$ , so  $g = f$ . We conclude that  $\eta : \vec{A} \rightarrow F$  satisfies the coreflection universal property.  $\square$

**Remark 3.30.** In contexts where pro-objects are defined as inverse systems of objects (with morphisms as in (3.3)), this coreflection operation is denoted simply by ‘lim’.

### 3.4 Strict functors

In this section we define *strict* objects of  $\vec{\mathcal{C}}$  which are roughly objects with well-behaved quotients in  $\mathcal{C}$ . When pro-objects are defined as inverse systems of objects, then the strict pro-objects are exactly those which are isomorphic to an inverse system consisting of epimorphisms (see Corollary 3.37). For this reason they are often called *epic* or *surjective* pro-objects (e.g [22]). Because in this thesis we primarily think of pro-objects as functors it would be somewhat confusing to use the word ‘surjective’. We therefore adopt the convention from [16] to call such pro-objects *strict*. Strict pro-objects are also sometimes called *Mittag-Leffler* (see Remark 3.41). If  $\mathcal{C} = \mathbf{Ab}$  then as shown in [22] the strict pro-objects are exactly the pro-objects representable by localic groups (generalisations of topological groups) and hence strictness is an important necessary condition for representing functors with topological groups. The main purpose of this section is to describe how the strictness of a pro-object relates to the cofiltered diagrams of which it is the limit.

**Definition 3.31.** Let  $F \in \vec{\mathcal{C}}$  and  $A \in \mathcal{C}$ . We will call an element  $s \in F(A)$  *epic* if  $\vec{s} : F \rightarrow \vec{A}$  is an epimorphism in  $\vec{\mathcal{C}}$ . We will refer to pairs  $(A, s)$  with  $A \in \mathcal{C}$  and  $s \in F(A)$  epic as  *$\mathcal{C}$ -valued quotients* of  $F$ . We say that  $F$  is *strict* if for all  $T \in \mathcal{C}$  and  $t \in F(T)$  there exists a  $\mathcal{C}$ -valued quotient  $(A, s)$  of  $F$  and a morphism  $f : A \rightarrow T$  so that  $f *_F s = t$ .

**Lemma 3.32.** Let  $F \in \vec{\mathcal{C}}$ ,  $A \in \mathcal{C}$  and  $s \in F(A)$ , then  $s$  is epic if and only if for any  $T \in \mathcal{C}$  and  $f, g : A \rightarrow T$  we have  $f *_F s = g *_F s \Rightarrow f = g$ .

**Proof.** The implication  $\Rightarrow$  follows from the fully faithfulness of the Yoneda embedding. Conversely assume the second statement holds and let  $G \in \vec{\mathcal{C}}$  with  $\eta, \epsilon : \vec{A} \rightarrow G$  so that  $\eta \circ \vec{s} = \epsilon \circ \vec{s}$ . Now for any  $T \in \mathcal{C}$  and  $t \in G(T)$  we have for  $r_1 = t * \eta, r_2 = t * \epsilon \in \text{Hom}(A, T)$  that

$$\overrightarrow{r_1 *_F s} = \vec{r_1} \circ \vec{s} = \vec{t} \circ \eta \circ \vec{s} = \vec{t} \circ \epsilon \circ \vec{s} = \vec{r_2} \circ \vec{s} = \overrightarrow{r_2 *_F s},$$

so  $r_1 *_F s = r_2 *_F s$ , so  $r_1 = r_2$ , so  $t * \eta = t * \epsilon$ . Since this holds for all  $T$  and  $t$  we conclude that  $\eta = \epsilon$ .  $\square$

**Example 3.33.** (Continuing Example 3.16) We will show that  $F : T \mapsto T[p^\infty]$  is strict. To this end consider a  $T \in \mathbf{Ab}$  with  $t \in T[p^\infty]$ . Write  $A$  for the subgroup of  $T$  generated by  $t$  with  $f : A \rightarrow T$  the inclusion. If we view  $t$  as an element of  $A[p^\infty] = F(A)$ , then  $f *_F t = t$ . It remains to show that  $t \in A[p^\infty]$  is epic which we will do using Lemma 3.32: If  $f, g : A \rightarrow B$  are homomorphisms so that  $f *_F t = g *_F t$ , then  $f(t) = f *_F t = g *_F t = g(t)$ , so  $f = g$  because  $t$  generates  $A$ . We conclude that  $F$  is strict.

**Lemma 3.34.** Let  $F \in \vec{\mathcal{C}}$  and  $D : \mathcal{I} \rightarrow \mathcal{C}$  a cofiltered diagram of which  $F$  is the limit in  $\vec{\mathcal{C}}$ , then all cone morphisms  $F \rightarrow \vec{D(i)}$  are epic if and only if all transition morphisms in  $D$  are epic.

**Proof.** For each  $i \in \mathcal{I}$  write  $\vec{s}_i : F \rightarrow \vec{D(i)}$  for the cone morphism. First assume that each  $\vec{s}_i$  is epic. Now let  $p : i \rightarrow j$  be a morphism, then we have  $\vec{D(p)} \circ \vec{s}_i = \vec{s}_j$ . Because  $\vec{s}_j$  is epic it follows that  $\vec{D(p)}$  is epic, so  $D(p)$  is epic. Now assume that  $D(p)$  is epic for any morphism  $p : i \rightarrow j$  in  $\mathcal{I}$ . Fix an  $i \in \mathcal{I}$  and let  $B \in \mathcal{C}$  and  $f, g : D(i) \rightarrow B$  so that  $f *_F s_i = g *_F s_i$ . We then have  $(f - g) *_F s_i = 0$ , so it follows from Lemma 3.13 that there exists a  $j \in \mathcal{I}$  and a morphism  $p : j \rightarrow i$  so that  $(f - g) \circ D(p) = 0$  and since  $D(p)$  is epic it follows that  $f - g = 0$ , so  $f = g$ . We conclude using Lemma 3.32 that  $\vec{s}_i$  is epic.  $\square$

**Lemma 3.35.** Assume  $\mathcal{C}$  has enough injectives. Let  $F \in \vec{\mathcal{C}}$  be strict, then the image of any morphism from  $F$  to a representable object of  $\vec{\mathcal{C}}$  is representable.

**Proof.** Note that because  $\mathcal{C}$  has enough injectives  $\vec{\mathcal{C}}$  is abelian, so we can indeed speak of the image of morphisms in  $\vec{\mathcal{C}}$ . Let  $A \in \mathcal{C}$  and  $s \in F(A)$ . By the strictness of  $F$  there exists a  $B \in \mathcal{C}$ , an  $f : B \rightarrow A$  and an epic  $t \in F(B)$  so that  $f *_F t = s$ , i.e.  $\vec{f} \circ \vec{t} = \vec{s}$ . If we factor  $f$  as  $B \xrightarrow{q} \text{im}(f) \xrightarrow{i} A$ , then by the

exactness of the abelian Yoneda embedding it follows that  $\vec{q} \circ \vec{t}$  is epic and  $\vec{t}$  is monic. Hence from the equality  $\vec{t} \circ (\vec{q} \circ \vec{t}) = \vec{f} \circ \vec{t} = \vec{s}$  it follows that the morphism  $\vec{q} \circ \vec{t} : F \rightarrow \overrightarrow{\text{im}}(\vec{f})$  is the image of  $\vec{s}$  in  $\vec{\mathcal{C}}$ .  $\square$

**Proposition 3.36.** *Assume  $\mathcal{C}$  has enough injectives. Let  $F \in \vec{\mathcal{C}}$  be strict and  $D : \mathcal{I} \rightarrow \mathcal{C}$  a cofiltered diagram of which  $F$  is the limit in  $\vec{\mathcal{C}}$ . Then  $F$  is also the limit of the induced diagram consisting of the images of the cone morphisms.*

**Proof.** For each  $i \in \mathcal{I}$  write  $\vec{s}_i : F \rightarrow \vec{D}(i)$  for the cone morphism, where  $s_i \in F(D(i))$ . Now using Lemma 3.35 factor each  $s_i$  as  $\iota_i *_F t_i$ , where  $Q(i) \in \mathcal{C}$ ,  $\iota_i : Q(i) \hookrightarrow D(i)$  is monic and  $t_i \in F(Q(i))$  is epic. As before  $\vec{t}_i : F \rightarrow Q(i)$  is the image of  $s_i$ . Now for any  $p : i \rightarrow j$  the equality  $D(\vec{p}) \circ \vec{s}_i = \vec{s}_j$  implies that there exists a unique  $Q(p) : Q(i) \rightarrow Q(j)$  so that  $Q(\vec{p}) \circ \vec{t}_i = \vec{t}_j$  (here we use that  $\vec{\mathcal{C}}$  is abelian and that the abelian Yoneda embedding is fully faithful). It follows that  $Q : \mathcal{I} \rightarrow \mathcal{C}$  is a diagram and that the  $\vec{t}_i$ 's form a cone over it in  $\vec{\mathcal{C}}$ . It remains to check the conditions from Lemma 3.13. To see 1 we can, for each  $T \in \mathbf{Ab}$  and  $t \in F(T)$ , simply apply the other direction of Lemma 3.13 to  $D$  to obtain  $i \in \mathcal{I}$ ,  $f : D(i) \rightarrow T$  so that  $f *_F s_i = t$ . Then precomposing  $f$  with  $\iota_i$  gives a morphism  $Q(i) \rightarrow T$  so that  $(f \circ \iota_i) *_F t_i = f *_F (\iota_i *_F t_i) = f *_F s_i = t$ . For 2, let  $i \in \mathcal{I}$  and  $f : Q(i) \rightarrow T$  so that  $f *_F t_i = 0$ . Since  $t_i$  is epic it follows that  $f = 0$ .  $\square$

**Corollary 3.37.** *Let  $F \in \vec{\mathcal{C}}$  be a pro-object. If  $F$  is the limit of a cofiltered diagram of representable objects in which all transition morphisms are epimorphisms, then  $F$  is strict. If  $\mathcal{C}$  has enough injectives, then the converse also holds.*

**Proof.** Assume  $F$  is the limit of a cofiltered diagram  $D : \mathcal{I} \rightarrow \mathcal{C}$  in  $\vec{\mathcal{C}}$  with all transition morphisms epimorphisms and write  $\vec{s}_i : F \rightarrow \vec{D}(i)$  the cone morphisms. Let  $T \in \mathcal{C}$  and  $t \in F(T)$ . By Lemma 3.13.1 there exists an  $i \in \mathcal{I}$  and a morphism  $f : D(i) \rightarrow T$  so that  $f *_F s_i = t$ . By Lemma 3.34  $s_i$  is epic, so we can conclude that  $F$  is strict. On the other hand if  $F$  is strict, then applying Proposition 3.36 (which assumes  $\mathcal{C}$  has enough injectives) to a cofiltered diagram of which  $F$  is the limit gives a diagram which by Lemma 3.34 has epic transition morphisms.  $\square$

**Remark 3.38.** Corollary 3.37 is also true assuming only that  $\mathcal{C}$  is well-powered. One can show this by first arguing that if  $F$  is  $\mathcal{A}$ -bounded and strict, then all  $\mathcal{C}$ -valued quotients are quotients of objects from  $\mathcal{A}$ . Then one can show if  $F$  is strict that the category of  $\mathcal{C}$ -valued quotients is cofiltered and that  $F$  is the limit of the diagram  $(A, s) \mapsto A$  over it. Thus if  $F$  is strict and  $\mathcal{A}$ -bounded, then it is the limit of a small cofiltered diagram with epic transition morphisms. We omit the details of this argument since for our applications  $\mathcal{C}$  will always have enough injectives.

**Definition 3.39.** Let  $D : \mathcal{I} \rightarrow \mathcal{C}$  be a cofiltered diagram. We say that  $D$  satisfies the *Mittag-Leffler condition* if for each  $i \in \mathcal{I}$  there exists a  $p : j \rightarrow i$  so that  $\text{im}(D(q)) = \text{im}(D(p))$  for all  $q : k \rightarrow i$  which factor through  $p$ .

**Lemma 3.40.** *Assume  $\mathcal{C}$  has enough injectives and let  $D : \mathcal{I} \rightarrow \mathcal{C}$  be a cofiltered diagram. If  $D$  satisfies the Mittag-Leffler condition, then the limit of  $D$  in  $\vec{\mathcal{C}}$  is strict.*

**Proof.** For each  $i \in \mathcal{I}$  write  $Q(i) = \text{im}(D(p)) \subseteq D(i)$  (here  $\subseteq$  denotes inclusion of subobjects) with  $p : j \rightarrow i$  as in the Mittag-Leffler condition (it is not hard to see using the cofilteredness of  $\mathcal{I}$  that this is independent of the choice of  $j$ ) and  $\iota_i : Q(i) \hookrightarrow D(i)$  for the inclusion. Now let  $p : i \rightarrow i'$  in  $\mathcal{I}$  and pick  $j$  and  $j'$  as in the Mittag-Leffler condition. Now choose a  $k \in \mathcal{I}$  with morphisms  $j \leftarrow k \rightarrow j'$  so that the compositions  $k \rightarrow j \rightarrow i \rightarrow i'$  and  $k \rightarrow j' \rightarrow i'$  are equal. Since the morphism  $k \rightarrow i$  factors through  $j \rightarrow i$  we have  $\text{im}(D(k \rightarrow i)) = \text{im}(D(j \rightarrow i)) = Q(i)$ . However since the morphism  $k \rightarrow i'$  factors through  $j' \rightarrow i'$  we have  $\text{im}(D(k \rightarrow i')) = \text{im}(D(j' \rightarrow i')) = Q(j)$ . Since  $\text{im}(D(k \rightarrow i'))$  is just the image of  $\text{im}(D(k \rightarrow i)) = Q(i)$  under  $D(i \rightarrow i')$ , it follows that the homomorphism  $D(i \rightarrow i')$  maps  $Q(i)$  onto  $Q(i')$ . We conclude that the morphism  $D(p) : D(i) \rightarrow D(i')$  restricts to an epimorphism

$Q(p) : Q(i) \rightarrow Q(i')$ , so  $Q : \mathcal{I} \rightarrow \mathcal{C}$  is diagram in which all transition morphisms are epic.

Now let  $F \in \vec{\mathcal{C}}$  be the limit of  $D$  in  $\vec{\mathcal{C}}$  with cone morphisms  $\vec{s}_i : F \rightarrow D(i)$  for  $i \in \mathcal{I}$ . Let  $i \in \mathcal{I}$  and pick  $p : j \rightarrow i$  as in the Mittag-Leffler condition, then because the abelian Yoneda embedding is exact, the embedding of  $Q(i) = \text{im}(D(p))$  in  $\vec{\mathcal{C}}$  will be the image of the morphism  $\vec{D}(p)$  in  $\vec{\mathcal{C}}$ . This means that the fact that  $\vec{s}_i$  factors through  $\vec{D}(p)$  implies that there exists a unique morphism  $\vec{t}_i : F \rightarrow \vec{Q}(i)$  so that  $\vec{t}_i \circ \vec{t}_i = \vec{s}_i$ , so the  $\vec{t}_i$  form a cone over  $(-) \circ D$ . Using Lemma 3.13 we will show through this cone  $F$  is the limit of  $Q$ . To see 1 we can, for each  $T \in \mathbf{Ab}$  and  $t \in F(T)$ , simply apply the other direction of Lemma 3.13 to  $D$  to obtain  $i \in \mathcal{I}, f : D(i) \rightarrow T$  so that  $f *_F s_i = t$ . Then precomposing  $f$  with  $\iota_i$  gives a morphism  $Q(i) \rightarrow T$  so that  $(f \circ \iota_i) *_F t_i = f *_F (\iota_i *_F t_i) = f *_F s_i = t$ . For 2 let  $i \in \mathcal{I}$  and  $f : Q(i) \rightarrow T$  a morphism so that  $f *_F t_i = 0$ . Let  $p : j \rightarrow i$  be as in the Mittag-Leffler condition, then  $D(p)$  factors as  $D(p) = \iota_i \circ g$  with  $g : D(j) \twoheadrightarrow Q(i)$  epic. Since  $\iota_i$  is monic the fact that  $\iota_i *_F (g *_F s_j) = (\iota_i \circ g) *_F s_j = D(p) *_F s_j = s_i = \iota_i *_F t_i$  implies that  $g *_F s_j = t_i$ , so  $(f \circ g) *_F s_j = f *_F (g *_F s_j) = f *_F t_i = 0$ . Thus we can apply the other direction of Lemma 3.13 to  $D$  to obtain a  $q : k \rightarrow j$  so that  $f \circ g \circ D(q) = 0$ . But then  $0 = f \circ g \circ D(q) \circ \iota_k = f \circ g \circ \iota_j \circ Q(q)$ , so  $f \circ g \circ \iota_j = 0$  since  $Q(q)$  is epic. Finally note that  $g \circ \iota_j = Q(p)$  since both are the restriction of  $D(p)$  to  $Q(j)$  and  $Q(i)$ . We conclude that  $f \circ Q(p) = 0$  and thus the proof is complete.  $\square$

**Remark 3.41.** Assuming all intersections of subobjects in  $\mathcal{C}$  exist, the converse also holds, see [32, section 1.3].

**Example 3.42.** The functor  $T \mapsto T[p^\infty]$  from Example 3.16 is strict since it is the limit of the diagram (3.1) which satisfies the Mittag-Leffler condition (in fact all transition morphisms are surjective). On the other hand the functor from Example 3.17 is the limit of the diagram (3.2) which does not satisfy the Mittag-Leffler condition, so by Remark 3.41 this functor is not strict.

### 3.5 Tensor products

Recall that for any abelian group  $A$  we have an adjunction between the functors  $A \otimes -$  and  $\text{Hom}(A, -)$  from  $\mathbf{Ab}$  to  $\mathbf{Ab}$  (here the tensor product is taken over  $\mathbb{Z}$ ). In particular this means that for every  $A, B \in \mathbf{Ab}$  we have a natural isomorphism  $\text{Hom}(A \otimes B, -) \simeq \text{Hom}(A, \text{Hom}(B, -))$ . Written using vec-notation this looks like  $A \otimes \vec{B} \simeq \vec{A} \circ \vec{B}$ . This motivates the following notation.

**Notation 3.43.** For functors  $F, G \in \vec{\mathbf{Ab}}$  we write  $F \otimes G := F \circ G$ .

**Lemma 3.44.** Let  $C_\bullet$  be a chain complex in  $\vec{\mathbf{Ab}}$  and  $F \in \vec{\mathbf{Ab}}$ . Then for any  $p \in \mathbb{Z}$  the homology object  $h_p(C_\bullet \otimes F)$  is the left-exactification of the functor  $\mathbf{Ab} \rightarrow \mathbf{Ab}, T \mapsto h^p(C_\bullet(F(T)))$ .

**Proof.** The proof of this is the exact same as Proposition 2.9 except that we replace  $\text{Hom}(-, T) : \mathbf{Ab}^{\text{op}} \rightarrow \mathbf{Ab}$  with the evaluation functor  $(-)(T) : \vec{\mathbf{Ab}}^{\text{op}} \rightarrow \mathbf{Ab}$  and the tensor-hom adjunction with the definition of  $\otimes$  in  $\vec{\mathbf{Ab}}$ . Note that the functor  $(-)(I) : \vec{\mathbf{Ab}}^{\text{op}} \rightarrow \mathbf{Ab}$  is exact for injective  $I \in \mathbf{Ab}$  by Proposition 3.6.  $\square$

**Remark 3.45.** This lemma generalises Proposition 2.9 and motivates in the exact same way the following mild generalisation of sheaf homology. While we will stick to abelian groups as coefficients in examples, this definition allows us to state many results, including the universal coefficient theorem below, much more cleanly.

**Definition 3.46.** Let  $X$  be a topological space and  $F \in \vec{\mathbf{Ab}}$ . We define the *sheaf homology of  $X$  with coefficients in  $F$* , denoted  $\vec{H}_p(X; F)$ , as the left-exactification of the functor  $H^p(X, -_X) \circ F$ .

**Remark 3.47.** Note that we need only consider left exact  $F : \mathbf{Ab} \rightarrow \mathbf{Ab}$  because for arbitrary additive

$F : \mathbf{Ab} \rightarrow \mathbf{Ab}$  the left-exactifications of  $H^p(X, -_X) \circ F$  and  $H^p(X, -_X) \circ \tilde{F}$  are the same (Lemma 2.11).

**Theorem 3.48.** (Universal Coefficient Theorem for Sheaf Homology) *Let  $X$  be a topological space with  $F \in \tilde{\mathbf{Ab}}$  and  $p \in \mathbb{Z}$ , then there is in  $\tilde{\mathbf{Ab}}$  a short exact sequence*

$$0 \rightarrow \tilde{H}_p(X) \otimes F \rightarrow \tilde{H}_p(X; F) \rightarrow \widetilde{R^1 \tilde{H}_{p-1}(X) \circ F} \rightarrow 0$$

**Proof.** This is Lemma 2.33 applied to  $H^{p-1}(X, -_X)$ ,  $H^p(X, -_X)$  and  $F$  where the connecting homomorphism follows from the exactness of  $-_X$  and the long exact sequence of sheaf cohomology.  $\square$

**Remark 3.49.** For this to reduce to the more familiar exact sequence using  $\text{Tor}_1$  if  $\tilde{H}_{p-1}(X)$  and  $\tilde{H}_p(X)$  are representable, we would need to have for  $A, B \in \mathbf{Ab}$  an isomorphism

$$\widetilde{R^1 \tilde{A} \circ \tilde{B}} \simeq \overline{\text{Tor}_1(A, B)}.$$

Indeed if  $A, B, C \in \mathbf{Ab}$  with  $C$  injective, then applying the tensor-hom adjunction to a projective resolution of  $A$  gives (with a few extra steps) an isomorphism  $\text{Ext}^1(A, \text{Hom}(B, C)) \simeq \text{Hom}(\text{Tor}_1(A, B), C)$  which, keeping in mind that  $R^p \tilde{A} = \text{Ext}^p(A, -)$ , implies the desired isomorphism.

**Corollary 3.50.** *If  $X$  is a topological space and  $p \in \mathbb{Z}$ , then  $\tilde{H}_p(X; \mathbb{Q}) \simeq \tilde{H}_p(X) \otimes \mathbb{Q}$ .*

**Proof.** Notice that for each  $T \in \mathbf{Ab}$  the abelian group  $\tilde{\mathbb{Q}}(T) = \text{Hom}(\mathbb{Q}, T)$  is a  $\mathbb{Q}$ -vector space, so divisible, so injective. Since  $R^1 \tilde{H}_{p-1}(X)$  is zero on injectives, the third term in the universal coefficient theorem is zero.  $\square$

The above corollary and the next two lemmas will be used in section 5.3 to show that  $\tilde{H}_p(X; \mathbb{C})$  is topologically representable if  $X$  is compact Hausdorff.

**Lemma 3.51.** *For any  $F \in \tilde{\mathbf{Ab}}$  the functor  $(-) \otimes F : \tilde{\mathbf{Ab}} \rightarrow \tilde{\mathbf{Ab}}$  commutes with cofiltered limits.*

**Proof.** Let  $D : \mathcal{I} \rightarrow \tilde{\mathbf{Ab}}$  be a cofiltered diagram with limit  $G$ . By Lemma 3.12 this means that  $G(T)$  is the colimit of the diagram  $i \mapsto D(i)(T)$  for each  $T \in \mathcal{C}$  and hence  $G(F(T))$  is the colimit of the diagram  $i \mapsto D(i)(F(T))$ . Using Lemma 3.12 again we conclude that  $G \otimes F = G \circ F$  is the limit of the diagram  $i \mapsto D(i) \circ F = D(i) \otimes F$ .  $\square$

**Proposition 3.52.** *Let  $F \in \tilde{\mathbf{Ab}}$  be pro-finitely generated, then  $F \otimes \tilde{\mathbb{Q}}$  is strict.*

**Proof.** By Proposition 3.23  $F$  is the limit in  $\tilde{\mathbf{Ab}}$  of a cofiltered diagram  $D : \mathcal{I} \rightarrow \mathbf{Ab}$  consisting of finitely generated abelian groups. According to Lemma 3.51  $F \otimes \tilde{\mathbb{Q}}$  will be the limit of the diagram  $i \mapsto D(i) \otimes \mathbb{Q}$ , then this is a diagram of finite-dimensional  $\mathbb{Q}$ -vector spaces and  $\mathbb{Q}$ -linear maps. Since finite dimensional  $\mathbb{Q}$ -vector spaces are artinian, the Mittag-Leffler condition is trivially satisfied, so applying Lemma 3.40 completes the proof.  $\square$

## 4 Topological representation

In this chapter we show that certain topological abelian groups, namely the *pro-discrete* abelian groups, embed fully faithfully in  $\overline{\mathbf{Ab}}$  as their functor of continuous homomorphisms to discrete groups. We characterise the topologically representable functors as the quotients of representable functors and give a different more practical sufficient condition for topological representability. Other than these results this chapter is fairly independent of the remaining chapters, so the reader can easily only take note of these results and continue with the next chapter.

### 4.1 Prodiscrete abelian groups

Recall that a *topological group* is a group  $G$  together with a topology on its underlying set so that group multiplication  $G \times G \rightarrow G$  and multiplicative inversion  $G \rightarrow G$  are continuous maps. A *morphism of topological groups*  $G \rightarrow H$  is a map which is simultaneously a continuous map and a group homomorphism. Of particular interest to us will be abelian topological groups, i.e. topological abelian groups.

**Notation 4.1.** We write  $\mathbf{TopAb}$  for the category consisting of topological abelian groups together with morphisms of topological groups between them. We consider  $\mathbf{Ab}$  as the subcategory of  $\mathbf{TopAb}$  consisting of discrete groups.

**Lemma 4.2.**  $\mathbf{TopAb}$  is complete with the limit of a diagram just being the limit of that diagram in  $\mathbf{Set}$  given the limit topology and the limit group structure.

**Proof.** Left to the reader. □

**Lemma 4.3.** Let  $A$  be a topological abelian group. Let  $H \subseteq A$  be a subgroup, then  $A/H$  is discrete if and only if  $H$  is open.

**Proof.** If  $q : A \rightarrow A/H$  is the quotient map then  $A/H$  is discrete  $\Leftrightarrow \{0\} \subseteq H$  is open  $\Leftrightarrow q^{-1}\{0\} = A/H$  is open in  $A$ . □

**Definition 4.4.** Let  $A$  be a topological abelian group. We define the *prodiscrete completion* of  $A$ , denoted by  $\overline{A}$ , to be the limit in  $\mathbf{TopAb}$  of the diagram  $D : \mathcal{I} \rightarrow \mathbf{TopAb}$  where  $\mathcal{I}$  is the poset of open subgroups of  $A$  and  $D$  maps a  $H \in \mathcal{I}$  to  $A/H$  and an inclusion  $H_1 \subseteq H_2$  to the obvious homomorphism  $A/H_1 \rightarrow A/H_2$ . We will call  $A$  *prodiscrete* if the canonical morphism  $A \rightarrow \overline{A}$  is an isomorphism.

**Lemma 4.5.** Let  $D : \mathcal{I} \rightarrow \mathbf{Top}$  be a diagram consisting of discrete spaces, then its limit in  $\mathbf{Top}$  is Hausdorff.

**Proof.** Let  $L$  be the limit of  $D$  with cone morphisms  $f_i : L \rightarrow D(i)$  for  $i \in \mathcal{I}$ . Let  $x, y \in L$  be two points with  $x \neq y$ . Let  $\{\star\}$  be the one-point space and let  $g, h : \{\star\} \rightarrow L$  be continuous maps mapping  $\star$  to  $x$  and  $y$  respectively. Applying the uniqueness part of the universal property of  $L$  to the fact that  $g \neq h$  shows that there is some  $i \in \mathcal{I}$  so that  $f_i \circ g \neq f_i \circ h$ . In other words  $f_i(g(\star)) \neq f_i(h(\star))$ , so  $f_i(x) \neq f_i(y)$ , so  $f_i^{-1}\{f_i(x)\}$  and  $f_i^{-1}\{f_i(y)\}$  are disjoint open neighbourhoods of  $x$  and  $y$  respectively. □

**Corollary 4.6.** Let  $A \in \mathbf{TopAb}$ , then  $\overline{A}$  is Hausdorff.

**Lemma 4.7.** Let  $D : \mathcal{I} \rightarrow \mathbf{Top}$  be a cofiltered diagram with limit  $L$ . Furthermore let  $X \in \mathbf{Top}$  together with continuous maps  $(f_i)_{i \in \mathcal{I}}$  forming a cone over  $D$  so that for each  $i \in \mathcal{I}$  the image of  $f_i$  is dense in  $D(i)$ , then the image of the induced morphism  $f : X \rightarrow L$  is dense in  $L$ .

**Proof.** For each  $i \in \mathcal{I}$  write  $p_i : L \rightarrow D(i)$  for the canonical morphism. Let  $C \subseteq X$  be a closed subset of  $X$  with  $\text{im}(f) \subseteq C$ . Then  $C$  is an intersection of closed sets of the form  $\bigcup_{i \in S} p_i^{-1}C_i$ , where  $S \subseteq \mathcal{I}$  is finite and each  $C_i \subseteq D(i)$  closed. Using the fact that  $D$  is cofiltered such a finite union is of the form  $p_j^{-1}C_j$  for a  $j \in \mathcal{I}$  and  $C_j \subseteq D(j)$  closed. Let  $p_j^{-1}C_j$  be such a term in the intersection, then from  $\text{im}(f) \subseteq p_j^{-1}C_j$  it follows that

$$\text{im}(f_j) = \text{im}(p_j \circ f) = p_j[\text{im}(f)] \subseteq p_j[p_j^{-1}C_j] \subseteq C_j.$$

Since  $\text{im}(f_j)$  is dense it follows that  $C_j = D(j)$ , so  $p^{-1}C_j = X$ . Thus  $C = X$ .  $\square$

**Corollary 4.8.** *Let  $A \in \mathbf{TopAb}$ , then the image of the canonical morphism  $A \rightarrow \bar{A}$  is dense in  $\bar{A}$ .*

**Proposition 4.9.** *Let  $D : \mathcal{I} \rightarrow \mathbf{Ab}$  be a cofiltered diagram, then its limit in  $\mathbf{TopAb}$  is prodiscrete.*

**Proof.** Let  $L$  be the limit of  $D$  in  $\mathbf{TopAb}$  with cone morphisms  $f_i : L \rightarrow D(i)$  for  $i \in \mathcal{I}$ . Also for each open subgroup  $H \subseteq L$  write  $g_H : \bar{L} \rightarrow L/H$  for the cone morphism. Let  $\varphi : L \rightarrow \bar{L}$  be the canonical morphism, of which we will construct an inverse  $\psi : \bar{L} \rightarrow L$ . Let  $i \in \mathcal{I}$  and write  $h_i : L/\ker f_i \rightarrow D(i)$  for the morphism obtained by applying the homomorphism theorem to  $f_i$ . Notice that  $\ker f_i$  is an open subgroup of  $L$ , so we can define  $\psi_i : \bar{L} \rightarrow D(i)$  to be  $h_i \circ g_{\ker f_i}$ . Also notice that if we write  $\pi : L \rightarrow L/\ker f_i$  for the quotient map, then

$$\psi_i \circ \varphi = h_i \circ g_{\ker f_i} \circ \varphi = h_i \circ \pi = f_i.$$

Now let  $p : i \rightarrow j$  be a morphism in  $\mathcal{I}$ , then it follows from the above equation and that  $D(p) \circ f_i = f_j$  that  $D(p) \circ \psi_i \circ \varphi = \psi_j \circ \varphi$ . By Corollary 4.8 the image of  $\varphi$  is dense. Combining this with the fact that discrete spaces are Hausdorff shows that  $D(p) \circ \psi_i = \psi_j$ . We conclude that  $(\psi_i)_{i \in \mathcal{I}}$  is a cone over  $D$ , so there exists a (unique)  $\psi : \bar{L} \rightarrow L$  so that  $\psi_i = f_i \circ \psi$  for all  $i \in \mathcal{I}$ . We claim that  $\psi$  is the inverse of  $\varphi$ . By Lemma 4.5 and Corollary 4.8 we know that  $\varphi$  is an epimorphism in the category of Hausdorff topological spaces, so it suffices to show that  $\psi$  is a left inverse of  $\varphi$ . To this end notice that for all  $i \in \mathcal{I}$  we have  $f_i \circ \psi \circ \varphi = \psi_i \circ \varphi = f_i = f_i \circ \text{id}$ , so  $\psi \circ \varphi = \text{id}$  by the universal property of the limit.  $\square$

**Corollary 4.10.** *If  $A \in \mathbf{TopAb}$ , then  $\bar{A}$  is prodiscrete.*

**Lemma 4.11.** *Let  $D : \mathcal{I} \rightarrow \mathbf{Ab}$  be a cofiltered diagram with limit  $L$  in  $\mathbf{TopAb}$  with cone morphisms  $f_i : L \rightarrow D(i)$  for  $i \in \mathcal{I}$ . Then each open subgroup of  $L$  is of the form  $f_i^{-1}K$  for some  $i \in \mathcal{I}$  and  $K \subseteq D(i)$  a subgroup.*

**Proof.** Let  $H \subseteq L$  be an open subgroup. Because  $H$  is an open subset it can be written in the form

$$H = \bigcup_{j \in J} \bigcap_{r=1}^{n_j} f_{i_{j,r}}^{-1}[U_{j,r}]$$

where  $J$  is some index set and for each  $j \in J$  we have  $n_j \in \mathbb{Z}_{\geq 1}$  and for each  $1 \leq r \leq n_j$  we have  $i_{j,r} \in \mathcal{I}$  and  $U_{j,r} \subseteq D(i_{j,r})$  is some subset. For each  $j \in J$  pick some  $i_j \in \mathcal{I}$  with morphisms  $p_r : i_j \rightarrow i_{j,r}$  for each  $1 \leq r \leq n_j$ . Then:

$$\bigcap_{r=1}^{n_j} f_{i_{j,r}}^{-1}[U_{j,r}] = \bigcap_{r=1}^{n_j} (D(p_r) \circ f_{i_j})^{-1}[U_{j,r}] = \bigcap_{r=1}^{n_j} f_{i_j}^{-1}[D(p_r)^{-1}[U_{j,r}]] = f_{i_j}^{-1} \left[ \bigcap_{r=1}^{n_j} D(p_r)^{-1}[U_{j,r}] \right]$$

Thus we can write

$$H = \bigcup_{j \in J} f_{i_j}^{-1}[U_j]$$

for subsets  $U_j \subseteq D(i_j)$ . By writing each  $U_j$  as the union of its singleton subsets we can re-index so that

$$H = \bigcup_{j \in J} f_{i_j}^{-1}\{x_j\}$$

Now pick some  $j \in J$  so that  $f_{i_j}^{-1}\{x_j\}$  is non-empty (this exists because  $H$  is non-empty) and pick a  $y \in f_{i_j}^{-1}\{x_j\} \subseteq H$ . Then for any  $z \in \ker f_{i_j}$  we still have  $y + z \in f_{i_j}^{-1}\{x_j\}$ , so  $y + z \in H$ , so  $z = (z + y) - y \in H$ . We conclude that  $\ker f_{i_j} \subseteq H$ , so  $H = f_{i_j}^{-1}[f_{i_j}[H]]$ .  $\square$

## 4.2 Topological representation

**Definition 4.12.** We define the functor  $(\vec{\phantom{x}}) : \mathbf{TopAb} \rightarrow \vec{\mathbf{Ab}}$  by

$$\begin{aligned} \mathbf{TopAb} &\longrightarrow \vec{\mathbf{Ab}} \\ A &\longmapsto \left( \begin{array}{ccc} \vec{A} : & \mathbf{Ab} & \rightarrow \mathbf{Ab} \\ & X & \mapsto \mathrm{Hom}_{\mathbf{TopAb}}(A, X) \\ (f : X \rightarrow Y) & \mapsto & \left( \begin{array}{c} \vec{A}(X) \rightarrow \vec{A}(Y) \\ \psi \mapsto f \circ \psi \end{array} \right) \end{array} \right) \\ (f : A \rightarrow B) &\longmapsto \vec{f} := \left( \begin{array}{ccc} \vec{B} & \Rightarrow & \vec{A} \\ X & \mapsto & \left( \begin{array}{c} \vec{B}(X) \rightarrow \vec{A}(X) \\ \psi \mapsto \psi \circ f \end{array} \right) \end{array} \right) \end{aligned}$$

Here in the expression  $\mathrm{Hom}_{\mathbf{TopAb}}(A, X)$  we give  $X$  the discrete topology. Note that for discrete topological abelian groups this definition agrees with the abelian Yoneda embedding. If  $F \in \vec{\mathbf{Ab}}$  is isomorphic to an object in the image of this functor we say  $F$  is *topologically representable*.

**Proposition 4.13.** *If  $A, B \in \mathbf{TopAb}$  with  $B$  pro-discrete, then the functor  $(\vec{\phantom{x}}) : \mathbf{TopAb} \rightarrow \vec{\mathbf{Ab}}$  induces an isomorphism  $\mathrm{Hom}_{\mathbf{TopAb}}(A, B) \simeq \mathrm{Hom}_{\vec{\mathbf{Ab}}}(\vec{A}, \vec{B})$  of abelian groups. In particular  $(\vec{\phantom{x}})$  is fully faithful on the full subcategory of pro-discrete topological abelian groups.*

**Proof.** Let  $f, g : A \rightarrow B$  be two morphisms and assume that  $\vec{f} = \vec{g}$ . Let  $H \subseteq B$  be an open subgroup with  $q : B \rightarrow B/H$  the quotient map, then  $q \circ f = q \circ \vec{f} = q \circ \vec{g} = q \circ g$ . Since  $B$  is the limit of its discrete quotients it follows that  $f = g$ . Now let  $\eta : \vec{B} \Rightarrow \vec{A}$  be a natural transformation. For each open subgroup  $H \subseteq B$  define  $f_H = q_H * \eta$ , where  $q_H : B \rightarrow B/H$  is the quotient morphism. Now let  $H, K \subseteq B$  be open subgroups with  $H \subseteq K$  and write  $g_{HK} : B/H \rightarrow B/K$  for the obvious homomorphism. Then by the naturality of  $\eta$  we have

$$g_{HK} \circ f_H = g_{HK} \circ (q_H * \eta) = g_{HK} *_{\vec{A}} (q_H * \eta) = (g_{HK} *_{\vec{B}} q_H) * \eta = (g_{HK} \circ q_H) * \eta = q_K * \eta = f_K.$$

Thus  $(f_H)_{H \subseteq B \text{ open subgrp.}}$  is a cone, so there exists a morphism  $f : A \rightarrow B$  so that  $q_H \circ f = f_H$  for all open subgroups  $H \subseteq B$ . We claim that  $\vec{f} = \eta$ . To this end let  $T \in \mathbf{Ab}$  and  $g : B \rightarrow T$ . Set  $H = \ker g$ ,  $q_H : B \rightarrow B/H$  the quotient map and  $\bar{g} : B/H \rightarrow T$  the induced homomorphism. Then again by the naturality of  $\eta$

$$g * \eta = (\bar{g} \circ q_H) * \eta = (\bar{g} *_{\vec{B}} q_H) * \eta = \bar{g} *_{\vec{A}} (q_H * \eta) = \bar{g} \circ f_H = \bar{g} \circ q_H \circ f = g \circ f = g * \vec{f}$$

so  $\eta = \vec{f}$ . We conclude that the homomorphism  $\mathrm{Hom}_{\mathbf{TopAb}}(A, B) \rightarrow \mathrm{Hom}_{\vec{\mathbf{Ab}}}(\vec{A}, \vec{B})$  induced by  $(\vec{\phantom{x}})$  is injective and surjective, so an isomorphism.  $\square$

**Proposition 4.14.** *If  $A \in \mathbf{TopAb}$ , then the canonical morphism  $A \rightarrow \vec{A}$  is mapped to an isomorphism  $\vec{A} \xrightarrow{\sim} \vec{A}$  under  $(\vec{\phantom{x}})$ .*

**Proof.** Let  $T \in \mathbf{Ab}$ . We have to show that the map  $\mathrm{Hom}_{\mathbf{TopAb}}(\vec{A}, T) \rightarrow \mathrm{Hom}_{\mathbf{TopAb}}(A, T)$  given by precomposition by the canonical morphism  $\varphi : A \rightarrow \vec{A}$  is an isomorphism. For surjectivity let  $f : A \rightarrow T$  be a homomorphism. Let  $H = \ker f$  and write  $\varphi_H : A \rightarrow A/H$  for the quotient homomorphism,  $\bar{f} : A/H \rightarrow T$  for the induced homomorphism, and  $q_H : \vec{A} \rightarrow A/H$  for the cone morphism. Then  $(\bar{f} \circ q_H) \circ \varphi = \bar{f} \circ \varphi_H = f$ , so  $\bar{f} \circ q_H$  is a pre-image for  $f$ . For injectivity suppose we



have homomorphisms  $f, g : \bar{A} \rightarrow T$  so that  $f \circ \varphi = g \circ \varphi$ . In particular  $f$  and  $g$  agree on the image of  $\varphi$ , which is dense in  $\bar{A}$  by Corollary 4.8, and their codomain is Hausdorff, so they agree on the whole of  $\bar{A}$ , so  $f = g$ .  $\square$

**Corollary 4.15.** *The full subcategory of  $\mathbf{TopAb}$  consisting of pro-discrete topological abelian groups is a reflective subcategory with the reflection of an object  $A$  being its pro-discrete completion together with the canonical morphism  $A \rightarrow \bar{A}$ .*

**Proof.** Let  $A, B \in \mathbf{TopAb}$  with  $B$  pro-discrete and  $f : A \rightarrow B$  a morphism. It suffices to show that there exists a unique morphism  $\bar{f} : \bar{A} \rightarrow B$  so that  $\bar{f} \circ \varphi = f$ , where  $\varphi : A \rightarrow \bar{A}$  is the canonical morphism. By Proposition 4.14 the morphism  $\vec{\varphi}$  in  $\vec{\mathbf{Ab}}$  is invertible, so we can construct the morphism  $\vec{f} \circ \vec{\varphi}^{-1} : \bar{A} \rightarrow B$ . By Proposition 4.13 this morphism is of the form  $\bar{f}$  for some  $\bar{f} : \bar{A} \rightarrow B$ . Now notice that  $\bar{f} \circ \varphi = \vec{f} \circ \vec{\varphi} = \vec{f} \circ \vec{\varphi}^{-1} \circ \vec{\varphi} = \vec{f}$ , so by Proposition 4.13 it follows that  $\bar{f} \circ \varphi = f$ . Suppose that there exists another morphism  $g : \bar{A} \rightarrow B$  so that  $g \circ \varphi = f$ , then  $\vec{g} \circ \vec{\varphi} = \vec{f}$ , so  $\vec{g} = \vec{f} \circ \vec{\varphi}^{-1} = \vec{f}$ , so  $g = \bar{f}$ .  $\square$

**Lemma 4.16.** *Let  $A \in \mathbf{TopAb}$ , then  $\bar{A}$  is a strict abelian pro-group.*

**Proof.** Let  $\kappa$  be the cardinality of  $A$ . Let  $T \in \mathbf{Ab}$  and  $f \in \vec{A}(T)$ , i.e.  $f : A \rightarrow T$  is a continuous homomorphism. Notice that we can factor  $f$  as  $\bar{f} \circ q$ , where  $q : A \rightarrow A/\ker f$  is the quotient map and  $\bar{f} : A/\ker f \rightarrow T$  the induced map. In other words we have  $q \in \vec{A}(A/\ker f)$  so that  $\bar{f} *_{\bar{A}} q = f$ . From the fact that  $A/\ker f$  has cardinality  $\leq \kappa$  and that  $q$  is epic as an element of  $\vec{A}(A/\ker f)$  (this follows immediately from the fact that  $q$  is surjective on the level of sets and Lemma 3.32), we conclude that  $\bar{A}$  is  $\kappa$ -bounded and strict. Applying Proposition 3.24 completes the proof.  $\square$

### 4.3 Sufficient conditions for topological representability

**Lemma 4.17.** *Let  $D : \mathcal{I} \rightarrow \mathbf{Ab}$  be a cofiltered diagram and let  $L$  be its limit in  $\mathbf{TopAb}$  with for each  $i \in \mathcal{I}$  the cone morphism written  $f_i : L \rightarrow D(i)$ , then the induced morphism from  $\bar{L}$  to the limit of  $D$  in  $\vec{\mathbf{Ab}}$  is monic. If we additionally assume that each  $f_i$  is surjective, it is an isomorphism.*

**Proof.** Let  $F \in \vec{\mathbf{Ab}}$  be the limit of  $D$  in  $\vec{\mathbf{Ab}}$ . Let  $\eta : \bar{L} \rightarrow F$  be the morphism induced by the universal property of the limit in  $\vec{\mathbf{Ab}}$ . Fix an object  $T \in \mathbf{Ab}$ . Recall from Lemma 3.12 that cofiltered limits in  $\vec{\mathbf{Ab}}$  correspond to taking filtered colimits object-wise. Thus  $F(T)$  consists of classes of pairs of the form  $(i, f)$  for  $i \in \mathcal{I}$  and  $f \in \text{Hom}_{\mathbf{Ab}}(D(i), T)$  where two pairs  $(i, f), (j, g)$  induce the same class if there exists a  $k \in \mathcal{I}$  and morphisms  $p : k \rightarrow i, q : k \rightarrow j$  so that  $f \circ D(p) = g \circ D(q)$ . Finally  $\eta_T : F(T) \rightarrow \text{Hom}_{\mathbf{TopAb}}(\bar{L}, T)$  is given by mapping a class  $[(i, f)]$  to  $f \circ f_i$ . We will first show that  $\eta_T$  is surjective when  $T$  is injective. Let  $f : L \rightarrow T$  be a continuous homomorphism, then by Lemma 4.11 there exists some  $i \in \mathcal{I}$  and a subgroup  $H \subseteq D(i)$  so that  $\ker f = f_i^{-1}H$ . Using this we obtain the following commutative diagram

$$\begin{array}{ccc}
 L & \xrightarrow{f} & T \\
 \searrow & & \uparrow \\
 & L/\ker(f) & \\
 \downarrow f_i & \nearrow & \uparrow \\
 \text{im}(f_i) & \xrightarrow{\iota} & D(i)
 \end{array}$$

The dotted line is from the first isomorphism theorem and the morphism from  $\text{im}(f_i)$  to  $L/\ker(f)$  exists due to the fact that  $\ker f_i \subseteq \ker f$ . Finally the dashed morphism from  $D(i)$  to  $T$  exists by the injectivity of  $T$ . Write  $h$  for this morphism, then  $\eta_T([(h, D(i))]) = h \circ f_i = h \circ \iota \circ f_i' = f$ . We conclude that  $\eta_T$  is surjective. Since this holds for all injective  $T \in \mathbf{Ab}$  it follows from Proposition 3.6 that  $\eta$  is monic. Now assume that each  $f_i$  is surjective. To see that  $\eta_T$  is injective, let  $[(i, f)] \in F(T)$  so that  $\eta_T([(i, f)]) = 0$ ,

i.e.  $f \circ f_i = 0$ . Since  $f_i$  is surjective it follows immediately that  $f = 0$ , so  $[(i, f)] = 0$ . We conclude that  $\eta_T$  is an isomorphism for all injective abelian groups  $T$ , so by Proposition 3.6  $\eta$  is an isomorphism.  $\square$

**Corollary 4.18.** *Let  $\mathcal{I}$  be the opposite poset of  $\mathbb{N}$  and let  $D : \mathcal{I} \rightarrow \mathbf{Ab}$  be a diagram so that each transition homomorphism is surjective. Let  $L$  be the limit of  $D$  in  $\mathbf{TopAb}$ , then  $\vec{L}$  (together with the obvious cone-morphisms) is the limit of  $D$  in  $\vec{\mathbf{Ab}}$ .*

**Proof.** Suppose  $D$  consists of the following objects and morphisms:

$$A_0 \xleftarrow{t_1} A_1 \xleftarrow{t_2} A_2 \xleftarrow{t_3} A_3 \xleftarrow{t_4} \dots$$

Now  $L$  consists of all sequences  $(a_i)_{i \geq 0}$  so that for all  $i \geq 1$  we have  $t_i(a_i) = a_{i-1}$ . By assumption each  $t_i$  is surjective. Now fix some  $n \in \mathbb{N}$  and an  $x \in A_n$ . Now using the fact that each  $t_i$  is surjective (and the axiom of choice) pick a sequence  $(a_i)_{i > n}$  so that for each  $i > n + 1$  we have  $t_i(a_i) = a_{i-1}$  and so that  $t_{n+1}(a_{n+1}) = x$ . Completing the sequence for  $i \leq n$  in the obvious way gives a full sequence  $(a_i)_{i \geq 0}$  in  $L$  whose  $i$ 'th index is  $x$ . Thus the cone morphism  $L \rightarrow A_n$  is surjective so we can apply Lemma 4.17.  $\square$

**Definition 4.19.** Let  $\mathcal{C}$  be a locally small abelian category and  $F \in \vec{\mathcal{C}}$ . We say that  $F$  is *sequential* if it is the limit in  $\vec{\mathcal{C}}$  of a diagram in  $\mathcal{C}$  indexed by  $\mathbb{N}^{\text{op}}$ .

**Theorem 4.20.** *Any strict and sequential abelian pro-group is topologically representable.*

**Proof.** Apply Proposition 3.35 to a diagram indexed by  $\mathbb{N}^{\text{op}}$  of which the pro-object is the limit and then apply Corollary 4.18.  $\square$

**Remark 4.21.** A similar condition holds for (not necessarily abelian) pro-groups, see [17, section 6.2].

**Proposition 4.22.** *If  $A \in \mathbf{TopAb}$ ,  $F \in \vec{\mathbf{Ab}}$  and  $\eta : F \rightarrow \vec{A}$  is a morphism in  $\vec{\mathbf{Ab}}$ , then  $\text{coker } \eta$  is topologically representable.*

**Proof.** Let  $\mathcal{I}$  be the partially ordered set of open subgroups  $H \subseteq A$  so that  $q_H * \eta = 0$ , where  $q_H : A \rightarrow A/H$  is the quotient map. If  $H_1, H_2 \in \mathcal{I}$  with quotient maps  $q_i : A \rightarrow A/H_i$  and  $q_{12} : A \rightarrow A/(H_1 \cap H_2)$ , then the map  $A/(H_1 \cap H_2) \xrightarrow{k} A/H_1 \times A/H_2$  induced by the quotient maps  $p_i : A/(H_1 \cap H_2) \rightarrow A/H_i$  is injective. Write  $\pi_i : A/H_1 \times A/H_2 \rightarrow A/H_i$  for the projection, then

$$\pi_i *_F (k *_F (q_{12} * \eta)) = (\pi_i \circ k) *_F (q_{12} * \eta) = p_i *_F (q_{12} * \eta) = (p_i \circ q_{12}) *_F \eta = q_i *_F \eta = 0$$

Since  $F$  preserves products it follows that  $k *_F (q_{12} * \eta) = 0$ . Since  $F$  preserves monics it follows from the injectivity of  $k$  that  $q_{12} * \eta = 0$  and hence  $H_1 \cap H_2 \in \mathcal{I}$ . Thus  $\mathcal{I}$  is cofiltered. Define  $B \in \mathbf{TopAb}$  to be the limit of the diagram  $H \mapsto A/H$  over  $\mathcal{I}$  with  $f_H : B \rightarrow A/H$  the cone maps for  $H \in \mathcal{I}$ . The universal property of the limit gives a continuous homomorphism  $\varphi : A \rightarrow B$  so that  $f_H \circ \varphi = q_H$  for all  $H \in \mathcal{I}$ . Note that because  $q_H$  is surjective each  $f_H$  must also be surjective. We can therefore apply Lemma 4.17 to see that  $B$  is also the limit of the diagram  $H \mapsto A/H$  in  $\vec{\mathbf{Ab}}$  with cone morphisms  $\vec{f}_H$ . Note that for each  $H \in \mathcal{I}$  we have  $\vec{f}_H \circ \vec{\varphi} \circ \eta = (\vec{f}_H \circ \varphi) \circ \eta = \vec{q}_H \circ \eta = \vec{q}_H * \vec{\eta} = 0$ , so by the universal property of the limit in  $\vec{\mathbf{Ab}}$  we have  $\vec{\varphi} \circ \eta = 0$ .

We claim that  $\vec{\varphi}$  is the cokernel of  $\eta$  in  $\vec{\mathbf{Ab}}$ . To this end let  $T \in \mathbf{Ab}$ . Using Proposition 3.6 we see that we must show that the homomorphism  $\vec{\varphi}_T : \vec{B}(T) \rightarrow \vec{A}(T)$  is the kernel of the homomorphism  $\eta_T : \vec{A}(T) \rightarrow F(T)$ . First of all by Lemma 4.7 the image of  $\varphi$  is dense, so for any  $f \in \vec{B}(T)$  we have  $\vec{\varphi}_T(f) = f \circ \varphi = 0$  if and only if  $f = 0$ , so  $\vec{\varphi}_T$  is injective. Secondly we have to show that  $\text{im}(\vec{\varphi}_T) = \ker(\eta_T)$ . Since  $\eta_T \circ \vec{\varphi}_T = (\vec{\varphi} \circ \eta)_T = 0$ , so  $\text{im}(\vec{\varphi}_T) \subseteq \ker(\eta_T)$ . Conversely suppose  $f : A \rightarrow T$  is a continuous homomorphism so that  $f * \eta = 0$ . Write  $\bar{f} : A/H \rightarrow T$  for the induced map and  $q = q_{\ker f} : A \rightarrow A/\ker(f)$  for the quotient map. Because  $F$  is left exact  $F(\bar{f})$  is injective, so we have  $f * \eta = (\bar{f} \circ q) * \eta = \bar{f} *_F (q * \eta) = 0 \Rightarrow q * \eta = 0$ , so we may write  $f = \bar{f} \circ q = \bar{f} \circ f_{\ker f} \circ \varphi = \vec{\varphi}_T(\bar{f} \circ f_{\ker f})$ . We conclude that  $\ker(\eta_T) \subseteq \text{im}(\vec{\varphi}_T)$ , so  $\vec{\varphi}$  is the cokernel of  $\eta$ .  $\square$

**Corollary 4.23.** *A functor  $F \in \vec{\mathbf{Ab}}$  is topologically representable if and only if it is a quotient of a representable functor.*

**Proof.** If there exists an epimorphism  $\vec{A} \twoheadrightarrow F$  for some  $A \in \mathbf{Ab}$ , then because  $\vec{\mathbf{Ab}}$  is abelian this morphism is the cokernel of some morphism, so applying the proposition shows that  $F$  is topologically representable. Conversely if  $A \in \mathbf{TopAb}$  then consider the continuous homomorphism  $A^{\text{discr}} \rightarrow A$  which is the identity on the level of sets but with  $A^{\text{discr}}$  being the underlying group of  $A$  given the discrete topology. The corresponding morphism  $\vec{A}^{\text{discr}} \rightarrow \vec{A}$  is an epimorphism since the induced map  $\text{Hom}_{\mathbf{TopAb}}(A, T) \rightarrow \text{Hom}_{\mathbf{TopAb}}(A^{\text{discr}}, T)$  is injective for all  $T \in \mathbf{Ab}$  (see Proposition 3.6). Thus  $\vec{A}$  is a quotient of a representable functor.  $\square$

**Example 4.24.** Let  $p$  be a prime and consider the topological abelian group of  $p$ -adic integers  $\mathbb{Z}_p$ , i.e. the limit in  $\mathbf{TopAb}$  of the diagram

$$0 \leftarrow \mathbb{Z}/p\mathbb{Z} \leftarrow \mathbb{Z}/p^2\mathbb{Z} \leftarrow \mathbb{Z}/p^3\mathbb{Z} \leftarrow \mathbb{Z}/p^4\mathbb{Z} \leftarrow \dots$$

Using Corollary 4.18 we see that  $\vec{\mathbb{Z}}_p$  is also the limit of this in  $\vec{\mathbf{Ab}}$ , so  $\vec{\mathbb{Z}}_p$  is the functor from Example 3.16 which maps each  $T \in \mathbf{Ab}$  to its  $p^\infty$ -torsion subgroup  $T[p^\infty]$ . Now consider for each  $T \in \mathbf{Ab}$  the inclusion  $T[p^\infty] \hookrightarrow T$ . Since the functor  $T \mapsto T$  is represented by  $\mathbb{Z}$  and because this inclusion is always injective it follows that we have an epimorphism  $\vec{\mathbb{Z}} \twoheadrightarrow \vec{\mathbb{Z}}_p$ .

**Remark 4.25.** It follows from Proposition 3.29 that if  $F \in \vec{\mathbf{Ab}}$  is the limit in  $\vec{\mathbf{Ab}}$  of a cofiltered diagram  $D : \mathcal{I} \rightarrow \mathbf{Ab}$ , then for the limit  $L_1$  of  $D$  the functor  $\vec{L}_1$  together with a canonical morphism  $\eta_1 : \vec{L}_1 \rightarrow F$  (induced by the cone morphisms of  $L_1$  over  $D$  embedded in  $\vec{\mathbf{Ab}}$ ) is the coreflection of  $F$  in the subcategory of representable functors. Using the exact same argument it can be shown that for the limit  $L_2$  of  $D$  taken in  $\mathbf{TopAb}$ , the functor  $\vec{L}_2$  together with the canonical morphism  $\eta_2 : \vec{L}_2 \rightarrow F$  is the coreflection of  $F$  in the subcategory of *topologically* representable functors. Consider the continuous homomorphism  $f : L_1 \rightarrow L_2$  which is just the identity on the level of sets. It is not too hard to see that  $\vec{f} \circ \eta_2 = \eta_1$  and that  $\vec{f}$  is an epimorphism (since  $f$  is surjective on the level of sets) and  $\eta_1$  is monic by Lemma 4.17. We conclude that the coreflection of  $F$  in the subcategory of topologically representable functors is the image of the coreflection morphism  $\vec{L}_1 \rightarrow F$ .

## 5 Pro-homology from cohomology

In this chapter we show that sheaf homology is the same as the cosheaf homology defined by Prasalov in [25], for paracompact Hausdorff spaces the same as various notions of Čech pro-homology and shape pro-homology and the same as singular homology for spaces which are in addition locally contractible. In particular this proves a conjecture of Prasalov. The main idea is to show that all of these pro-homology theories are obtained from their corresponding cohomology functors in the same way using left-exactification. It then suffices to apply known results relating these cohomology theories.

In section 5.1 we show that, for a locally small abelian category  $\mathcal{C}$ , the category of cosheaves with values in  $\mathbf{pro}\text{-}\mathcal{C}$  may be viewed as a subcategory of the category  $[\mathbf{Ab}, \mathbf{Sh}(X, \mathbf{Ab})]_{\text{LE}}^{\text{op}}$  of left exact functors from  $\mathbf{Ab}$  to the category of sheaves of abelian groups on  $X$ . This then reduces the theory of cosheaves with values in  $\mathbf{pro}\text{-}\mathcal{C}$  (namely cosheafification and exact sequences) to the theory of abelian sheaves on  $X$  which is well-understood. Using this equivalence we show that the cosheaf homology of a cosheaf  $\mathcal{A}$  of pro-abelian groups is the same as a straight forward generalisation of sheaf homology applied to the corresponding functor. In particular this shows that the sheaf homology with coefficients in a functor corresponding to a cosheaf of abelian pro-groups is always a pro-group. In section 5.2 we give a proof of this fact independent of the theory of cosheaves.

In section 5.3 we show that cosheaf Čech pro-homology, nerve-based Čech pro-homology and shape pro-homology can all be obtained in the same way from their corresponding cohomology theory. We then use existing theorems comparing sheaf and Čech cohomology to give comparison morphisms between these homology pro-groups, in particular showing they are all the same for paracompact Hausdorff spaces. This proves Conjecture 1.0.3.1 of [25]. We then combine these results with the theory from chapters 3 and 4 to give sufficient conditions for the sheaf homology of compact Hausdorff spaces to be topologically representable.

In section 5.4 we review literature on when sheaf cohomology agrees with singular cohomology and use it to show that singular and sheaf homology are naturally (in the space) isomorphic for paracompact Hausdorff locally contractible spaces. We also show that, up to a technical detail ( $\star$ ) that remains to be checked, more recent literature can be used to show that the paracompact Hausdorff assumption can be dropped. This more general isomorphism applies to locally finite spaces, proving (assuming ( $\star$ ) holds) Conjecture 1.0.5 of [25]. We then use this to give examples of finite spaces for which each of the homology pro-groups from section 5.3 are different.

Finally in section 5.5 we briefly discuss another family of homology theories which attempts to generalise algebraic topology beyond locally contractible spaces, the Steenrod-style theories, and discuss why it will likely be hard to compare these theories to sheaf homology.

### 5.1 Cosheaves

We begin this section by considering the following straight forward generalisation of sheaf homology. Instead of precomposing the  $p$ -th sheaf cohomology functor  $H^p(X, -) : \mathbf{Sh}(X) \rightarrow \mathbf{Ab}$  with functors of the form  $(-)_X \circ F$  with  $F : \mathbf{Ab} \rightarrow \mathbf{Ab}$  left exact, we can precompose  $H^p(X, -)$  with an arbitrary left exact functor  $F : \mathbf{Ab} \rightarrow \mathbf{Sh}(X)$ . Note again we need only consider left exact  $F$  by Lemma 2.11. This gives the following definition.

**Definition 5.1.** Let  $X$  be a topological space and  $F : \mathbf{Ab} \rightarrow \mathbf{Sh}(X)$  be a left exact functor. We define the *sheaf homology of  $X$  with coefficients in  $F$* , denoted  $\tilde{H}_p(X; F)$ , to be the left-exactification of the functor  $H^p(X, -) \circ F$ . If the distinction is necessary we will refer to the sheaf homology from Definition 3.46 as *sheaf homology with constant coefficients*.

Just as how left exact functors  $\mathbf{Ab} \rightarrow \mathbf{Ab}$  correspond (up to a mild size condition) to abelian pro-groups, one may wonder which objects correspond to left exact functors  $\mathbf{Ab} \rightarrow \mathbf{Sh}(\mathbf{Ab})$ . As we will see below they correspond to cosheaves of abelian pro-groups. Under this correspondence the functors of the form  $(-)_X \circ F$  for  $F : \mathbf{Ab} \rightarrow \mathbf{Ab}$  left exact correspond to constant cosheaves. Before we get to the details let us first fix our terminology surrounding sheaves and cosheaves in general categories.

**Definition 5.2.** Let  $X$  be a topological space and  $\mathcal{C}$  a category with all coproducts. We define a (pre)cosheaf on  $X$  with values in  $\mathcal{C}$  to be a (pre)sheaf with values in  $\mathcal{C}^{\text{op}}$ . Equivalently a precosheaf is a covariant functor  $\mathcal{A} : \text{Op}(X) \rightarrow \mathcal{C}$  (here  $\text{Op}(X)$  is the poset of open subsets of  $X$ ) and it is a cosheaf if for any open  $U \in \text{Op}(X)$  and open cover  $(U_i)_{i \in I}$  of  $U$  the diagram

$$\bigoplus_{(i,j) \in I^2} \mathcal{A}(U_i \cap U_j) \rightrightarrows \bigoplus_{i \in I} \mathcal{A}(U_i) \rightarrow \mathcal{A}(U)$$

is a coequaliser diagram. Here the rightmost arrow is induced by the corestriction morphisms  $\mathcal{A}(U_i) \rightarrow \mathcal{A}(U)$  and the leftmost two arrows are constructed using, for each  $(i, j) \in I^2$ , either the corestriction morphism  $\mathcal{A}(U_i \cap U_j) \rightarrow \mathcal{A}(U_i)$  or  $\mathcal{A}(U_i \cap U_j) \rightarrow \mathcal{A}(U_j)$ . If  $\mathcal{C}$  is a category we write  $\text{PSh}(X, \mathcal{C})$  and  $\text{PCoSh}(X, \mathcal{C})$  for the category of presheaves and precosheaves in  $\mathcal{C}$  on  $X$  respectively. If  $\mathcal{C}$  has all products we write  $\text{Sh}(X, \mathcal{C})$  for the category of sheaves in  $\mathcal{C}$  and if  $\mathcal{C}$  has all coproducts  $\text{CoSh}(X, \mathcal{C})$  for the category of cosheaves in  $\mathcal{C}$ .

Since this definition is perfectly dual to that of a sheaf one would expect cosheaves to be just as well-behaved as sheaves. Unfortunately if  $\mathcal{C}$  is an arbitrary abelian category, then the proof that  $\text{Sh}(X, \mathcal{C})$  is a reflective subcategory of  $\text{PSh}(X, \mathcal{C})$  and that it is abelian requires additional assumptions. For example one of the requirements is that  $\mathcal{C}$  needs to be AB5, i.e. it needs to have all filtered colimits and filtered colimits need to be exact [15, Thm. 17.4.9]. This is satisfied by  $\mathbf{Ab}$ , but not by its dual category  $\mathbf{Ab}^{\text{op}}$  so the same arguments won't apply to cosheaves in  $\mathbf{Ab}$ . This forced authors such as Bredon in [7, ch. VI] to resort to defining complicated notions such as *smooth* precosheaves (precosheaves for which a well-behaved reflector did exist). In [24] Prasolov showed that, on a Grothendieck site  $X$  and for a category  $\mathcal{C}$  with is either locally presentable or co-locally presentable, there exists a coreflection  $\text{PCoSh}(X, \mathcal{C})$  to  $\text{CoSh}(X, \mathcal{C})$ . However most enlightening is his idea to (if  $\mathcal{C}$  is cocomplete) first embed  $\mathcal{C}$  in the category of pro-objects,  $\mathbf{pro}\text{-}\mathcal{C}$ . The coreflection  $\text{PCoSh}(X, \mathbf{pro}\text{-}\mathcal{C}) \rightarrow \text{CoSh}(X, \mathbf{pro}\text{-}\mathcal{C})$  can then be explicitly constructed and turns out to be well behaved (in particular it is *exact*). He also showed that the smooth precosheaves defined by Bredon are exactly those cosheaves of which the coreflection into  $\text{CoSh}(X, \mathbf{pro}\text{-}\mathcal{C})$  lies in  $\text{CoSh}(X, \mathcal{C})$ . In [25] he elaborated on this idea in the case that  $\mathcal{C} = k\text{-}\mathbf{Mod}$  for  $k$  a (quasi-noetherian) ring. Here Prasolov shows that  $\text{CoSh}(X, \mathbf{pro}\text{-}k\text{-}\mathbf{Mod})$  is an abelian category satisfying Grothendieck's AB5\* axiom and he constructs a notion of cosheaf homology using so called quasi-projective resolutions.

In the proposition below we show that, in the case that  $\mathcal{C}$  is abelian, taking cosheaves in the slightly larger category  $\vec{\mathcal{C}}$  yields just the category of additive and left exact functors from  $\mathcal{C}$  to the category of sheaves of abelian groups on  $X$ . Using this perspective and the power of left-exactification we show that this allows one to construct the coreflector and prove the abelian properties of  $\text{CoSh}(X, \vec{\mathcal{C}})$  (assuming  $\mathcal{C}$  has enough injectives) from known properties of  $\text{Sh}(X, \mathbf{Ab})$ . Most importantly we show that (if  $k = \mathbb{Z}$ ) the *cosheaf homology* of a cosheaf defined by Prasolov is exactly the sheaf homology of  $X$  with coefficients in the corresponding functor. However the author hopes that this chapter and chapter 6 will convince the reader that a theory of cosheaves is not at all necessary to understand this homology theory.

**Remark 5.3.** In what follows we will only work with topological spaces, but any reader with more background in sites should have no problem generalising this to sites generated by a pretopology (or arbitrary Grothendieck sites).

**Proposition 5.4.** *Let  $\mathcal{C}$  be a locally small abelian category. Then there is an equivalence*

$$\mathrm{PCoSh}(X, \vec{\mathcal{C}}) \xrightarrow{\sim} [\mathcal{C}, \mathrm{PSh}(X, \mathbf{Ab})]_{\mathrm{LE}}^{\mathrm{op}}$$

*between the category of precosheaves on  $X$  with values in  $\vec{\mathcal{C}}$  and the category of left exact functors from  $\mathcal{C}$  to the category of presheaves of abelian groups on  $X$ . This restricts to an equivalence*

$$\mathrm{CoSh}(X, \vec{\mathcal{C}}) \xrightarrow{\sim} [\mathcal{C}, \mathrm{Sh}(X, \mathbf{Ab})]_{\mathrm{LE}}^{\mathrm{op}}$$

*between the category of cosheaves on  $X$  with values in  $\vec{\mathcal{C}}$  and the category of left exact functors from  $\mathcal{C}$  to the category of sheaves of abelian groups on  $X$ .*

**Proof.** Note that we have the following elementary sequence of equivalences (even isomorphisms) of functor categories

$$[\mathrm{Op}(X), [\mathcal{C}, \mathbf{Ab}]^{\mathrm{op}}] \simeq [\mathrm{Op}(X)^{\mathrm{op}}, [\mathcal{C}, \mathbf{Ab}]]^{\mathrm{op}} \simeq [\mathcal{C}, [\mathrm{Op}(X)^{\mathrm{op}}, \mathbf{Ab}]]^{\mathrm{op}} = [\mathcal{C}, \mathrm{PSh}(X, \mathbf{Ab})]^{\mathrm{op}}$$

which in total simply maps a functor  $F = (U \mapsto (T \mapsto F(T)(U)))$  to  $(T \mapsto (U \mapsto F(T)(U)))$  and swaps the direction of some morphisms. This then easily restricts to an equivalence

$$[\mathrm{Op}(X), [\mathcal{C}, \mathbf{Ab}]_{\mathrm{add}}^{\mathrm{op}}] \simeq [\mathcal{C}, \mathrm{PSh}(X, \mathbf{Ab})]_{\mathrm{add}}^{\mathrm{op}}.$$

Now given a functor  $F$  in the rightmost category, it is left exact if and only if it maps an exact sequence  $0 \rightarrow A \rightarrow B \rightarrow C$  in  $\mathcal{C}$  to an exact sequence  $0 \rightarrow F(A) \rightarrow F(B) \rightarrow F(C)$  in  $\mathrm{PSh}(X, \mathbf{Ab})$ . This sequence is exact if and only if it is exact on each open  $U \subseteq X$ . If we write  $G$  for the corresponding functor in the leftmost category, this is equivalent to stating that for each  $U \in \mathrm{Op}(X)$  and exact sequence  $0 \rightarrow A \rightarrow B \rightarrow C$  in  $\mathcal{C}$  the sequence  $0 \rightarrow G(U)(A) \rightarrow G(U)(B) \rightarrow G(U)(C)$  is exact, i.e. that  $G(U)$  is left exact for all  $U \in \mathrm{Op}(X)$ . We conclude that the equivalence restricts to an equivalence

$$\mathrm{PCoSh}(X, \vec{\mathcal{C}}) \xrightarrow{\sim} [\mathcal{C}, \mathrm{PSh}(X, \mathbf{Ab})]_{\mathrm{LE}}^{\mathrm{op}}.$$

Now recall that a precosheaf  $F \in \mathrm{PCoSh}(X, \vec{\mathcal{C}})$  is a cosheaf if and only if for each open  $U \in \mathrm{Op}(X)$  and  $(U_i)_{i \in I}$  an open cover of  $U$  the diagram

$$\bigoplus_{(i,j) \in I^2} F(U_i \cap U_j) \rightrightarrows \bigoplus_{i \in I} F(U_i) \rightarrow F(U)$$

(with morphisms as in the definition of cosheaves) is a coequaliser diagram in  $\vec{\mathcal{C}}$ . From the fact that the inclusion functor  $\vec{\mathcal{C}} \rightarrow [\mathcal{C}, \mathbf{Ab}]_{\mathrm{add}}^{\mathrm{op}}$  preserves colimits (see Proposition 2.29) we find that this is exact if and only if the diagram

$$F(U)(T) \rightarrow \prod_{i \in I} F(U_i)(T) \rightrightarrows \prod_{(i,j) \in I^2} F(U_i \cap U_j)(T)$$

is an equaliser diagram in  $\mathbf{Ab}$  for all  $T \in \mathcal{C}$ . Now this is exactly the condition that  $G(T)$  is a sheaf for all  $T \in \mathcal{C}$  (where  $G$  is the functor corresponding to  $F$  in  $[\mathcal{C}, \mathrm{PSh}(X, \mathbf{Ab})]_{\mathrm{LE}}^{\mathrm{op}}$ ), so we can restrict the equivalence to an equivalence

$$\mathrm{CoSh}(X, \vec{\mathcal{C}}) \xrightarrow{\sim} [\mathcal{C}, \mathrm{Sh}(X, \mathbf{Ab})]_{\mathrm{LE}}^{\mathrm{op}}. \quad \square$$

**Remark 5.5.** Note that in [25] for a cosheaf  $\mathcal{A}$  in  $\mathbf{pro}\text{-}\mathcal{C}$  the sheaves  $F(T)$  for  $T \in \mathcal{C}$  where  $F$  is the corresponding functor are used a lot in proofs. However the full equivalence as in Proposition 5.4 is never mentioned. As we will see this equivalence greatly clarifies the situation.

**Proposition 5.6.** *Let  $\mathcal{C}$  be a locally small abelian category with enough injectives. Then  $\mathrm{CoSh}(X, \vec{\mathcal{C}})$  is an abelian category in which a sequence  $\mathcal{A}_1 \rightarrow \mathcal{A}_2 \rightarrow \mathcal{A}_3$  is exact if and only if the corresponding sequence of functors  $F_1 \rightarrow F_2 \rightarrow F_3$  gives an exact sequence  $F_3(I) \rightarrow F_2(I) \rightarrow F_1(I)$  in  $\mathrm{Sh}(X, \mathbf{Ab})$  for all injective  $I \in \mathcal{C}$ .*



**Proof.** Apply the equivalence  $\text{CoSh}(X, \vec{\mathcal{C}}) \simeq [\mathcal{C}, \text{Sh}(X, \mathbf{Ab})]_{\text{LE}}^{\text{op}}$  and Proposition 2.29.  $\square$

**Proposition 5.7.** *Let  $\mathcal{C}$  be a locally small abelian category, then  $\text{CoSh}(X, \vec{\mathcal{C}})$  is a reflective subcategory of  $\text{PCoSh}(X, \vec{\mathcal{C}})$ . Under the equivalences from Proposition 5.4 the coreflection  $\text{PCoSh}(X, \vec{\mathcal{C}}) \rightarrow \text{CoSh}(X, \vec{\mathcal{C}})$  is given by composition with the sheafification functor  $\text{PSh}(X, \mathbf{Ab}) \rightarrow \text{Sh}(X, \mathbf{Ab})$ . If  $\mathcal{C}$  has enough injectives then this coreflection is exact.*

**Proof.** Let  $F : \mathcal{C} \rightarrow \text{PSh}(X, \mathbf{Ab})$  and  $G : \mathcal{C} \rightarrow \text{Sh}(X, \mathbf{Ab})$  be left exact and  $\eta : F \Rightarrow G$  a natural transformation. Write  $\bar{F} = (-)^+ \circ F$  for the composition of  $F$  with sheafification and write  $\epsilon : F \Rightarrow \bar{F}$  for the natural transformation giving the sheafification morphism at each  $T \in \mathcal{C}$ . Since for each  $T \in \mathcal{C}$  there exists a unique morphism  $\bar{\eta}_T : \bar{F}(T) \rightarrow G(T)$  so that  $\bar{\eta}_T \circ \epsilon_T = \eta_T$ , there will exist at most one natural transformation  $\bar{\eta} : \bar{F} \Rightarrow G$  so that  $\bar{\eta} \circ \epsilon = \eta$ . For the existence of such a  $\bar{\eta}$  we can apply the universal property of sheafification object-wise, and then by the naturality of this universal property obtain a natural transformation  $\bar{\eta} : \bar{F} \Rightarrow G$  so that  $\bar{\eta} \circ \epsilon = \eta$ . We conclude that composition with sheafification gives a reflection from the category  $[\mathcal{C}, \text{PSh}(X, \mathbf{Ab})]_{\text{LE}}$  to  $[\mathcal{C}, \text{Sh}(X, \mathbf{Ab})]_{\text{LE}}$ , so a coreflection from  $\text{PCoSh}(X, \vec{\mathcal{C}})$  to  $\text{CoSh}(X, \vec{\mathcal{C}})$ . For exactness, let  $F \rightarrow G \rightarrow H$  be an exact sequence in  $[\mathcal{C}, \text{PSh}(X, \mathbf{Ab})]_{\text{LE}}^{\text{op}}$ . By Proposition 2.29 this means that for each injective  $I \in \mathcal{C}$  the sequence  $H(I) \rightarrow G(I) \rightarrow F(I)$  is exact in  $\text{PSh}(X, \mathbf{Ab})$ . Because sheafification is exact it follows that  $\bar{H}(I) \rightarrow \bar{G}(I) \rightarrow \bar{F}(I)$  is exact in  $\text{Sh}(X, \mathbf{Ab})$  (with the bar-notation as above), so by Proposition 5.6 the sequence  $\bar{F} \rightarrow \bar{G} \rightarrow \bar{H}$  is exact.  $\square$

**Example 5.8.** Let  $F \in \vec{\mathcal{C}}$ , then analogous to sheaves we can define the *constant cosheaf*  $F^X$  to be the cosheafification of the precosheaf  $U \mapsto F$ . The corresponding functor  $\mathcal{C} \rightarrow \text{PSh}(X, \mathbf{Ab})$  of this precosheaf is  $(-)_X^{\text{presheaf}} \circ F$  (here  $T_X^{\text{presheaf}}$  denotes the constant presheaf of an object  $T \in \mathcal{C}$ ), so by Proposition 5.7 the constant cosheaf corresponds to the functor  $(-)^+ \circ (-)_X^{\text{presheaf}} \circ F = (-)_X \circ F$ . Specifically if  $G \in \mathbf{Ab}$ , then the constant cosheaf of  $\vec{G}$  corresponds to the functor  $\mathbf{Ab} \rightarrow \text{Sh}(X, \mathbf{Ab}), T \mapsto \text{Hom}(G, T)_X$  and hence the constant coheaf of  $\vec{\mathbb{Z}}$  corresponds to  $T \mapsto T_X$ . Compare to Definition 1.1.

To properly compare to Prasolov's work we need to consider the categories of (pre)cosheaves in  $\mathbf{pro}\text{-}\mathcal{C}$  instead of  $\vec{\mathcal{C}}$ . Note that because  $\mathbf{pro}\text{-}\mathcal{C}$  is a full subcategory of  $\vec{\mathcal{C}}$  we can immediately see  $\text{PCoSh}(X, \mathbf{pro}\text{-}\mathcal{C})$  as a full subcategory of  $\text{PCoSh}(X, \vec{\mathcal{C}})$ . Under the equivalence of Proposition 5.4 this subcategory consists of the functors  $F$  so that  $\Gamma(U, -) \circ F$  is bounded for all open  $U \subseteq X$  (see Definition 5.9 below). If  $\mathcal{C}$  is well-powered and has enough injectives, then by Theorem 3.27 a precosheaf in  $\mathbf{pro}\text{-}\mathcal{C}$  is a cosheaf if and only if it is a cosheaf considered as a precosheaf in  $\vec{\mathcal{C}}$ . Therefore we can view  $\text{CoSh}(X, \mathbf{pro}\text{-}\mathcal{C})$  as a subcategory of  $\text{CoSh}(X, \vec{\mathcal{C}})$  as well. We will proceed by showing for  $\mathcal{C} = \mathbf{Ab}$  that Proposition 5.6 and 5.7 restrict to the subcategory  $\text{CoSh}(X, \vec{\mathcal{C}})$ . This shows that Theorem 2.2.6.4, Theorem 3.3.1.1 (partially), Theorem 3.3.1.5-7 of [25] are consequences of thinking about cosheaves of pro-objects as functors  $\mathcal{C} \rightarrow \text{Sh}(X, \mathbf{Ab})$ . For simplicity we restrict to the case that  $\mathcal{C} = \mathbf{Ab}$ , but the reader should have no difficulty generalising to  $R\text{-Mod}$  or (with a bit more bookkeeping) an arbitrary abelian category which is locally small, well-powered, has all coproducts and has enough injectives.

**Definition 5.9.** Let  $F$  be a functor from a locally small abelian category  $\mathcal{C}$  to  $\text{Sh}(X, \mathbf{Ab})$  or  $\text{PSh}(X, \mathbf{Ab})$  on a topological space  $X$ . We say that  $F$  is *bounded* if  $\Gamma(U, -) \circ F$  is bounded for all open  $U \subseteq X$ .

**Lemma 5.10.** *Let  $F$  be a functor from  $\mathbf{Ab}$  to  $\text{Sh}(X)$  or  $\text{PSh}(X)$ . Then  $F$  is bounded if and only there exists a cardinal  $\kappa$  with the following property: For all  $T \in \mathbf{Ab}$  and collections  $((U_i, t_i))_{i \in I}$  of opens in  $X$  together with a section of  $F(T)$  over them, there exists a subgroup  $\iota : A \hookrightarrow T$  of cardinality  $\leq \kappa^{|I|}$  and for each  $i \in I$  a section  $s_i \in F(A)(U_i)$  so that  $F(\iota)_{U_i}(s_i) = t_i$ .*

**Proof.** The  $\Leftarrow$ -direction follows immediately by applying the property to a single open and section. For the  $\Rightarrow$ -direction assume that  $\Gamma(U, -) \circ F$  is  $\kappa$ -bounded for all open  $U \subseteq X$  with  $\kappa$  a cardinal. Let  $T$  and  $((U_i, t_i))_{i \in I}$  be as in the statement of the lemma. Now for each  $i \in I$  apply the boundedness of  $\Gamma(U_i, -) \circ F$  to obtain an  $A_i \in \mathbf{Ab}$  of cardinality  $\leq \kappa$ , a homomorphism  $f_i : A_i \rightarrow T$  and a section

$r_i \in F(A_i)(U_i)$  so that  $F(f_i)_{U_i}(r_i) = t_i$ . Now let  $B = \bigoplus_{i \in I} A_i$  with  $g_i : A_i \rightarrow B$  the canonical map for each  $i \in I$ . Let  $f : B \rightarrow T$  be the map induced by the  $f_i$ 's and set  $A = \text{im}(f)$  with  $\tilde{f} : B \rightarrow A$  the restriction and  $\iota : A \rightarrow T$  the inclusion. Now define for  $i \in I$  the section  $s_i = F(\tilde{f} \circ g_i)_{U_i}(r_i)$ , then

$$F(\iota)_{U_i}(s_i) = F(\iota)_{U_i}(F(\tilde{f} \circ g_i)_{U_i}(r_i)) = (F(\iota) \circ F(\tilde{f} \circ g_i))_{U_i}(r_i) = F(\iota \circ \tilde{f} \circ g_i)_{U_i}(r_i) = F(f_i)_{U_i}(r_i) = t_i$$

Finally note that since  $\tilde{f}$  is surjective we have  $|A| \leq |B| \leq \kappa^{|I|}$ , so  $\kappa$  has the desired property.  $\square$

**Lemma 5.11.** *Let  $F : \mathbf{Ab} \rightarrow \text{PSh}(X)$  be left exact and bounded, then  $(-)^+ \circ F$  is bounded, where  $(-)^+ : \text{PSh}(X) \rightarrow \text{Sh}(X)$  denotes sheafification.*

**Proof.** Let  $\kappa$  be as in Lemma 5.10. Fix an open  $U \subseteq X$  and write  $\lambda$  for the cardinality of the set of open subsets of  $U$ . Now let  $T \in \mathbf{Ab}$  and  $t \in F(T)^+(U)$ , then  $t$  is represented by an open cover  $(U_i)_{i \in I}$  of  $U$  with for each  $i \in I$  a section  $t_i \in F(T)(U_i)$  that agree on overlaps. Assume without loss of generality that each open subset of  $U$  occurs at most once, then  $|I| \leq \lambda$ . Applying Lemma 5.10 gives us a subgroup  $\iota : A \hookrightarrow T$  of cardinality  $\leq \kappa^\lambda$  and for each  $i \in I$  a section  $s_i \in F(T)(U_i)$  so that  $F(\iota)_{U_i}(s_i) = t_i$ . Because  $F$  is left exact,  $F(\iota)$  is injective on each open, so the  $s_i$ 's also agree on overlaps, so correspond to an element  $s \in F(T)^+(U)$  which is mapped to  $t$  by  $(F(\iota)^+)_U$ . We conclude that  $\Gamma(U, -) \circ (-)^+ \circ F$  is  $\kappa^\lambda$ -bounded.  $\square$

**Example 5.12.** If  $F \in \vec{\mathbf{Ab}}$  is an abelian pro-group, then the functor  $(-)_X^{\text{presh}} \circ F$  from Example 5.8 is bounded since each composition with  $\Gamma(U, -)$  gives  $F$ . It follows that the constant cosheaf  $F^X$  is bounded. In particular the functors  $\mathbf{Ab} \rightarrow \text{Sh}(X), T \mapsto \text{Hom}(G, T)_X$  for  $G \in \mathbf{Ab}$  and  $T \mapsto T_X$  are bounded.

**Proposition 5.13.** *The category  $\text{CoSh}(X, \mathbf{pro-Ab})$  is abelian with the inclusion  $\text{CoSh}(X, \mathbf{pro-Ab}) \hookrightarrow \text{CoSh}(X, \vec{\mathbf{Ab}})$  exact and the coreflection from Proposition 5.7 restricts to an exact coreflection from  $\text{PCoSh}(X, \mathbf{pro-Ab})$  to  $\text{CoSh}(X, \mathbf{pro-Ab})$ . It follows that a sequence  $\mathcal{A}_1 \rightarrow \mathcal{A}_2 \rightarrow \mathcal{A}_3$  of cosheaves in  $\mathbf{pro-Ab}$  is exact if and only if the corresponding sequence of functors  $\mathbf{Ab} \rightarrow \text{Sh}(X, \mathbf{Ab})$  is exact on injectives.*

**Proof.** Let  $f : \mathcal{A}_1 \rightarrow \mathcal{A}_2$  be a morphism in  $\text{CoSh}(X, \mathbf{pro-Ab})$ . We first consider this as a morphism in  $\text{PCoSh}(X, \vec{\mathbf{Ab}})$ . Since  $\text{PCoSh}(X, \vec{\mathbf{Ab}})$  is just the functor category  $[\text{Op}(X), \vec{\mathbf{Ab}}]$ , kernels and cokernels in this category can be computed object-wise. It follows using Theorem 3.27 that kernel and cokernel in  $\text{PCoSh}(X, \vec{\mathbf{Ab}})$  of  $f$  are bounded. Applying Proposition 5.7 and 5.11 then shows that the kernel and cokernel of  $f$  in  $\text{CoSh}(X, \vec{\mathbf{Ab}})$  are bounded and hence objects in  $\text{CoSh}(X, \mathbf{pro-Ab})$ . A similar argument shows that biproducts of objects in  $\text{CoSh}(X, \mathbf{pro-Ab})$  taken in  $\text{CoSh}(X, \vec{\mathbf{Ab}})$  lie in  $\text{CoSh}(X, \mathbf{pro-Ab})$ . This shows that  $\text{CoSh}(X, \mathbf{pro-Ab})$  is an abelian category where sequences are exact if and only if they are exact in  $\text{CoSh}(X, \vec{\mathbf{Ab}})$ . Combining Proposition 5.7 and 5.11 then shows that there is an exact coreflection  $\text{PCoSh}(X, \mathbf{pro-Ab}) \rightarrow \text{CoSh}(X, \mathbf{pro-Ab})$ .  $\square$

**Remark 5.14.** It would be interesting to generalise the above to the setting where  $\mathcal{C}$  is any locally small category by considering the cosheaves with values in  $[\mathcal{C}, \mathbf{Set}]_{\text{LE}}^{\text{op}}$  (where left exact is defined as preserving finite limits). We expect this reduces the theory of cosheaves in  $\mathbf{pro-}\mathcal{C}$  to the theory of sheaves of sets similar to the above.

We conclude this section by showing that cosheaf homology of a cosheaf of abelian pro-groups as defined in [25] is the same as the sheaf homology with coefficients the corresponding functor. We first adopt some notation and terminology from [25].

**Definition 5.15.** If  $\mathcal{A} \in \text{CoSh}(X, \mathbf{pro-Ab})$  and  $T \in \mathbf{Ab}$  we write  $\text{Hom}(\mathcal{A}, T) \in \text{Sh}(X, \mathbf{Ab})$  for the sheaf given on each open  $U \subseteq X$  by  $\text{Hom}(\mathcal{A}(U), T)$ . This is exactly  $F(T)$  where  $F$  is the functor corresponding to  $\mathcal{A}$ .  $\mathcal{A}$  is called *quasi-projective* if  $\text{Hom}(\mathcal{A}, I)$  is an injective sheaf for all injective  $I \in \mathbf{Ab}$ . In [25,



Theorem 3.4.1.1] is shown that each cosheaf is a quotient of a quasi-projective cosheaf (in fact in a functorial manner). It follows that each  $\mathcal{A} \in \text{CoSh}(X, \mathbf{pro}\text{-}\mathbf{Ab})$  has a resolution

$$\cdots \rightarrow \mathcal{P}_3 \rightarrow \mathcal{P}_2 \rightarrow \mathcal{P}_1 \rightarrow \mathcal{P}_0 \rightarrow \mathcal{A} \rightarrow 0$$

consisting of quasi-projectives. The *cosheaf homology* of  $\mathcal{A}$ , denoted  $L_p\Gamma(X, \mathcal{A})$ , is then defined as  $h_p(\Gamma(X, \mathcal{P}_\bullet))$ .

**Theorem 5.16.** *Let  $X$  be a topological space and  $\mathcal{A} \in \text{CoSh}(X, \mathbf{pro}\text{-}\mathbf{Ab})$ , then the cosheaf homology of  $X$  with coefficients in  $\mathcal{A}$  is canonically isomorphic to  $\vec{H}_p(X, F)$  where  $F : \mathbf{Ab} \rightarrow \text{Sh}(X)$  is the functor associated to  $\mathcal{A}$ . In particular this shows that  $\vec{H}_p(X, F)$  is an abelian pro-group for all  $F : \mathbf{Ab} \rightarrow \text{Sh}(X)$  corresponding to a cosheaf in  $\mathbf{pro}\text{-}\mathbf{Ab}$ .*

**Proof.** In [25, Theorem 3.4.1.3] it has already been noted that

$$\text{Hom}_{\mathbf{pro}\text{-}\mathbf{Ab}}(L_p\Gamma(X, \mathcal{A}), T) \simeq H^p(X, \text{Hom}(\mathcal{A}, T)) = H^p(X, F(T))$$

for a cosheaf  $\mathcal{A}$  and any injective  $T \in \mathbf{Ab}$ . Up to arguing that this isomorphism is natural in  $T$  this implies immediately that  $L_p\Gamma(X, \mathcal{A}) \simeq \vec{H}_p(X; F)$  by Lemma 2.5. We will go through the construction of this isomorphism again to verify this naturality in  $T$ .

Let  $\mathcal{A} \in \text{CoSh}(X, \mathbf{pro}\text{-}\mathbf{Ab})$  and  $\mathcal{A}[0] \leftarrow \mathcal{P}_\bullet$  a resolution of  $\mathcal{A}$  by quasi-projective cosheaves. Fix an injective  $T \in \mathbf{Ab}$ , then by Proposition 5.13 the functor  $\text{Hom}(-, T) : \text{CoSh}(X, \mathbf{pro}\text{-}\mathbf{Ab}) \rightarrow \text{Sh}(X, \mathbf{Ab})$  is exact. It follows that  $\text{Hom}(\mathcal{A}, T)[0] \rightarrow \text{Hom}(\mathcal{P}_\bullet, T)$  is an injective resolution of  $\text{Hom}(\mathcal{A}, T)$ . Thus for each  $p \in \mathbb{Z}$  we have  $h^p(\Gamma(X, \text{Hom}(\mathcal{P}_\bullet, T))) \simeq H^p(X, \text{Hom}(\mathcal{A}, T)) = H^p(X, F(T))$ . Note that this isomorphism is natural in  $T$ , since if  $f : S \rightarrow T$  is a homomorphism, then this induces a chain map  $\text{Hom}(\mathcal{P}_\bullet, S) \rightarrow \text{Hom}(\mathcal{P}_\bullet, T)$  which induces the map  $f \circ (-) : \text{Hom}(\mathcal{A}, S) \rightarrow \text{Hom}(\mathcal{A}, T)$  on 0-th cohomology. Taking the induced map on cohomology then gives the induced map  $H^p(X, \text{Hom}(\mathcal{A}, S)) \rightarrow H^p(X, \text{Hom}(\mathcal{A}, T))$  by the definition of the functoriality of  $H^p(X, -) : \text{Sh}(X, \mathbf{Ab}) \rightarrow \mathbf{Ab}$ . On the other hand  $L_p\Gamma(X, \mathcal{A})$  is defined as the  $p$ -th homology of  $\Gamma(X, \mathcal{P}_\bullet)$ , so using the exactness of  $\text{Hom}(-, T) : \mathbf{pro}\text{-}\mathbf{Ab} \rightarrow \mathbf{Ab}$  we have

$$\begin{aligned} \text{Hom}(L_p\Gamma(X, \mathcal{A}), T) &= \text{Hom}(h_p(\Gamma(X, \mathcal{P}_\bullet)), T) \simeq^* h^p(\text{Hom}(\Gamma(X, \mathcal{P}_\bullet), T)) \simeq^{**} h^p(\Gamma(X, \text{Hom}(\mathcal{P}_\bullet, T))) \\ &\simeq H^p(X, F(T)). \end{aligned}$$

The isomorphism marked with  $*$  is natural by Lemma 2.1 and the isomorphism marked with  $**$  is natural by the definition of the functor  $\text{Hom}(-, T) : \text{CoSh}(X, \mathbf{pro}\text{-}\mathbf{Ab}) \rightarrow \text{Sh}(X, \mathbf{Ab})$ .  $\square$

## 5.2 Boundedness of sheaf homology

In this section we show that for bounded left exact functors  $F : \mathbf{Ab} \rightarrow \text{Sh}(X)$ , i.e. those corresponding to cosheaves in  $\mathbf{pro}\text{-}\mathbf{Ab}$ , the sheaf homology functors  $\vec{H}_p(X; F)$  are bounded, i.e. abelian pro-groups. We already know this to be true from Theorem 5.16, but here we present a proof independent from Prasolov's work. Even though in later sections we will mostly study sheaf homology with constant coefficients, the added generality of using functors  $F : \mathbf{Ab} \rightarrow \text{Sh}(X)$  as coefficients is useful in this proof as it allows us to perform induction on  $p$ . Note that apart from avoiding foundational issues the boundedness of  $\vec{H}_p(X; F)$  will not be used in the remainder of this thesis. This is because, as we have shown in chapter 3 the category  $\mathbf{Ab}$  is a perfectly good abelian category with a very explicit description of exact sequences.

For the rest of this section  $X$  is a fixed topological space.

**Lemma 5.17.** *Let  $F : \mathbf{Ab} \rightarrow \text{Sh}(X)$  be an additive functor, then there exists an additive functor  $G : \mathbf{Ab} \rightarrow \text{Sh}(X)$  such that for all  $p \geq 2$  we have  $H^p(X, F(-)) \simeq H^{p-1}(X, G(-))$  and so that if  $F$  is bounded then  $\vec{G}$  is bounded.*

**Proof.** Recall that we have a functor  $(-)^0 : \text{Sh}(X) \rightarrow \text{Sh}(X)$  which maps a sheaf  $\mathcal{F}$  to the sheaf  $\mathcal{F}^0$  given by  $\mathcal{F}^0(U) = \prod_{x \in U} \mathcal{F}_x$  for all  $x \in X$  and we have a natural transformation  $\eta : \text{id}_{\text{Sh}(X)} \Rightarrow (-)^0$  given in all  $\mathcal{F} \in \text{Sh}(X)$  by the morphism  $\eta_{\mathcal{F}} : \mathcal{F} \rightarrow \mathcal{F}^0$  given on any open set  $U \subseteq X$  by  $s \mapsto (s_x)_{x \in U}$ . Now define the functor  $G : \mathbf{Ab} \rightarrow \text{Sh}(X)$  by  $G(A) = \text{coker } \eta_{F(A)}$  (with it mapping morphisms to the induced morphism on cokernels). Clearly  $G$  is an additive functor. Finally for  $A \in \mathbf{Ab}$  applying the long exact sequence to the short exact sequence  $0 \rightarrow F(A) \xrightarrow{\eta_A} F(A)^0 \rightarrow G(A) \rightarrow 0$  and the fact that  $F(A)^0$  is flasque gives isomorphisms  $H^{p-1}(X, G(A)) \simeq H^p(X, F(A))$  for  $p \geq 2$  and by the naturality of the long exact sequence these are in fact natural isomorphisms.

Now assume that  $F$  is bounded. First of all define  $H : \mathbf{Ab} \rightarrow \text{PSh}(X, \mathbf{Ab})$  by  $H(T)(U) = \text{coker } \eta_{F(T)}$ , then  $(-)^+ \circ H \simeq G$ . By Lemma 2.11 we have  $\tilde{G} \simeq (-)^+ \circ \tilde{H}$ , so by Lemma 5.11 it suffices to show that  $\tilde{H}$  is bounded. For each open  $U \subseteq X$  we have by Lemma 2.11 that  $\Gamma(U, -) \circ \tilde{H} = \Gamma(\overline{U}, -) \circ H$ , so by Lemma 3.26 it suffices to show that  $\Gamma(U, -) \circ H$  is bounded for all open  $U \subseteq X$ . For each  $T \in \mathbf{Ab}$  we have  $(\Gamma(U, -) \circ H)(T) = \text{coker}(\eta_{F(T)})_U$ , so by Lemma 3.25 it suffices to show that the functor  $K : \mathbf{Ab} \rightarrow \mathbf{Ab}, T \mapsto F(T)^0(U)$  is bounded.

First of all pick a cardinal  $\kappa$  as in Lemma 5.10 for  $F$ . Let  $T \in \mathbf{Ab}$  and  $(s^x)_{x \in U} \in F(T)^0(U) = K(T)$ . For each  $x \in U$  pick an open  $x \in U_x \subseteq U$  and a section  $t^x \in F(T)(U_x)$  so that  $t^x_x = s^x$ . By the assumed property of  $\kappa$  there exists a subgroup  $\iota : A \rightarrow T$  of cardinality  $\leq \kappa^{|U|}$  and for each  $x \in U$  a section  $r^x \in F(A)(U_x)$  so that  $F(\iota)_{U_x}(r^x) = t^x$ . This means in particular for each  $x \in U$  that  $F(\iota)_x(r^x_x) = t^x_x = s^x$  and hence  $K(\iota)((r^x_x)_{x \in U}) = (s^x)_{x \in U}$ . We conclude that  $K$  is  $\kappa^{|U|}$ -bounded and the proof is complete.  $\square$

**Lemma 5.18.** *Let  $F : \mathbf{Ab} \rightarrow \text{Sh}(X)$  be left exact and bounded, then  $H^1(X, F(-))$  is bounded.*

**Proof.** Recall in degree 1 sheaf cohomology agrees with Čech cohomology (see the upcoming Proposition 5.27), so it suffices to show that  $\check{H}^1(X, F(-))$  is bounded. Let  $\mu$  be the cardinality of the set of open sets of  $X$  and let  $\kappa$  be as in Lemma 5.10. Now let  $A \in \mathbf{Ab}$  and  $s \in \check{H}^1(X, F(A))$ , then  $s$  is represented by a class  $[\alpha] \in \check{H}^1(\mathcal{U}, F(A))$  for an open cover  $\mathcal{U} = (U_i)_{i \in I}$  of  $X$  and

$$\alpha = (\alpha_{ij})_{i,j \in I} \in \prod_{(i,j) \in I^2} F(A)(U_i \cap U_j)$$

a representative. Assume without generality that  $\mathcal{U}$  contains no duplicate sets and hence  $|I| \leq \mu$ . Now apply Lemma 5.10 to  $((U_i \cap U_j, \alpha_{ij}))_{i,j \in I}$  to obtain a subgroup  $B \subseteq A$  of cardinality  $\leq \mu^2 \kappa$  with inclusion  $\iota$  and for each  $i, j \in I$  a  $\beta_{ij} \in F(B)$  so that  $F(\iota)_{U_i \cap U_j}(\beta_{ij}) = \alpha_{ij}$ . Now because  $F$  is left exact the sheaf homomorphism  $F(\iota)$  is injective, so for all  $i, j, k \in I$  it follows from

$$\begin{aligned} F(\iota)_{U_{ijk}}(\beta_{ij}|_{U_{ijk}} - \beta_{ik}|_{U_{ijk}} + \beta_{jk}|_{U_{ijk}}) &= F(\iota)_{U_{ij}}(\beta_{ij})|_{U_{ijk}} - F(\iota)_{U_{ik}}(\beta_{ik})|_{U_{ijk}} + F(\iota)_{U_{jk}}(\beta_{jk})|_{U_{ijk}} \\ &= \alpha_{ij} - \alpha_{ik} + \alpha_{jk} = 0 \end{aligned}$$

(here  $U_{ijk} = U_i \cap U_j \cap U_k$ ) that  $\beta_{ij} - \beta_{ik} + \beta_{jk} = 0$ . Thus  $(\beta_{ij})_{i,j \in I}$  represents a class  $t \in \check{H}^1(X, F(B))$  which is mapped to  $s$  under the morphism  $\check{H}^1(X, F(B)) \rightarrow \check{H}^1(X, F(A))$  induced by  $F(\iota)$ . We conclude that  $H^1(X, F(-)) = \check{H}^1(X, F(-))$  is  $\mu^2 \kappa$ -bounded.  $\square$

**Theorem 5.19.** *Let  $X$  be a topological space,  $F : \mathbf{Ab} \rightarrow \text{Sh}(X)$  left exact and bounded, then for any  $p \in \mathbb{Z}$  the functor  $\tilde{H}_p(X; F)$  is an abelian pro-group. In particular  $\tilde{H}_p(X)$  is an abelian pro-group and for any coefficient group  $G \in \mathbf{Ab}$  the functor  $\tilde{H}_p(X; G)$  is an abelian pro-group.*

**Proof.** For  $p = 0$  the functor  $H^0(X, F(-)) \simeq \Gamma(X, -) \circ F$  is bounded by the boundedness of  $F$ , so applying Lemma 3.26 shows  $\tilde{H}_0(X; F)$  is bounded. Lemma 5.18 proves the case  $p = 1$ . We proceed by induction on  $p$ . Let  $p \geq 1$  and assume that for all left exact and bounded functors  $F : \mathbf{Ab} \rightarrow \text{Sh}(X)$  the left-exactification of  $H^p(X, F(-))$  is bounded. Let  $F : \mathbf{Ab} \rightarrow \text{Sh}(X)$  be left exact and bounded. Lemma 5.17 gives an additive  $G : \mathbf{Ab} \rightarrow \text{Sh}(X)$  so that  $H^{p+1}(X, F(-)) \simeq H^p(X, G(-))$  and with  $\tilde{G}$  bounded. By induction  $H^p(X, \tilde{G}(-))$  is bounded, so by Lemma 2.11  $H^p(X, G(-)) \simeq H^{p+1}(X, F(-))$  is bounded. This proves the induction step. The final statement follows from applying Example 5.12.  $\square$

### 5.3 Čech and shape homology

We begin this section by reviewing shape homology and various notions of Čech homology and cohomology. Since each definition is simply called ‘Čech (co)homology’ in the context where it is used we give each theory distinguished notation and names.

**Definition 5.20.** An *abstract simplicial complex* is a pair  $(S, \Delta)$  where  $S$  is a set and  $\Delta$  is a collection of non-empty finite subsets of  $S$  closed under taking non-empty subsets. The elements of  $S$  are called *vertices* and the elements of  $\Delta$  are called *faces*. If  $F \in \Delta$  is a face, then the *dimension* of  $F$  is defined as  $|F| - 1$ . Given an abstract simplicial complex  $(S, \Delta)$  we define for each  $p \geq 0$  the abelian group  $C_p((S, \Delta))$  to be the free group on all  $(p+1)$ -tuples  $(v_0, \dots, v_p)$  so that  $\{v_0, \dots, v_p\} \in \Delta$ , but modulo the subgroup generated by

$$(v_0, \dots, v_p) - \text{sgn}(f) \cdot (v_{f(0)}, \dots, v_{f(p)})$$

for all such tuples  $(v_0, \dots, v_p)$  and a permutation  $f$  of  $\{0, 1, \dots, p\}$ . Given a  $v_0, \dots, v_p \in S$  with  $\{v_0, \dots, v_p\} \in \Delta$  we write  $\langle v_0, \dots, v_p \rangle$  for the class of the tuple  $(v_0, \dots, v_p)$ . Notice how  $\langle v_0, \dots, v_p \rangle = 0$  if any  $v_i$  occurs multiple times in  $v_0, \dots, v_p$ . For each  $p \in \mathbb{Z}_{\geq 1}$  we have a boundary map  $\partial : C_p((S, \Delta)) \rightarrow C_{p-1}((S, \Delta))$  given for all  $v_0, \dots, v_p \in S$  so that  $\{v_0, \dots, v_p\} \in \Delta$  by:

$$\partial \langle v_0, \dots, v_p \rangle = \sum_{i=0}^p (-1)^i \langle v_0, \dots, v_{i-1}, v_{i+1}, \dots, v_p \rangle$$

These boundary maps give the  $C_p((S, \Delta))$ -groups the structure of a chain complex  $C_\bullet((S, \Delta))$ . A *simplicial map*  $f : (S, \Delta) \rightarrow (T, \Gamma)$  between simplicial complexes is a map  $f : S \rightarrow T$  so that  $f(F) \in \Gamma$  for all  $F \in \Delta$ . This induces a chain map  $f_\# : C_\bullet((S, \Delta)) \rightarrow C_\bullet((T, \Gamma))$  given for all  $v_0, \dots, v_p \in S$  with  $\{v_0, \dots, v_p\} \in \Delta$  by

$$f_\#(\langle v_0, \dots, v_p \rangle) = \langle f(v_0), \dots, f(v_p) \rangle.$$

**Definition 5.21.** A *open cover* (or just *cover*) of a topological space  $X$  is a family  $\mathcal{U} = (U_i)_{i \in I}$  of open subsets  $X$  indexed by an index set  $I$  so that  $\bigcup_{i \in I} U_i = X$ . We say that a cover  $\mathcal{U} = (U_i)_{i \in I}$  is a *refinement* of another cover  $\mathcal{V} = (V_j)_{j \in J}$  if for any  $i \in I$  there exists a  $j \in J$  so that  $U_i \subseteq V_j$ . A *refinement projection* of such a refinement is a map  $\varphi : I \rightarrow J$  so that  $U_i \subseteq V_{\varphi(i)}$  for all  $i \in I$ . We define the category  $\text{Cov}(X)$  to be the posetal category of all open covers of  $X$  with there being a morphism  $\mathcal{U} \rightarrow \mathcal{V}$  if  $\mathcal{U}$  is a refinement of  $\mathcal{V}$ . This category is cofiltered and essentially small (each cover is isomorphic to one with no duplicate opens).

Let  $X$  be a topological space with  $\mathcal{U} = (U_i)_{i \in I}$  an open cover. The *nerve* of  $\mathcal{U}$ , written  $N(\mathcal{U})$ , is the simplicial set  $(I, \Delta)$  where  $\Delta$  consists of all non-empty finite subsets  $F \subseteq I$  so that  $\bigcap_{i \in F} U_i \neq \emptyset$ . If  $\mathcal{V} = (V_j)_{j \in J}$  is another open cover and  $\varphi : I \rightarrow J$  a refinement projection, then  $\varphi$  is a simplicial map between  $N(\mathcal{U})$  and  $N(\mathcal{V})$ . We will call  $C_\bullet(N(\mathcal{U}))$  the *Čech nerve chain complex* and write  $\check{C}_\bullet^N(\mathcal{U})$  for it. If  $\psi : I \rightarrow J$  is another refinement projection, then  $\varphi_\#$  and  $\psi_\#$  are homotopic as can be seen by the homotopy  $h$  given for all  $p \geq 0$  and  $v_0, \dots, v_p \in S$  with  $\{v_0, \dots, v_p\} \in \Delta$  by

$$h(\langle v_0, \dots, v_p \rangle) = \sum_{n=0}^p (-1)^n \langle \varphi(i_0), \varphi(i_1), \dots, \varphi(i_n), \psi(i_n), \psi(i_{n+1}), \dots, \psi(i_p) \rangle.$$

Therefore for each refinement  $\mathcal{U} \rightarrow \mathcal{V}$  we have a unique homotopy class of chain maps  $\check{C}_\bullet^N(\mathcal{U}) \rightarrow \check{C}_\bullet^N(\mathcal{V})$ . This makes  $\check{C}_\bullet^N(-)$  into a covariant functor from  $\text{Cov}(X)$  to  $K_h(\mathbf{Ab})$ , where  $K_h(\mathbf{Ab})$  denotes the homotopy category of chain complexes in  $\mathbf{Ab}$ .

- Composing this functor with  $- \otimes G$  for a  $G \in \mathbf{Ab}$  and taking  $p$ -th homology for a  $p \in \mathbb{Z}$  gives for each  $\mathcal{U}$  the  $p$ -th Čech nerve homology group of  $\mathcal{U}$ , written  $\check{H}_p^N(\mathcal{U}; G)$ .

- Taking the limit of this over  $\mathcal{U}$  in  $\mathbf{Ab}$  yields the *Čech nerve homology groups of  $X$*  with coefficients in  $G$ .
- Taking the limit in  $\tilde{\mathbf{Ab}}$  instead gives the *Čech nerve homology pro-groups of  $X$*  with coefficients in  $G$  which we denote by  $\check{H}_p^N(X; G)$ . By Proposition 3.29 the Čech nerve homology group of  $X$  is simply the coreflection of  $\check{H}_p^N(X; G)$  in  $\mathbf{Ab}$ .

For each  $T \in \mathbf{Ab}$  and  $\mathcal{U} \in \text{Cov}(X)$  we can apply the functor  $\text{Hom}(-, T)$  to  $\check{C}_\bullet^N(\mathcal{U})$  to obtain the *Čech nerve cochain complex*  $\check{C}_\bullet^N(\mathcal{U}, T)$ .

- Taking  $p$ -th cohomology of this complex gives the *Čech nerve cohomology groups of  $\mathcal{U}$* , written  $\check{H}_N^p(\mathcal{U}, T)$ .
- This is a bifunctor in  $\mathcal{U}$  and  $T$ , so we can take the colimit (in  $\mathbf{Ab}$ ) over all open covers to obtain the *Čech nerve cohomology groups*  $\check{H}_N^p(X, T)$  which are additively functorial in  $T$ .

A cover  $\mathcal{U}$  of  $X$  is called *normal* if it has a partition of unity subordinate to it consisting of continuous functions  $X \rightarrow [0, 1]$ . Note that the subcategory of  $\text{Cov}(X)$  consisting of normal covers is also cofiltered, see [19, App. 1, Thm. 4].

- For  $G \in \mathbf{Ab}$  restricting the diagram  $\check{H}_p^N(-; G) : \text{Cov}(X) \rightarrow \mathbf{Ab}$  to normal covers and taking the limit in  $\tilde{\mathbf{Ab}}$  gives the *shape homology pro-groups*  $\tilde{H}_p^{\text{shp}}(X; G)$ .
- Similarly for each  $T \in \mathbf{Ab}$  restricting the diagram  $\check{H}_N^p(-, T) : \text{Cov}(X)^{\text{op}} \rightarrow \mathbf{Ab}$  to normal covers and taking the colimit in  $\mathbf{Ab}$  gives the *shape cohomology groups*  $H_{\text{shp}}^p(X, T)$  with coefficients in  $T$ .

**Remark 5.22.** The shape homology pro-group originates from shape theory. Shape is a variation of homotopy which works much better for spaces with complicated local structure. A motivating example is the Warsaw circle: The closed topologist's sine curve with the ends connected by a path (see the upcoming Example 6.23). This space is not homotopy equivalent to a circle, but it does have the same shape as a circle. Write  $\mathbf{HTop}$  for the homotopy category of topological spaces and  $\mathbf{HPol}$  for the full subcategory of objects homotopy equivalent to a polyhedron. One way of defining the *shape* of a topological space  $X$  is as the pro-object in  $\mathbf{HPol}$  representing the functor  $\text{Hom}_{\mathbf{HTop}}(X, -) : \mathbf{HPol} \rightarrow \mathbf{Set}$  ([19, Ch. I, 2.4]). Note that the existence of such a pro-object is not at all trivial, but when it exists this characterises it up to isomorphism. Let  $(X_i)_{i \in I}$  be an inverse system in  $\mathbf{HPol}$  representing the shape of  $X$ , then one can apply various functors from algebraic topology to it (homotopy groups, (co)homology groups, etc.) and take the (co)limit to obtain the shape theoretic analogue. It turns out that  $(X_i)_{i \in I}$  can be taken to be the system of nerves (realised as topological spaces) of *normal* open covers of  $X$  which explains the definition of shape homology above ([19, App. 1, 3.1]).

**Remark 5.23.** The Čech nerve cohomology groups are quite well-behaved. In particular they satisfy the Eilenberg-Steenrod axioms for a cohomology theory. Čech nerve homology groups on the other hand are less well-behaved in the sense that they fail to satisfy the *exactness axiom* which requires the existence of the long exact sequence of relative homology. This problem comes from the fact that cofiltered limits in  $\mathbf{Ab}$  are *not* exact (see [9, p. 233]). Only during the advent of shape theory was it realised that by taking the limit in **pro-Ab**, an abelian category where cofiltered limits are exact, a homology theory is obtained which satisfies all the Eilenberg-Steenrod axioms (except that it takes values in **pro-Ab**).

**Lemma 5.24.** For any small cofiltered category  $\mathcal{I}$ , any  $G \in \mathbf{Ab}$  and  $p \in \mathbb{Z}$  the diagram

$$\begin{array}{ccc}
 & & \vec{\mathbf{Ab}} \\
 & \nearrow \text{left-exactification} & \uparrow \text{limit} \\
 & [\mathbf{Ab}, \mathbf{Ab}]_{\text{add}} & [\mathcal{I}, \vec{\mathbf{Ab}}] \\
 & \uparrow (-) \circ \vec{G} & \uparrow \text{index-wise } (\vec{-}) \\
 & [\mathbf{Ab}, \mathbf{Ab}]_{\text{add}} & [\mathcal{I}, \mathbf{Ab}] \\
 & \uparrow \text{group-wise colimit} & \uparrow \text{index-wise } h_p(-) \\
 [\mathcal{I}^{\text{op}}, [\mathbf{Ab}, \mathbf{Ab}]_{\text{add}}] & & [\mathcal{I}, K_h(\mathbf{Ab})] \\
 \uparrow \text{index-wise, group-wise } h^p(-) & & \uparrow \text{index-wise } (-) \otimes G \\
 [\mathcal{I}^{\text{op}}, [\mathbf{Ab}, K^h(X)]_{\text{add}}] & \leftarrow \text{---} & [\mathcal{I}, K_h(\mathbf{Ab})]
 \end{array} \tag{5.1}$$

of categories and functors commutes, in the sense that there is a natural isomorphism between the two compositions  $[\mathcal{I}, K_h(\mathbf{Ab})] \rightarrow \vec{\mathbf{Ab}}$ . Here  $K_h(\mathbf{Ab})$  denotes the homotopy category of chain complexes in  $\mathbf{Ab}$ ,  $K^h(\mathbf{Ab})$  the homotopy category of cochain complexes in  $\mathbf{Ab}$  and the lower horizontal functor maps a functor  $D : \mathcal{I} \rightarrow K_h(\mathbf{Ab})$  to the functor  $i \mapsto (T \mapsto \text{Hom}(D(i), T))$ . Dashed arrows indicate contravariant functors.

**Proof.** First of all notice that we can replace the segment

$$[\mathcal{I}^{\text{op}}, [\mathbf{Ab}, \mathbf{Ab}]_{\text{add}}] \xrightarrow{\text{group-wise colimit}} [\mathbf{Ab}, \mathbf{Ab}]_{\text{add}} \xrightarrow{(-) \circ \vec{G}} [\mathbf{Ab}, \mathbf{Ab}]_{\text{add}}$$

with the segment

$$[\mathcal{I}^{\text{op}}, [\mathbf{Ab}, \mathbf{Ab}]_{\text{add}}] \xrightarrow{\text{group-wise } (-) \circ \vec{G}} [\mathcal{I}^{\text{op}}, [\mathbf{Ab}, \mathbf{Ab}]_{\text{add}}] \xrightarrow{\text{group-wise colimit}} [\mathbf{Ab}, \mathbf{Ab}]_{\text{add}}$$

since colimits in  $[\mathbf{Ab}, \mathbf{Ab}]_{\text{add}}$  are given by taking the colimit object-wise. We split the resulting diagram in two by drawing a dashed horizontal arrow  $[\mathcal{I}^{\text{op}}, [\mathbf{Ab}, \mathbf{Ab}]_{\text{add}}] \rightarrow [\mathcal{I}, \vec{\mathbf{Ab}}]$  (the upper most  $[\mathcal{I}^{\text{op}}, [\mathbf{Ab}, \mathbf{Ab}]_{\text{add}}]$ ) given by index-wise left-exactification. It then suffices to prove that the upper square and the lower heptagon commute. Note that in the heptagon all categories are functor categories on  $\mathcal{I}$  with the functors acting by applying a functor for each fixed  $i \in \mathcal{I}$ , so it suffices to show that the diagrams

$$\begin{array}{ccc}
 [\mathbf{Ab}, \mathbf{Ab}]_{\text{add}} & \xrightarrow{\text{left-exactification}} & \vec{\mathbf{Ab}} \\
 \uparrow \text{group-wise colimit} & & \uparrow \text{limit} \\
 [\mathcal{I}^{\text{op}}, [\mathbf{Ab}, \mathbf{Ab}]_{\text{add}}] & \xrightarrow{\text{---}} & [\mathcal{I}, \vec{\mathbf{Ab}}]
 \end{array} \text{ and } \begin{array}{ccc}
 & & \vec{\mathbf{Ab}} \\
 & \nearrow \text{left-exactification} & \uparrow (\vec{-}) \\
 & [\mathbf{Ab}, \mathbf{Ab}]_{\text{add}} & \mathbf{Ab} \\
 & \uparrow (-) \circ \vec{G} & \uparrow h_p \\
 & [\mathbf{Ab}, \mathbf{Ab}]_{\text{add}} & K_h(\mathbf{Ab}) \\
 & \uparrow \text{group-wise } h^p & \uparrow (-) \otimes G \\
 [\mathbf{Ab}, K^h(\mathbf{Ab})]_{\text{add}} & \leftarrow \text{---} & K_h(\mathbf{Ab})
 \end{array} \tag{5.2}$$

commute. The diagram on the left commutes by Lemma 3.14 and the one on the right by Proposition 2.9.  $\square$

**Theorem 5.25.** For any topological space  $X$  and  $G \in \mathbf{Ab}$  the functors  $\vec{H}_p^N(X; G)$  and  $\vec{H}_p^{\text{shp}}(X; G)$  are the left-exactifications of the additive functors  $\mathbf{Ab} \rightarrow \mathbf{Ab}$  given by  $T \mapsto \vec{H}_N^p(X, \text{Hom}(G, T))$  and  $T \mapsto H_{\text{shp}}^p(X, \text{Hom}(G, T))$  respectively.

**Proof.** Apply the above Lemma with  $\mathcal{I} = \text{Cov}(X)$  (resp.  $\mathcal{I}$  the subcategory of normal covers).

Then in diagram (5.1) the functor  $\mathcal{U} \mapsto \check{C}^N_\bullet(X)$  in  $[\mathcal{I}, K_h(\mathbf{Ab})]$  gets mapped to  $\check{H}^N_p(X, G)$  (resp.  $H_p^{\text{shp}}(X, G)$ ) through the rightmost path and to the left-exactification of  $T \mapsto \check{H}^p_N(X, \text{Hom}(G, T))$  (resp.  $T \mapsto H_{\text{shp}}^p(X, \text{Hom}(G, T))$ ) in the leftmost path.  $\square$

**Definition 5.26.** Let  $X$  be a topological space,  $\mathcal{F}$  a presheaf of abelian groups on  $X$  and  $\mathcal{U} = (U_i)_{i \in I}$ . If  $p \in \mathbb{Z}_{\geq 0}$  then we write  $S_p(\mathcal{U})$  for the set of tuples  $(i_0, \dots, i_p)$  of  $p + 1$  elements in  $I$  so that  $U_{i_0} \cap \dots \cap U_{i_p} \neq \emptyset$ . For  $p \in \mathbb{Z}_{\geq 0}$  we define the  $p$ -th Čech (pre)sheaf cochain group, denoted  $\check{C}^p(\mathcal{U}, \mathcal{F})$ , as the group of functions  $\alpha$  which map an element  $\sigma = (i_0, \dots, i_p) \in S_p(\mathcal{U})$  to an element  $\alpha(\sigma) \in \mathcal{F}(U_\sigma)$  so that  $\alpha(\sigma) = 0$  whenever  $\sigma$  contains duplicate elements and so that  $\alpha(f(\sigma)) = \text{sgn}(f)\alpha(\sigma)$  for any permutation  $f$  of  $0, 1, \dots, p$ . Here  $U_\sigma := U_{i_0} \cap U_{i_1} \cap \dots \cap U_{i_p}$  and  $f(\sigma) = (i_{f(0)}, \dots, i_{f(p)})$  if  $\sigma = (i_0, \dots, i_p)$ . Many authors replace  $S_p(\mathcal{U})$  with the set of all  $p + 1$ -tuples  $I^{p+1}$  when working only with sheaves, but for presheaves (in particular for Proposition 5.28) the distinction is important. For each  $p \geq 0$  we define a coboundary map  $d : \check{C}^p(\mathcal{U}, \mathcal{F}) \rightarrow \check{C}^{p+1}(\mathcal{U}, \mathcal{F})$  so that for all  $\alpha \in \check{C}^p(\mathcal{U}, \mathcal{F})$  and  $\sigma = (i_0, i_1, \dots, i_{p+1}) \in S_{p+1}(\mathcal{U})$  it holds that

$$(d\alpha)(i_0, i_1, \dots, i_{p+1}) = \sum_{n=0}^{p+1} (-1)^n \alpha(i_0, \dots, i_{n-1}, i_{n+1}, \dots, i_{p+1})|_{U_\sigma}. \quad (5.3)$$

Setting  $\check{C}^p(\mathcal{U}, \mathcal{F}) = 0$  and  $d = 0$  for  $p < 0$  gives us the Čech (pre)sheaf cochain complex  $\check{C}^\bullet(\mathcal{U}, \mathcal{F})$ . Notice that a morphism  $\mathcal{F} \rightarrow \mathcal{G}$  induces a morphism of complexes  $\check{C}^\bullet(\mathcal{U}, \mathcal{F}) \rightarrow \check{C}^\bullet(\mathcal{U}, \mathcal{G})$  making  $\check{C}^\bullet(\mathcal{U}, -)$  functorial in the presheaf. In fact it is an additive functor in  $\mathcal{F}$ . Also if  $(U_i)_{i \in I} = \mathcal{U} \rightarrow \mathcal{V} = (V_j)_{j \in J}$  (i.e.  $\mathcal{U}$  is a refinement of  $\mathcal{V}$ ) and  $\varphi : I \rightarrow J$  a refinement projection, then this allows us to define a chain map  $\varphi^* : \check{C}^\bullet(\mathcal{V}, \mathcal{F}) \rightarrow \check{C}^\bullet(\mathcal{U}, \mathcal{F})$  which is given in all  $p \geq 0$ ,  $\alpha \in \check{C}^p(\mathcal{V}, \mathcal{F})$  and  $(i_0, \dots, i_p) \in I^{p+1}$  by

$$(\varphi^*\alpha)(i_0, i_1, \dots, i_p) = \alpha(\varphi(i_0), \varphi(i_1), \dots, \varphi(i_p))|_{U_{(i_0, \dots, i_p)}} \quad (5.4)$$

If  $\psi : I \rightarrow J$  is another refinement projection, then this gives a potentially different chain map  $\psi^* : \check{C}^\bullet(\mathcal{V}, \mathcal{F}) \rightarrow \check{C}^\bullet(\mathcal{U}, \mathcal{F})$ . However between these two chain maps there is a cochain homotopy  $h$  given in all  $p \geq 0$ ,  $\alpha \in \check{C}^{p+1}(\mathcal{V}, \mathcal{F})$  and  $(i_0, \dots, i_p) \in I^{p+1}$  by

$$(h\alpha)(i_0, \dots, i_p) = \sum_{n=0}^p (-1)^n \alpha(\varphi(i_0), \varphi(i_1), \dots, \varphi(i_n), \psi(i_n), \psi(i_{n+1}), \dots, \psi(i_p))|_{U_{(i_0, \dots, i_p)}}.$$

Therefore if we consider  $\check{C}^\bullet(-, \mathcal{F})$  as a contravariant functor to  $K^h(\mathbf{Ab})$  it is functorial in the cover. Finally one can check that these two functor structures are compatible and thus turn  $\check{C}^\bullet(-, -)$  into a bifunctor from  $\text{Cov}(X)^{\text{op}} \times \text{Sh}(X)$  to  $K^h(\mathbf{Ab})$ . Now for a presheaf  $\mathcal{F}$  and a cover  $\mathcal{U}$  we can take the  $p$ -th cohomology of  $\check{C}^\bullet(\mathcal{U}, \mathcal{F})$  to obtain the Čech (pre)sheaf cohomology groups of  $\mathcal{U}$  denoted  $\check{H}^p(\mathcal{U}, \mathcal{F})$ , which are bifunctorial in  $\mathcal{U}$  and  $\mathcal{F}$ . While these groups have their uses we will mostly be interested in the Čech (pre)sheaf cohomology groups of  $X$ , denoted  $\check{H}^p(X, \mathcal{F})$ , which is the colimit of  $\mathcal{U} \mapsto \check{H}^p(\mathcal{U}, \mathcal{F})$  over  $\text{Cov}(X)^{\text{op}}$ . Because  $\check{H}^p(-, -)$  is a bifunctor and additive in the presheaf it follows that  $\check{H}^p(X, -)$  is a functor which is additive in the presheaf.

Before we start comparing these notions of (co)homology we first establish some well known properties of sheaf Čech cohomology and use these to prove a result regarding  $\check{H}_1(X)$  when  $X$  is locally connected.

**Proposition 5.27.** *Let  $X$  be a topological space and  $\mathcal{F}$  a sheaf on  $X$ , then*

1. *if  $\mathcal{U}$  is an open cover of  $X$  then  $\check{H}^0(\mathcal{U}, \mathcal{F}) \simeq \mathcal{F}(X)$ ,*
2. *if  $\mathcal{U} \rightarrow \mathcal{V}$  are open covers, then the induced map  $\check{H}^1(\mathcal{V}, \mathcal{F}) \rightarrow \check{H}^1(\mathcal{U}, \mathcal{F})$  is injective,*
3. *for each  $p \geq 0$  there is a morphism  $\check{H}^p(X, \mathcal{F}) \rightarrow H^p(X, \mathcal{F})$  natural in  $\mathcal{F}$ ,*
4. *if  $p = 0, 1$  or if  $X$  is paracompact Hausdorff, then this is an isomorphism.*

**Proof.** 1 follows directly from the sheaf property. By [13, Lemma III.4.4] there is for each  $\mathcal{U}$  a canonical map  $\check{H}^p(\mathcal{U}, \mathcal{F}) \rightarrow H^p(X, \mathcal{F})$  natural in  $\mathcal{F}$  compatible with maps  $\check{H}^p(\mathcal{U}, \mathcal{F}) \rightarrow \check{H}^p(\mathcal{V}, \mathcal{F})$  induced by refinements  $\mathcal{U} \rightarrow \mathcal{V}$  (see [13, Ex. III.4.4b]). The universal property of the colimit gives a map  $\check{H}^p(X, \mathcal{F}) \rightarrow H^p(X, \mathcal{F})$  natural in  $\mathcal{F}$ . The injectivity of the transition maps  $\check{H}^1(\mathcal{V}, \mathcal{F}) \rightarrow \check{H}^1(\mathcal{U}, \mathcal{F})$  follows from the fact that for each  $\mathcal{U} \in \text{Cov}(X)$  the map  $\check{H}^1(\mathcal{U}, \mathcal{F}) \rightarrow H^1(X, \mathcal{F})$  is injective. For this see for example Exercise III.15 of [7] with  $n = m = 1$ . For 4 see [12, pp. 227, 230].  $\square$

**Proposition 5.28.** *Let  $X$  be a topological space and  $\mathcal{F}$  a presheaf on  $X$ . If  $X$  is Hausdorff paracompact, then for all  $p \in \mathbb{Z}$  the homomorphism  $\check{H}^p(X, \mathcal{F}) \rightarrow \check{H}^p(X, \mathcal{F}^+)$  induced by the sheafification morphism  $\mathcal{F} \rightarrow \mathcal{F}^+$  is an isomorphism.*

**Proof.** See [7, Corollary III.4.5].  $\square$

**Theorem 5.29.** *Let  $X$  be a locally connected topological space, then  $\vec{H}_1(X)$  is strict.*

**Proof.** Notice that because  $X$  is locally connected, for any open  $U$  of  $X$  and  $T \in \mathbf{Ab}$  the group  $T_X(U)$  is isomorphic to the product of  $T$  over all connected components of  $U$ . Thus the functor  $T \mapsto T_X(U)$  is represented by  $\bigoplus_{s \in S} \mathbb{Z}$ , where  $S$  is the set of connected components of  $U$ . Let  $\mathcal{U} = (U_i)_{i \in I}$  be an open cover of  $X$ . If we pick a linear order on  $I$  then  $\check{C}^p(\mathcal{U}, T_X)$  can be thought of as the product of  $T_X(U_\sigma)$  where  $\sigma \in S_p(\mathcal{U})$  ranges over all *strictly increasing* tuples. Hence the functor  $T \mapsto \check{C}^p(\mathcal{U}, T_X)$  is represented by  $\bigoplus_{s \in S'} \mathbb{Z}$  for some set  $S'$ . Since the abelian Yoneda embedding is exact it follows that the homology objects of the complex  $(T \mapsto \check{C}^p(\mathcal{U}, -_T))_{p \in \mathbb{Z}}$  in  $\mathbf{Ab}$  are representable. By the exactness of left-exactification these homology objects are the left-exactification of  $T \mapsto \check{H}^p(\mathcal{U}, T_X)$ , so  $\check{H}^p(\mathcal{U}, -_X)$  is representable for each  $\mathcal{U} \in \text{Cov}(X)$ . Now by Lemma 3.14 the left-exactification of the colimit  $\mathcal{U} \mapsto (T \mapsto \check{H}^p(\mathcal{U}, T_X))$  over  $\text{Cov}(X)^{\text{op}}$ , which is the left-exactification of  $\check{H}^p(X, -_X)$ , is the limit of the left-exactifications of  $T \mapsto \check{H}^p(\mathcal{U}, T_X)$  over  $\text{Cov}(X)$ . Now notice that in the case  $p = 1$  we know from Proposition 5.27.3 that for a  $T \in \mathbf{Ab}$  each transition morphism  $\check{H}^1(\mathcal{V}, T_X) \rightarrow \check{H}^1(\mathcal{U}, T_X)$  is injective, so by Proposition 3.6.5 the corresponding morphism on left-exactifications is an epimorphism in  $\mathbf{Ab}$ . We conclude that the left-exactification of  $\check{H}^1(X, -_X)$ , i.e.  $\vec{H}_1(X)$  by Proposition 5.27.4, is the limit of a diagram of representable objects in  $\mathbf{Ab}$  with surjective transition morphisms. By Corollary 3.37 we may conclude that  $\vec{H}_1(X)$  is strict.  $\square$

**Definition 5.30.** Let  $X$  be a topological space and  $\mathcal{C}$  an abelian category with all products and filtered colimits. We can extend the definition of Čech (pre)sheaf homology to a presheaf  $\mathcal{F}$  on  $X$  with values in  $\mathcal{C}$  by altering the definition slightly. If  $\mathcal{U} = (U_i)_{i \in I}$  is an open cover pick a linear order on  $I$  and for each  $p \in \mathbb{Z}$  define  $\check{C}^p(\mathcal{U}, T)$  as the product in  $\mathcal{C}$  of  $\mathcal{F}(U_{i_0} \cap \cdots \cap U_{i_p})$  where  $(i_0, \dots, i_p)$  ranges over all *strictly increasing* sequences in  $I$  so that  $U_{i_0} \cap \cdots \cap U_{i_p} \neq \emptyset$ . Then using the universal property of products and the additive structure on hom-sets in  $\mathcal{C}$  the coboundary homomorphisms, isomorphisms showing this definition does not depend on the choice of ordering on  $I$ , and transition morphisms induced by refinement projections can be defined using the restriction morphisms of  $\mathcal{F}$ . For the sake of brevity we omit the details of this definition. Finally  $\check{H}^p(\mathcal{U}, \mathcal{F})$  can then be defined as the cohomology of this complex and  $\check{H}^p(X, \mathcal{F})$  as the colimit of these objects in  $\mathcal{C}$ . See [25, Def. B.2.3] for more details and even further generalisations.

The advantage of this generalisation is that if  $\mathcal{C}$  is abelian, has all coproducts and cofiltered limits then we can apply this definition to a presheaf in  $\mathcal{C}^{\text{op}}$ , i.e. a precosheaf  $\mathcal{A}$  in  $\mathcal{C}$ . This gives for  $\mathcal{U}$  a chain complex  $\check{C}_\bullet(\mathcal{U}, \mathcal{A})$  in  $\mathcal{C}$  which is functorial in  $\mathcal{U}$  and  $\mathcal{A}$  up to homotopy and thus Čech (pre)cosheaf homology groups  $\check{H}_p(\mathcal{U}, \mathcal{A})$  and  $\check{H}_p(X, \mathcal{A})$ . If  $\mathcal{C} = \mathbf{Ab}$  and  $G \in \mathbf{Ab}$  we write (for consistency with the previous notation for pro-homology groups)  $\check{H}_p(X, G) := \check{H}_p(X, G^X)$  where  $G^X$  is the constant cosheaf with coefficients in  $G$  (see Example 5.8).



**Remark 5.31.** Note that [7, VI.4] defines the Čech homology of a presheaf with values in  $\mathbf{Ab}$ , which (after unfolding the above abstract definition) can be seen to be the same as the above definition with  $\mathcal{C} = \mathbf{Ab}$ . The following lemma tells us that we lose no information by instead working with presheaves in  $\tilde{\mathbf{Ab}}$ . Combining this with the upcoming Theorem 5.33 this shows that this notion of Čech homology can also be derived from Čech (pre)sheaf homology.

**Lemma 5.32.** *Let  $X$  be a topological space and let  $\mathcal{A} \in \text{PCoSh}(X, \mathbf{Ab})$ . Let  $\tilde{\mathcal{A}} \in \text{PCoSh}(X, \tilde{\mathbf{Ab}})$  be the presheaf given by  $\tilde{\mathcal{A}}(U) = \overrightarrow{\mathcal{A}(U)}$ , then  $\check{H}_p(X, \mathcal{A})$  is the coreflection in  $\mathbf{Ab}$  of  $\check{H}_p(X, \tilde{\mathcal{A}})$ .*

**Proof.** The abelian Yoneda embedding preserves coproducts, so we have  $\overrightarrow{\check{C}_\bullet(\mathcal{U}, \mathcal{A})} \simeq \check{C}_\bullet(\mathcal{U}, \tilde{\mathcal{A}})$ . Since the abelian Yoneda embedding is exact it follows that

$$\overrightarrow{\check{H}_p(\mathcal{U}, \mathcal{A})} = \overrightarrow{h_p(\check{C}_\bullet(\mathcal{U}, \mathcal{A}))} \simeq h_p(\overrightarrow{\check{C}_\bullet(\mathcal{U}, \mathcal{A})}) \simeq h_p(\check{C}_\bullet(\mathcal{U}, \tilde{\mathcal{A}})) = \check{H}_p(\mathcal{U}, \tilde{\mathcal{A}}).$$

Since  $\check{H}_p(X, \mathcal{A})$  is by definition the limit in  $\mathbf{Ab}$  of  $\check{H}_p(\mathcal{U}, \mathcal{A})$  over  $\mathcal{U}$  and  $\check{H}_p(X, \tilde{\mathcal{A}})$  is by definition the limit in  $\tilde{\mathbf{Ab}}$  of  $\check{H}_p(\mathcal{U}, \tilde{\mathcal{A}})$  over  $\mathcal{U}$  it follows from Proposition 3.29 that  $\check{H}_p(X, \mathcal{A})$  is the coreflection in  $\mathbf{Ab}$  of  $\check{H}_p(X, \tilde{\mathcal{A}})$ .  $\square$

**Theorem 5.33.** *Let  $X$  be a topological space,  $p \in \mathbb{Z}$  and  $\mathcal{A} \in \text{PCoSh}(X, \tilde{\mathbf{Ab}})$ , then  $\check{H}_p(X, \mathcal{A})$  is the left-exactification of the functor  $\mathbf{Ab} \rightarrow \mathbf{Ab}$  given by  $T \mapsto \check{H}^p(X, -) \circ F$  where  $F : \mathbf{Ab} \rightarrow \text{PSh}(X)$  is the functor corresponding to  $\mathcal{A}$ . In particular if  $G \in \mathbf{Ab}$  then  $\check{H}_p(X; G)$  is the left-exactification of  $T \mapsto \check{H}^p(X, \text{Hom}(G, T)_X)$ .*

**Proof.** The main observation to make is that  $\check{C}_\bullet(\mathcal{U}, \mathcal{A})(T) \simeq \check{C}^\bullet(\mathcal{U}, F(T))$  for  $\mathcal{U} \in \text{Cov}(X)$  and  $T \in \mathbf{Ab}$ . This is because the contravariant functor  $(-)(T) : \tilde{\mathbf{Ab}} \rightarrow \mathbf{Ab}$  maps colimits to limits (see Lemma 3.12) and because  $\mathcal{A}(U)(T) \simeq F(T)(U)$  by definition for open  $U \subseteq X$ . Since the boundary homomorphisms of  $\check{C}_\bullet(\mathcal{U}, \mathcal{A})$  are defined using the universal property of colimits and the boundary homomorphisms are defined dually using the universal properties of limits, it also follows that this isomorphism respects the boundary homomorphisms. Since if  $T \in \mathbf{Ab}$  is injective the functor  $(-)(T) : \tilde{\mathbf{Ab}} \rightarrow \mathbf{Ab}$  is exact (see Proposition 3.6) it follows that

$$\check{H}_p(\mathcal{U}, \mathcal{A})(T) = (h_p(\check{C}_\bullet(\mathcal{U}, \mathcal{A}))) (T) \simeq h^p(\check{C}_\bullet(\mathcal{U}, \mathcal{A})(T)) \simeq h^p(\check{C}^\bullet(\mathcal{U}, F(T))) = \check{H}^p(\mathcal{U}, F(T))$$

and hence  $\check{H}_p(\mathcal{U}, \mathcal{A})$  is the left-exactification of  $T \mapsto \check{H}^p(\mathcal{U}, F(T))$ . Again by a similar proof sketch as with the boundary homomorphisms this isomorphism respects the transition morphisms. Now again because left-exactification, considered as a contravariant functor  $\text{LE} : [\mathbf{Ab}, \mathbf{Ab}]_{\text{add}} \rightarrow \tilde{\mathbf{Ab}}$  maps colimits to limits it follows that

$$\check{H}_p(X, \mathcal{A}) \simeq \lim_{\mathcal{U}, \text{in } \tilde{\mathbf{Ab}}} \text{LE}(T \mapsto \check{H}^p(\mathcal{U}, F(T))) \simeq \text{LE}(T \mapsto \text{colim}_{\mathcal{U}} \check{H}^p(\mathcal{U}, F(T))) = \text{LE}(\check{H}^p(X, -) \circ F). \quad \square$$

**Lemma 5.34.** *Let  $X$  be a topological space and  $T \in \mathbf{Ab}$ . Consider the constant presheaf  $T_X^{\text{presheaf}}$  on  $X$  given by  $T_X^{\text{presheaf}}(U) = T$ . Then there is an isomorphism  $\check{H}^p(X, T_X^{\text{presheaf}}) \simeq H_N^p(X, T)$  natural in  $T$ .*

**Proof.** It suffices to show that for any open cover  $\mathcal{U}$  of  $X$  there is an isomorphism  $\check{C}^\bullet(\mathcal{U}, T_X^{\text{presheaf}}) \simeq \check{C}_N^\bullet(\mathcal{U}, T)$  compatible with the transition morphisms induced by refinement projections. For a fixed  $p \in \mathbb{Z}_{\geq 0}$  it is not hard to see that  $\text{Hom}(\check{C}_p^N(\mathcal{U}), T) \simeq \check{C}^p(\mathcal{U}, T_X^{\text{presheaf}})$  by mapping an  $f : \check{C}_p^N(\mathcal{U}) \rightarrow T$  to  $(i_0, \dots, i_p) \mapsto f(\langle i_0, \dots, i_p \rangle)$ . Note here that  $T_X^{\text{presheaf}}$  gives  $T$  on all opens, so this is well-defined. Also the construction of  $\check{C}_p^N(\mathcal{U})$  as the subset of functions  $S_p(\mathcal{U})$  which are antisymmetric and are zero on tuples with duplicates corresponds exactly to the construction of  $C_p^N(\mathcal{U})$  as the free group on  $S_p(\mathcal{U})$  modulo certain relations. Simply by inspecting formula (5.3) and (5.4) and ignoring the restrictions to open subsets in them (they do nothing for  $T_X^{\text{presheaf}}$ ) one also sees that this isomorphism respects the boundary morphisms and the transition morphisms induced by refinement projections.  $\square$

**Theorem 5.35.** *Let  $X$  be a topological space, then for each  $p \in \mathbb{Z}$  and  $T \in \mathbf{Ab}$  there are canonical*



morphisms

$$H^p(X, T_X) \leftarrow \check{H}^p(X, T_X) \leftarrow \check{H}_N^p(X, T) \leftarrow H_{\text{shp}}^p(X, T)$$

in  $\mathbf{Ab}$  natural in  $T$ . Consequently for all  $G \in \mathbf{Ab}$  there are canonical morphisms

$$\vec{H}_p(X; G) \rightarrow \vec{H}_p(X; G) \rightarrow \vec{H}_p^N(X; G) \rightarrow \vec{H}_p^{\text{shp}}(X; G)$$

in  $\vec{\mathbf{Ab}}$ . If  $X$  is paracompact Hausdorff then all of these morphisms are isomorphisms.

**Proof.** The morphism  $H^p(X, T_X) \leftarrow \check{H}^p(X, T_X)$  and that it is an isomorphism when  $X$  is Hausdorff paracompact follows from Proposition 5.27.2& 3. The morphism  $\check{H}_N^p(X, T) \rightarrow \check{H}^p(X, T_X)$  is given by applying  $\check{H}^p(X, -)$  to the sheafification morphism  $T_X^{\text{presheaf}} \rightarrow T_X$  and composing with the isomorphism from Lemma 5.34. If  $X$  is paracompact Hausdorff then this gives an isomorphism by Proposition 5.28. The morphism  $\check{H}_N^p(X, T) \leftarrow \check{H}_{\text{shp}}^p(X, T)$  follows from the fact that  $\check{H}_{\text{shp}}^p(X, T)$  is the colimit of a subdiagram of the one defining  $\check{H}_N^p(X, T)$ . If  $X$  is Hausdorff paracompact, then all open covers of  $X$  are normal (see [19, App. 1, 3.1]), so this morphism is an isomorphism. The statement for the homology pro-groups follows from the above by applying the functor  $(-) \circ \tilde{G} : [\mathbf{Ab}, \mathbf{Ab}]_{\text{add}} \rightarrow [\mathbf{Ab}, \mathbf{Ab}]_{\text{add}}$  and the left-exactification functor  $[\mathbf{Ab}, \mathbf{Ab}]_{\text{add}} \rightarrow [\mathbf{Ab}, \mathbf{Ab}]_{\text{LE}}$  and applying Theorem 5.25 and Theorem 5.33. Since functors map isomorphisms to isomorphisms the obtained morphisms will also be isomorphisms when  $X$  is paracompact Hausdorff.  $\square$

**Remark 5.36.** This theorem settles Conjecture 1.0.3.1 of [25] in the positive. Conjecture 1.0.3.2 states that shape homology agrees with cosheaf homology with constant coefficients on *all* topological spaces provided we work on the site with objects the open sets of  $X$  and coverings given by *normal* open coverings. A similar proof technique may also work for this conjecture, but this would require adapting Čech sheaf cohomology and its relation to sheaf cohomology to this different site.

**Intuition 5.37.** Here we give some intuition as to why paracompact Hausdorffness is necessary for the isomorphism between Čech sheaf cohomology and Čech nerve cohomology. First of all for each cover  $\mathcal{U}$ , any  $T \in \mathbf{Ab}$  and  $p \geq 0$  we can think of  $\check{C}_N^\bullet(\mathcal{U}, T)$  as the subgroup of  $\check{C}^\bullet(\mathcal{U}, T_X)$  consisting of all cochains which give a *constant* (not just locally constant) function with values in  $T$  on each non-empty  $(p+1)$ -tuple intersection. The question is then when the inclusion gives an isomorphism after we take the induced map on cohomology and then the induced maps between the colimits over  $\mathcal{U}$ . One easily verifies that for this it suffices to show the following: Given  $\alpha \in \check{C}^p(\mathcal{U}, T_X)$  there exists a cover  $\mathcal{V}$  and a refinement projection  $\varphi : \mathcal{V} \rightarrow \mathcal{U}$  so that  $\varphi^* \alpha$  is in  $\check{C}_N^p(\mathcal{V}, T)$ . Consider the example of a space  $X$  with  $\mathcal{U} = \{U_1, U_2\}$  a cover so that  $U_1 \cap U_2$  consists of 2 clopen connected components  $C_1, C_2$  and  $\alpha \in C^1(\mathcal{U}; T_X)$  with  $\alpha_{12} \in T_X(U_1 \cap U_2)$  taking two different values on these two components. This situation is illustrated in Figure 1a for  $X$  a circle. Now if  $X$  is paracompact Hausdorff then we can apply the so called *shrinking lemma* which states that for any locally finite open cover  $\mathcal{U} = (U_i)_{i \in I}$  of  $X$  there exists another open cover  $(W_i)_{i \in I}$  so that  $\overline{W_i} \subseteq U_i$  for all  $i \in I$ . In this case it gives us open sets  $W_1, W_2$  so that  $X = W_1 \cup W_2$  and  $\overline{W_i} \subseteq U_i$ . We can then construct a cover  $\mathcal{V} = \{V_1, V_2, V_3, V_4\}$  where  $V_1 = W_1 \setminus \overline{W_2}, V_2 = W_2 \setminus \overline{W_1}, V_3 = C_1, V_4 = C_2$  with refinement projection  $\varphi : \mathcal{V} \rightarrow \mathcal{U}, 1 \mapsto 1, 2 \mapsto 2, 3 \mapsto 1, 4 \mapsto 1$ . This construction is illustrated in Figure 1b. Now the only non-empty intersections are  $W_1 \cap W_3, W_1 \cap W_4, W_2 \cap W_3, W_2 \cap W_4$ , on which  $\varphi^* \alpha$  takes values  $\alpha_{12}|_{C_1}, \alpha_{12}|_{C_2}, 0, 0$  respectively, which are all constant. In Example 5.47 we will see a similar situation for a non-Hausdorff space where such a refinement is not possible. The author originally tried to prove the equivalence of Čech nerve and Čech sheaf cohomology by generalising the above construction to arbitrary covers (which worked only for  $p = 1$  and was very complicated), before realising it was an immediate corollary of Proposition 5.28. The proof of Proposition 5.28 is a little more abstract, but also relies heavily on the shrinking lemma.

**Corollary 5.38.** Let  $X$  be a compact Hausdorff space, then  $\vec{H}_p(X)$  is pro-finitely generated for all  $p \in \mathbb{Z}$ .

**Proof.** Clearly  $X$  is Hausdorff paracompact so by the theorem we can work with Čech nerve pro-homology instead. Since  $X$  is compact, the poset of open covers of  $X$  has a final subset consisting of

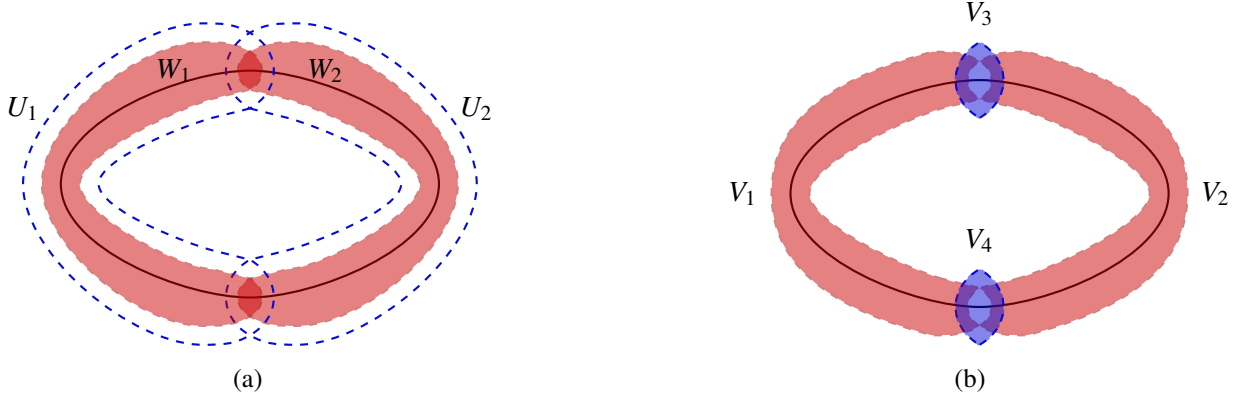


Figure 1: Illustrations of the construction from Intuition 5.37.

finite covers. For a finite  $\mathcal{U} \in \text{Cov}(X)$  the Čech nerve complex  $\check{C}_\bullet^N(\mathcal{U})$  consists of finitely generated groups, so its homology groups are also finitely generated. It follows that  $\vec{H}_p^N(X)$  is a cofiltered limit of finitely generated groups and hence pro-finitely generated by Proposition 3.23.  $\square$

**Corollary 5.39.** *Let  $X$  be a compact metrizable space, then  $\vec{H}_p(X)$  is sequential for all  $p \in \mathbb{Z}$ .*

**Proof.** Since compact metrizable spaces are Hausdorff paracompact, we can use Čech nerve pro-homology to compute  $\vec{H}_p(X)$ . This expresses  $\vec{H}_p(X)$  as a cofiltered limit in  $\mathbf{Ab}$  of abelian groups indexed by the open covers of  $X$ . However by the Lebesgue number lemma the open covers of  $X$  have a final subset consisting of the covers consisting of all open balls of radius  $2^{-n}$  for a fixed  $n \in \mathbb{N}$ . This shows that  $\vec{H}_p(X)$  is the limit in  $\mathbf{Ab}$  of a diagram of abelian groups indexed by  $\mathbb{N}$ .  $\square$

**Theorem 5.40.** *Let  $X$  be a compact metrizable space, then for each  $p \in \mathbb{Z}$  the pro-group  $\vec{H}_p(X; \mathbb{Q})$  is topologically representable. If in addition  $X$  is locally connected, then  $\vec{H}_1(X)$  is topologically representable.*

**Proof.**  $\vec{H}_p(X)$  is pro-finitely generated by Corollary 5.38, so  $\vec{H}_p(X) \otimes \vec{\mathbb{Q}}$  is strict by Proposition 3.52. Using Corollary 3.50 we conclude that  $\vec{H}_p(X; \mathbb{Q})$  is strict. For the sequentiality we take a diagram in  $\mathbf{Ab}$  indexed by  $\mathbb{N}$  with limit  $\vec{H}_p(X)$  which exists by Corollary 5.39. By Corollary 3.50 and Lemma 3.51, tensoring this diagram with  $\mathbb{Q}$  gives a diagram in  $\mathbf{Ab}$  whose limit is  $\vec{H}_p(X; \mathbb{Q})$  in  $\mathbf{Ab}$ , which shows that  $\vec{H}_p(X; \mathbb{Q})$  is sequential. Using Theorem 4.20 we conclude that  $\vec{H}_p(X; \mathbb{Q})$  is topologically representable. If  $X$  is locally connected, then  $\vec{H}_1(X)$  is sequential by Corollary 5.39 and strict by Theorem 5.29, so topologically representable by Theorem 4.20.  $\square$

## 5.4 Singular homology and finite spaces

Recall from the motivation of sheaf homology that if a topological space  $X$  is nice enough for the singular cohomology functors  $H_{\text{sing}}^p(X, -)$  to be isomorphic to the constant coefficient sheaf cohomology functor  $H^p(X, -_X)$ , then the singular homology groups of  $X$  represent the sheaf homology functors of  $X$  for any coefficient group. We now review some literature on exactly when  $H_{\text{sing}}^p(X, -) \simeq H^p(X, -_X)$ .

**Definition 5.41.** A topological space  $X$  is said to be *locally cohomologically connected with respect to  $T$*  ( $T$ -LCHC) if for any  $x \in X$  and  $p \in \mathbb{Z}$  the colimit of the relative singular cohomology  $H_{\text{sing}}^p(U, \{x\}, T)$  vanishes as  $U$  ranges over the open neighbourhoods of  $x$ . We say that  $X$  is *locally cohomologically connected* (LCHC) if  $X$  is  $T$ -LCHC for all  $T \in \mathbf{Ab}$ . We say that  $X$  is *strongly locally homologically connected* (SLHC) if for any neighbourhood  $U$  of a point  $x \in X$  there exists a neighbourhoods  $x \in V \subseteq U$

so that the inclusion  $V \rightarrow U$  induces the zero map  $H_p^{\text{sing}}(V, \mathbb{Z}) \rightarrow H_p^{\text{sing}}(U, \mathbb{Z})$  for all  $p \in \mathbb{Z}$ . Both of these conditions are satisfied if  $X$  is *locally contractible*, meaning that each open neighbourhood of a point  $x \in X$  contains a contractible neighbourhood of  $x$ .

**Theorem 5.42.** *Let  $X$  be a paracompact Hausdorff SLHC (e.g. locally contractible) topological space. Then for all  $p \in \mathbb{Z}$  and  $G \in \mathbf{Ab}$  there is an isomorphism*

$$\overrightarrow{H_p^{\text{sing}}(X; G)} \simeq \tilde{H}_p(X; G)$$

*natural in  $X$ .*

**Proof.** In [7, Theorem III.1.1] it is shown that for a paracompact Hausdorff SLHC space  $X$  and  $T \in \mathbf{Ab}$  there is an isomorphism  $H_p^{\text{sing}}(X, T) \simeq H^p(X, T_X)$  natural in  $X$  and  $T$ . If  $G \in \mathbf{Ab}$ , then by the functoriality of precomposing with  $\tilde{G} \in \tilde{\mathbf{Ab}}$  and left-exactification we obtain an isomorphism

$$\tilde{H}_p(X; G) \simeq \overline{T \mapsto H^p(X, \text{Hom}(G, T))}$$

natural in  $X$ . Applying Proposition 2.9 then gives us an isomorphism  $\text{Hom}(H_p^{\text{sing}}(X; G), -) \simeq \tilde{H}_p(X; G)$ . If  $f : X \rightarrow Y$  is a continuous map, then the induced morphisms on singular homology and cohomology are simply the maps induced by the chain map between the singular chain complexes  $C_\bullet^{\text{sing}}(X) \rightarrow C_\bullet^{\text{sing}}(Y)$ . Hence to show that the isomorphism  $\text{Hom}(H_p^{\text{sing}}(X; G), -) \simeq \tilde{H}_p(X; G)$  is natural in  $X$  it suffices to show that the isomorphism  $\text{Hom}(h_p(C_\bullet) \otimes G, -) \simeq h^p(X, \text{Hom}(G, -))$  from Proposition 2.9 is natural in  $C_\bullet$ . We leave the verification of this to the reader.  $\square$

**Corollary 5.43.** *Let  $X$  be a locally contractible compact Hausdorff space, then the integral singular homology and cohomology groups of  $X$  are finitely generated.*

**Proof.** By Corollary 5.38  $\tilde{H}_p(X)$  is pro-finitely generated for all  $p \in \mathbb{Z}$ , so  $\tilde{H}$  is pro-finitely generated where  $H := H_p^{\text{sing}}(X)$ . Recall that this means that any element  $\tilde{H}(T) = \text{Hom}(H, T)$  for  $T \in \mathbf{Ab}$  factors through a finitely generated group. Applying this to the element  $\text{id}_H \in \tilde{H}(H)$  we see that  $\text{id}_H$  factors through a finitely generated group which can only happen if  $H$  is finitely generated. Using the universal coefficient theorem for singular cohomology and the fact that  $\text{Ext}^1(T, \mathbb{Z})$  is finitely generated for  $T \in \mathbf{Ab}$  finitely generated, we conclude that the integral singular cohomology groups of  $X$  are also finitely generated.  $\square$

While the naturality of the isomorphism from [7, Theorem III.1.1] in both  $X$  and  $T$  is very useful, this and other classical comparison theorems all require  $X$  to be paracompact Hausdorff. This prevents us from comparing sheaf and singular (co)homology for non-Hausdorff finite spaces which, as shown in [20], have very interesting singular homology. Surprisingly recently it was shown in [26] that assuming only that  $X$  is CHLC, the singular and sheaf cohomology groups are isomorphic. In [23] an  $\infty$ -category theoretic argument was given for the same result. Unfortunately to apply Proposition 2.9 it is essential that the isomorphism is natural in the coefficient group and neither of these papers check this. Checking that Sella's isomorphism is natural in  $T$  would require us to recap the entire construction which we unfortunately do not have time for. Also the author's understanding of  $\infty$ -category theory is too limited to be able to say anything about Petersen's construction. At least a quick scan of Sella's construction shows that nothing particularly 'unnatural' is ever done to the coefficient group, so it is almost certainly natural in  $T$ . Nonetheless we will mark this as an assumption when we use it.

( $\star$ ) The isomorphisms from either [26] or [23] are natural in the coefficient group.

Assuming this holds and applying Proposition 2.9 again gives the following theorem.

**Theorem 5.44.** (Assuming ( $\star$ )) *Let  $X$  be a LCHC (e.g. locally contractible) topological space. Then for all  $p \in \mathbb{Z}$  and  $G \in \mathbf{Ab}$  the sheaf homology functor  $\tilde{H}_p(X; G)$  is represented by the singular homology group  $H_p^{\text{sing}}(X; G)$ .*

**Corollary 5.45.** (Assuming  $(\star)$ ; Conjecture 1.0.5 of [25]) *If  $X$  is a locally finite space and  $G \in \mathbf{Ab}$ , then  $\vec{H}_p(X; G)$  is represented by the  $p$ -th singular homology group of  $X$  with coefficients in  $G$ .*

**Proof.** It suffices to show that  $X$  is locally contractible. Let  $x \in X$  and  $W$  the smallest neighbourhood of  $x$ . It suffices to show that  $W$  is contractible, since then each neighbourhood of  $x$  contains the contractible neighbourhood  $W$ . Define the homotopy  $H : [0, 1] \times W \rightarrow W$  in all  $w \in W$  by  $H(t, w) = w$  for  $t \in [0, \frac{1}{2})$  and  $H(t, w) = x$  for  $x \in [\frac{1}{2}, 1]$ . This is continuous because for every open  $U \subseteq W$  not containing  $x$  we have  $H^{-1}[U] = [0, \frac{1}{2}) \times U$ . By the minimality of  $W$  the only open containing  $x$  is  $W$  itself and for this we have  $H^{-1}[W] = [0, 1] \times W$ . We conclude that  $\text{id}_W$  is homotopic to  $w \mapsto x$ , so  $W$  is contractible.  $\square$

Using this corollary we give, for each of the 3 morphisms between pro-homologies in Theorem 5.35, an example of a non-Hausdorff finite space where it is not an isomorphism. In the examples below we will calculate the sheaf homology of the spaces using singular homology, but this does assume  $(\star)$ . Alternatively one could obtain the same result by directly calculating the sheaf homology using the homotopy invariance and open Mayer–Vietoris theorems which we will prove in the next chapter.

**Remark 5.46.** Recall from Remark 5.22 that the shape of a space  $X$  is constructed from its continuous maps to polyhedra. Notice that if  $x, x' \in X$  are two points which cannot be separated by disjoint neighbourhoods, then for any continuous map  $f : X \rightarrow Y$  to a Hausdorff space  $Y$  we have  $f(x) = f(x')$  (otherwise take the inverse image of disjoint neighbourhoods of  $f(x)$  and  $f(x')$ ). Using this it is not hard to see that all continuous maps from any of the spaces in the upcoming examples to a Hausdorff space are constant. Hence all continuous maps from them to a polyhedron are constant, so they have the shape of a point and thus the shape homology of a point.

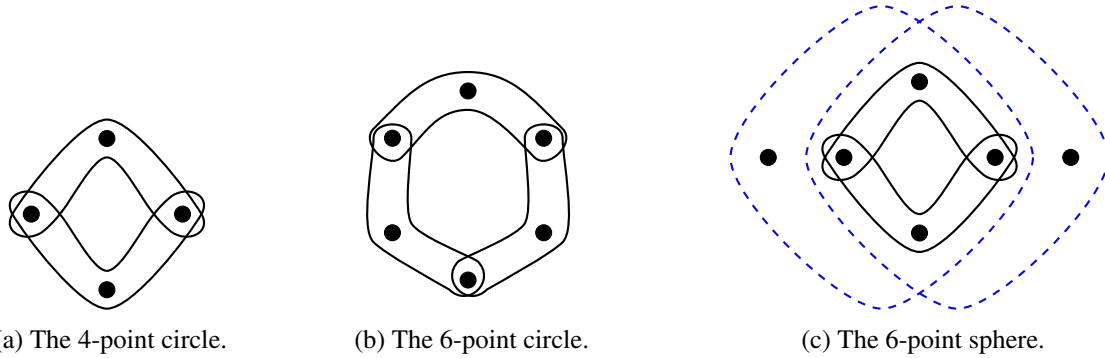


Figure 2: Illustrations of the finite topological spaces from Examples 5.47-5.49. Points enclosed by a closed curve indicate open sets which generate the topology.

**Example 5.47.** (4-point circle, see also [25, Ex. 4.0.2]) Consider the space  $X = \{a, b, c, d\}$  with open sets

$$\emptyset, \{a\}, \{c\}, \{a, c\}, \{a, b, c\}, \{c, d, a\}, X$$

See Figure 2a. This space has a minimal (or most refined) cover  $\mathcal{U}$  consisting of  $\{a, b, c\}$  and  $\{c, d, a\}$ , so both Čech sheaf cohomology and Čech nerve pro-homology of the space reduce to the (co)homology of  $\mathcal{U}$ . First of all notice that the nerve of  $\mathcal{U}$  is just a pair of points connected by an edge, so  $\check{H}_1^N(X) = 0$ . Secondly for all  $T \in \mathbf{Ab}$  the Čech complex  $\check{C}^\bullet(\mathcal{U}, T_X)$  reduces to

$$0 \rightarrow T \times T \rightarrow T \times T \rightarrow 0$$

with the middle arrow given by  $(a, b) \mapsto (a + b, a + b)$ . From this we see that  $\check{H}^1(X, T_X) \simeq T$ , so  $\check{H}_1(X) \simeq \mathbb{Z}$  by Theorem 5.33. The crucial difference between this and the nerve complex is that here it is detected that the intersection  $\{a, b, c\} \cap \{c, d, a\}$  consists of two separate points. It follows that the morphism  $\check{H}_1(X) \rightarrow \check{H}_1^N(X)$  is not an isomorphism.

**Example 5.48.** (6-point circle) Consider the space  $X = \{a, b, c, d, e, f\}$  with topology generated by

$$\{a\}, \{c\}, \{e\}, \{a, b, c\}, \{c, d, e\}, \{e, f, a\}.$$

See Figure 2b. Again this space has a minimal cover given by  $\mathcal{U}$  consisting of  $\{a, b, c\}$ ,  $\{c, d, e\}$  and  $\{e, f, a\}$ , so both Čech sheaf cohomology and Čech nerve pro-homology of the space reduce to the (co)homology of  $\mathcal{U}$ . Notice that the nerve of  $\mathcal{U}$  is a non-filled triangle, so  $X$  has first Čech nerve pro-homology  $\mathbb{Z}$ . By Remark 5.46  $X$  has the shape of a point, so its shape homology will be zero for  $p > 0$ . This shows that the morphism  $\check{H}_1^N(X) \rightarrow \check{H}_1^{\text{shp}}(X)$  is not an isomorphism. Using a Mayer–Vietoris calculation the first singular homology of  $X$  can be calculated to be  $\mathbb{Z}$ , so the first sheaf homology of  $X$  is  $\mathbb{Z}$ . Since the sheaf homology of a point is 0, this example also shows that sheaf homology is not a shape invariant.

**Example 5.49.** (6-point sphere) Consider the space  $X = \{a, b, c, d, x, y\}$  with topology generated by

$$\{a\}, \{c\}, \{a, b, c\}, \{c, d, a\}, \{a, b, c, d, x\}, \{a, b, c, d, y\}.$$

See Figure 2c. This can also be seen as the *finite suspension* of the 4-point circle. Again this space has a minimal cover given by  $\mathcal{U}$  consisting of  $\{a, b, c, d, x\}$  and  $\{a, b, c, d, y\}$ . Since this cover consists of only two sets, we have  $\check{H}_2(X) = 0$ . However applying Mayer–Vietoris to  $\{a, b, c, d, x\}$  and  $\{a, b, c, d, y\}$ , keeping in mind that  $\{a, b, c, d, x\}$  and  $\{a, b, c, d, y\}$  are contractible as in the proof of Corollary 5.45, we see that  $H_2^{\text{sing}}(X) \simeq H_1^{\text{sing}}(\{a, b, c, d\})$ . Another simple Mayer–Vietoris calculation shows that  $H_1^{\text{sing}}(\{a, b, c, d\}) \simeq \mathbb{Z}$ , so  $\check{H}_2(X)$  is represented by  $\mathbb{Z}$ . We conclude that the morphism  $\check{H}_2(X) \rightarrow \check{H}_2(X)$  is not an isomorphism.

**Remark 5.50.** In some sense the singular homology and the shape of a topological space  $X$  are defined in opposite ways. Singular (co)homology is defined entirely using maps  $\Delta_n \rightarrow X$ , where the  $\Delta_n$  are the standard simplices, which are in particular polyhedra. On the other hand the shape of  $X$  is defined using the maps  $X \rightarrow P$  for polyhedra  $P$  (see Remark 5.22). We also see this in the assumptions for singular and shape (co)homology to agree with sheaf (co)homology: Local cohomological connectedness is clearly a statement about maps from simplices to  $X$ . Also if  $X$  is  $T_1$ , then  $X$  is paracompact if and only if all open coverings of  $X$  are normal (see [19, Appendix. 1, 3.1] for the case that  $X$  is Hausdorff). Then notice that an open cover being normal is a statement about continuous maps from  $X$  to polyhedra (in this case the unit interval). The definition of sheaf (co)homology on the other hand has absolutely nothing to do with polyhedra, so it should not be surprising that it is able to interpolate between these two.

## 5.5 Steenrod-style homology

For locally compact Hausdorff spaces there is also another sheaf based homology theory: Borel–Moore homology. Due to its close relation with sheaves one might think that it is the easiest to relate to sheaf homology. Here we briefly outline some of the properties of Borel–Moore homology and other Steenrod-style homology theories that show they are fundamentally different from sheaf homology, making a comparison theorem necessarily difficult.

First of all Borel–Moore (with base ring  $\mathbb{Z}$ ) satisfies for all locally compact Hausdorff spaces  $X$  a split short exact sequence

$$0 \rightarrow \text{Ext}_{\mathbb{Z}}(H_c^{p+1}(X, \mathbb{Z}_X), \mathbb{Z}) \rightarrow H_p^{\text{BM}}(X; \mathbb{Z}) \rightarrow \text{Hom}(H_c^p(X, \mathbb{Z}), \mathbb{Z}) \rightarrow 0$$

in **Ab** where  $H_c^p$  denotes sheaf cohomology with compact support [7, p. 292]. Note that this is the opposite of the more familiar universal coefficient theorem for sheaf homology (Theorem 2.17) where sheaf cohomology is in the middle and  $\check{H}_{p-1}(X)$  and  $\check{H}_p(X)$  are on the sides. If  $C_{\bullet}$  is a chain

complex of finite rank free abelian groups (e.g. the cellular complex of a finite CW-complex), then one could obtain a similar short exact sequence relating  $h^{p+1}(C_\bullet, \mathbb{Z})$  and  $h^p(C_\bullet, \mathbb{Z})$  to the homology of  $\text{Hom}(\text{Hom}(C_\bullet, \mathbb{Z}), \mathbb{Z})$  by applying the universal coefficient theorem for free chain complexes to  $(\text{Hom}(C_{-n}, \mathbb{Z}))_{n \in \mathbb{Z}}$ . Hence Borel–Moore homology behaves more like a co-cohomology theory than a homology theory. One consequence of this sequence is that ext terms from higher homology can drop down to make 0th homology more complicated. For example [7, p. V.5.12] gives an example of a connected compact metric space  $X$  with  $H_0^{\text{BM}}(X) \simeq \mathbb{Z} \oplus \text{Ext}(\mathbb{Q}, \mathbb{Z})$ . On the other hand, as we will see in Corollary 6.6,  $\tilde{H}_0(X) \simeq \tilde{\mathbb{Z}}$  for all connected spaces  $X$ . Another major difference is that Borel–Moore homology generalises to have coefficients in a sheaf while sheaf homology generalises to have coefficients in cosheaves.

Borel–Moore homology is a part of a larger class of abelian group valued homology theories: The Steenrod-style homology theories [4, p. 148]. These are abelian group valued homology theories which extend the Steenrod homology groups beyond compact metric spaces. The Steenrod homology groups can be axiomatised on compact metric pairs and in particular satisfy the Eilenberg–Steenrod axioms [21]. One application of Steenrod homology is that it extends Alexander duality to arbitrary subspaces of  $S^n$  [21, p. 80]. Another common theme of Steenrod-homology theories that sets them apart from sheaf homology is their close relation to derived limits. For example if  $(X_i)_{i \geq 0}$  is an inverse system of compact metric spaces indexed by  $\mathbb{Z}_{\geq 0}$  with limit  $X$ , then there exists a short exact sequence

$$0 \rightarrow \lim_i^1 H_{p+1}(X_i) \rightarrow H_p(X) \rightarrow \lim_i H_p(X_i) \rightarrow 0$$

where  $H_p$  denotes the Steenrod homology [21, Theorem 4]. (See also [7, p. V.5.15] for the analogous statement for Borel–Moore.) On the other hand, as we will see in section 6.6, the induced map  $\tilde{H}_p(X) \rightarrow \lim_i \tilde{H}_p(X_i)$  is an isomorphism.

Additionally *strong shape homology*, a variation of shape homology, extends Steenrod homology to arbitrary topological spaces [2, p. 2]. However the close relation to higher derived limits causes elementary properties such as mapping disjoint unions of spaces to coproducts depend on your underlying set theoretic axioms [2]. On the other hand  $\tilde{H}_p(\coprod_i X_i) \simeq \bigoplus_i \tilde{H}_p(X_i)$  for all families of spaces  $(X_i)_{i \in I}$ . (This follows from the fact that  $H^p(\coprod_i X_i, \mathcal{F}) \simeq \prod_i H^p(X_i, \mathcal{F}|_{X_i})$  for all sheaves  $\mathcal{F}$  on  $\coprod_i X_i$ .)



## 6 Calculation theorems

In this chapter we use existing theorems for sheaf cohomology from [14] and [7] to obtain corresponding theorems for sheaf homology. We use these to calculate the sheaf homology of various spaces that illustrate the general theorems from the previous chapters. In these calculations we attempt to use as little of the existing theory of singular and shape (co)homology as possible in order to illustrate that sheaf (co)homology functions as an independent theory. In particular we realise the infinite topological product  $\prod_{i \geq 0}^{\text{Top}} \mathbb{Z}$  as the first sheaf homology of the Hawaiian earring and the topological group of 2-adic integers  $\mathbb{Z}_2$  as the first sheaf homology of a quotient of the solid torus. We also show that the sheaf homology of the topologist's sine curve is not topologically representable, but can still be expressed as the limit of an explicit inverse system of abelian groups.

As in section 5 we see that sheaf homology contains no new information compared to sheaf cohomology. Nonetheless we hope that these examples illustrate that sheaf homology can be used to summarize the entire sheaf cohomology functor using very concrete mathematical objects (topological groups, inverse systems of groups) and that these objects accurately reflect the geometry of the space.

In section 6.1 we describe the functoriality of sheaf homology with constant coefficients in the topological space. In section 6.2 we show that the 0th homology of a space  $X$  can be concretely described using locally constant functions on  $X$  and topologically represented by the free pro-discrete abelian group on the space of quasi-components of  $X$ . In section 6.3 we state the Vietoris–Begle mapping theorem for sheaf (co)homology and use it to prove a very general notion of homotopy invariance for sheaf (co)homology. In sections 6.4 and 6.5 we prove Mayer–Vietoris theorems for sheaf (co)homology and use it to calculate the sheaf homology of various spaces. In particular closed Mayer–Vietoris allows us to cut straight through the singularities of spaces. In section 6.6 we give the continuity theorem for the sheaf (co)homology of inverse systems of compact Hausdorff spaces which allows us to directly obtain inverse systems representing the sheaf homology of various spaces. In section 6.7 we discuss how relative sheaf cohomology gives rise to relative sheaf homology and use it to give an example of a space  $X$  with  $\tilde{H}_1(X)$  topologically represented by the 2-adic integers  $\mathbb{Z}_2$  and with a non-trivial universal coefficient theorem for sheaf homology.

For the rest of this chapter all sheaves take values in **Ab** and we denote  $\text{Sh}(X)$  for the category of sheaves on a space  $X$ .

### 6.1 Functoriality in the space

**Definition 6.1.** Let  $f : X \rightarrow Y$  be a continuous map of topological spaces. Recall that the functor  $f_* : \text{Sh}(X) \rightarrow \text{Sh}(Y)$ , which maps a sheaf  $\mathcal{F}$  to the sheaf  $V \mapsto \Gamma(f^{-1}V, \mathcal{F})$  on  $Y$ , has a left adjoint  $f^{-1} : \text{Sh}(Y) \rightarrow \text{Sh}(X)$ . In particular this means that we have for all open  $\mathcal{F} \in \text{Sh}(Y)$  a morphism  $\mathcal{F} \rightarrow f_* f^{-1} \mathcal{F}$  and hence for all  $V \subseteq Y$  a map  $\Gamma(V, \mathcal{F}) \rightarrow \Gamma(f^{-1}V, f^{-1} \mathcal{F})$  natural in  $\mathcal{F}$ .  $f^{-1}$  is best characterised by the fact that for  $x \in X$  taking the colimit of the above map (composed with the canonical map  $\Gamma(f^{-1}V, f^{-1} \mathcal{F}) \rightarrow (f^{-1} \mathcal{F})_x$ ) as  $V$  ranges over the open neighbourhoods of  $f(x)$  gives an isomorphism  $\mathcal{F}_{f(x)} \xrightarrow{\sim} (f^{-1} \mathcal{F})_x$ . In particular this shows that  $f^{-1}$  is exact. If  $T \in \mathbf{Ab}$ , then there is a unique isomorphism  $f^{-1}T_Y \simeq T_X$  so that the composition  $T \simeq (T_X)_{f(x)} \simeq (f^{-1}T_Y)_x \simeq T$  is the identity on  $T$  for all  $x \in X$ . By looking at stalks and using the naturality of the isomorphism  $\mathcal{F}_{f(x)} \xrightarrow{\sim} (f^{-1} \mathcal{F})_x$  we see that this isomorphism is natural in  $T$ . [14, II.4]

**Lemma 6.2.** Let  $X, Y$  be topological spaces with  $f : X \rightarrow Y$  continuous. For any  $p \in \mathbb{Z}$  and  $\mathcal{F} \in \text{Sh}(Y)$  there exists a canonical map  $H^p(Y, \mathcal{F}) \rightarrow H^p(X, f^{-1} \mathcal{F})$ . Furthermore this map is natural in  $\mathcal{F}$ .

**Proof.** Let  $T^\bullet$  be an injective resolution of  $\mathcal{F}$ . By the above discussion we have a morphism of

complexes  $\Gamma(Y, \mathcal{I}^\bullet) \rightarrow \Gamma(X, f^{-1}\mathcal{I}^\bullet)$  and hence maps  $h^p(\Gamma(Y, \mathcal{I}^\bullet)) \rightarrow h^p(\Gamma(X, f^{-1}\mathcal{I}^\bullet))$ . Because  $f^{-1}$  is exact  $f^{-1}\mathcal{I}^\bullet$  is a resolution of  $f^{-1}\mathcal{F}$ , so if  $\mathcal{J}^\bullet$  is an injective resolution of  $f^{-1}\mathcal{F}$ , then by Lemma 2.3 the isomorphism  $h^0(f^{-1}\mathcal{I}^\bullet) \simeq f^{-1}\mathcal{F} \simeq h^0(\mathcal{J}^\bullet)$  extends to a chain map  $\varphi : f^{-1}\mathcal{I}^\bullet \rightarrow \mathcal{J}^\bullet$ . Applying  $\Gamma(X, -)$  and taking the induced map on cohomology gives a map  $h^p(\Gamma(X, f^{-1}\mathcal{I}^\bullet)) \rightarrow h^p(X, \mathcal{J}^\bullet)$ . Composing these maps and keeping in mind that  $h^p(\Gamma(Y, \mathcal{I}^\bullet)) = H^p(Y, \mathcal{F})$  and  $h^p(\Gamma(X, \mathcal{J}^\bullet)) = H^p(X, f^{-1}\mathcal{F})$  by definition gives the desired morphism.

One can, with some effort, argue that this morphism is independent of the choice of  $\mathcal{I}^\bullet$  and  $\mathcal{J}^\bullet$  and natural in  $\mathcal{F}$ . Alternatively we can use the fact that the functors  $H^p(Y, -) : \mathbf{Sh}(Y) \rightarrow \mathbf{Ab}$  for  $p \in \mathbb{Z}$  form a universal  $\delta$ -functor (see [13, III.1]). Because  $f^{-1}$  is exact the functors  $H^p(X, f^{-1}-)$  for  $p \in \mathbb{Z}$  also form a delta functor. Now by the universal property of universal delta functors the natural transformation  $\Gamma(Y, -) \Rightarrow \Gamma(X, f^{-1}-)$  extends uniquely to a morphism of delta functors  $H^*(Y, -) \Rightarrow H^*(X, f^{-1}-)$  which immediately gives the desired natural transformations.  $\square$

**Proposition 6.3.** *Let  $X, Y$  be topological spaces with  $f : X \rightarrow Y$  continuous. For all  $p \in \mathbb{Z}$  and  $T \in \mathbf{Ab}$  there is a canonical morphism  $f^* : H^p(Y, T_Y) \rightarrow H^p(X, T_X)$  induced by  $f$  which is natural in  $T$ . This makes  $X \mapsto H^p(X, -_X)$  into a contravariant functor  $\mathbf{Top} \rightarrow [\mathbf{Ab}, \mathbf{Ab}]_{\text{add}}$ .*

**Proof.** The isomorphism  $f^{-1}T_Y \xrightarrow{\sim} T_X$  is natural in  $T$ , so the induced isomorphism  $H^p(X, f^{-1}T_Y) \rightarrow H^p(X, T_X)$  is natural in  $T$ . By Lemma 6.2 the morphism  $H^p(Y, T_Y) \rightarrow H^p(X, f^{-1}T_Y)$  is natural in  $T$ , so the composition  $H^p(Y, T_Y) \rightarrow H^p(X, T_X)$  is natural in  $T$ . For the last statement it suffices to show that for a fixed  $T \in \mathbf{Ab}$  the construction  $X \mapsto H^p(X, T_X)$  together with the above functoriality satisfies the functor axioms. This is a purely sheaf theoretic exercise which we leave to the reader.  $\square$

**Corollary 6.4.** *Let  $X, Y$  be topological spaces with  $f : X \rightarrow Y$  continuous. Then for each  $p \in \mathbb{Z}$  and  $F \in \vec{\mathbf{Ab}}$  there is a canonical morphism  $f_* : \vec{H}_p(X; F) \rightarrow \vec{H}_p(Y; F)$ . Furthermore this makes  $X \mapsto \vec{H}_p(X; F)$  into a covariant functor  $\mathbf{Top} \rightarrow \vec{\mathbf{Ab}}$ .*

**Proof.** Compose the functor  $X \mapsto H^p(X, -_X)$  with the precomposition functor  $- \circ F : [\mathbf{Ab}, \mathbf{Ab}]_{\text{add}} \rightarrow [\mathbf{Ab}, \mathbf{Ab}]_{\text{add}}$  and the contravariant left-exactification functor  $[\mathbf{Ab}, \mathbf{Ab}]_{\text{add}} \rightarrow \vec{\mathbf{Ab}}$  to obtain the covariant functor  $\mathbf{Top} \rightarrow \vec{\mathbf{Ab}}$ .  $\square$

## 6.2 0-th Homology

**Proposition 6.5.** *If  $X$  is a topological space and  $T \in \mathbf{Ab}$ , then  $\vec{H}_0(X)(T) \simeq H^0(X, T_X)$ . Furthermore  $H^0(X, T_X)$  is naturally isomorphic to the abelian group of locally constant functions on  $X$  with values in  $T$ . If  $f : X \rightarrow Y$  is a continuous map, then under this identification  $f^* : H^0(Y, T_Y) \rightarrow H^0(X, T_X)$  (and hence also  $(f_*)_T : \vec{H}_0(Y)(T) \rightarrow \vec{H}_0(X)(T)$ ) is given by precomposition with  $f$ .*

**Proof.** First of all  $H^0(X, -_X) : \mathbf{Ab} \rightarrow \mathbf{Ab}$  is already left exact since  $\Gamma(X, -)$  and  $-_X$  are, so it is isomorphic to its left-exactification  $\vec{H}_0(X)$ . Secondly if  $T \in \mathbf{Ab}$  then  $\Gamma(X, T_X)$  is naturally (with respect to  $T$ ) isomorphic to the group of locally constant functions on  $X$  with values in  $T$ , and this isomorphism maps a section  $s \in \Gamma(X, T_X)$  to  $x \mapsto s_x$  using the identification  $(T_X)_x \simeq T$ . Let  $f : X \rightarrow Y$  be a continuous map,  $x \in X$  and  $s \in \Gamma(Y, T_Y)$ . Recall that  $f^*(s)$  is the image of  $s$  under the composition of the canonical map  $\Gamma(Y, A_Y) \rightarrow \Gamma(X, f^{-1}A_Y)$  and the isomorphism  $f^{-1}A_Y \simeq A_X$ . By the definition of the isomorphism  $A_X \simeq f^{-1}A_Y$  we gave in Definition 6.1  $s_{f(x)}$  and  $f^*(s)_x$  correspond to the same element in  $T$ . Hence under the identification of sections of  $T_X$  and  $T_Y$  with locally constant functions,  $f^* : \Gamma(Y, T_Y) \rightarrow \Gamma(X, T_X)$  is given by precomposition with  $f$ .  $\square$

**Corollary 6.6.** *If  $X$  is connected, then  $\vec{H}_0(X) \simeq \vec{\mathbb{Z}}$ .*

**Convention 6.7.** From now on we identify  $\vec{H}_0(X)(T)$  and  $H^0(X, T_X)$  with the set of locally constant



functions on  $X$  with values in  $T$  as in the above.

**Proposition 6.8.** *Let  $X$  be topological space, then  $\vec{H}_0(X)$  is topologically representable.*

**Proof.** Let  $\mathcal{I}$  be the category consisting of partitions  $\mathcal{U}$  of  $X$  into non-empty clopen (= closed and open) subsets, where between partitions  $\mathcal{U}, \mathcal{V}$  there is a unique morphism  $\mathcal{U} \rightarrow \mathcal{V}$  if  $\mathcal{U}$  is a refinement of  $\mathcal{V}$ . Then  $\mathcal{I}$  is cofiltered (one can take pairwise intersections of partitions) and we have a diagram  $D : \mathcal{I} \rightarrow \mathbf{Ab}$  given by  $\mathcal{U} \mapsto \mathbb{Z}^{(\mathcal{U})}$ , where we map a morphism  $\mathcal{U} \rightarrow \mathcal{V}$  to the map  $\mathbb{Z}^{(\mathcal{U})} \rightarrow \mathbb{Z}^{(\mathcal{V})}$  which maps a basis vector  $e_U$  for  $U \in \mathcal{U}$  to  $e_V$  where  $V \in \mathcal{V}$  is the unique element so that  $U \subseteq V$ . We will first show that  $\vec{H}_0(X)$  is the limit of this diagram in  $\mathbf{Ab}$ . First for each  $\mathcal{U} \in \mathcal{I}$  we define the element  $s_{\mathcal{U}} \in \vec{H}_0(X)(\mathbb{Z}^{(\mathcal{U})})$  to be the function which maps each point  $x \in X$  to  $e_U$  where  $U \in \mathcal{U}$  is so that  $x \in U$ . If  $\mathcal{U}$  is a refinement of  $\mathcal{V}$ , then it is not hard to see that  $s_{\mathcal{U}} *_{\vec{H}_0(X)} D(\mathcal{U} \rightarrow \mathcal{V}) = s_{\mathcal{V}}$ . Thus we can use Lemma 3.13 to prove that  $\vec{H}_0(X)$  is the limit of  $D$  in  $\mathbf{Ab}$ . For condition 1 let  $f \in \vec{H}_0(X)(T)$ . This gives a clopen partition  $\mathcal{U}$  which consists of sets of the form  $f^{-1}(t)$  for  $t \in T$  when  $f^{-1}(t) \neq \emptyset$ . If we define the homomorphism  $g : \mathbb{Z}^{(\mathcal{U})} \rightarrow T$  by mapping each basis vector  $e_{f^{-1}(t)}$  to  $t$ , it follows fairly immediately from the definitions that  $g *_{\vec{H}_p(X)} s_{\mathcal{U}} = f$ , so condition 1 is satisfied. For condition 2 let  $\mathcal{U} \in \mathcal{I}$  and  $f : \mathbb{Z}^{(\mathcal{U})} \rightarrow T$  so that  $f *_{\vec{H}_0(X)} s_{\mathcal{U}} = 0$ . For each  $U \in \mathcal{U}$  we can then pick a point  $x \in U$  to see that  $f(e_U) = f(s(x)) = (f *_{\vec{H}_0(X)} s_{\mathcal{U}})(x) = 0$ , so  $f = 0$ . We conclude that  $\vec{H}_0(X)$  is the limit of  $D$  in  $\mathbf{Ab}$ .

By Lemma 4.17 in order to show that  $\vec{H}_0(X)$  is topologically representable, it suffices to show that for the limit  $L$  of  $D$  in  $\mathbf{TopAb}$  each cone morphism is surjective. For this fix some  $\mathcal{U} \in \mathcal{I}$  and a  $U \in \mathcal{U}$ . We need to extend  $e_U$  to a family  $(\alpha_{\mathcal{V}})_{\mathcal{V} \in \mathcal{I}}$  with  $\alpha_{\mathcal{V}} \in \mathbb{Z}^{(\mathcal{V})}$  for all  $\mathcal{V} \in \mathcal{I}$  consistent with the transition morphisms. Pick a point  $x \in U$  and then set for each  $\mathcal{V} \in \mathcal{I}$ :  $\alpha_{\mathcal{V}} = e_V$  where  $V \in \mathcal{V}$  is the unique clopen so that  $x \in V$ . Now if  $\mathcal{W}$  is a refinement of  $\mathcal{V}$ , then  $\alpha_{\mathcal{W}} = e_W$  gets mapped by  $D(\mathcal{W} \rightarrow \mathcal{V})$  to  $e_V$ , where  $V \in \mathcal{V}$  and  $W \subseteq V$ , but then  $x \in V$  so  $e_V = \alpha_{\mathcal{V}}$ . Thus this family is consistent with the transition morphisms. Finally since  $x \in U$  we have  $\alpha_{\mathcal{U}} = e_U$ .  $\square$

**Example 6.9.** Let  $X = \{x_0, x_1, x_2, \dots\} \cup \{x_{\infty}\}$  be a topological space with open sets being any set not containing  $x_{\infty}$  and any set containing  $x_{\infty}$  and all but finitely many points. (This can also be seen as the subspace  $\{1, 1/2, 1/3, \dots\} \cup \{0\}$  of  $\mathbb{R}$ .) By looking at neighbourhoods of  $x_{\infty}$  we see that locally constant functions on  $X$  are exactly the functions which are eventually constant on the sequence  $x_0, x_1, \dots$  and take the limit value on  $x_{\infty}$ . Thus we have a natural isomorphism between the functor  $\vec{H}_0(X)$  and the functor  $T \mapsto \bigoplus_{i=-1}^{\infty} T$  given at each  $T \in \mathbf{Ab}$  by

$$\begin{aligned} \vec{H}_0(X)(T) &\longrightarrow \bigoplus_{i=-1}^{\infty} T \\ f &\longmapsto (f(x_{\infty}), (f(x_i) - f(x_{\infty}))_{i \geq 0}) \end{aligned}$$

Now compare this to the topological group  $\prod_{i \geq -1}^{\mathbf{Top}} \mathbb{Z}$  with the product topology. It is pro-discrete as it can be realized as the limit of the diagram  $0 \leftarrow \mathbb{Z} \leftarrow \mathbb{Z}^2 \leftarrow \mathbb{Z}^3 \leftarrow \dots$  where each arrow  $\mathbb{Z}^{n-1} \leftarrow \mathbb{Z}^n$  is given by dropping the  $n$ -th coordinate. Notice that for each  $T \in \mathbf{Ab}$  any continuous homomorphism  $f : \prod_{i \geq -1}^{\mathbf{Top}} \mathbb{Z} \rightarrow T$  must have open kernel, thus a kernel which contains  $\{(a_i)_{i=-1}^{\infty} : a_i = 0 \text{ for } -1 \leq i \leq N\}$  for some  $N$ . Thus  $f$  is determined by the values  $f(e_i)$  for  $-1 \leq i \leq N$ , where each  $e_i$  is sequence with a 1 at index  $i$  and 0 elsewhere. In fact we have for each  $T \in \mathbf{Ab}$  an isomorphism

$$\begin{aligned} \mathrm{Hom}_{\mathbf{TopAb}} \left( \prod_{i \geq -1}^{\mathbf{Top}} \mathbb{Z}, T \right) &\longrightarrow \bigoplus_{i=-1}^{\infty} T \\ f &\longmapsto (f(e_i))_{i \geq -1} \end{aligned}$$

natural in  $T$ . Composing these two isomorphisms gives us a natural isomorphism  $\eta : \vec{H}_0(X) \Rightarrow \overrightarrow{\prod_{i \geq -1}^{\mathbf{Top}} \mathbb{Z}}$  given at each  $T \in \mathbf{Ab}$  by mapping a locally constant function  $f : X \rightarrow T$  to the homomorphism

$g : \prod_{i \geq -1}^{\text{Top}} \mathbb{Z} \rightarrow T$  which maps  $e_{-1}$  to  $f(x_\infty)$  and  $e_i$  to  $f(x_i) - f(x_\infty)$ . We conclude that  $\vec{H}_0(X)$  is represented by  $\prod_{i \geq -1}^{\text{Top}} \mathbb{Z}$ .

**Notation 6.10.** By Proposition 4.13 an element of a pro-discrete group  $A$ , i.e. a homomorphism  $\mathbb{Z} \rightarrow A$ , is the same as a natural transformation  $\vec{A} \Rightarrow \text{id}_{\mathbf{Ab}}$  given by evaluation in that element. If  $X$  is a topological space and  $x \in X$  we define  $[x]$  to be the element of  $\vec{H}_0(X)$  (seen as a topological group) corresponding to the natural transformation  $\vec{H}_0(X) \Rightarrow \text{id}_{\mathbf{Ab}}$  given in each  $T \in \mathbf{Ab}$  by  $f \mapsto f(x)$ . Alternatively this can be seen as the image of 1 under the isomorphism  $\vec{\mathbb{Z}} \simeq \vec{H}_0(\{\star\})$  and the morphism  $i_* : \vec{H}_0(\{\star\}) \rightarrow \vec{H}_0(X)$  where  $i(\star) = x$ .

**Lemma 6.11.** *Let  $X$  be a topological space. Then the elements of the form  $[x]$  for  $x \in X$  topologically generate  $\vec{H}_0(X)$ , that is the subgroup generated by them is dense.*

**Proof.** We first identify  $\vec{H}_0(X)$  with the limit  $L$  from the proof of Proposition 6.8. The most effort will be to see what the element  $[x]$  for  $x \in X$  looks like under this identification. Define  $(\alpha_U)_{U \in \mathcal{I}} \in L$  by setting for each  $U \in \mathcal{I}$ :  $\alpha_U = e_U$  where  $U \in \mathcal{U}$  so that  $x \in U$ . We claim that under the identification between  $\vec{H}_0(X)$  and  $\vec{L}$ , the natural transformation corresponding to  $(\alpha_U)_{U \in \mathcal{I}}$  is the same as that for  $[x]$ . Fix a  $T \in \mathbf{Ab}$  and  $f \in \vec{H}_0(X)(T)$ . For each  $U \in \mathcal{I}$  write  $\pi_U : L \rightarrow \mathbb{Z}^{(U)}$  for the corresponding projection. Recall that the corresponding homomorphism  $f' : L \rightarrow T$  will be given by  $\pi_U \circ g$  where

$$\mathcal{U} = \{f^{-1}(t) : t \in T, f^{-1}(t) \neq \emptyset\}$$

and  $g : \mathbb{Z}^{(U)} \rightarrow T$  maps  $e_{f^{-1}(t)}$  to  $t$ . Thus  $f'((\alpha_V)_{V \in \mathcal{I}}) = g(\alpha_U) = g(e_{f^{-1}(f(x))}) = f(x)$ . We conclude that  $[x] = (\alpha_U)_{U \in \mathcal{I}}$ .

Write  $A = \bigoplus_{x \in X} \mathbb{Z}$  (with the discrete topology) with for each  $x \in X$  the corresponding generator denoted  $e_x$  and write  $\varphi : A \rightarrow L$  for the (continuous) homomorphism sending  $e_x$  to  $[x]$ . Let  $U \in \mathcal{I}$ , then the cone morphism  $\pi_U \circ \varphi$  is given by mapping  $e_x$  to  $e_U$  where  $U \in \mathcal{U}$  is so that  $x \in U$ . Since each  $U \in \mathcal{U}$  is non-empty this cone morphism is surjective. Applying Lemma 4.7 shows that the image of  $\varphi$  is dense in  $L$ , so the subgroup generated by the  $[x]$ 's for  $x \in X$  is dense in  $L$ .  $\square$

**Example 6.12.** (Continuing Example 6.9.) Observe the point  $x_\infty \in X$ . Notice that under the isomorphism  $\eta_T$  for  $T \in \mathbf{Ab}$ , evaluating an  $f \in \vec{H}_0(X)(T)$  in  $x_\infty$  is the same as evaluating the corresponding homomorphism in  $e_{-1}$ . It follows that  $[x_\infty] = e_{-1}$ . However for  $i \in \{0, 1, \dots\}$  evaluating an  $f \in \vec{H}_0(X)(T)$  in  $x_i$  is the same as evaluating the corresponding homomorphism in  $e_i + e_{-1}$ , so  $[x_i] = e_i + e_{-1}$ . Notice that the sequence  $([x_i])_{i \geq 0} = (e_i + e_{-1})_{i \geq 0}$  in  $\prod_{i \geq -1}^{\text{Top}} \mathbb{Z}$  converges to  $e_{-1} = [x_\infty]$  (the sequence consists of sequences of 0's with two 1's, one at the start and one moving away to infinity). This reflects the fact that the sequence  $(x_i)_{i \geq 0}$  converges to  $x_\infty$  in  $X$ . Notice how we are able to recover this topological fact from the purely algebraic object  $\vec{H}_0(X)$ .

**Definition 6.13.** A topological space  $X$  is called *totally separated* if for any two distinct points  $x, y \in X$  there exists a clopen subset  $U \subseteq X$  so that  $x \in U$  and  $y \notin U$ . Any totally separated space is totally disconnected but not necessarily the other way around (see Example 6.41.) On a topological space  $X$  we have an equivalence relation  $\sim$  which relates two elements if and only if they are contained in the same clopen sets. The equivalence class of a point  $x \in X$  under  $\sim$  is called its *quasi-component*, denoted  $[x]_q$ . We will denote the quotient space by  $X_{qc} := X/\sim$ .

**Lemma 6.14.** *Let  $X$  be a topological space, then  $X_{qc}$  is totally separated and the quotient map  $q : X \rightarrow X_{qc}$  induces a bijection between the clopen sets of  $X$  and  $X_{qc}$ .*

**Proof.** Exercise.  $\square$

**Lemma 6.15.** *Let  $X$  be a topological space with  $x, y \in X$ , then  $[x]_q = [y]_q \Leftrightarrow [x] = [y]$ .*

**Proof.** If  $[x]_q = [y]_q$ , then for  $T \in \mathbf{Ab}$  and  $f \in \vec{H}_0(X)(T)$  the set  $f^{-1}\{f(x)\}$  is a clopen set containing

$x$  and hence also  $y$ , so  $f(x) = f(y)$  and thus  $[x] = [y]$ . If  $[x]_q \neq [y]_q$ , then there exists a clopen set  $U \subseteq X$  with  $x \in U$  and  $y \notin U$ . Now define the locally constant function  $f : X \rightarrow \mathbb{Z}$  by  $f(z) = 1$  for  $z \in U$  and  $f(z) = 0$  for  $z \notin U$ , then  $f(x) \neq f(y)$  and hence  $[x] \neq [y]$ .  $\square$

**Proposition 6.16.** *Let  $X$  be a topological space, then the quotient map  $q : X \rightarrow X_{qc}$  induces an isomorphism  $\vec{H}_0(X) \xrightarrow{\sim} \vec{H}_0(X_{qc})$ .*

**Proof.** If  $T \in \mathbf{Ab}$  then a locally constant function  $f : X \rightarrow T$  simply partitions  $X$  into non-empty clopen sets and takes a single value on each. By Lemma 6.14.1 it follows that precomposing with  $q$  gives a bijection between the locally constant functions  $X \rightarrow T$  and  $X_{qc} \rightarrow T$ , and hence  $q_* : \vec{H}_0(X) \xrightarrow{\sim} \vec{H}_0(X_{qc})$  is an isomorphism.  $\square$

**Theorem 6.17.** *Let  $X$  be a topological space and write  $f$  for the map from  $X$  to  $\vec{H}_0(X)$  (considered as a topological abelian group) which maps a point  $x \in X$  to  $[x]$ . Then for any pro-discrete topological abelian group  $A$  and continuous map  $g : X \rightarrow A$  there exists a unique continuous homomorphism  $h : \vec{H}_0(X) \rightarrow A$  so that  $h \circ f = g$ .*

**Proof.** We first prove the case that  $A$  is discrete. Let  $A \in \mathbf{Ab}$  and let  $g : X \rightarrow A$  be continuous, i.e. locally constant. By Lemma 6.11  $h$  is completely determined by where it sends the elements of the form  $[x]$ , so there exists at most one such  $h$ . For existence let  $h : \vec{H}_0(X) \rightarrow A$  be the homomorphism corresponding to  $g \in \vec{H}_0(X)(A)$ , then by definition we have for each  $x \in X$ :  $h([x]) = g(x)$ . It follows that  $h \circ f = g$ . Now the case for pro-discrete groups will follow by expressing  $A$  as a limit of a cofiltered diagram of discrete groups and combining the universal property of the limit and the above universal property of each discrete abelian group and invoking Lemma 6.11 a few more times.  $\square$

**Remark 6.18.** Combining Proposition 6.16 with Theorem 6.17 shows that  $\vec{H}_0(X)$  is a sort of *free pro-discrete abelian group* on the space of quasi-components of  $X$ . This is analogous to how  $H_0^{\text{sing}}(X)$  is the free abelian group on the set of path connected components of  $X$ .

### 6.3 Vietoris–Begle mapping theorem

In this section we state a general form of the Vietoris–Begle mapping theorem for sheaf cohomology as presented in [14, IV.1] and show how it implies a Vietoris–Begle mapping theorem for sheaf homology. In particular it shows that sheaf homology is a homotopy invariant even for a very general form of homotopy.

**Definition 6.19.** Let  $f : X \rightarrow Y$  be a continuous map. We say that  $f$  is *strongly proper* if

1.  $f$  is closed,
2. for any  $y \in Y$  and  $x, x' \in f^{-1}\{y\}$  there exist disjoint open neighbourhoods of  $x$  and  $x'$  in  $X$ ,
3. all fibres of  $f$  are compact.

If  $X$  and  $Y$  are locally compact Hausdorff this agrees with the standard definition of a proper map. (See [14, IV.1.3].) Furthermore we say that a topological space  $X$  is *acyclic* if  $H^0(X, \mathbb{Z}_X) \simeq \mathbb{Z}$  and  $H^p(X, \mathbb{Z}_X) = 0$  for all  $p > 0$ . By Theorem 2.17 this is in particular the case if  $\vec{H}_0(X) \simeq \mathbb{Z}$  and  $\vec{H}_p(X) = 0$  for  $p > 0$ .

**Proposition 6.20.** *Let  $f : X \rightarrow Y$  be a strongly proper map with acyclic fibres. Then for any  $p \in \mathbb{Z}$  and sheaf  $\mathcal{F}$  on  $Y$  the homomorphism  $f^* : H^p(Y, \mathcal{F}) \rightarrow H^p(X, f^{-1}\mathcal{F})$  as in Lemma 6.2 is an isomorphism.*

**Proof.** This proof is sketched in [14, IV.1].  $\square$

**Remark 6.21.** In [7, Thm II.11.7] a more general version of this theorem is proven weakening the

assumption on the fibres of  $f$  from compact and relatively Hausdorff in  $X$  to taut in  $X$  (see Definition 6.54).

**Theorem 6.22.** (Vietoris–Begle mapping theorem) *Let  $f : X \rightarrow Y$  be a strongly proper map with acyclic fibres. Then for any  $p \in \mathbb{Z}$  and  $T \in \mathbf{Ab}$  the induced homomorphism  $f^* : H^p(Y, T_Y) \rightarrow H^p(X, T_X)$  is an isomorphism. Consequently for any  $F \in \mathbf{Ab}$  the induced morphism  $f_* : \vec{H}_p(X; F) \rightarrow \vec{H}_p(Y; F)$  is an isomorphism.*

**Proof.** Recall that the induced homomorphism  $f^* : H^p(Y, T_Y) \rightarrow H^p(X, T_X)$  is the composition of the homomorphism  $f^* : H^p(Y, T_Y) \rightarrow H^p(X, f^{-1}T_Y)$  as in Lemma 6.2 and the isomorphism  $H^p(X, f^{-1}T_Y) \rightarrow H^p(X, T_X)$  coming from the canonical isomorphism  $f^{-1}T_Y \xrightarrow{\sim} T_X$ , so the first statement follows immediately from Proposition 6.20. For the second statement recall that taking the left-exactification and precomposing with  $F$  are functorial, so applying these operations to the natural isomorphism  $f^* : H^p(Y, -_Y) \Rightarrow H^p(X, -_X)$  gives an isomorphism.  $\square$

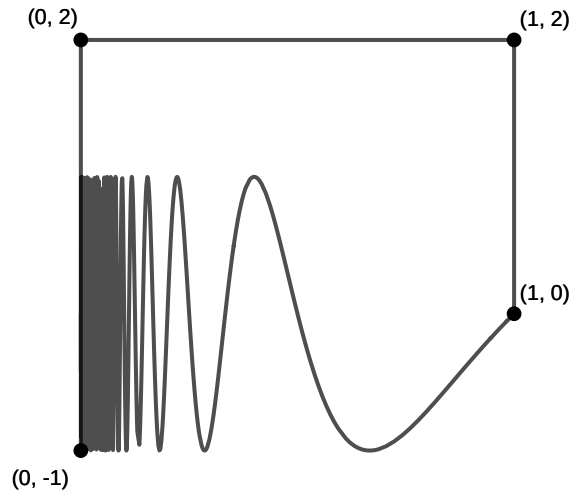


Figure 3: The Warsaw circle from Example 6.23.

**Example 6.23.** (Warsaw circle) Consider the space

$$X = \{(x, y) \in \mathbb{R}^2 : y = \sin(\pi/x), 0 < x \leq 1\} \cup \{0\} \times [-1, 2] \cup [0, 1] \times \{2\} \cup \{1\} \times [0, 2],$$

see Figure 3. Consider the continuous map  $p : \mathbb{R}^2 \rightarrow \mathbb{R}^2$  which maps a point  $(x, y)$  to  $(x, y)$  if  $y \geq 1$  and to  $(x, 1)$  if  $y \leq 1$ .  $Y := p[X]$  is a rectangle and hence homeomorphic to  $S^1$ . The restriction of  $p$  to  $X$  and  $Y$  has fibres homeomorphic either to a point or  $[0, 1]$  and it is (strongly) proper as it is a continuous map between compact Hausdorff spaces. Thus by Theorem 6.22  $X$  has the same sheaf homology pro-groups as  $S^1$  which are represented by the singular homology groups of  $S^1$  by Theorem 5.42. On the other hand  $X$  has the homotopy groups of a point [19, p. xiii], so by the Whitehead theorem also the homology groups of a point. Hence  $X$  is an example of a space whose sheaf homology is representable, but not by its singular homology. This space is one of the motivating examples of shape theory: It is not homotopic to a circle, but it does have the shape of a circle [19, p. 67].

**Definition 6.24.** Let  $Z$  be a topological space. We say that two continuous maps  $f, g : X \rightarrow Y$  are *homotopic along  $Z$*  if there exist points  $z_0, z_1 \in Z$  and a continuous map  $H : X \times Z \rightarrow Y$  so that  $H(x, z_0) = f(x)$  and  $H(x, z_1) = g(x)$  for all  $x \in X$ .

**Theorem 6.25.** *Let  $f, g : X \rightarrow Y$  be continuous maps and suppose that  $f$  and  $g$  are homotopic along some acyclic compact Hausdorff space, then the morphisms induced by  $f$  and  $g$  on sheaf cohomology*

and sheaf homology with constant coefficients are equal.

**Proof.** Suppose  $f, g$  are homotopic along an acyclic compact Hausdorff space  $Z$  with points  $z_0, z_1 \in Z$  and  $H : X \times Z \rightarrow Y$  as in the definition. Write  $\pi : X \times Z \rightarrow X$  for the projection and  $\iota_i : X \rightarrow X \times Z, x \mapsto (x, z_i)$  for  $i \in \{0, 1\}$ .  $\pi$  is closed since  $Z$  is compact (see for example [6, p. 103, Cor. 5]). Also  $\pi$  has fibres homeomorphic to  $Z$  which are compact and acyclic. Finally if  $(x, z) \neq (x', z') \in X \times Z$  are so that  $\pi(x, z) = \pi(x', z')$ , then  $z \neq z'$ , so there exist disjoint neighbourhoods  $U, V$  of  $z, z'$  in  $Z$  and hence  $X \times U$  and  $X \times V$  are disjoint neighbourhoods of  $(x, z), (x', z')$  in  $X \times Z$ . Hence  $\pi$  is strongly proper with acyclic fibres. Fix a  $p \in \mathbb{Z}$  and a  $T \in \mathbf{Ab}$ , then by Theorem 6.22  $\pi^* : H^p(X, T_X) \rightarrow H^p(X \times Z, T_{X \times Z})$  is an isomorphism. From the equation  $i_0^* \circ \pi^* = (\pi \circ i_0)^* = (\pi \circ i_1)^* = i_1^* \circ \pi^*$  it then follows that  $i_0^* = i_1^*$ , so  $f^* = (H \circ i_0)^* = i_0^* \circ H^* = i_1^* \circ H^* = (H \circ i_1)^* = g^*$ . From the fact that the natural transformations  $f^*, g^* : H^p(Y, -_Y) \Rightarrow H^p(X, -_X)$  are equal it follows immediately that the induced morphisms on sheaf homology are equal.  $\square$

**Example 6.26.** (Long ray) For each ordinal  $\alpha$  let  $L_\alpha = \alpha \times [0, 1)$  (here  $[0, 1) \subseteq \mathbb{R}$ ) with the lexicographical order and let  $L_\alpha^* = L_\alpha \cup \{e\}$  where we let  $e \notin L_\alpha$  be an upper bound for  $L_\alpha$ . We give each of these ordered sets the order topology, i.e. the topology generated by the open rays.  $L_{\omega_1}$  and  $L_{\omega_1}^*$ , where  $\omega_1$  is the first uncountable ordinal, are called the long ray and extended long ray respectively. Set  $X = L_{\omega_1}$ . This space is locally contractible, so using Theorem 5.44 (assuming  $(\star)$ ) one can simply use singular homology to compute the sheaf homology of  $X$ . Note though that because  $X$  is not paracompact (see [28, p. 71]) one cannot use the weaker Theorem 5.42 based on more classical results. In Example 6.53 we will show using the continuity theorem that  $L_{\omega_1}^*$  is acyclic. We now show that  $X$  is contractible along  $L_{\omega_1}^*$  and hence that  $X$  is acyclic. Let  $H : L_{\omega_1}^* \times X \rightarrow X$  be the map given by  $H(e, x) = x$  for all  $x \in X$  and for all  $t, x \in L_{\omega_1} = X$  we take

$$H(t, x) = \begin{cases} x & \text{if } t \geq x \\ t & \text{if } t \leq x \end{cases}$$

To see that  $H$  is continuous notice that for all  $a \in X$  we have

$$H^{-1}\{x \in X : x < a\} = \{(t, x) \in L_{\omega_1}^* \times X : t < a\} \cup \{(t, x) \in L_{\omega_1}^* \times X : x < a\}$$

and

$$H^{-1}\{x \in X : x > a\} = \{(t, x) \in L_{\omega_1}^* \times X : t > a, x > a\}.$$

We conclude that  $\text{id}_X$  is homotopic to  $x \mapsto (0, 0)$  along  $L_{\omega_1}^*$ , so  $X$  is contractible along  $L_{\omega_1}^*$ , so  $X$  is also acyclic. Note that  $X$  is not contractible in the usual sense, see [11, Prop. 1.22].

## 6.4 Closed Mayer–Vietoris

In this section we give the construction for the closed Mayer–Vietoris theorem for sheaf cohomology as presented in [14, II.5], from which the corresponding theorem for sheaf homology follows. At first sight this seems only like a mild variation of Mayer–Vietoris with open sets, but the use of closed sets allows us to ‘cut through’ the singularities of spaces.

We first establish some basic facts about closed subspaces and sheaves. Let  $X$  be a topological space with  $K \subseteq X$  closed and  $i : K \rightarrow X$  the inclusion. It is not hard to see that if  $\mathcal{F}$  is a sheaf on  $K$  then the stalk of  $i_*\mathcal{F}$  at a point  $x \in X$  is given by  $\mathcal{F}_x$  if  $x \in K$  and 0 if  $x \notin K$  (and of course these isomorphisms are natural in  $\mathcal{F}$ ). Consequently the functor  $i_* : \text{Sh}(K) \rightarrow \text{Sh}(X)$  is exact. Because the left adjoint  $i^{-1}$  of  $i_*$  is exact,  $i_*$  maps injectives to injectives ([14, Corr. II.4.13]). Therefore if  $\mathcal{I}^\bullet$  is an injective resolution of  $\mathcal{F}$  then  $i_*(\mathcal{I}^\bullet)$  is an injective resolution of  $i_*\mathcal{F}$ , so for each  $p \in \mathbb{Z}$  the group  $h^p(\Gamma(X, i_*(\mathcal{I}^\bullet)))$  is canonically isomorphic to  $H^p(X, i_*\mathcal{F})$ . However  $\Gamma(X, i_*(\mathcal{I}^\bullet)) = \Gamma(K, \mathcal{I}^\bullet)$ , so  $h^p(\Gamma(X, i_*(\mathcal{I}^\bullet))) = h^p(\Gamma(K, \mathcal{I}^\bullet))$  is

also isomorphic to  $H^p(K, \mathcal{F})$ . The uniqueness and the naturality in  $\mathcal{F}$  of this construction follows from the functoriality and uniqueness of injective resolutions (up to homotopy). We conclude that we have a canonical isomorphism  $H^p(X, i_*\mathcal{F}) \xrightarrow{\sim} H^p(K, \mathcal{F})$  natural in  $\mathcal{F}$ .

**Lemma 6.27.** *Let  $X$  be a topological space with  $K$  a closed subspace with inclusion  $i : K \rightarrow X$  and  $\mathcal{F}$  a sheaf on  $X$ . Then any morphism  $\mathcal{F} \rightarrow \mathcal{G}$  where  $\mathcal{G}$  has zero stalks outside  $K$  factors uniquely through the adjunction morphism  $\mathcal{F} \rightarrow i_*i^{-1}\mathcal{F}$ .*

**Proof.** First of all notice that the adjunction morphism  $\mathcal{F} \rightarrow i_*i^{-1}\mathcal{F}$  is an epimorphism (as can be seen by looking at the induced maps on stalks), so there is at most one such factorisation. To construct the factorisation apply the naturality of the adjunction morphism to  $\mathcal{F} \rightarrow \mathcal{G}$  to obtain

$$\begin{array}{ccc} \mathcal{F} & \longrightarrow & i_*i^{-1}\mathcal{F} \\ \downarrow & & \downarrow \\ \mathcal{G} & \longrightarrow & i_*i^{-1}\mathcal{G} \end{array}$$

By looking at stalks we see that the lower horizontal morphism is an isomorphism, so the required factorisation exists.  $\square$

**Lemma 6.28.** *Let  $X$  be a topological space with closed subsets  $K_1, K_2$  so that  $K_1 \cup K_2 = X$ . Write  $K_1 \cap K_2 = K_{12}$  and label the inclusions as*

$$\begin{array}{ccccc} K_1 & \xrightarrow{j_1} & X & & \\ i_1 \uparrow & & \nearrow k & & \uparrow j_2 \\ K_{12} & \xrightarrow{i_2} & K_2 & & \end{array}$$

Then for any sheaf  $\mathcal{F}$  on  $X$  there exists a long exact sequence

$$\cdots \rightarrow H^p(X, \mathcal{F}) \xrightarrow{j_1^* \times j_2^*} H^p(K_1, j_1^{-1}\mathcal{F}) \times H^p(K_2, j_2^{-1}\mathcal{F}) \xrightarrow{i_1^* - i_2^*} H^p(K_{12}, k^{-1}\mathcal{F}) \xrightarrow{\delta} H^{p+1}(X, \mathcal{F}) \rightarrow \cdots$$

natural in  $\mathcal{F}$ . Here  $f^*$  for a continuous map  $f$  denotes the homomorphisms as in Lemma 6.2.

**Proof.** For a sheaf  $\mathcal{F}$  we first construct a sequence

$$0 \rightarrow \mathcal{F} \rightarrow j_{1*}j_1^{-1}\mathcal{F} \times j_{2*}j_2^{-1}\mathcal{F} \rightarrow k_*k^{-1}\mathcal{F} \rightarrow 0 \quad (6.1)$$

The first morphism is simply the product of the adjunction morphisms. The second morphism is the difference of the morphisms  $j_{n*}j_n^{-1}\mathcal{F} \rightarrow k_*k^{-1}\mathcal{F}$  for  $n = 1, 2$  constructed by applying Lemma 6.27. Notice how we defined all of the morphisms in this sequence using adjunction natural transformations and by applying Lemma 6.27 (which you can easily check is functorial), so the sequence (6.1) is natural in  $\mathcal{F}$ . Therefore taking the long exact sequence of cohomology gives a long exact sequence natural in  $\mathcal{F}$ . Using the natural isomorphisms  $H^p(X, i_*\mathcal{G}) \simeq H^p(K, \mathcal{G})$  for a sheaf  $\mathcal{G}$  on  $K$  with  $i : K \rightarrow X$  a closed subspace we obtain from this long exact sequence a long exact sequence with the desired objects. We omit the verification that the morphisms in this long exact sequence are indeed those from Lemma 6.2.  $\square$

**Proposition 6.29.** *Let  $X, K_1, K_2, i_1, \dots$  be as in Lemma 6.28, then for any  $T \in \mathbf{Ab}$  we can construct a long exact sequence*

$$\cdots \rightarrow H^p(X, T_X) \xrightarrow{j_1^* \times j_2^*} H^p(K_1, T_{K_1}) \times H^p(K_2, T_{K_2}) \xrightarrow{i_1^* - i_2^*} H^p(K_{12}, T_{K_{12}}) \xrightarrow{\delta} H^{p+1}(X, T_X) \rightarrow \cdots$$

natural in  $T$ . Here  $f^*$  for a continuous map  $f$  denotes the homomorphisms as in Proposition 6.3.

**Proof.** This follows immediately from Lemma 6.28 by taking  $\mathcal{F} = T_X$  and moving the sequence along

the canonical isomorphisms  $j_1^{-1}T_X \xrightarrow{\sim} T_{K_1}, j_2^{-1}T_X \xrightarrow{\sim} T_{K_2}, \dots$ . It is not hard to see using the naturality of the morphism from Lemma 6.2 and the definition of the morphism from Proposition 6.3 that this gives the required long exact sequence.  $\square$

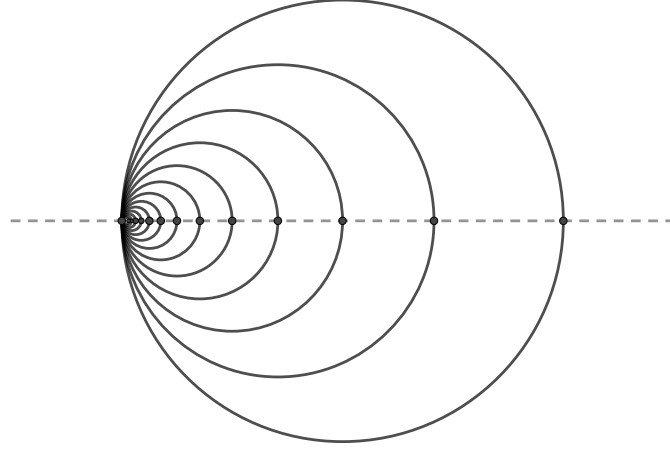


Figure 4: The Hawaiian Earring and the line along which we cut it in Example 6.31.

**Lemma 6.30.** *Let  $X$  be a compact space with a basis  $\mathcal{B}$  consisting of clopen sets, then every sheaf on  $X$  is  $\Gamma$ -acyclic.*

**Proof.** It suffices to show that the functor  $\Gamma(X, -) : \mathbf{Sh}(X) \rightarrow \mathbf{Ab}$  is exact, i.e. right exact. Let  $\varphi : \mathcal{F} \rightarrow \mathcal{G}$  be an epimorphism of sheaves and  $s \in \Gamma(X, \mathcal{G})$  a global section. Pick an open cover  $\mathcal{U} = (B_i)_{i \in I}$ , without loss of generality consisting of opens in  $\mathcal{B}$ , together with sections  $t_i \in \mathcal{G}(B_i)$  for  $i \in I$  so that  $\varphi_{B_i}(t_i) = s|_{B_i}$ . By the compactness of  $X$  we may assume that  $I = \{1, \dots, n\}$ . Then set for each  $1 \leq i \leq n$ :  $B'_i = B_i \setminus (B_1 \cup \dots \cup B_{i-1})$ , then  $(B'_i)_{i=1}^n$  is a partition of  $X$  into open sets. Now glue the elements  $t_i|_{B'_i}$  to give an element  $t \in \mathcal{F}(X)$  so that  $t|_{B'_i} = t_i|_{B'_i}$  for  $1 \leq i \leq n$ . Then

$$\varphi(t)|_{B'_i} = \varphi(t|_{B'_i}) = \varphi(t_i|_{B'_i}) = \varphi(t_i|_{B_i})|_{B'_i} = s|_{B_i}|_{B'_i} = s|_{B'_i}$$

for all  $1 \leq i \leq n$ , so  $\varphi_X(t) = s$ , so  $\varphi_X$  is surjective.  $\square$

**Example 6.31.** (Hawaiian Earring) Let  $X = \bigcup_{n \geq 0} \{(x, y) \in \mathbb{R}^2 : (x - 2^{-n-1})^2 + y^2 = 4^{-n-1}\}$  be the Hawaiian earring. We will calculate its sheaf (co)homology using the closed Mayer–Vietoris sequence applied to

$$K_1 = X \cap \{(x, y) \in \mathbb{R}^2 : x \geq 0\} \text{ and } K_2 = X \cap \{(x, y) \in \mathbb{R}^2 : x \leq 0\}.$$

This ‘cutting the Hawaiian earring in half’ is illustrated in Figure 4. Fix a  $T \in \mathbf{Ab}$ . Notice that each  $K_i$  is contractible (just contract each arc simultaneously to  $(0, 0)$ ), so by Theorem 6.25 we have  $H^0(K_i, T_{K_i}) \simeq T$  and  $H^p(K_i, T_{K_i}) = 0$  for  $p > 0$ . Furthermore  $K_1 \cap K_2$  is the space from Example 6.9 and 6.12 so  $H^0(K_1 \cap K_2, T_{K_1 \cap K_2})$  is naturally isomorphic to the group  $F(T)$  consisting of sequences  $(t_i)_{i \geq 0}$  of elements in  $t$  which are eventually constant. Lemma 6.30 shows that  $H^p(K_1 \cap K_2, T_{K_1 \cap K_2}) = 0$  for  $p > 0$ . Plugging all of this into the Mayer–Vietoris sequence gives

$$0 \rightarrow H^0(X, T_X) \rightarrow T \times T \xrightarrow{f} F(T) \rightarrow H^1(X, T_X) \rightarrow 0 \quad (6.2)$$

and  $0 \rightarrow H^p(X, T_X) \rightarrow 0$  for  $p > 1$ . Thus  $H^0(X, T_X) \simeq \ker f$ ,  $H^1(X, T_X) \simeq \operatorname{coker} f$  and  $H^p(X, T_X) = 0$  for  $p > 1$ . Recall that  $i_1^* : H^0(K_1, T_{K_1}) \rightarrow H^0(K_1 \cap K_2, T_{K_1 \cap K_2})$  simply restricts a locally constant function on  $K_1$  to  $K_1 \cap K_2$ . Since all locally constant functions on  $K_1$  are constant,  $i_1^*$  looks like  $x \mapsto (x)_{i \geq 0}$  under the isomorphisms  $H^0(K_1, T_{K_1}) \simeq T$  and  $H^0(K_1 \cap K_2, T_{K_1 \cap K_2}) \simeq F(T)$ . Doing the



same for  $K_2$  shows that  $f$  is given by  $(x, y) \mapsto (x - y)_{i \geq 0}$ . Therefore the exact sequence (6.2) is naturally (in  $T$ ) isomorphic to:

$$0 \longrightarrow T \xrightarrow{x \mapsto (x, x)} T \times T \xrightarrow{(x, y) \mapsto (x - y)_{i \geq 0}} F(T) \xrightarrow{(x_i)_{i \geq 0} \mapsto (x_i - \lim_{i \rightarrow \infty} x_i)} \bigoplus_{i \geq 0} T \longrightarrow 0 \quad (6.3)$$

We conclude that  $H^0(X, -_X) \simeq \text{id}_{\mathbf{Ab}} \simeq \mathbb{Z}$  and that  $H^1(X, -_X) \simeq (T \mapsto \bigoplus_{i \geq 0} T)$  which is topologically represented by  $\prod_{n \geq 0}^{\text{Top}} \mathbb{Z}$  as seen in Example 6.9. Since  $H^0(X, -_X)$  and  $H^1(X, -_X)$  are always left exact, these are actually 0th and 1st sheaf homology functors. We conclude that  $\vec{H}_0(X)$  is represented by  $\mathbb{Z}$ ,  $\vec{H}_1(X)$  is topologically represented by  $\prod_{n \geq 0}^{\text{Top}} \mathbb{Z}$  and that  $\vec{H}_p(X) = 0$  for  $p > 1$ . Comparing with Example 6.9 we see that in some sense  $X$  ‘has the same number of holes as there are points in the space  $\{0\} \cup \{1/n : n \in \mathbb{Z}_{\geq 1}\}$ ’. Contrastingly the singular homology of the Hawaiian earring, computed in [8], is much more complicated.

**Remark 6.32.** Note that  $\prod_{n \geq 0}^{\text{Top}} \mathbb{Z}$  can be seen as the limit in  $\mathbf{TopAb}$  of the diagram  $\mathbb{Z}^0 \leftarrow \mathbb{Z}^1 \leftarrow \mathbb{Z}^2 \leftarrow \mathbb{Z}^3 \leftarrow \dots$  where each map removes the last coordinate. By Corollary 4.18 it represents the limit of this diagram in  $\vec{\mathbf{Ab}}$  and hence this inverse system represents  $\vec{H}_1(X)$  with  $X$  as in the above example. In [17, 6.3] it was shown that the fundamental pro-group of  $X$  (or rather its topos of sheaves of sets) is represented by the inverse system  $F_0 \leftarrow F_1 \leftarrow F_2 \leftarrow \dots$  where  $F_n$  is the non-abelian free group on  $n$  variables and each map  $F_n \rightarrow F_{n-1}$  maps the  $n$ -th variable to the identity. This is clearly the non-abelian analogue of the above inverse system.

**Theorem 6.33.** Let  $X, K_1, K_2, i_1, \dots$  be as in Proposition 6.29. Then for all  $F \in \vec{\mathbf{Ab}}$  there is a long exact sequence

$$\dots \leftarrow \vec{H}_p(X; F) \xleftarrow{j_{1*} + j_{2*}} \vec{H}_p(K_1; F) \oplus \vec{H}_p(K_2; F) \xleftarrow{i_{1*} \times (-i_{2*})} \vec{H}_p(K_1 \cap K_2; F) \xleftarrow{\partial} \vec{H}_{p+1}(X; F) \leftarrow \dots$$

in  $\vec{\mathbf{Ab}}$ . Here  $f_*$  for a continuous map  $f$  denotes the morphism as in Corollary 6.4.

**Proof.** This follows from applying the exact functors  $(-) \circ F : [\mathbf{Ab}, \mathbf{Ab}]_{\text{add}} \rightarrow [\mathbf{Ab}, \mathbf{Ab}]_{\text{add}}$  and left-exactification  $[\mathbf{Ab}, \mathbf{Ab}]_{\text{add}} \rightarrow \vec{\mathbf{Ab}}$  to the long exact sequence for cohomology from Proposition 6.29, seen as a sequence in  $[\mathbf{Ab}, \mathbf{Ab}]_{\text{add}}$ .  $\square$

**Example 6.34.** (Continuing Example 6.31) Recall from Example 6.9 that  $\vec{H}_0(K_1 \cap K_2)$  is represented by  $\prod_{i \geq -1}^{\text{Top}} \mathbb{Z}$ . We claim that using this and the representations of  $\vec{H}_0(X)$  and  $\vec{H}_1(X)$  found earlier, the Mayer-Vietoris sequence for homology becomes the following sequence in  $\mathbf{TopAb}$ :

$$\begin{array}{ccccccc} x + y & \longleftarrow & (x, y) & & (0, (x_i)_{i \geq 0}) & \longleftarrow & (x_i)_{i \geq 0} \\ 0 & \longleftarrow & \mathbb{Z} & \longleftarrow & \mathbb{Z} \oplus \mathbb{Z} & \longleftarrow & \prod_{i \geq -1}^{\text{Top}} \mathbb{Z} \longleftarrow \prod_{i \geq 0}^{\text{Top}} \mathbb{Z} \longleftarrow 0 \\ & & & & (x_{-1}, -x_{-1}) & \longleftarrow & (x_i)_{i \geq -1} \end{array} \quad (6.4)$$

For each  $T \in \mathbf{Ab}$ , combining the exact sequence (6.3) with the correct isomorphisms gives

$$\begin{array}{ccccccc} 0 & \longrightarrow & T & \xrightarrow{x \mapsto (x, x)} & T \times T & \xrightarrow{(x, y) \mapsto (x - y)_{i \geq 0}} & F(T) \xrightarrow{(x_i)_{i \geq 0} \mapsto (x_i - \lim_j x_j)_{i \geq 0}} \bigoplus_{i \geq 0} T \longrightarrow 0 \\ & & \uparrow & & \uparrow & & \downarrow (x_i)_{i \geq 0} \mapsto (\lim_i x_i, (x_i - \lim_j x_j)_{i \geq 0}) \\ & & f \mapsto f(1) & & f \mapsto (f(e_1), f(e_2)) & & \bigoplus_{i \geq -1} T \\ & & & & & & \uparrow f \mapsto (f(e_i))_{i \geq -1} \\ 0 & \longrightarrow & \text{Hom}(\mathbb{Z}, T) & \longrightarrow & \text{Hom}(\mathbb{Z} \oplus \mathbb{Z}, T) & \longrightarrow & \text{Hom}(\prod_{i \geq -1}^{\text{Top}} \mathbb{Z}, T) \longrightarrow \text{Hom}(\prod_{i \geq 0}^{\text{Top}} \mathbb{Z}, T) \longrightarrow 0 \end{array}$$

where the lower hom-sets are taken in  $\mathbf{TopAb}$ . Carefully walking across these isomorphisms shows that the lower horizontal homomorphisms are given by precomposition with the maps in (6.4).

**Example 6.35.** (Barratt–Milnor Spheres) Consider the space

$$X = \bigcup_{n \geq 0} \{(x, y, z) \in \mathbb{R}^3 : (x - 2^{-n-1})^2 + y^2 + z^2 = 4^{-n-1}\}$$

and set  $K_1 = \{(x, y, z) \in X : z \geq 0\}$  and  $K_2 = \{(x, y, z) \in X : z \leq 0\}$ . Now  $K_1 \cap K_2$  is just homomorphic to the Hawaiian earring and the  $K_i$  are contractible, so the Mayer–Vietoris sequence for sheaf homology becomes

$$0 \leftarrow \vec{H}_0(X) \leftarrow \vec{\mathbb{Z}} \oplus \vec{\mathbb{Z}} \leftarrow \vec{\mathbb{Z}} \leftarrow 0 \leftarrow 0 \leftarrow \vec{H}_1(K_1 \cap K_2) \leftarrow \vec{H}_2(X) \leftarrow 0$$

So  $\vec{H}_0(X) \simeq \vec{\mathbb{Z}}$ ,  $\vec{H}_2(X)$  is represented by  $\prod_{i \geq 0}^{\text{Top}} \mathbb{Z}$  and  $\vec{H}_p(X) = 0$  for  $p \neq 0, 2$ . Compare to the fact that the  $p$ -th singular homology of  $X$  is non-trivial for *infinitely* many  $p$  (see [3]), while  $X$  is clearly at most 3-dimensional for any reasonable notion of dimension. This indicates a serious inability of singular homology to capture the geometry of spaces with locally anomalous behaviour. Additionally this serves as an example to show that singular homology does not satisfy closed Mayer–Vietoris:  $K_1$  and  $K_2$  are contractible and  $K_1 \cap K_2$  is known to have  $H_p^{\text{sing}}(K_1 \cap K_2) = 0$  for  $p > 1$  (see [8]), so if closed Mayer–Vietoris were to hold for singular homology then  $H_p^{\text{sing}}(X) = 0$  for  $p > 2$ .

**Proposition 6.36.** (Partial naturality of closed Mayer–Vietoris) *Let  $X$  be a topological space with closed subsets  $K_1, K_2, A$  so that  $K_1 \cup K_2 = X$ . Write  $K_1 \cap K_2 = K_{12}$ ,  $K'_1 = A \cap K_1$ ,  $K'_2 = K_2 \cap A$ ,  $K'_{12} = A \cap K_{12}$ , then for any sheaf  $\mathcal{F}$  on  $X$  there is a morphism*

$$\begin{array}{ccccccc} \cdots & \rightarrow & H^p(X, \mathcal{F}) & \rightarrow & H^p(K_1, \mathcal{F}) \times H^p(K_2, \mathcal{F}) & \rightarrow & H^p(K_{12}, \mathcal{F}) \rightarrow H^{p+1}(X, \mathcal{F}) \rightarrow \cdots \\ & & \downarrow & & \downarrow & & \downarrow \\ \cdots & \rightarrow & H^p(A, \mathcal{F}) & \rightarrow & H^p(K'_1, \mathcal{F}) \times H^p(K'_2, \mathcal{F}) & \rightarrow & H^p(K'_{12}, \mathcal{F}) \rightarrow H^{p+1}(A, \mathcal{F}) \rightarrow \cdots \end{array}$$

between the long exact sequences from Lemma 6.28 applied to  $X, K_1, K_2$  and to  $A, K'_1, K'_2$ , where the vertical arrows are given by the induced morphism by the various inclusions as in Lemma 6.2. Here for a closed subset  $K$  of  $X$  with inclusion  $i : K \rightarrow X$  we write  $H^p(K, \mathcal{F})$  for  $H^p(K, i^{-1}\mathcal{F})$ . Furthermore this morphism is natural in  $\mathcal{F}$ . Consequently for any  $T$  there is a morphism of long exact sequences

$$\begin{array}{ccccccc} \cdots & \rightarrow & H^p(X, T_X) & \rightarrow & H^p(K_1, T_{K_1}) \times H^p(K_2, T_{K_2}) & \rightarrow & H^p(K_{12}, T_{K_{12}}) \rightarrow H^{p+1}(X, T_X) \rightarrow \cdots \\ & & \downarrow & & \downarrow & & \downarrow \\ \cdots & \rightarrow & H^p(A, T_A) & \rightarrow & H^p(K'_1, T_{K'_1}) \times H^p(K'_2, T_{K'_2}) & \rightarrow & H^p(K'_{12}, T_{K'_{12}}) \rightarrow H^{p+1}(A, T_A) \rightarrow \cdots \end{array}$$

between the long exact sequences from Proposition 6.29 applied to  $X, K_1, K_2$  and to  $A, K'_1, K'_2$ , where each vertical arrow is given by the morphisms from Proposition 6.3. Finally from this it follows that for all  $F \in \tilde{\mathbf{Ab}}$  there is a morphism of long exact sequences

$$\begin{array}{ccccccc} \cdots & \leftarrow & \vec{H}_p(X; F) & \leftarrow & \vec{H}_p(K_1; F) \oplus \vec{H}_p(K_2; F) & \leftarrow & \vec{H}_p(K_{12}; F) \leftarrow \vec{H}_{p+1}(X; F) \leftarrow \cdots \\ & & \uparrow & & \uparrow & & \uparrow \\ \cdots & \leftarrow & \vec{H}_p(A; F) & \leftarrow & \vec{H}_p(K'_1; F) \oplus \vec{H}_p(K'_2; F) & \leftarrow & \vec{H}_p(K'_{12}; F) \leftarrow \vec{H}_{p+1}(A; F) \leftarrow \cdots \end{array}$$

between the long exact sequences from Theorem 6.33 applied to  $X, K_1, K_2$  and to  $A, K'_1, K'_2$ , where each vertical arrow is given by the morphisms from Corollary 6.4.

**Proof.** If we temporarily write  $\mathcal{F}_K := i_* i^{-1} \mathcal{F}$  for  $i : K \rightarrow X$  the inclusion of a closed subset  $K$ , then we have the following morphism of exact sequences of sheaves:

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathcal{F} & \longrightarrow & \mathcal{F}_{K_1} \times \mathcal{F}_{K_2} & \longrightarrow & \mathcal{F}_{K_{12}} \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & \mathcal{F}_A & \longrightarrow & \mathcal{F}_{K'_1} \times \mathcal{F}_{K'_2} & \longrightarrow & \mathcal{F}_{K'_{12}} \longrightarrow 0 \end{array}$$

Here as in the proof of Lemma 6.28, for closed subsets  $A \subseteq B$  the morphism  $\mathcal{F}_B \rightarrow \mathcal{F}_A$  is constructed using Lemma 6.27. One can check that this diagram commutes and is exact in the rows as before using stalks. Also all morphisms here are natural in  $\mathcal{F}$  for the exact same reason as before. Taking the corresponding morphism of long exact sequences and applying the isomorphism  $H^p(X, \mathcal{F}_K) \simeq H^p(K, i^{-1}\mathcal{F})$  gives the required morphism of long exact sequences. Note that there are quite a lot of things that need to be checked here, for example that the morphisms in the lower long exact sequence are indeed those from Lemma 6.28 applied to  $A, K'_1, K'_2$ . We will omit these verifications here. The morphisms for constant coefficients and for sheaf homology follow similar to before (with again a lot of details to be checked).  $\square$

**Example 6.37.** (Continuing Example 6.31) Fix an  $n \in \mathbb{Z}_{\geq 0}$  and let  $A = \{(x, y) \in \mathbb{R}^2 : (x - 2^{-n-1})^2 + y^2 = 4^{n-1}\}$  be the  $n$ -th circle in the Hawaiian earring  $X$ . Notice that each  $K'_i = K_i \cap A$  is simply a contractible circular arc and that  $K'_{12} = A \cap K_1 \cap K_2$  consists of two points. Thus applying Proposition 6.36 gives the following morphism of exact sequences

$$\begin{array}{ccccccc} 0 & \longrightarrow & H^0(X, T_X) & \longrightarrow & T \times T & \xrightarrow{(x,y) \mapsto (x-y)_{i \geq 0}} & F(T) \longrightarrow H^1(X, T_X) \longrightarrow 0 \\ & & \downarrow & & \text{id} \downarrow & & (x_i)_{i \geq 0} \mapsto \downarrow \\ 0 & \longrightarrow & H^0(A, T_A) & \longrightarrow & T \times T & \xrightarrow{(\lim_i x_i, x_n) \downarrow} & T \times T \longrightarrow H^1(A, T_A) \longrightarrow 0 \\ & & & & & & (x,y) \mapsto (x-y, x-y) \end{array}$$

Here we identified  $H^0(K'_{12}, T_{K'_{12}})$  with  $T \times T$  by setting the first entry to be the value at the singularity and the second at the other point of  $K'_{12}$ . The descriptions of the maps here can be found by tracking how locally constant functions restrict to the various subsets. Identifying the upper exact sequence with (6.3) and identifying the kernel and cokernel of the lower map  $T^2 \rightarrow T^2$  with  $T$  by setting the quotient map to  $T \times T \rightarrow T, (x, y) \mapsto (y - x)$  gives

$$\begin{array}{ccccccc} 0 & \longrightarrow & T & \longrightarrow & T \times T & \xrightarrow{(x,y) \mapsto (x-y)_{i \geq 0}} & F(T) \xrightarrow{(x_i)_{i \geq 0} \mapsto (x_i - \lim_j x_j)_i} \bigoplus_{i \geq 0} T \longrightarrow 0 \\ & & \downarrow & & \text{id} \downarrow & & (x_i)_{i \geq 0} \mapsto \downarrow \\ 0 & \longrightarrow & T & \longrightarrow & T \times T & \xrightarrow{(\lim_i x_i, x_n) \downarrow} & T \times T \xrightarrow{(x,y) \mapsto y-x} T \longrightarrow 0 \\ & & & & & & (x,y) \mapsto (x-y, x-y) \end{array}$$

From this we can see that the morphism  $\bigoplus_{i \geq 0} T \rightarrow T$  is given by the projection on the  $n$ -th coordinate. Finally identifying  $T \simeq \text{Hom}(\mathbb{Z}, T)$  and  $\bigoplus_{i \geq 0} T \simeq \text{Hom}(\prod_{i \geq 0}^{\text{Top}} \mathbb{Z}, T)$  we see that this morphism is given by precomposition with the (continuous) homomorphism  $\mathbb{Z} \rightarrow \prod_{i \geq 0}^{\text{Top}} \mathbb{Z}$  which maps 1 to  $e_n = (0, 0, \dots, 0, 1, 0, \dots)$  (with the 1 at index  $n$ ). Thus, under the isomorphism  $\tilde{H}_1(X) \simeq \prod_{i \geq 0}^{\text{Top}} \mathbb{Z}$  from Example 6.9 and the isomorphism  $\tilde{H}_1(A) \simeq \mathbb{Z}$  we found here, we conclude that the inclusion  $A \rightarrow X$  corresponds to the map  $\mathbb{Z} \rightarrow \prod_{i \geq 0}^{\text{Top}} \mathbb{Z}$  which maps 1 to  $e_n$ .

This shows that each  $e_n$  corresponds to the  $n$ -th loop of the earring. Notice that due to the topology on  $\prod_{i \geq 0}^{\text{Top}} \mathbb{Z}$  we have  $\lim_{n \rightarrow \infty} e_n = 0$  which reflects the geometric fact that the loops in the earring shrink to a point. Furthermore notice that while the  $e_i$  don't generate the group  $\prod_{i \geq 0}^{\text{Top}} \mathbb{Z}$ , they do topologically generate  $\prod_{i \geq 0}^{\text{Top}} \mathbb{Z}$ . This is because infinite sums such as  $\sum_{i=0}^{\infty} e_i$  converge in  $\tilde{H}_1(X)$ . This aligns with the topological fact that it is possible to define a path  $\gamma : [0, 1] \rightarrow X$  which circles around each loop of the earring once.

## 6.5 Open Mayer–Vietoris

In this section we give the construction for the open Mayer–Vietoris theorem for sheaf cohomology, from which the corresponding theorem for sheaf homology follows. We give a slightly rephrased version of

the construction from [14, p. 104] which shows that the Mayer–Vietoris long exact sequence arises from a certain short exact sequence in  $[\text{Sh}(X), \mathbf{Ab}]_{\text{LE}}$ . For the rest of this section let  $X$  be a topological space with  $U_1, U_2 \subseteq X$  open sets so that  $U_1 \cup U_2 = X$ . Write  $U_{12} = U_1 \cap U_2$  and label the various inclusions as

$$\begin{array}{ccc} U_1 & \xrightarrow{j_1} & X \\ i_1 \uparrow & \nearrow k & \uparrow j_2 \\ U_{12} & \xrightarrow{i_2} & U_2 \end{array}$$

**Lemma 6.38.** ([5, Lemma 1.13]) *Any injective sheaf is flasque.*

**Proof.** Let  $\mathcal{I} \in \text{Sh}(X)$  be an injective sheaf. Consider the embedding  $\mathcal{I} \hookrightarrow \mathcal{I}^0$  of  $\mathcal{I}$  into its sheaf of discontinuous sections. Since  $\mathcal{I}$  is injective this morphism splits, so we have an isomorphism  $\mathcal{I} \oplus \mathcal{F} \simeq \mathcal{I}^0$  for some sheaf  $\mathcal{F}$ . Since  $\mathcal{I}^0$  is flasque, for each inclusion of opens  $V \subseteq U \subseteq X$  the restriction map  $(\mathcal{I} \oplus \mathcal{F})(U) \rightarrow (\mathcal{I} \oplus \mathcal{F})(V)$ , which is just the direct sum of the maps  $\mathcal{I}(U) \rightarrow \mathcal{I}(V)$  and  $\mathcal{F}(U) \rightarrow \mathcal{F}(V)$ , is surjective. It follows that  $\mathcal{I}(U) \rightarrow \mathcal{I}(V)$  is surjective, so  $\mathcal{I}$  is flasque.  $\square$

**Lemma 6.39.** *For any sheaf  $\mathcal{F}$  on  $X$  there is a long exact sequence*

$$\cdots \rightarrow H^p(X, \mathcal{F}) \xrightarrow{j_1^* \times j_2^*} H^p(U_1, j_1^{-1} \mathcal{F}) \times H^p(U_2, j_2^{-1} \mathcal{F}) \xrightarrow{i_1^* - i_2^*} H^p(U_{12}, k^{-1} \mathcal{F}) \xrightarrow{\delta} H^{p+1}(X, \mathcal{F}) \rightarrow \cdots$$

*natural in  $\mathcal{F}$ . Here  $f^*$  for a continuous map  $f$  denotes the homomorphism as in Lemma 6.2.*

**Proof.** Firstly for opens  $U \subseteq X$  we write  $\Gamma_U : \text{Sh}(X) \rightarrow \mathbf{Ab}$  for the functor given by taking  $U$ -sections. If  $V \subseteq U$  we have a morphism of functors  $\rho_{UV} : \Gamma_U \rightarrow \Gamma_V$  given on each object by the restriction map. Now we claim that the sequence

$$0 \longrightarrow \Gamma_X \xrightarrow{\rho_{XU_1} \times \rho_{XU_2}} \Gamma_{U_1} \times \Gamma_{U_2} \xrightarrow{\rho_{U_1 U_{12}} - \rho_{U_2 U_{12}}} \Gamma_{U_{12}} \longrightarrow 0$$

is exact in  $[\text{Sh}(X), \mathbf{Ab}]_{\text{LE}}$ . First of all for all sheaves  $\mathcal{F}$  on  $X$  the sequence  $0 \rightarrow \Gamma(X, \mathcal{F}) \rightarrow \Gamma(U_1, \mathcal{F}) \times \Gamma(U_2, \mathcal{F}) \rightarrow \Gamma(U_{12}, \mathcal{F})$  is exact by the sheaf property. Secondly by Lemma 6.38 for all injective sheaves  $\mathcal{I} \in \text{Sh}(X)$  the restriction map  $\Gamma(U_1, \mathcal{I}) \rightarrow \Gamma(U_{12}, \mathcal{I})$  is surjective, so in particular the map  $\Gamma(U_1, \mathcal{I}) \times \Gamma(U_2, \mathcal{I}) \rightarrow \Gamma(U_{12}, \mathcal{I})$  is surjective. By Proposition 2.29 the claimed exactness follows, so using Lemma 2.31 we obtain a long exact sequence

$$0 \rightarrow \Gamma_X \rightarrow \Gamma_{U_1} \times \Gamma_{U_2} \rightarrow \Gamma_{U_{12}} \rightarrow R^1 \Gamma_X \rightarrow R^1 \Gamma_{U_1} \times R^1 \Gamma_{U_2} \rightarrow R^1 \Gamma_{U_{12}} \rightarrow R^2 \Gamma_X \rightarrow \cdots$$

Of course  $R^p \Gamma_X = H^p(X, -)$  and if  $U \subseteq X$  is open with inclusion  $i$ , then  $R^p \Gamma_U(\mathcal{F}) \simeq H^p(U, i^{-1} \mathcal{F})$  for  $\mathcal{F} \in \text{Sh}(X)$ . To obtain this isomorphism note that  $i^{-1} : \text{Sh}(X) \rightarrow \text{Sh}(U)$ , which simply restricts a sheaf to  $U$ , is exact and maps flasque (and hence injective) sheaves to flasque sheaves. Therefore if  $\mathcal{I}^\bullet$  is an injective resolution of  $\mathcal{F}$  in  $\text{Sh}(X)$  then  $i^{-1} \mathcal{I}^\bullet$  is a flasque resolution of  $i^{-1} \mathcal{F}$ , so  $R^p \Gamma_U(\mathcal{F}) \simeq h^p(\Gamma(U, \mathcal{I}^\bullet)) \simeq h^p(\Gamma(U, i^{-1} \mathcal{I}^\bullet)) \simeq H^p(U, i^{-1} \mathcal{F})$ . We conclude that we have an exact sequence with the right objects. To see that the morphisms in the long exact sequence are of the required form one can use the fact that the morphism from Lemma 6.2 applied to the inclusion of an open subset is just the restriction morphism (under the isomorphism  $\mathcal{F}(U) \simeq (i^{-1} \mathcal{F})(U)$ ) and then applying a lot of definition unfolding.  $\square$

**Theorem 6.40.** *For any  $T \in \mathbf{Ab}$  there exists a long exact sequence*

$$\cdots \rightarrow H^p(X, T_X) \xrightarrow{j_1^* \times j_2^*} H^p(U_1, T_{U_1}) \times H^p(U_2, T_{U_2}) \xrightarrow{i_1^* - i_2^*} H^p(U_{12}, T_{U_{12}}) \xrightarrow{\delta} H^{p+1}(X, T_X) \rightarrow \cdots$$

*natural in  $T$ . Consequently for any  $F \in \vec{\mathbf{Ab}}$  there is a long exact sequence in  $\vec{\mathbf{Ab}}$*

$$\cdots \leftarrow \vec{H}_p(X; F) \xleftarrow{j_1^* + j_2^*} \vec{H}_p(U_1; F) \oplus \vec{H}_p(U_2; F) \xleftarrow{i_1^* \times (-i_2^*)} \vec{H}_p(U_1 \cap U_2; F) \xleftarrow{\partial} \vec{H}_{p+1}(X) \rightarrow \cdots$$

**Proof.** This follows from applying Lemma 6.39 to  $T_X$ , keeping in mind the natural isomorphisms  $j_1^{-1}T_X \simeq T_{U_1}$ . We omit the verification that the morphisms in the sequence are indeed those from Lemma 6.3. The long exact sequence for sheaf homology follows from applying the *exact* functors  $(-) \circ F : [\mathbf{Ab}, \mathbf{Ab}]_{\text{add}} \rightarrow [\mathbf{Ab}, \mathbf{Ab}]_{\text{add}}$  and left-exactification  $[\mathbf{Ab}, \mathbf{Ab}]_{\text{add}} \rightarrow \tilde{\mathbf{Ab}}$  to the sequence for sheaf cohomology.  $\square$

**Example 6.41.** (Double limit sequence) Let  $X = \{x_0, x_1, x_2, \dots\} \cup \{x_\infty, x'_\infty\}$  be the topological space with a basis consisting of the sets  $\{x_i\}$  for  $i \in \mathbb{Z}_{\geq 0}$  and  $\{x_\infty\} \cup \{x_N, x_{N+1}, \dots\}$  and  $\{x'_\infty\} \cup \{x_N, x_{N+1}, \dots\}$  for  $N \geq 0$ . This space can also be seen as the subset  $\{1/n : n \geq 1\} \cup \{0, 0'\}$  of the line with two origins. Now set  $U_1 = \{x_0, x_1, \dots\} \cup \{x_\infty\}$  and  $U_2 = \{x_0, x_1, \dots\} \cup \{x'_\infty\}$ . Notice that  $U_1$  and  $U_2$  are homeomorphic to the space from Example 6.9 and that  $U_{12} = U_1 \cap U_2$  is simply a countable discrete space, so locally constant functions on it will be just sequences with values in  $T$ . Thus for  $T \in \mathbf{Ab}$  we have the following exact sequence natural in  $T$ :

$$0 \rightarrow H^0(X, T_X) \rightarrow F(T) \times F(T) \xrightarrow{f} \prod_{i=0}^{\infty} T \rightarrow H^1(X, T_X) \rightarrow 0$$

and  $0 \rightarrow H^p(X, T_X) \rightarrow 0$  for  $p > 0$ . Here  $F$  is the functor as in Example 6.31. Notice that the homomorphism  $f$  is given by  $((x_i)_{i \geq 0}, (y_i)_{i \geq 0}) \mapsto (x_i - y_i)_{i \geq 0}$ . The kernel of  $f$  is given by the set of pairs of the form  $((x_i)_{i \geq 0}, (x_i)_{i \geq 0})$  for  $(x_i)_{i \geq 0} \in F(T)$ , so  $H^0(X, -_X) \simeq F$  and  $\tilde{H}_0(X)$  is therefore represented by  $\prod_{i \geq 0}^{\text{Top}} \mathbb{Z}$ . The image of  $f$  consists of all eventually constant sequences in  $T$ , so the cokernel of  $f$ , i.e.  $H^1(X, T_X)$ , is given by the group  $G(T)$  consisting of *tails* of sequences in  $T$  (i.e. equivalence classes under the relation of being eventually equal) modulo constant sequences. Since the Mayer–Vietoris sequence is natural in  $T$  the functor  $H^1(X, -_X)$ , and hence  $\tilde{H}_1(X)$  ( $H^1(X, -_X)$  is always left exact), is isomorphic to the functor  $G : \mathbf{Ab} \rightarrow \mathbf{Ab}$ . We conclude that the Mayer–Vietoris sequence for sheaf homology becomes

$$0 \leftarrow \prod_{i \geq 0}^{\text{Top}} \mathbb{Z} \leftarrow \prod_{i \geq 0}^{\text{Top}} \mathbb{Z} \oplus \prod_{i \geq 0}^{\text{Top}} \mathbb{Z} \leftarrow \bigoplus_{i \geq 0} \mathbb{Z} \leftarrow G \leftarrow 0$$

In order to understand the functor  $G$  better first consider the functor  $H : \mathbf{Ab} \rightarrow \mathbf{Ab}$  from Example 3.17 which maps a  $T \in \mathbf{Ab}$  to the group of tails of sequences in  $T$  (with the obvious functor structure). Now for each  $T \in \mathbf{Ab}$  we have a homomorphism  $\eta_T : H(T) \rightarrow G(T)$  given by mapping the class of a sequence  $(x_i)_{i \geq 0}$  to the class of the sequence  $(\sum_{i=0}^{n-1} x_i)_{n \geq 0}$  and a homomorphism  $\epsilon_T : G(T) \rightarrow H(T)$  which maps the class of a sequence  $(x_i)_{i \geq 0}$  to the class of  $(x_{i+1} - x_i)_{i \geq 0}$ . It is not hard to see that these morphisms are natural in  $T$  and inverses of each other, so  $G \simeq H$ . We know from Example 3.42 that  $H$  is not strict and hence not topologically representable. This shows that the local connectivity assumption of Theorem 5.29 is necessary. We also know from Example 3.22 that  $H$  is not pro-finitely generated. Since  $X$  is compact this shows that the Hausdorff assumption of Corollary 5.38 is necessary.

**Example 6.42.** (Topologist's sine curve) Consider the subspace  $\{(0, 0)\} \cup \{(x, \sin(\pi/x)) : x \in (0, 1]\}$  of  $\mathbb{R}^2$ . For simplicity we will work with the homeomorphic space

$$X = \{(0, 0)\} \cup \{(x, |\sin(\pi/x)|) : x \in (0, 1]\}.$$

(The continuous map  $p : \mathbb{R}^2 \rightarrow \mathbb{R}^2, (x, y) \mapsto (x, |y|)$  restricts to a bijection between these two sets whose inverse can also be seen to be continuous.) See Figure 5. Let  $U_1 = \{(x, y) \in X : y < 2/3\}$  and  $U_2 = \{(x, y) \in X : y > 1/3\}$ . Notice that  $U_2$  and  $U_{12} = U_1 \cap U_2$  are a countable disjoint union of open unit intervals, so they are homotopy equivalent to a countable disjoint union of points and therefore have trivial higher sheaf (co)homology and 0th cohomology  $\prod_{i=0}^{\infty} T$  with coefficients in  $T \in \mathbf{Ab}$ . See Figure 5 for which connected component corresponds to what index of this product.  $U_1$  on the other hand has a retract  $\{(0, 0)\} \cup \{(1/n, 0) : n \in \mathbb{Z}_{\geq 1}\}$  which we know from Example 6.9 to have trivial higher

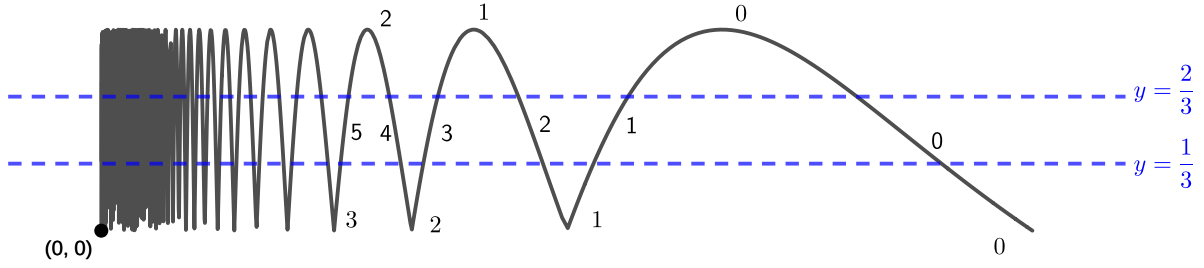


Figure 5: The Topologist's sine curve. The numbers indicate what index we give to each connected component of  $U_1$  (lower numbers),  $U_2$  (top numbers) and  $U_1 \cap U_2$  (middle numbers). Each arc intersects the x-axis in reality, but some do not appear to in this picture due to numerical imprecision.

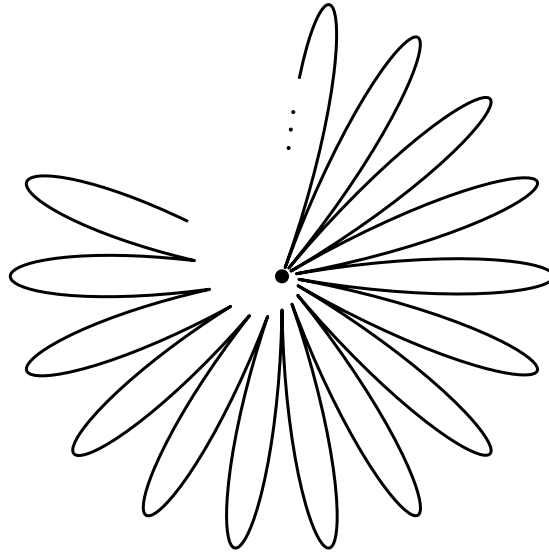


Figure 6: The Topologist's sine curve wrapped around the point  $(0, 0)$ .

sheaf cohomology 0 and  $H^0(X, -_X) \simeq F$  where  $F$  maps a  $T \in \mathbf{Ab}$  to the group of eventually constant sequences in  $T$ . What entry of an element of  $F(T)$  corresponds to which value of the corresponding locally constant function is visualised in Figure 5. Putting this together we see that for each  $T \in \mathbf{Ab}$  the Mayer–Vietoris sequence for sheaf cohomology becomes

$$0 \longrightarrow T \longrightarrow F(T) \times \prod_{i=0}^{\infty} T \xrightarrow{f} \prod_{i=0}^{\infty} T \longrightarrow H^1(X, T_X) \longrightarrow 0$$

and  $0 \rightarrow H^p(X, T_X) \rightarrow 0$  for  $p > 1$ . By carefully looking at Figure 5 to see how locally constant functions restrict we see that  $f$  maps a pair  $((x_i)_{i \geq 0}, (y_i)_{i \geq 0})$  to the sequence  $(z_i)_{i \geq 0}$  where  $z_{2i} = x_i - y_i$  and  $z_{2i+1} = x_{i+1} - y_i$  for  $i \geq 0$ . Now the image of  $f$  consists of all sequences  $(z_i)_{i \geq 0}$  so that  $z_{2i+1} - z_{2i}$  is eventually constant as  $i \rightarrow \infty$ . It follows that  $\text{coker } f \simeq G(T)$  with  $G$  as in the previous example. We conclude that  $\vec{H}_1(X)$  again the functor from Example 3.17.

**Intuition 6.43.** In the previous two examples we saw two completely different spaces both with first sheaf homology the functor from Example 3.17 which is the limit of the diagram

$$\bigoplus_{i \geq 0} \mathbb{Z} \leftarrow \bigoplus_{i \geq 1} \mathbb{Z} \leftarrow \bigoplus_{i \geq 2} \mathbb{Z} \leftarrow \bigoplus_{i \geq 3} \mathbb{Z} \leftarrow \dots \quad (6.5)$$

in  $\tilde{\mathbf{Ab}}$ . At first sight there do not appear to be any holes in the topologist's sine curve, so it is surprising that it has a non-trivial first homology at all. Here we take some time to develop intuition for why the limit of this diagram is a reasonable first homology of both spaces.

First of all notice that in the Topologist's sine curve there are no limit points to 'witness' that the top of each arc in Figure 5 gets closer and closer to each other. This is not the case in the Warsaw circle (see Figure 3), where the points on the set  $\{0\} \times [-1, 1]$  are limit points of the sine curve. This means that we could just as well draw each arc at the same size and with the same distance apart. The exception is the point  $(0, 0)$  which is a limit point of the points  $(1/n, 0)$  for  $n \in \mathbb{Z}_{>0}$ . We should therefore draw each arc at the same size, with the points  $(1/n, 0)$  converging to  $(0, 0)$ . In Figure 6 this is done by wrapping the sine curve around the circle. Note that if this figure were to continue, the 'flower petals' would eventually start overlapping each other, but this effect should be ignored (using a covering of  $\mathbb{R}^2$  branched at  $(0, 0)$  for example.) Note that from a distance this space looks like a point with an infinite number of loops attached to it. The further we zoom into the point  $(0, 0)$ , the more loops we see are actually not connected to  $(0, 0)$ , but at each scale an infinite number of loops continue to seem attached. This behaviour is reflected in the inverse system (6.5): at each stage there are an infinite number of generators, but each individual generator is lost eventually.

We can apply a similar style of reasoning to the non-Hausdorff space from Example 6.41. It can be thought of as two points  $x_\infty, x'_\infty$  connected to each other by an infinite number of lines (each actually consisting of one point just as in the 4 point circle), but the closer you look (i.e. the smaller neighbourhoods of  $x_\infty$  and  $x'_\infty$  you consider) the more of these points are actually disconnected. Nonetheless at each 'distance' an infinite number of lines and hence loops remain. Hopefully this convinces the reader that it is reasonable that these two quite different spaces have the same first homology.

**Remark 6.44.** In [16, p. 6.2] the fundamental pro-group of the topologist's sine curve is calculated to be the (non-abelian) pro-group represented by the inverse system

$$F(x_1, x_2, \dots) \leftarrow F(x_2, x_3, \dots) \leftarrow F(x_3, x_4, \dots) \leftarrow \dots$$

where  $F(s_1, s_2, \dots)$  denotes the free group on the symbols  $s_1, s_2, \dots$  and the maps are given by mapping each  $x_i$  to itself. Clearly this is the non-abelian analogue of (6.5).

## 6.6 Limits of spaces

**Theorem 6.45.** (Continuity theorem for sheaf cohomology) *Let  $D : I \rightarrow \mathbf{Top}$  be a diagram of compact Hausdorff topological spaces indexed by a codirected set  $I$  with limit  $X$ . Then for each  $A \in \mathbf{Ab}$  and  $p \in \mathbb{Z}$  applying the functor  $H^p(-, A_-)$  (with functoriality as in section 6.1) to  $X$  and its cone over  $D$  gives the colimit of the diagram  $H^p(-, A_-) \circ D$ .*

**Proof.** This is a special case of Corollary 14.6 from [7]. This theorem applies to locally compact spaces, proper maps and for sheaf cohomology with compact support. For compact Hausdorff spaces this is equivalent to working with arbitrary continuous maps and sheaf cohomology. Showing that the transition morphisms of Bredon's Corollary 14.6 are indeed the morphisms from section 6.1 requires only some simple (but tedious) unfolding of definitions.  $\square$

**Theorem 6.46.** (Continuity theorem for sheaf homology) *Let  $D : I \rightarrow \mathbf{Top}$  be a diagram of compact Hausdorff topological spaces indexed by a codirected set  $I$  with limit  $X$ . Then for each  $F \in \tilde{\mathbf{Ab}}$  and  $p \in \mathbb{Z}$  applying the functor  $\tilde{H}_p(-; F)$  to  $X$  and its cone over  $D$  gives the limit of  $\tilde{H}_p(-; F) \circ D$  in  $\tilde{\mathbf{Ab}}$ . Written more coarsely:*

$$\tilde{H}_p(\lim_i X_i; F) \simeq \lim_i \tilde{H}_p(X_i; F)$$

**Proof.** Theorem 6.45 tells that the functor  $G : \mathbf{Top} \rightarrow [\mathbf{Ab}, \mathbf{Ab}]_{\text{add}}, Y \mapsto (T \mapsto H^p(Y, T_X))$  maps the



colimit of  $D$  to the colimit of  $G \circ D$  (recall that colimits in  $[\mathbf{Ab}, \mathbf{Ab}]_{\text{add}}$  are computed object-wise). The result then follows from the fact that the functor  $- \circ F : [\mathbf{Ab}, \mathbf{Ab}]_{\text{add}} \rightarrow [\mathbf{Ab}, \mathbf{Ab}]_{\text{add}}$  preserves colimits (again because colimits in  $[\mathbf{Ab}, \mathbf{Ab}]_{\text{add}}$  are computed object-wise) and because by Lemma 3.14 left-exactification considered as a functor  $[\mathbf{Ab}, \mathbf{Ab}]_{\text{add}} \rightarrow \vec{\mathbf{Ab}}$  maps colimits to limits.  $\square$

**Corollary 6.47.** *Let  $(F_i)_{i \in I}$  be a decreasing directed family of closed subspaces of a compact Hausdorff space and write  $F_\infty = \bigcap_{i \in I} F_i$ . Then  $\vec{H}_p(F_\infty)$ , together with the cone morphisms induced by the inclusions, is the limit of the diagram  $(\vec{H}_p(F_i))_{i \in I}$  with transition morphisms induced by the inclusions.*

**Example 6.48.** (Hawaiian earring, continuing Example 6.31) Notice that we can write  $X$  as the intersection of the subsets  $X_n \subseteq \mathbb{R}^2$  for  $n \in \mathbb{Z}_{\geq 1}$  given by  $X_n = X \cup B((2^{-n-1}, 0), 2^{-n-1})$ , where  $B(p, r)$  denotes the closed ball centered at  $p$  with radius  $r$ . All of these spaces are locally contractible, so by Theorem 5.42 we can use singular homology to see that their first sheaf homology and the homomorphisms induced by the inclusions form a diagram

$$0 \leftarrow \mathbb{Z} \leftarrow \mathbb{Z} \oplus \mathbb{Z} \leftarrow \mathbb{Z} \oplus \mathbb{Z} \oplus \mathbb{Z} \leftarrow \dots$$

where each morphism drops the last coordinate, seen as a diagram in  $\vec{\mathbf{Ab}}$ . This has limit  $\prod_{i \geq 0}^{\text{Top}} \mathbb{Z}$  in  $\mathbf{TopAb}$  and using Corollary 4.18 we see that it represents the limit in  $\vec{\mathbf{Ab}}$ . Furthermore, if we consider the inclusion  $f : S^1 \rightarrow X$  of the  $i$ -th circle, then the composition  $S^1 \rightarrow X \hookrightarrow X_n$  (for  $n \geq i$ ) is given on homology by the homomorphism  $\mathbb{Z} \rightarrow \mathbb{Z}^n$  which maps 1 to the  $i$ -th standard basis vector. By the universal property of the limit it follows that  $f : S^1 \rightarrow X$  is given on homology by the continuous homomorphism  $\mathbb{Z} \rightarrow \prod_{i \geq 0}^{\text{Top}} \mathbb{Z}$  mapping 1 to  $e_i$ , just as in Example 6.37. Example 6.35 can similarly be calculated much faster with the continuity theorem. Note though that this calculation relies more heavily on the theory of singular homology while the method using Mayer–Vietoris does not.

**Example 6.49.** Note that many bounded fractals in  $\mathbb{R}^n$  can, similarly to the above example, be naturally expressed as a countable intersection of locally contractible compact subsets. Examples are the Sierpinski Triangle, the Sierpinski Pyramid and the Menger Sponge. Unfortunately in all of these examples the diagram of first homology becomes  $\mathbb{Z}^{n_1} \leftarrow \mathbb{Z}^{n_2} \leftarrow \mathbb{Z}^{n_3} \leftarrow \dots$  for some strictly increasing sequence  $(n_i)_{i \geq 0}$  in  $\mathbb{N}$ , which always has limit represented by  $\prod_{i \geq 0}^{\text{Top}} \mathbb{Z}$ , and the zero sequence on higher homology. Thus the isomorphism class of sheaf homology is not capable of distinguishing these spaces.

**Example 6.50.** (Warsaw circle, continuing Example 6.23, see also [7, Ex. 10.9]) The Warsaw circle  $X$  can be written as the intersection of the closed sets  $X_n = X \cup [0, 2^{-n}] \times [-1, 1]$  for  $n \in \mathbb{Z}_{\geq 0}$ . Each of these  $X_n$  are locally contractible and homotopic to a circle with the inclusions being a homotopy equivalence, so for each  $p$  the diagram of  $p$ -th sheaf homology is a sequence of identity maps. Using Corollary 6.47 it follows that  $X$  has the sheaf homology of a circle. On the other hand  $X$  has the singular homology of a point.

**Remark 6.51.** Note that compactness is a necessary assumption for Corollary 6.47. To see this consider the space  $X = \mathbb{Z}_{\geq 1} \subseteq \mathbb{R}$  written as an intersection  $X = \bigcap_{n \geq 1} X_n$  where  $X_n = X \cup [n, \infty)$  and the space  $Y = \{0\} \cup \{1/n : n \in \mathbb{Z}_{\geq 1}\}$  written as an intersection  $Y = \bigcap_{n \geq 1} Y_n$  where  $Y_n = Y \cup [0, 1/n]$ . Both families of subsets give on 0th homology the diagram  $\mathbb{Z} \leftarrow \mathbb{Z}^2 \leftarrow \mathbb{Z}^3 \leftarrow \dots$  (where each morphism adds the last two coordinates and leaves the rest unchanged). However we know from Example 6.9 that  $Y$  has 0-th homology represented by  $\prod_{n \geq 0}^{\text{Top}} \mathbb{Z}$  and  $X$  has 0th homology represented by  $\bigoplus_{n \geq 0} \mathbb{Z}$ .

**Example 6.52.** (2-adic solenoid) Let  $X$  be the limit in  $\mathbf{Top}$  of the diagram  $S^1 \leftarrow S^1 \leftarrow S^1 \leftarrow \dots$  where each map is the 2-fold covering. By the continuity theorem  $X$  has  $\vec{H}_0(X) \simeq \vec{\mathbb{Z}}$ ,  $\vec{H}_p(X) = 0$  for  $p > 1$  and  $\vec{H}_1$  given by the limit of

$$\mathbb{Z} \xleftarrow{\cdot 2} \mathbb{Z} \xleftarrow{\cdot 2} \mathbb{Z} \xleftarrow{\cdot 2} \mathbb{Z} \xleftarrow{\cdot 2} \mathbb{Z} \xleftarrow{\cdot 2} \mathbb{Z} \xleftarrow{\cdot 2} \mathbb{Z} \leftarrow \dots$$



in  $\mathbf{Ab}$ , which we will denote by  $K_2$ . Notice that this diagram does not satisfy the Mittag-Leffler condition, so by Remark 3.41  $K_2$  is not strict and hence not topologically representable. Notice that if we tensor this diagram with  $\mathbb{Q}$ , then we simply get a diagram consisting of copies of  $\mathbb{Q}$  and isomorphisms, so the limit in  $\mathbf{Ab}$  is  $\hat{\mathbb{Q}}$ . Using Corollary 3.50 and Lemma 3.51 it follows that  $\tilde{H}_p(X; \mathbb{Q}) \simeq \hat{\mathbb{Q}}$ , so this example fits with Theorem 5.40 (note that  $X$  is metrizable as it can be embedded in  $\mathbb{R}^3$ , see Example 6.59).

While  $K_2$  is not topologically representable, we can actually give a good characterisation of it in terms of topological groups. If  $T \in \mathbf{Ab}$ , then  $K_2(T)$  is the colimit of the diagram  $T \xrightarrow{\cdot 2} T \xrightarrow{\cdot 2} T \xrightarrow{\cdot 2} \dots$ , so  $K_2(T)$  consists of classes of pairs  $(k, t)$  with  $k \in \mathbb{Z}_{\geq 0}$  and  $t \in T$ , where  $(k, t)$  and  $(k', t')$  are considered equivalent if there exists a  $k'' \geq k, k'$  so that  $2^{k''-k}t = 2^{k''-k'}t'$ . Now let  $\eta : K_2 \rightarrow \hat{\mathbb{Z}}$  be the cone-morphism to the very first  $\mathbb{Z}$  in the diagram, then for each  $T \in \mathbf{Ab}$  the map  $\eta_T : T \rightarrow K_2(T)$  (here we identify  $\hat{\mathbb{Z}}$  with the identity functor) is given by  $t \mapsto [(0, t)]$ . Notice that the kernel of  $\eta_T$  consists of all  $2^\infty$ -torsion elements of  $T$  and we know from Example 4.24 that this functor is represented by  $\hat{\mathbb{Z}}_2$ . Furthermore notice that if  $T$  is injective, then  $\eta_T$  is surjective. Indeed if  $[(k, t)] \in K_2(T)$  then because  $T$  is divisible we can choose a  $t' \in T$  so that  $2^k t' = t$  and then  $[(0, t')] = [(k, t)]$ . We conclude using Proposition 3.6 that we have the following short exact sequence in  $\mathbf{Ab}$ :

$$0 \longrightarrow K_2 \xrightarrow{\eta} \hat{\mathbb{Z}} \longrightarrow \hat{\mathbb{Z}}_2 \longrightarrow 0. \quad (6.6)$$

**Example 6.53.** (Closed long ray, continuing Example 6.26) For each ordinal  $\alpha$  let  $L_\alpha = \alpha \times [0, 1)$  (here  $[0, 1) \subseteq \mathbb{R}$ ) with the lexicographical order and let  $L_\alpha^* = L_\alpha \cup \{e\}$  where we let  $e \notin L_\alpha$  be an upper bound for  $L_\alpha$ . Again we give each of these linearly ordered sets the order topology. Using a simple ordinal induction argument we can see that for all *countable* ordinals  $\alpha \neq 0$  the ordered set  $L_\alpha$  is order isomorphic to  $[0, 1) \subseteq \mathbb{R}$ . For ordinals  $\alpha < \beta$  we have a continuous map  $f_{\alpha\beta} : L_\beta^* \rightarrow L_\alpha^*$  which maps  $e$  and all pairs  $(\gamma, r)$  with  $\alpha \leq \gamma < \beta$  to  $e$  and the remaining elements to themselves. Now let  $X = L_{\omega_1}^*$  where  $\omega_1$  is the first uncountable ordinal, then the maps  $f_{\omega_1\alpha}$  give a cone over the diagram of  $L_\alpha^*$  for countable  $\alpha < \omega_1$ . Clearly  $X$  together with these cone morphisms is the limit of the  $X_\alpha$  on the level of sets. To see that it is the limit as a topological space we need to show that each open ray in  $X$  is a union of sets of the form  $f_{\omega_1\alpha}^{-1}U$  for  $\alpha < \omega_1$  and  $U \subseteq L_\alpha^*$  open. Indeed for  $a = (\alpha, r) \in L_{\omega_1}$  we have  $\{x \in X : x < a\} = f_{\omega_1(\alpha+1)}^{-1}\{x \in L_{\alpha+1}^* : x < a\}$  and  $\{x \in X : x > a\} = f_{\omega_1(\alpha+1)}^{-1}\{x \in L_{\alpha+1}^* : x > a\}$  and finally

$$\{x \in X : x < e\} = \bigcup_{\alpha < \omega_1} f_{\omega_1\alpha}^{-1}\{x \in L_\alpha^* : x < e\}.$$

Since each  $L_\alpha^*$  for  $\alpha < \omega_1$  is homeomorphic to the unit interval we conclude using the continuity theorem that  $X$  has the sheaf (co)homology of the unit interval, i.e. a point. Contrastingly the 0th singular homology of  $X$  is  $\mathbb{Z} \oplus \mathbb{Z}$  since there is no path in  $X$  connecting  $(0, 0)$  and  $e$  (see [28, p. 71]), but  $L_{\omega_1}$  is path connected.

## 6.7 Relative homology

As in [7, II.12] relative sheaf cohomology of a space  $X$ , an arbitrary subspace  $A \subseteq X$  and a sheaf  $\mathcal{F}$  on  $X$  can be defined. By precomposing with a left exact  $F : \mathbf{Ab} \rightarrow \mathbf{Sh}(X)$  and taking left-exactification one immediately obtains a notion of relative sheaf homology in the same generality. Here we present the equivalent construction in the case that  $A$  is a *taut* subspace. This is sufficient for our applications and it is interesting to see that the construction of the long exact sequence of relative cohomology in this case can (just as with open Mayer–Vietoris) be viewed as arising from a certain short exact sequence in  $[\mathbf{Sh}(X), \mathbf{Ab}]_{\text{LE}}$ .

**Definition 6.54.** Let  $X$  be a topological space and  $A \subseteq X$  a subspace with inclusion  $i : A \rightarrow X$ . We say that  $A$  is a *taut* subspace of  $X$  if for any flasque sheaf  $\mathcal{F}$  on  $X$  it holds that (1) the canonical map

$\Gamma(X, \mathcal{F}) \rightarrow \Gamma(A, i^{-1}\mathcal{F})$  is surjective and (2)  $i^{-1}\mathcal{F}$  is acyclic. Sufficient conditions for  $A$  being taut include

1.  $A$  is open,
2.  $A$  compact and relatively Hausdorff in  $X$  (e.g.  $A$  is a point),
3.  $X$  is Hausdorff paracompact and  $A$  is closed,
4.  $X$  is Hausdorff hereditarily paracompact (e.g. metric) and  $A$  is arbitrary

(see [7, Def. II.10.5]). If  $A$  is taut we write  $\Gamma_{X,A} : \text{Sh}(X) \rightarrow \mathbf{Ab}$  for the functor given in each  $\mathcal{F}$  by the kernel of the canonical map  $\Gamma(X, \mathcal{F}) \rightarrow \Gamma(A, i^{-1}\mathcal{F})$ . For  $p \in \mathbb{Z}$  we define the *relative sheaf cohomology* functor  $H^p(X, A, -) : \text{Sh}(X) \rightarrow \mathbf{Ab}$  to be the  $p$ -th right derived functor of  $\Gamma_{X,A}$ . If  $A$  is open this is also known as the *local cohomology* of  $X$  and  $\mathcal{F}$  with support in  $X \setminus A$  (see [14, II.9]). For  $F \in \mathbf{Ab}$  we define the *relative sheaf homology* with coefficients in  $F$ , written,  $\vec{H}_p(X, A; F)$ , as the left-exactification of  $H^p(X, A, (-)_X) \circ F$ . More generally for  $F : \mathbf{Ab} \rightarrow \text{Sh}(X)$  additive and left exact we define  $\vec{H}_p(X, A; F)$ , the relative sheaf homology of  $X$  with coefficients in  $F$ , to be left-exactification of  $H^p(X, A, -) \circ F$ .

**Proposition 6.55.** *Let  $X$  be a topological space with  $A \subseteq X$  a taut subspace with inclusion  $i$ . For any sheaf  $\mathcal{F}$  there is a long exact sequence*

$$\cdots \rightarrow H^p(X, A, \mathcal{F}) \rightarrow H^p(X, \mathcal{F}) \rightarrow H^p(A, i^{-1}\mathcal{F}) \rightarrow H^{p+1}(X, A, \mathcal{F}) \rightarrow \cdots$$

*natural in  $\mathcal{F}$ . Consequently for any  $T \in \mathbf{Ab}$  there is a long exact sequence*

$$\cdots \rightarrow H^p(X, A, T_X) \rightarrow H^p(X, T_X) \xrightarrow{i^*} H^p(A, T_A) \rightarrow H^{p+1}(X, A, T_X) \rightarrow \cdots$$

**Proof.** Write  $\Gamma_X, \Gamma_A \in [\text{Sh}(X), \mathbf{Ab}]_{\text{LE}}$  for the functors which map a sheaf  $\mathcal{F} \in \text{Sh}(X)$  to  $\Gamma(X, \mathcal{F})$  and  $\Gamma(A, i^{-1}\mathcal{F})$  respectively. By Lemma 6.2 we have a morphism  $\Gamma_X \rightarrow \Gamma_A$  and by definition  $\Gamma_{X,A}$  is the kernel of this morphism. If  $\mathcal{F}$  is injective, then  $\mathcal{F}$  is flasque (Lemma 6.38), so by requirement (1) of the tautness of  $A$  the map  $\Gamma_X(\mathcal{F}) \rightarrow \Gamma_A(\mathcal{F})$  surjective. It follows that we have an exact sequence  $0 \rightarrow \Gamma_{X,A} \rightarrow \Gamma_X \rightarrow \Gamma_A \rightarrow 0$  in  $[\text{Sh}(X), \mathbf{Ab}]_{\text{LE}}$ , so by Lemma 2.31 we get a long exact sequence

$$0 \rightarrow \Gamma_{X,A} \rightarrow \Gamma_X \rightarrow \Gamma_A \rightarrow R^1\Gamma_{X,A} \rightarrow R^1\Gamma_X \rightarrow R^1\Gamma_A \rightarrow R^2\Gamma_{X,A} \rightarrow \cdots \quad (6.7)$$

in  $[\text{Sh}(X), \mathbf{Ab}]_{\text{add}}$ . By property (2) of the tautness of  $A$  the functor  $i^{-1}$  maps flasque sheaves to acyclic sheaves (and hence injectives to acyclics), so the derived functors of  $\Gamma_A$  are the composition of the derived functors of  $\Gamma(A, -) : \text{Sh}(A) \rightarrow \mathbf{Ab}$  with  $i^{-1}$ . Plugging this isomorphism into (6.7) gives the first of the desired long exact sequences. Finally taking  $\mathcal{F} = T_X$  and using the natural isomorphism  $i^{-1}T_X \simeq T_A$  gives the second.  $\square$

**Corollary 6.56.** *Let  $X$  be a topological space with  $A \subseteq X$  a taut subspace with inclusion  $i$ . For any functor  $F \in [\mathbf{Ab}, \text{Sh}(X)]_{\text{LE}}$  there is a long exact sequence*

$$\cdots \leftarrow \vec{H}_p(X, A; F) \leftarrow \vec{H}_p(X; F) \leftarrow \vec{H}_p(A; i^{-1} \circ F) \leftarrow \vec{H}_{p+1}(X, A; F) \leftarrow \cdots$$

*in  $\mathbf{Ab}$ . Also for any  $F \in \mathbf{Ab}$  there is a long exact sequence*

$$\cdots \leftarrow \vec{H}_p(X, A; F) \leftarrow \vec{H}_p(X; F) \xleftarrow{i_*} \vec{H}_p(A; F) \leftarrow \vec{H}_{p+1}(X, A; F) \leftarrow \cdots$$

**Remark 6.57.** Note that there are many questions left to answer regarding relative sheaf (co)homology such as functoriality in the space, are the morphisms in the long exact sequence the same as from the functoriality and excision. For this we invite the reader to read [7, II.12] and take left-exactification to obtain the corresponding facts about sheaf homology. For the upcoming example we will only need the following very strong version of the excision theorem.

**Lemma 6.58.** *Let  $(X, A), (Y, B)$  be pairs of topological spaces with  $X$  and  $Y$  paracompact Hausdorff and  $A, B$  closed in  $X$  and  $Y$  respectively. If  $f : (X, A) \rightarrow (Y, B)$  is a map of pairs so that  $f$  restricts*

to a homeomorphism between  $X \setminus A$  and  $Y \setminus B$ , then for all  $p \in \mathbb{Z}$  and  $\mathcal{F} \in \text{Sh}(Y)$  the induced map  $f^* : H^p(Y, B, \mathcal{F}) \rightarrow H^p(X, A, f^{-1}\mathcal{F})$  is an isomorphism. Consequently the induced morphisms on constant coefficient sheaf (co)homology is an isomorphism.

**Proof.** See [7, Corollary II.12.5]. □

**Example 6.59.** In this example we will construct a compact Hausdorff space whose first sheaf homology (and by Theorem 5.35 also shape homology) is represented by the topological group  $\mathbb{Z}_2$  of 2-adic integers. Most importantly this is an example where it is easier to work with sheaf homology directly instead of with sheaf cohomology.

We first realise  $\mathbb{Z}_2$  as the relative homology of a pair of spaces. We do this using the construction of the 2-adic solenoid as a subspace of the solid torus. More precisely we use the Smale-Williams attractor (see [27] and [31]). View  $S^1$  and  $D^2$  as the unit circle and closed unit disk in  $\mathbb{C}$  respectively, then we view the solid torus  $X$  as the product  $S^1 \times D^2$ . Define the continuous map  $f : X \rightarrow X$  by  $(z, w) \mapsto (z^2, \frac{1}{4}w + \frac{1}{2}z)$ . This is a topological embedding and under the projection  $\pi : X \rightarrow S^1$  (which is a homotopy equivalence) it is given by the 2-fold covering  $z \mapsto z^2$  and hence  $f$  is given on first (sheaf) homology by multiplication by 2. If we fix an isomorphism  $H_1(S^1) \simeq \mathbb{Z}$ , then  $\pi$  induces an isomorphism  $H_1(X) \simeq \mathbb{Z}$  (here  $H_1$  denotes the singular homology, but it could just as well denote a representative of  $\vec{H}_1$  by Theorem 5.42). If we factor  $f$  as  $f = i \circ g$  where  $g : X \rightarrow f[X]$  is the corestriction and  $i : f[X] \rightarrow X$  the inclusion, then  $g$  is a homeomorphism. If we then identify  $\mathbb{Z} \simeq H_1(X)$  with  $H_1(f[X])$  through  $g_*$  we see that  $i_* : H_1(f[X]) \rightarrow H_1(X)$  is given by multiplication by  $\mathbb{Z} \xrightarrow{\cdot 2} \mathbb{Z}$ . Set  $A_0 = X$  and  $A_1 = f[X]$ , then under the homeomorphism  $g : A_0 \xrightarrow{\sim} A_1$  we can apply the above construction again to  $A_1$  to obtain a closed subspace  $A_2 \subseteq A_1$ , a homeomorphism  $A_1 \simeq A_2$  under which the inclusion  $A_2 \rightarrow A_1$  induces the map  $\mathbb{Z} \rightarrow \mathbb{Z}$  on first homology given by multiplication by 2. Continuing inductively we obtain a sequence of closed subspaces  $X = A_0 \supseteq A_1 \supseteq \dots$  together with isomorphisms  $H_1(A_i) \simeq \mathbb{Z}$  so that the inclusion  $A_{i+1} \rightarrow A_i$  induces the map  $\mathbb{Z} \rightarrow \mathbb{Z}$  given by multiplication by 2. We can then set  $A = \bigcap_{n=0}^{\infty} A_n$ . Notice that in fact  $A_n = f^n[X]$  for all  $n \geq 0$ , so if we think of  $(X, f)$  as a dynamical system, then  $A$  is an attractor (hence the name Smale-Williams attractor). These subsets are depicted in Figure 7.

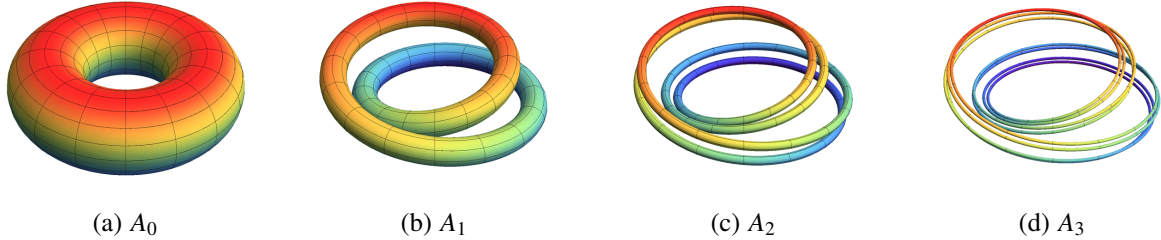


Figure 7: First few stages of the construction of the 2-adic solenoid as a subspace of the solid torus. Made by Wikimedia Commons user Jim.belk, see <https://commons.wikimedia.org/wiki/File:Solenoid.gif>.

By Corollary 6.47 we have  $\vec{H}_0(A) = \vec{\mathbb{Z}}$  and that  $\vec{H}_1(A)$  is the limit in  $\mathbf{Ab}$  of the diagram  $\mathbb{Z} \xleftarrow{\cdot 2} \mathbb{Z} \xleftarrow{\cdot 2} \mathbb{Z} \xleftarrow{\cdot 2} \dots$  with in particular the inclusion  $i : A \rightarrow A_0 = X$  given by the cone morphism to the first copy of  $\mathbb{Z}$ . We conclude that  $\vec{H}_1(A) = K_2$  and  $i_* = \eta$  as in the short exact sequence (6.6) from Example 6.52. Now observe the long exact sequence of relative homology for  $X$  and  $A$ :

$$0 \leftarrow \vec{H}_0(X, A) \leftarrow \vec{\mathbb{Z}} \leftarrow \vec{\mathbb{Z}} \leftarrow \vec{H}_1(X, A) \leftarrow \vec{\mathbb{Z}} \xleftarrow{i_*} K_2 \leftarrow \vec{H}_2(X, A) \leftarrow 0$$

Since the map  $\vec{\mathbb{Z}} \rightarrow \vec{\mathbb{Z}}$  is just the identity and because  $i_* = \eta$  is monic we see that  $\vec{H}_0(X, A) = \vec{H}_2(X, A) = 0$  and  $\vec{H}_1(X, A) = \text{coker } \eta$ . By the short exact sequence (6.6) we conclude that  $\vec{H}_1(X, A) \simeq \vec{\mathbb{Z}}_2$ .

Now consider the quotient space  $X'$  of  $X$  where we glue the entirety of  $A$  to a point and write  $A'$  for the singleton set consisting of this point. Since  $X$  is compact Hausdorff and hence normal, any  $x \in X \setminus A$

can be separated from  $A$  using a neighbourhood of  $x$  and of  $A$  and thus  $X'$  is again Hausdorff. Also since  $X$  is compact  $X'$  is also compact. Finally note that the quotient map  $q : X \rightarrow X'$  restricts to a homeomorphism between  $X \setminus A$  and  $X' \setminus A'$ , so by Lemma 6.58 we have  $\vec{H}_p(X, A) \simeq \vec{H}_p(X', A')$ . Since  $A'$  is just a point and  $X'$  is connected we see that the long exact sequence of relative homology of  $X'$  and  $A'$  takes the form

$$0 \leftarrow 0 \leftarrow \vec{\mathbb{Z}} \xleftarrow{\text{id}} \vec{\mathbb{Z}} \leftarrow \vec{\mathbb{Z}}_2 \leftarrow \vec{H}_1(X') \leftarrow 0$$

and  $0 \leftarrow \vec{H}_p(X') \leftarrow 0$  for  $p > 1$ . Hence  $\vec{H}_0(X') = \vec{\mathbb{Z}}$ ,  $\vec{H}_1(X') \simeq \vec{\mathbb{Z}}_2$  and  $\vec{H}_p(X') = 0$  for  $p > 1$ .

Additionally  $X'$  gives us a non-trivial example of the universal coefficient theorem (Theorem 2.17). Namely Theorem 2.17 tells us (note that  $R^1\vec{\mathbb{Z}} = 0$ ) that  $H^0(X', T_{X'}) \simeq T$ ,  $H^1(X', T_{X'}) \simeq \vec{\mathbb{Z}}_2(T) = \text{Hom}_{\mathbf{TopAb}}(\mathbb{Z}_2, T)$  and  $H^2(X', T_{X'}) \simeq R^1\vec{\mathbb{Z}}_2(T)$ . To compute  $R^1\vec{\mathbb{Z}}_2$  we can apply Lemma 2.31 to the short exact sequence (6.6) to obtain, for all  $T \in \mathbf{Ab}$ , a long exact sequence

$$0 \rightarrow \vec{\mathbb{Z}}_2(T) \rightarrow \vec{\mathbb{Z}}(T) \rightarrow K_2(T) \rightarrow R^1\vec{\mathbb{Z}}_2(T) \rightarrow R^1\vec{\mathbb{Z}}(T) \rightarrow R^1K_2(T) \rightarrow 0.$$

Since  $R^1\vec{\mathbb{Z}} = 0$  we conclude that  $R^1\vec{\mathbb{Z}}_2(T)$  is the cokernel of the map  $\vec{\mathbb{Z}}(T) \rightarrow K_2(T)$ . Using the description of  $K_2$  and  $\eta$  from Example 6.52 we see that  $R^1\vec{\mathbb{Z}}_2$  maps a  $T \in \mathbf{Ab}$  to the set of pairs  $(k, t)$  for  $k \in \mathbb{Z}_{\geq 0}$  and  $t \in T$  where we identify pairs  $(k, t), (k', t')$  if there exists a  $k'' \geq k, k'$  and an  $r \in T$  so that  $2^{k''-k}t = 2^{k''-k'}t' + 2^{k''}r$ . If one would have calculated the sheaf cohomology of  $X'$  directly using cohomology, then this calculation would be much more complicated since this rather complicated functor has to appear in degree 2. Hence in this case it is easier to work with sheaf homology than to work with sheaf cohomology directly. In particular sheaf homology allows us to summarise the entire, rather complicated, sheaf cohomology functors of  $X'$  using the concrete mathematical objects  $\mathbb{Z}$  and  $\mathbb{Z}_2$ .

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